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FITTED EMPIRICAL REFERENCE CROSS SECTIONS FOR *K*-SHELL IONIZATION BY ALPHA PARTICLES*

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On the basis of our collection of experimental x-ray and Auger production cross-section data for H and He ions, we present a table of best-fitted cross sections for *K*-shell vacancy production (direct ionization plus electron capture) by ^{4}He ions on all elements from $_6$C to $_{92}$U. Experimental values are first converted (if necessary) to vacancy production cross sections using fluorescence yields with an approximate correction for the effect of multiple ionization. These values are then normalized, i.e., divided, by an improved version (due to Benka et al.) of the ECPSSR theory by Brandt and Lapicki, to which a correction for electron capture by the projectile (following Lapicki and McDaniel) has been added. Since we have found empirically that the normalized cross sections (which describe the deviation of theory from experiment), at a certain scaled projectile speed, depend only on the ratio of projectile and target atomic numbers, we can use the data for both He and H ions as input. The normalized values are averaged, fitted by a polynomial, and multiplied by theory to produce best reference cross sections. Discrepant data sets are rejected using a statistical criterion which compares the deviations found to the errors stated in the original publications. The error assigned to the cross section consists of a calculated random contribution and an estimated systematic contribution that describes the limitations of the method. A list of the experimental input data (for ^{1}H, ^{2}H, ^{3}He, and ^{4}He projectiles) and a table showing the consistency or inconsistency of these data are also given. © 1993 Academic Press, Inc.

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INTRODUCTION

Some time ago, we reviewed[1] the experimental cross sections for K-shell ionization by light ions that are available in the literature. Using this wealth of information, we then produced[2] a table of fitted empirical reference cross sections for protons on all targets from beryllium to uranium. This table has since been used for practical applications[3,4] and to test theories.[5,6] Our results have been reviewed by Cohen[7] and compared to other fits by Rosato.[8] Also, Lapicki[9] has produced a careful review of cross sections for K x-ray production by hydrogen and helium ions; his review includes tables of the data and identifies discrepant data. In this paper, we present a table of fitted empirical cross sections for K-shell ionization of all elements from $_6$C to $_{92}$U by ^{4}He ions, based on our data file[1] (with recent additions); we think that these values are the best presently available.

In our previous papers,[1,2] we introduced normalized cross sections

$$s = \sigma_I/\sigma_{th} \qquad (1)$$

for every projectile, where σ_I is the experimental and σ_{th} the theoretical ionization cross section. For theory, we used the convenient and well-tested ECPSSR theory for direct K-shell ionization of Brandt and Lapicki.[10] The normalized cross sections were considered as a function of scaled projectile speed ξ (defined below), and weighted averages were then calculated for various regions of target atomic number Z_2. The curves thus obtained showed certain discrepancies between the theory and the data; in particular, the theoretical cross section appears too large for heavy targets at small projectile speeds. The same result was obtained by Lapicki[9] using unweighted averages (see the figures in his paper).

In the present paper, we take advantage of our empirical finding[11] that, for a certain scaled projectile speed ξ, the quantity s that describes the deviation of theory from experiment turns out to be a function only of $z = Z_2/Z_1$, if data for different projectiles with different atomic numbers Z_1 and from different targets are considered. We can therefore use the entire data set from measurements with protons and deuterons (for $Z_2 \leq 50$) and with ^{3}He and ^{4}He ions as input. Also, we now define s in a slightly different way: we keep Eq. (1), but we use our MECPSSR theory[11] for σ_{th}.

Like the ECPSSR theory,[10] our theory is based on the plane-wave Born approximation with hydrogenic tar-

get wave functions and is corrected for energy loss of the projectile during the collision (E), for deceleration and deflection of the projectile in the Coulomb field (C), for perturbation of the stationary target electron-state energies and for polarization of the wave function by the passing projectile (PSS), and for relativistic electron motion (R). In our modified theory (M) we take into account, in addition, the increased target electron speed due to the passing projectile, and—what is more important for small Z_1—we reduce the relativistic correction for slow projectiles since these projectiles will not come close to the target nucleus due to Coulomb deflection. As an example, Table A compares the cross sections for 4-MeV ^{4}He ions on Pb. The reduction of the cross section compared to ECPSSR reduces the discrepancy with theory found for heavy targets[1,11] since s is now closer to unity. Like Lapicki,[9] we add a correction for electron capture to σ_{th} (see below).

As in our earlier papers,[1,2] we consider the normalized cross sections s as functions of the scaled projectile speed[10] $\xi = (2/\theta)(v_1/v_{2K})$, where $\theta = I_K/(Z_2 - 0.3)^2R$ is the outer-screening parameter, v_1 is the laboratory-frame speed of the projectile, $v_{2K} = (Z_2 - 0.3)(2R/m_e)^{0.5}$ is the hydrogenic Bohr velocity of the target K-shell electron, I_K is the experimental ionization energy,[15] R is the Rydberg energy, and m_e is the electron mass. The data are lumped into eight partially overlapping regions: $3 \leqslant z \leqslant 15$, $10.5 \leqslant z \leqslant 20$, $15.5 \leqslant z \leqslant 25$, etc., up to $40.5 \leqslant z \leqslant 50$. In every region, we subdivide the entire ξ range into intervals of equal $\Delta \log \xi = 0.1$ (where log means $_{10}$log), and we calculate[1] a weighted average $\bar{s} \pm \delta$ [using Eq. (1)] and corresponding weighted averages $\log \bar{\xi}$ and $\bar{z}$ for every interval. The weights are based on the experimental errors stated in the various publications. Within each publication, and within every interval $\Delta \log \xi$, we use only one data point for every value z within the z-group considered, in order to avoid bias in favor of authors who measured at many closely spaced energies. An average $\bar{s}$ is calculated only if there are data in the interval from at least two publications or, if not, from at least two targets.

Our data file, described in Ref. 1, has been augmented with newer data found in the literature until the end of 1991. Five recent papers with proton data not used for various reasons are listed in Ref. 2. The experimental data from our file are listed in Table II, except for those already given in Ref. 2. All the publications containing data are given in the References for Table II; the list contains 131 papers. Of these, 11 contain only proton data for $Z_2 > 50$ (CF84, CK79, EA87, GO83, GU89, HO81, KA77, PF84, PP90, SZ87, and ZE86) that are not used here. To check the compatibility of the data (on the basis of the authors' stated errors), we again use the "rejection ratio" r defined earlier,[1]

$$r = \frac{\chi^2/m - 1}{\sqrt{2/m}}, \tag{2}$$

whose expectation value is 0 ± 1 for compatible data and which assumes large positive values for discrepant data. Here, χ^2 is the sum of the squared deviations (divided by the experimental error) from the mean, and m is the number of data points taken from the publication in question. Table B shows the rejection ratios for all the publications used, in three nonoverlapping regions of z: $3 \leqslant z \leqslant 15$, $15.5 \leqslant z \leqslant 30$, and $30.5 \leqslant z \leqslant 50$. We reject the data from a publication whenever $r \geqslant 6$ in any one of these three regions. On this basis, the data from 28 publications (i.e., AV81, AV83, BA77, BD78, BH82A, BO75, BS78, EI90, HA73, HW73, KL76, KN77, KO68, LE53, MA79, ME58, MI76, MK74, OR89, RI81, RS77, SE69, SH71, SI57, ST73, ST75, SU84, and TA76) were rejected, starting with the publications with highest r, recalculating the rejection ratios, rejecting again, and so on, until no publications with $r \geqslant 6$ were left. A comparison with Ref. 1 shows that exclusion of proton data for $Z_2 > 50$ and inclusion of deuteron and He ion data have changed a few of the rejections. It should be noted that we always exclude entire publications, whereas Lapicki[9] determines discrepant data on a point-by-point basis (by comparison to ECPSSR).

Table B also shows the average normalized deviation from the fitted curve

$$d = \sum [(s_i - \bar{s})/\epsilon_i]/m \tag{3}$$

for every publication. Here, s_i is a normalized cross section, ϵ_i is the corresponding error derived from the error stated in the publication, and the sum runs over all the data points from the publication used, within the stated limits of $z = Z_2/Z_1$.

Figure 1 shows the averages $\bar{s}$ obtained for the eight data groups, versus $\log \bar{\xi}$. To produce a value σ_{ref} for a certain laboratory-frame energy E' and a certain target atomic number Z'_2, we calculate the corresponding scaled

TABLE A

Comparison of ECPSSR and MECPSSR Cross Sections for 4-MeV ^{4}He Ions on Pb

σ_{ECPSSR} (Brandt and Lapicki[10])	1.451 mb
σ_{ECPSSR} (Cohen and Harrigan[12])	1.483 mb
$\sigma_{MECPSSR}$ (Benka et al.[11])	0.973 mb
σ_1 (AN78)	0.911 mb

Note. The value from the published table by Cohen and Harrigan[12] is higher than the original,[10] due to an unfortunate change[13,14] of the relativistic correction. An experimental value σ_1, taken from Table II, is also shown (experimental error, 25%).

TABLE B

Average Normalized Deviation d [Eq. (3)] of the Measured Value from the Fitted Curve and Rejection Ratio r [Eq. (2)], for Every Publication, in Three Regions of z

	$3 \leq z \leq 15$		$15.5 \leq z \leq 30$		$30.5 \leq z \leq 50$	
	d	r	d	r	d	r
AK74			0.53	−1.2	−0.69	1.4
AN78			0.15	−2.0	−0.06	−4.0
AV81*	2.57	19.9	0.59	3.2		
AV83*			−3.34	27.3		
AV84			0.94	0.3	0.81	1.3
BA77*	1.06	9.2	2.65	27.8		
BA80			0.92	−0.1		
BA82			0.36	−1.3	0.33	0.2
BB73	−0.44	−1.0				
BB78	−1.38	3.5	−1.09	0.6		
BD78*			0.68	3.4	0.77	6.8
BE73			−1.46	3.0	−0.85	0.2
BE78					−0.82	0.8
BG80	−0.37	−1.9	−0.26	−2.3	−0.67	−0.9
BH82	1.32	1.3	0.04	5.3		
BH82A*			−0.53	0.9	0.20	7.0
BI70			−0.11	−0.4		
BI72					0.76	0.3
BM78	0.16	−1.0				
BO75*			2.35	21.2		
BR78	−0.59	−0.8	−0.30	−1.3	−0.17	−1.5
BR82			0.44	−1.8		
BS78*	−1.63	2.4	0.49	2.8	−2.08	16.6
CH77	−0.49	−0.6				
CR74					−0.11	4.4
CZ75			0.06	−1.0		
DE77					0.66	1.8
DI84					0.62	−1.6
DO81A			−1.30	1.2	0.56	0.0
DO81B			1.32	2.9	0.94	1.0
DU72			−0.13	−1.9		
EI90*			−1.01	6.5	1.67	13.7
FB74	0.36	2.0	−1.04	0.6	−0.39	1.1
FR85					−0.87	−0.1
GB86	−0.09	−1.2	−0.08	−1.2		
GE90	0.69	0.2	−0.12	−0.4	0.14	−0.7
HA57			−0.47	−0.8	−0.08	−1.5
HA73*			2.38	8.5		
HO75A			0.26	−1.9	0.33	−1.0
HO75B			0.36	−0.9		
HW73*	3.03	20.4	−0.56	2.1	−0.44	11.4
IS74					0.56	−0.8
IS82	1.13	1.0				
JA72			−0.00	−1.0	0.14	−1.1
JL80			0.18	2.1	−0.88	0.2
JL82					−1.14	0.6
JO62			−0.47	−1.0	−0.28	−1.0
KB76	−0.31	3.9				
KB79	−0.34	−0.3				
KH64	−1.59	4.4	−0.63	−0.7		
KH65	−0.45	−2.1				
KH75					−1.15	1.3
KL76*	5.55	66.2	−0.20	0.4	4.39	24.0
KN77*			−2.52	13.6		
KO68*	−11.00	393.9	14.76	636.5		
LA62			−0.65	1.2	−0.90	−0.1
LA76	−1.22	0.8	−0.24	−1.3		

TABLE B—*Continued*

	$3 \leq z \leq 15$		$15.5 \leq z \leq 30$		$30.5 \leq z \leq 50$	
	d	*r*	*d*	*r*	*d*	*r*
LA79			0.29	−1.0	0.43	−0.7
LD73	0.74	−1.3	0.43	−0.7	0.05	−2.0
LE53*					5.96	48.5
LE73			−0.34	4.0	−0.75	1.3
LI73			0.20	−2.5	−0.82	2.0
LJ82	0.05	−2.3	−0.29	−2.6		
LL80	0.03	−0.7	−0.15	−0.9	0.80	−0.3
LN75					−0.59	−0.7
LO78			−0.45	−1.6		
LO79A			−0.87	1.6		
LO79B			−0.03	−1.2	−0.32	−1.8
LO80			−0.81	0.0	−1.29	2.2
MA79*			−3.58	19.4	−2.33	24.1
MD75A	0.26	0.8	−0.95	1.4		
ME58*			−1.73	7.0	6.00	184.3
ME87	0.65	3.0				
MI76*			−1.78	11.6		
MK74*	−0.30	2.3	−1.60	13.3		
MK76	−0.06	2.2				
OG64	0.13	−0.7	−0.13	−1.2		
OR89*	−0.05	10.7	0.01	0.3	−0.09	2.4
PA90	0.29	3.4	−0.26	−1.4		
PE79	0.64	0.2	0.79	−0.2	0.84	0.9
PO77	−0.96	−0.1				
RI81*	−1.28	1.1	0.32	5.3	1.59	7.7
RN76			−1.45	5.4		
RS76			0.63	1.7	−0.40	0.4
RS77*			−0.97	10.5	−2.33	15.9
SE69*	0.95	24.8				
SE80	−1.05	0.2	−0.36	−1.7		
SH71*	−4.30	25.3	4.14	23.3		
SI57*	−3.30	9.9	4.62	46.5	−3.13	9.1
SI84	0.13	−2.1				
SI89	0.16	−3.2	0.02	−1.1		
SI91					0.90	0.0
SN81	−0.33	−1.3	−0.68	−0.6	−0.23	−1.4
SO76	−0.40	−0.4	−0.99	2.5		
ST72	0.71	3.5				
ST73*	−0.19	8.1				
ST75*	−1.46	7.8				
SU84*					3.68	15.5
TA74			1.44	4.0		
TA76*	−1.54	7.6				
TO72B	0.24	−2.3				
TO77	0.12	−1.0				
TO83	0.15	−0.7				
WA73	−0.92	0.7				
WL76			0.35	1.1	−0.27	5.1
WN73A			−0.39	−1.2		
WO74	1.12	0.7				
WO76	−0.24	−1.5				
YU91A	−1.33	8.0				
YU91B	0.42	−0.5				
ZA79			0.05	0.5		
ZE84					1.29	2.7

Note. The sign of *d* shows if the data are predominantly too large (+) or too small (−). A negative or small positive *r* means good consistency and a large positive *r* large discrepancy. If *r* is large and positive and *d* is small in value, this means large positive *and* negative deviations. Rejected publications are indicated by an asterisk.

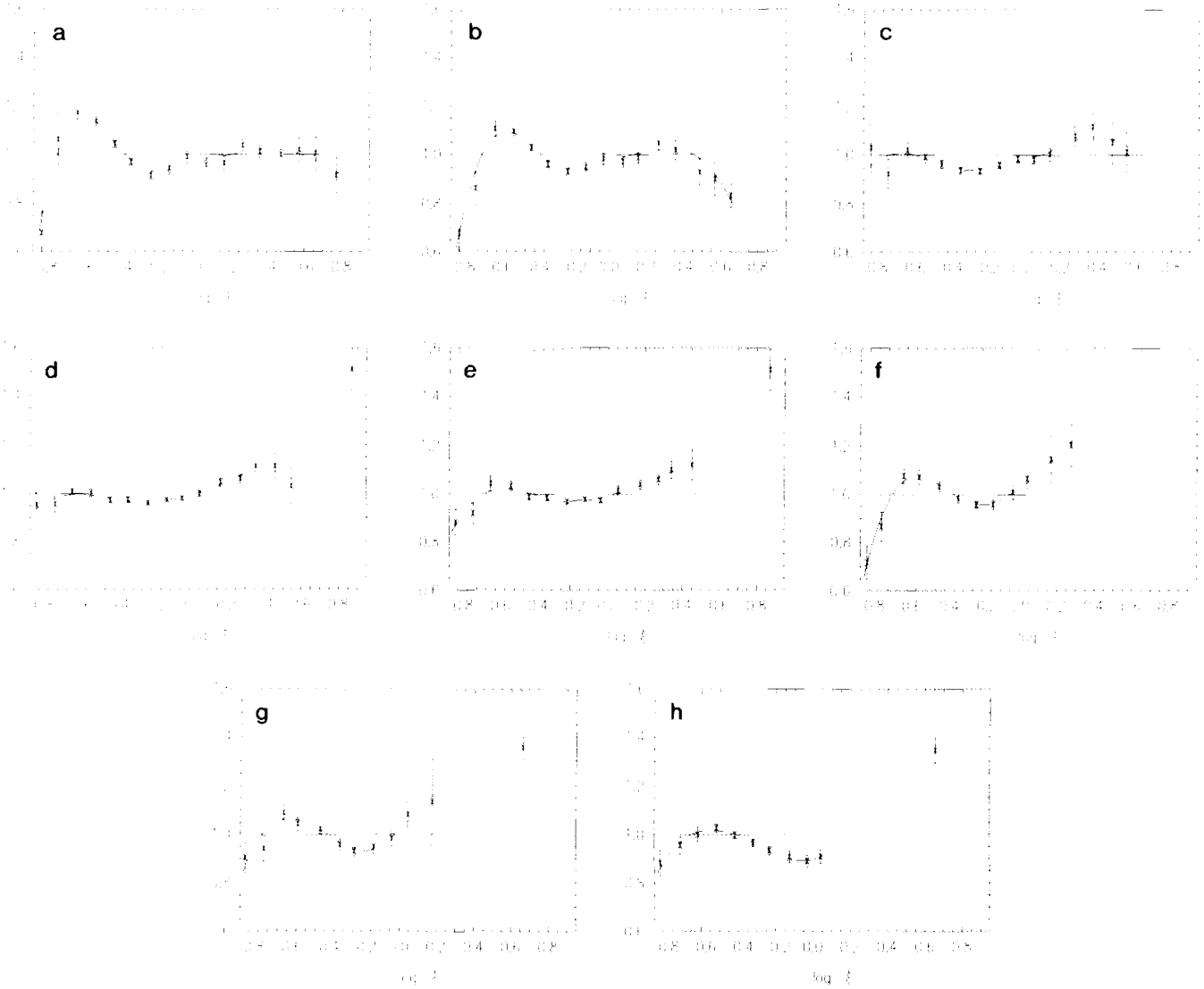


Figure 1. The average normalized cross section $\bar{s} \pm \delta$ versus the average logarithm (to the base 10) of the scaled velocity $\bar{\xi}$. Normalization was carried out using σ_{th} according to Eq. (5). The normalized data were averaged in equal intervals $\Delta \log \xi = 0.1$, separately for different regions of $z = Z_2/Z_1$: (a) z = 3–15, (b) z = 10.5–20, (c) z = 15.5–25, (d) z = 20.5–30, (e) z = 25.5–35, (f) z = 30.5–40, (g) z = 35.5–45, (h) z = 40.5–50. The curves show the results obtained by carrying out the interpolation for 100 equidistant abscissas and connecting the resulting points by straight lines. The isolated averages appearing in (d), (e), (g), and (h) for $\log \bar{\xi} > 0.6$ are all based on data by JA72 for 160-MeV protons; they are not used here.

velocity ξ'. We select that set of values $\bar{s}$ from Fig. 1 in which the center of the region of $z = Z_2/Z_1$ is closest to $z' = Z_2'/2$. We then interpolate the selected function $\bar{s}$ versus $\log \bar{\xi}$ using a polynomial of the third degree; this yields a value $s' \pm \delta$. Here the error (standard deviation) is taken from the point ($\bar{s}$, $\log \bar{\xi}$) closest to $\log \xi'$. The polynomial is fitted to 4 neighboring points if δ is small; the number of points is increased with increasing δ up to a maximum of 13. Each of the curves in Fig. 1 is the result of carrying out the interpolation for 100 equidistant points. The extreme ξ values for which the fit is carried out are chosen close to the smallest and largest ξ of available experimental data, within each data group. The reference cross section is then obtained as

$$\sigma_{ref}(E') = \sigma_{th}(E') \times (s' \pm \delta). \quad (4)$$

Influence of Electron Capture

The K-shell hole can be produced not only by direct ionization (DI), as described by the theories[10,11] mentioned, but can also be due to electron capture by the projectile (EC). The total theoretical cross section for vacancy production is therefore a sum of two terms:

$$\sigma_{th} = \sigma_{DI} + \sigma_{EC} = \sigma_{MECPSSR} + \sigma_{EC}. \quad (5)$$

Although electron capture is important only for light targets, we take its influence into account for all the targets, both in the determination of normalized cross sections s and in the calculation of reference cross sections σ_{ref}. To determine σ_{EC}, we follow Lapicki[9] and use the theory by Lapicki and McDaniel[16] which describes electron capture by fully stripped ions. Unlike Lapicki[9] we do not always assume, however, that the ions are fully stripped.

Normalized cross sections. To determine normalized cross sections from measurements on gas targets, we estimate (using the charge-changing cross sections for nitrogen gas due to Pivovar et al.[17]) that the ionic charge remains essentially unchanged while the projectile passes the target volume. We therefore use the full Lapicki–McDaniel cross section for H^+ and for He^{2+} ions (data references WA73, KB76, and KB79; see References for Table II). For He^+ ions (ST73, ST75, MK76, and ME87) we take half the EC contribution for He^{2+} ions, as a rough approximation. The largest correction factors $(\sigma_{DI} + \sigma_{EC})/\sigma_{DI}$ applicable to the theoretical values σ_{th} for our gas data then amount to 1.10 for 0.3-MeV protons on boron and to 1.71 for 0.6-MeV He^{2+} on carbon, respectively. These corrections agree with the corrections given by Lapicki[9] in his Table 7.

To evaluate measurements on solid targets, we assume that the projectile carries an equilibrium charge $Q_1 = (Z_1 - N)e$, where N is the average number of electrons bound to the ion. We estimate the fractional equilibrium charge $qe = Q_1/Z_1$ using Eqs. (3-39), (3-37), (3-32), and (3-33) from the book by Ziegler et al.,[18] which are based on the work by Brandt and Kitagawa,[19] although these equations are not meant for ions as light as H or He. We then multiply the Lapicki–McDaniel cross section by q (which does not differ much from unity in most cases). Here, the largest correction of the cross section due to EC amounts to only 7% (for 4-MeV ^{4}He ions on Mg).

Including electron capture in σ_{th} according to Eq. (5) has the effect that the rather low results by ST73 and ST75 must now be rejected, but the (higher) results due to ME87 are now acceptable. Since the correction is small for most of the cross sections, the values $\bar{s}$ are changed by a few percent at most, and only for the smallest values z (Fig. 1a).

Reference cross sections. For the calculation of reference cross sections (which include direct ionization and electron capture) we assume that all He ions are fully ionized in gas targets, and that they carry an equilibrium charge in solids. Otherwise, the corrections are calculated in the same way as described above.

Fluorescence Yields

The experimental ionization cross section σ_I is deduced from Auger cross sections σ_A by $\sigma_I = \sigma_A/(1 - \omega)$ and from x-ray cross sections by $\sigma_I = \sigma_X/\omega$. Here, ω is the single-hole K-shell fluorescence yield taken from Krause's[20] Table 3 for target atomic numbers $Z_2 \geq 16$ and from his Table 1 (for solid targets) for $11 \leq Z_2 \leq 15$. Also if the authors themselves deduced σ_I from σ_X, we recalculate the value from σ_X, where necessary, using Krause's ω. Since for light targets, chemical effects may well influence the fluorescence yield,[1,21] we exclude x-ray data for $Z_2 < 11$ from our averages. A comparison[22] of various tables of ω shows that these fluorescence yields may well be in error by 3–4% for $Z_2 < 35$ (see also Ref. 23).

These fluorescence yields ω can be used to describe K-shell vacancy production by protons, but for heavier ions, the probability of simultaneous production of L holes becomes important, and this effect increases ω beyond the single-hole value, especially for small values Z_2/Z_1. Benka[24] has used a scaling law due to Watson et al.[25] to estimate the change. For α's on Al, he predicts a 13% increase of ω at 1 MeV (where the L-shell ionization is highest). To calculate the effect for other α energies (where it is smaller), one can average the multiple-hole fluorescence yields of Hopkins et al.[26] using the partial ionization cross sections due Richard et al.[27]

It seems hardly possible, however, to do the same for all other light targets and for all projectile energies. But the effect is small and it affects only a small part of the data, so it should suffice to introduce an approximate correction: we apply 80% of Benka's correction[24] to all energies where the average L ionization probability is more than one-half of its maximum value, judging from Fig. 7 of Watson et al.,[25] and no correction to the other energies. We do this for all the targets for which the correction as given in Fig. 3 of Ref. 24 exceeds 2%, i.e., for $12 \leq Z_2 \leq 15$ for H ions, and for $12 \leq Z_2 \leq 22$ for He ions. The correction decreases rapidly with increasing Z_2; the largest correction (for ^{4}He ions on Mg) amounts to 16% from 0.5 to 2.5 MeV (and 0% outside this range). Since the correction is not available for Na, we exclude all the Na data (all of which are x-ray data contained in the recent papers YU91A and YU91B) from our averages.

Due to the averaging of many data points, the change of $\bar{s}$ due to this correction is indeed small: it affects only six values of the region $3 \leq z \leq 15$ (Fig. 1a) and the decrease is at most as large as the error δ. The data from YU91A are the only ones that would have to be rejected due to the fluorescence yield correction (since $r = +8$).

Because of the approximate nature of the correction, we do not, however, reject this publication.

Estimation of Errors

To obtain the error δ of $\bar{s}$ mentioned above (see also Fig. 1), we have chosen[1] to take the larger of the internal error (based on the measurement errors stated in the various publications) and the external error (which describes the compatibility of the numbers averaged). Also, we do not use the laws of error propagation in polynomial fitting, but we simply take the error of the closest anchor point.

Since this "random" error δ_r turns out to be unreasonably small in several cases, we add a "systematic" error δ_s. The fact that the eight curves in Fig. 1 are not identical may indicate a slightly incorrect Z_2 dependence of the theory. Since we can ascribe an average value $\bar{z}$ to every point in Fig. 1, we can reinterpret these curves as $\bar{s}$ versus $\bar{z}$ (with $\log \bar{\xi}$ as a parameter) and hence determine $\Delta\bar{s}/\Delta\bar{z}$ as a function of $\log \xi$. The result (partly found by calculation, partly a conservative estimate) is shown in Table C. We calculate $\delta_s = (z - \bar{z}) \times |\Delta\bar{s}/\Delta\bar{z}|$ and add this systematic error linearly to obtain the total error $\delta_t = \delta_r + \delta_s$. This automatically produces a large error if the atomic number of the element of interest is far from the elements actually measured, as it should (see also Ref. 2).

Discussion

It is of interest to check how well our reference cross sections agree with the experimental input data. Figure 2 shows a histogram of the relative deviations $s/\bar{s} - 1$ of the ^{4}He ion data from the fitted curve, normalized by the relative experimental error as given in the publication from which the data are taken (cf. Table II). X-ray data for $Z_2 < 12$ and rejected data were omitted. Sixty percent of all the data points lie within ± 1 and 93% within ± 2. This is close to what is expected for a Gaussian distribution of standard deviation 1 and hence quite satisfactory.

TABLE C

Estimated Value of $|\Delta\bar{s}/\Delta\bar{z}|$ as a Function of $\log \bar{\xi}$, for Two Regions of $z = Z_2/Z_1$

Range of $\log \bar{\xi}$	$3 \leq z \leq 23$	$23.5 \leq z \leq 50$
(−0.9, −0.8)	0.05	0.005
(−0.8, −0.6)	0.03	0.005
(−0.6, −0.5)	0.02	0.005
(−0.5, −0.3)	0.01	0.005
(−0.3, 0.1)	0.002	0.005
(0.1, 0.2)	0.005	0.01
(0.2, 0.3)	0.005	0.02
(0.3, 0.6)	0.01	0.02
(0.6, 0.8)	0.05	—

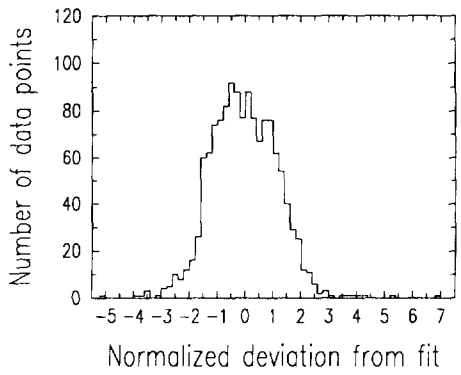


Figure 2. Histogram of normalized deviations $(s/\bar{s} - 1)/\epsilon_r$ of the He ion data, where s is given by Eq. (1) and where ϵ_r is the relative error of the data point taken from the publication (cf. Table II). X-ray data for $Z_2 < 12$ and data from rejected publications are omitted.

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EXPLANATION OF TABLES

TABLE I. Empirical Reference Cross Sections for *K*-Shell Ionization by α Particles

This table presents vacancy production cross sections [i.e., direct ionization plus electron capture (EC) by the projectile] for ^{4}He ions, obtained by averaging the available experimental data for H and He ions (cf. Table II) published until 1991. All target elements from $_6$C to $_{92}$U are covered. The energy values are approximately equidistant on a logarithmic scale. The extreme energy values are chosen approximately corresponding to the smallest and largest values of the scaled projectile velocity ξ for which experimental data exist, within each of the eight ranges of Z_2/Z_1 (see Figs. 1a–1h). Theoretical values $\sigma_{\mathrm{MECPSSR}} + \sigma_{\mathrm{EC}}$ and the correction factors $1 + \sigma_{\mathrm{EC}}/\sigma_{\mathrm{MECPSSR}}$ are also presented.

Targets with Z_2 = 6, 7, 8, 9, 10, 17, 18, 36, 54, and 86 are assumed to be gaseous, and for these, σ_{EC} is calculated assuming fully ionized He projectiles; for the remaining (solid) targets, the ions are assumed to be in charge equilibrium. This holds for the electron capture contributions contained in σ_{th}, and also for those contained in σ_{ref}.

It should be noted that the large correction factors found for light gaseous targets and small energy indicate that the corresponding cross sections are not realistic; this is because slow $^4He^{2+}$ projectiles will immediately capture electrons. The uncertainty of this correction is not contained in our error estimate.

In addition to the random error [based on the errors stated in the source publications (see Table II) and on the discrepancies between the data], we introduce an estimated systematic error. The total error is the sum of the two. The rather large changes of error from one line to the next which occur at low- and high-energy values in the table are a consequence of the discrete steps used in estimating the systematic error.

The uncertainty of the fluorescence yield (cf. the discussion in the text) is not included in our error estimate. Hence, for values based essentially on x-ray measurements (i.e., the majority), our relative error is really the uncertainty of the x-ray production cross section.

If interpolation between energy values in the table is necessary, we suggest using linear interpolation of log σ_{ref} vs log E_1. A FORTRAN subroutine that performs this interpolation within the table is available on floppy disk to interested users.

TARGET	Atomic number and element symbol of target
ENERGY	Projectile energy in MeV (in laboratory frame)
LOG XI	$_{10}\log \xi$, where ξ is the scaled projectile velocity
SIGMA	
THEORY	Theoretical ionization cross section according to the MECPSSR approach,[11] with electron capture added according to Lapicki and McDaniel,[16] in barns (1 b = 100 fm^2)
REFERENCE	Empirical reference cross section for vacancy production σ_{ref}, in barns (1.132E + 03 = $1.132 \times 10^{+3}$)
ERROR	
RND	Random error of σ_{ref} (standard deviation), as a percentage
TOT	Total (random plus systematic) error of σ_{ref} (standard deviation), as a percentage
COR	Correction factor $1 + \sigma_{\mathrm{EC}}/\sigma_{\mathrm{MECPSSR}}$ (listed only if COR > 1.00)

EXPLANATION OF TABLES continued

TABLE IIa. Experimental Cross Sections for *K*-Shell X-Ray Production, Auger Electron Production, and Ionization by ^{1}H Ions

TABLE IIb. Experimental Cross Sections for *K*-Shell X-Ray Production, Auger Electron Production, and Ionization by ^{2}H Ions

TABLE IIc. Experimental Cross Sections for *K*-Shell X-Ray Production, Auger Electron Production, and Ionization by ^{3}He Ions

TABLE IId. Experimental Cross Sections for *K*-Shell X-Ray Production, Auger Electron Production, and Ionization by ^{4}He Ions

Table II lists the data file which forms the basis of our fitted cross sections (except for the proton data already listed in Table II of Ref. 2). It contains data for H and He ions taken from the compilations by Rutledge and Watson[28] and by Gardner and Gray (Table I)[29] and from more recent original publications. Information on misprints discovered in the original tables can be found in the footnotes to Table 1 of Ref. 1 and in the References for Table II. As explained in the text, the data from 28 publications were rejected using a statistical criterion.

In Ref. 1 and in an earlier publication,[30] the normalized data were also compared graphically.

TARGET	Atomic number and element symbol of target
ENERGY	Projectile energy in MeV (in laboratory frame)
SIGMA	Total *K*-shell x-ray production, Auger electron production, or ionization cross section as identified by the first letter of the SOURCE below, in barns (1 b = 100 fm^2; 1.132E + 03 = $1.132 \times 10^{+3}$)
ERROR	Error (standard deviation, as a percentage) as given by the authors of the publication; where no error was stated, 80% has been arbitrarily assumed
REFERENCE	Publication source, coded as follows:

First letter

X	X-ray cross section
A	Auger cross section
I	For gaseous targets: equivalent to A (only for $Z_2 < 11$) For solid targets: ionization cross section as given by authors, based on x-ray measurement

Second letter

G	Gaseous target
N	Thin solid target
K	Thick solid target (infinitely thick for the projectile in question)

Remaining letters and numbers	Reference code for publication (see References for Table II)

TABLE I. Empirical Reference Cross Sections for *K*-Shell Ionization by α Particles

See page 84 for Explanation of Tables

TARGET	ENERGY [MeV]	LOG XI	SIGMA [barn] THEORY	SIGMA [barn] REFERENCE	ERROR [%] RND	ERROR [%] TOT	COR
6-C	0.008	-0.811	5.330E-05	3.800E-05	13.4	89.2	5.27
	0.009	-0.785	2.440E-04	1.990E-04	13.4	53.2	3.61
	0.010	-0.762	8.710E-04	7.790E-04	13.4	49.7	2.71
	0.011	-0.741	2.600E-03	2.480E-03	9.7	37.7	2.19
	0.012	-0.723	6.690E-03	6.700E-03	9.7	36.4	1.89
	0.014	-0.689	3.140E-02	3.350E-02	9.7	34.7	1.58
	0.016	-0.660	1.040E-01	1.150E-01	3.0	32.4	1.45
	0.018	-0.634	2.680E-01	3.040E-01	3.0	31.7	1.39
	0.020	-0.612	5.820E-01	6.670E-01	3.0	31.4	1.37
	0.023	-0.581	1.490E+00	1.710E+00	3.0	21.8	1.35
	0.026	-0.555	3.150E+00	3.630E+00	1.6	19.3	1.36
	0.030	-0.524	7.110E+00	8.090E+00	1.6	19.5	1.38
	0.035	-0.490	1.610E+01	1.800E+01	1.6	10.7	1.43
	0.040	-0.461	3.170E+01	3.470E+01	1.4	10.3	1.49
	0.045	-0.436	5.670E+01	6.070E+01	1.4	10.5	1.56
	0.050	-0.413	9.480E+01	9.870E+01	1.4	10.8	1.63
	0.055	-0.392	1.510E+02	1.540E+02	1.4	10.9	1.71
	0.060	-0.373	2.300E+02	2.320E+02	1.4	10.7	1.78
	0.070	-0.340	4.900E+02	4.810E+02	1.4	11.0	1.91
	0.080	-0.311	9.550E+02	9.250E+02	1.4	11.1	2.02
	0.090	-0.285	1.740E+03	1.660E+03	1.4	3.4	2.09
	0.100	-0.262	2.990E+03	2.840E+03	1.8	3.6	2.13
	0.110	-0.241	4.910E+03	4.630E+03	1.8	3.6	2.12
	0.120	-0.223	7.730E+03	7.260E+03	1.8	3.6	2.11
	0.140	-0.189	1.720E+04	1.610E+04	1.8	3.6	2.05
	0.160	-0.160	3.360E+04	3.160E+04	2.7	4.5	1.99
	0.180	-0.134	5.930E+04	5.580E+04	2.7	4.5	1.95
	0.200	-0.112	9.620E+04	8.950E+04	2.7	4.5	1.92
	0.230	-0.081	1.760E+05	1.650E+05	2.7	4.5	1.89
	0.260	-0.055	2.870E+05	2.700E+05	2.5	4.2	1.86
	0.300	-0.024	4.860E+05	4.620E+05	2.5	4.1	1.84
	0.350	0.010	8.050E+05	7.720E+05	2.5	4.1	1.81
	0.400	0.039	1.180E+06	1.140E+06	2.9	4.6	1.79
	0.450	0.064	1.590E+06	1.550E+06	2.9	4.6	1.77
	0.500	0.087	2.020E+06	1.990E+06	2.9	4.6	1.75
	0.550	0.108	2.440E+06	2.420E+06	2.9	7.0	1.73
	0.600	0.127	2.850E+06	2.830E+06	2.8	6.6	1.71
	0.700	0.160	3.590E+06	3.580E+06	2.8	6.5	1.68
	0.800	0.189	4.210E+06	4.190E+06	2.8	6.5	1.64
	0.900	0.215	4.700E+06	4.710E+06	2.8	6.5	1.60
	1.000	0.238	5.090E+06	5.110E+06	2.2	6.0	1.57
	1.100	0.259	5.380E+06	5.410E+06	2.2	6.0	1.53
	1.200	0.277	5.590E+06	5.610E+06	2.2	6.0	1.50
	1.400	0.311	5.830E+06	5.890E+06	2.2	9.6	1.45
	1.600	0.340	5.900E+06	6.000E+06	2.7	8.9	1.39
	1.800	0.366	5.880E+06	6.010E+06	2.7	8.9	1.35
	2.000	0.388	5.800E+06	5.930E+06	2.7	8.9	1.31
	2.300	0.419	5.610E+06	5.730E+06	3.3	9.8	1.26
	2.600	0.445	5.390E+06	5.490E+06	3.3	9.9	1.22
	3.000	0.476	5.080E+06	5.170E+06	3.3	9.9	1.18
	3.500	0.510	4.730E+06	4.790E+06	3.3	9.9	1.14
	4.000	0.539	4.410E+06	4.450E+06	4.6	11.2	1.11
	4.500	0.564	4.130E+06	4.140E+06	4.6	11.3	1.09
	5.000	0.587	3.890E+06	3.880E+06	4.6	11.3	1.07
	5.500	0.608	3.670E+06	3.640E+06	4.6	38.2	1.06
	6.000	0.627	3.480E+06	3.430E+06	6.6	44.0	1.05
	7.000	0.660	3.160E+06	3.080E+06	6.6	44.5	1.03
	8.000	0.689	2.900E+06	2.800E+06	6.6	44.9	1.02
	9.000	0.715	2.680E+06	2.560E+06	7.7	40.6	1.02
	10.000	0.738	2.500E+06	2.360E+06	7.7	40.9	1.01
	11.000	0.759	2.340E+06	2.190E+06	7.7	41.2	1.01
	12.000	0.777	2.200E+06	2.040E+06	7.7	41.5	1.01
7-N	0.012	-0.811	1.330E-04	9.460E-05	13.4	85.7	2.03
	0.014	-0.777	8.690E-04	7.330E-04	13.4	50.0	1.62
	0.016	-0.748	3.840E-03	3.590E-03	9.7	36.7	1.42
	0.018	-0.723	1.260E-02	1.270E-02	9.7	34.9	1.32
	0.020	-0.700	3.340E-02	3.510E-02	9.7	33.8	1.26
	0.023	-0.669	1.070E-01	1.170E-01	9.7	32.7	1.21
	0.026	-0.643	2.650E-01	2.980E-01	3.0	30.6	1.19
	0.030	-0.612	6.870E-01	7.870E-01	3.0	30.1	1.18
	0.035	-0.578	1.730E+00	2.000E+00	3.0	20.9	1.18
	0.040	-0.549	3.620E+00	4.160E+00	1.6	18.5	1.19
	0.045	-0.524	6.660E+00	7.590E+00	1.6	18.6	1.20
	0.050	-0.501	1.120E+01	1.270E+01	1.6	18.8	1.22
	0.055	-0.480	1.780E+01	1.980E+01	1.6	10.4	1.23
	0.060	-0.461	2.690E+01	2.940E+01	1.4	9.9	1.25
	0.070	-0.428	5.510E+01	5.850E+01	1.4	10.1	1.28
	0.080	-0.399	1.020E+02	1.050E+02	1.4	10.4	1.31
	0.090	-0.373	1.770E+02	1.780E+02	1.4	10.3	1.34
	0.100	-0.350	2.900E+02	2.870E+02	1.4	10.4	1.36
7-N	0.110	-0.330	4.550E+02	4.450E+02	1.4	10.5	1.38
	0.120	-0.311	6.900E+02	6.690E+02	1.4	10.6	1.39
	0.140	-0.277	1.460E+03	1.390E+03	1.8	3.5	1.41
	0.160	-0.248	2.790E+03	2.640E+03	1.8	3.5	1.41
	0.180	-0.223	4.950E+03	4.650E+03	1.8	3.5	1.41
	0.200	-0.200	8.200E+03	7.690E+03	1.8	3.5	1.41
	0.230	-0.169	1.570E+04	1.470E+04	2.7	4.4	1.41
	0.260	-0.143	2.720E+04	2.550E+04	2.7	4.4	1.40
	0.300	-0.112	4.990E+04	4.640E+04	2.7	4.4	1.40
	0.350	-0.078	9.160E+04	8.570E+04	2.5	4.1	1.40
	0.400	-0.049	1.490E+05	1.400E+05	2.5	4.0	1.40
	0.450	-0.024	2.210E+05	2.100E+05	2.5	4.0	1.41
	0.500	-0.001	3.060E+05	2.930E+05	2.5	4.0	1.41
	0.550	0.020	4.010E+05	3.860E+05	2.5	4.0	1.41
	0.600	0.039	5.040E+05	4.880E+05	2.9	4.5	1.41
	0.700	0.072	7.230E+05	7.110E+05	2.9	4.5	1.41
	0.800	0.101	9.440E+05	9.330E+05	2.9	6.8	1.40
	0.900	0.127	1.160E+06	1.150E+06	2.8	6.3	1.40
	1.000	0.150	1.350E+06	1.350E+06	2.8	6.3	1.39
	1.100	0.170	1.530E+06	1.520E+06	2.8	6.3	1.38
	1.200	0.189	1.690E+06	1.680E+06	2.8	6.3	1.37
	1.400	0.223	1.940E+06	1.950E+06	2.2	5.7	1.35
	1.600	0.252	2.130E+06	2.140E+06	2.2	5.7	1.33
	1.800	0.277	2.260E+06	2.270E+06	2.2	5.7	1.31
	2.000	0.300	2.350E+06	2.370E+06	2.2	9.2	1.28
	2.300	0.331	2.420E+06	2.460E+06	2.7	8.5	1.25
	2.600	0.357	2.440E+06	2.490E+06	2.7	8.4	1.23
	3.000	0.388	2.430E+06	2.480E+06	2.7	8.4	1.19
	3.500	0.422	2.370E+06	2.420E+06	3.3	9.4	1.16
	4.000	0.451	2.290E+06	2.330E+06	3.3	9.4	1.13
	4.500	0.476	2.200E+06	2.240E+06	3.3	9.4	1.11
	5.000	0.499	2.110E+06	2.140E+06	3.3	9.4	1.09
	5.500	0.520	2.030E+06	2.050E+06	3.3	9.4	1.08
	6.000	0.539	1.950E+06	1.960E+06	4.6	10.7	1.07
	7.000	0.572	1.800E+06	1.800E+06	4.6	10.8	1.05
	8.000	0.601	1.680E+06	1.660E+06	4.6	35.6	1.04
	9.000	0.627	1.570E+06	1.540E+06	6.6	41.5	1.03
	10.000	0.650	1.470E+06	1.440E+06	6.6	41.8	1.02
	11.000	0.670	1.390E+06	1.350E+06	6.6	42.0	1.02
	12.000	0.689	1.320E+06	1.270E+06	6.6	42.3	1.01
	14.000	0.723	1.190E+06	1.130E+06	7.7	38.1	1.01
	16.000	0.752	1.090E+06	1.020E+06	7.7	38.4	1.01
	18.000	0.777	1.000E+06	9.310E+05	7.7	38.8	1.00
8-O	0.016	-0.810	1.870E-04	1.340E-04	13.4	81.9	1.47
	0.018	-0.784	7.580E-04	6.200E-04	13.4	49.4	1.34
	0.020	-0.761	2.410E-03	2.160E-03	13.4	46.2	1.26
	0.023	-0.731	9.700E-03	9.510E-03	9.7	33.9	1.20
	0.026	-0.705	2.890E-02	3.000E-02	9.7	32.5	1.16
	0.030	-0.673	8.990E-02	9.810E-02	9.7	31.5	1.14
	0.035	-0.640	2.650E-01	2.990E-01	3.0	29.2	1.13
	0.040	-0.611	6.150E-01	7.050E-01	3.0	28.8	1.12
	0.045	-0.585	1.220E+00	1.400E+00	3.0	20.1	1.12
	0.050	-0.563	2.160E+00	2.490E+00	1.6	17.6	1.12
	0.055	-0.542	3.550E+00	4.070E+00	1.6	17.6	1.13
	0.060	-0.523	5.490E+00	6.260E+00	1.6	17.8	1.13
	0.070	-0.489	1.160E+01	1.290E+01	1.6	9.8	1.14
	0.080	-0.460	2.170E+01	2.370E+01	1.4	9.4	1.16
	0.090	-0.435	3.750E+01	4.010E+01	1.4	9.6	1.17
	0.100	-0.412	6.100E+01	6.350E+01	1.4	9.8	1.17
	0.110	-0.391	9.470E+01	9.690E+01	1.4	10.0	1.18
	0.120	-0.372	1.420E+02	1.430E+02	1.4	9.8	1.19
	0.140	-0.339	2.900E+02	2.850E+02	1.4	10.0	1.20
	0.160	-0.310	5.410E+02	5.240E+02	1.4	10.1	1.21
	0.180	-0.284	9.380E+02	8.980E+02	1.4	3.2	1.21
	0.200	-0.261	1.530E+03	1.460E+03	1.8	3.4	1.22
	0.230	-0.231	2.920E+03	2.750E+03	1.8	3.4	1.22
	0.260	-0.205	5.090E+03	4.770E+03	1.8	3.4	1.22
	0.300	-0.173	9.530E+03	8.940E+03	2.7	4.3	1.23
	0.350	-0.140	1.820E+04	1.710E+04	2.7	4.3	1.23
	0.400	-0.111	3.080E+04	2.870E+04	2.7	4.3	1.23
	0.450	-0.085	4.790E+04	4.470E+04	2.7	4.3	1.24
	0.500	-0.063	6.940E+04	6.510E+04	2.5	3.9	1.24
	0.550	-0.042	9.510E+04	8.970E+04	2.5	3.9	1.24
	0.600	-0.023	1.250E+05	1.190E+05	2.5	3.9	1.24
	0.700	0.011	1.930E+05	1.850E+05	2.5	3.9	1.25
	0.800	0.040	2.710E+05	2.620E+05	2.9	4.4	1.25
	0.900	0.065	3.530E+05	3.440E+05	2.9	4.4	1.26
	1.000	0.088	4.360E+05	4.310E+05	2.9	4.4	1.26
	1.100	0.109	5.180E+05	5.130E+05	2.9	6.5	1.26
	1.200	0.128	5.970E+05	5.930E+05	2.8	6.0	1.26
	1.400	0.161	7.410E+05	7.390E+05	2.8	6.0	1.25

TABLE I. Empirical Reference Cross Sections for *K*-Shell Ionization by α Particles

See page 84 for Explanation of Tables

TARGET	ENERGY [MeV]	LOG XI	SIGMA [barn] THEORY	SIGMA [barn] REFERENCE	ERROR [%] RND	ERROR [%] TOT	COR
8-O	1.600	0.190	8.630E+05	8.600E+05	2.8	6.0	1.24
	1.800	0.216	9.640E+05	9.650E+05	2.8	6.0	1.24
	2.000	0.239	1.050E+06	1.050E+06	2.2	5.5	1.23
	2.300	0.269	1.140E+06	1.140E+06	2.2	5.5	1.21
	2.600	0.295	1.200E+06	1.210E+06	2.2	5.4	1.20
	3.000	0.327	1.250E+06	1.270E+06	2.7	8.0	1.18
	3.500	0.360	1.280E+06	1.300E+06	2.7	7.9	1.15
	4.000	0.389	1.280E+06	1.310E+06	2.7	7.9	1.13
	4.500	0.415	1.260E+06	1.290E+06	2.7	7.9	1.12
	5.000	0.437	1.240E+06	1.270E+06	3.3	8.9	1.10
	5.500	0.458	1.210E+06	1.240E+06	3.3	8.9	1.09
	6.000	0.477	1.180E+06	1.200E+06	3.3	8.9	1.08
	7.000	0.511	1.120E+06	1.140E+06	3.3	8.9	1.06
	8.000	0.540	1.060E+06	1.070E+06	4.6	10.2	1.05
	9.000	0.565	1.010E+06	1.010E+06	4.6	10.3	1.04
	10.000	0.588	9.580E+05	9.540E+05	4.6	10.3	1.03
	11.000	0.609	9.110E+05	9.030E+05	4.6	33.1	1.02
	12.000	0.628	8.690E+05	8.560E+05	6.6	39.0	1.02
	14.000	0.661	7.960E+05	7.760E+05	6.6	39.3	1.01
	16.000	0.690	7.340E+05	7.080E+05	6.6	39.7	1.01
	18.000	0.716	6.820E+05	6.510E+05	7.7	35.4	1.01
	20.000	0.739	6.370E+05	6.020E+05	7.7	35.6	1.00
	23.000	0.769	5.800E+05	5.400E+05	7.7	36.0	1.00
	26.000	0.795	5.330E+05	4.900E+05	7.7	36.4	1.00
9-F	0.020	-0.817	1.800E-04	1.240E-04	13.4	81.2	1.27
	0.023	-0.786	8.970E-04	7.270E-04	13.4	47.8	1.20
	0.026	-0.760	3.190E-03	2.870E-03	9.7	34.4	1.16
	0.030	-0.729	1.200E-02	1.190E-02	9.7	32.2	1.12
	0.035	-0.695	4.230E-02	4.470E-02	9.7	30.7	1.10
	0.040	-0.666	1.110E-01	1.220E-01	9.7	29.9	1.09
	0.045	-0.641	2.420E-01	2.720E-01	3.0	27.9	1.09
	0.050	-0.618	4.600E-01	5.250E-01	3.0	27.5	1.09
	0.055	-0.597	7.940E-01	9.140E-01	3.0	19.2	1.09
	0.060	-0.578	1.280E+00	1.470E+00	3.0	19.2	1.09
	0.070	-0.545	2.840E+00	3.260E+00	1.6	16.8	1.09
	0.080	-0.516	5.490E+00	6.230E+00	1.6	16.9	1.09
	0.090	-0.490	9.650E+00	1.080E+01	1.6	9.4	1.10
	0.100	-0.467	1.580E+01	1.740E+01	1.6	9.5	1.10
	0.110	-0.446	2.460E+01	2.660E+01	1.4	9.0	1.11
	0.120	-0.428	3.680E+01	3.910E+01	1.4	9.2	1.11
	0.140	-0.394	7.460E+01	7.660E+01	1.4	9.5	1.12
	0.160	-0.365	1.380E+02	1.380E+02	1.4	9.3	1.12
	0.180	-0.339	2.360E+02	2.310E+02	1.4	9.5	1.12
	0.200	-0.317	3.800E+02	3.700E+02	1.4	9.6	1.13
	0.230	-0.286	7.140E+02	6.840E+02	1.4	3.1	1.13
	0.260	-0.260	1.240E+03	1.170E+03	1.8	3.3	1.13
	0.300	-0.229	2.320E+03	2.170E+03	1.8	3.3	1.14
	0.350	-0.195	4.470E+03	4.190E+03	1.8	3.3	1.14
	0.400	-0.166	7.730E+03	7.250E+03	2.7	4.2	1.14
	0.450	-0.141	1.230E+04	1.160E+04	2.7	4.2	1.15
	0.500	-0.118	1.830E+04	1.700E+04	2.7	4.2	1.15
	0.550	-0.097	2.580E+04	2.410E+04	2.7	4.2	1.15
	0.600	-0.078	3.480E+04	3.260E+04	2.5	3.8	1.15
	0.700	-0.045	5.720E+04	5.390E+04	2.5	3.8	1.16
	0.800	-0.016	8.450E+04	8.050E+04	2.5	3.8	1.16
	0.900	0.010	1.160E+05	1.110E+05	2.5	3.8	1.17
	1.000	0.033	1.490E+05	1.440E+05	2.9	4.3	1.17
	1.100	0.054	1.850E+05	1.790E+05	2.9	4.3	1.17
	1.200	0.072	2.210E+05	2.170E+05	2.9	4.3	1.17
	1.400	0.106	2.920E+05	2.890E+05	2.9	6.2	1.18
	1.600	0.135	3.590E+05	3.570E+05	2.8	5.8	1.18
	1.800	0.161	4.190E+05	4.180E+05	2.8	5.8	1.17
	2.000	0.183	4.730E+05	4.710E+05	2.8	5.8	1.17
	2.300	0.214	5.410E+05	5.410E+05	2.8	5.8	1.17
	2.600	0.240	5.950E+05	5.970E+05	2.2	5.2	1.16
	3.000	0.271	6.480E+05	6.500E+05	2.2	5.2	1.15
	3.500	0.305	6.920E+05	6.980E+05	2.2	8.2	1.13
	4.000	0.334	7.170E+05	7.280E+05	2.7	7.5	1.12
	4.500	0.359	7.300E+05	7.430E+05	2.7	7.5	1.11
	5.000	0.382	7.340E+05	7.510E+05	2.7	7.4	1.10
	5.500	0.403	7.330E+05	7.490E+05	2.7	7.4	1.09
	6.000	0.422	7.270E+05	7.430E+05	3.3	8.4	1.08
	7.000	0.455	7.080E+05	7.220E+05	3.3	8.4	1.06
	8.000	0.484	6.850E+05	6.960E+05	3.3	8.4	1.05
	9.000	0.510	6.600E+05	6.680E+05	3.3	8.4	1.04
	10.000	0.533	6.350E+05	6.400E+05	4.6	9.7	1.03
	11.000	0.554	6.110E+05	6.130E+05	4.6	9.8	1.03
	12.000	0.572	5.880E+05	5.880E+05	4.6	9.8	1.02
	14.000	0.606	5.460E+05	5.410E+05	4.6	30.6	1.02
	16.000	0.635	5.090E+05	5.000E+05	6.6	36.5	1.01
	18.000	0.661	4.760E+05	4.640E+05	6.6	36.8	1.01
9-F	20.000	0.683	4.480E+05	4.330E+05	6.6	37.0	1.01
	23.000	0.714	4.110E+05	3.920E+05	6.6	37.4	1.00
	26.000	0.740	3.800E+05	3.590E+05	7.7	33.0	1.00
	30.000	0.771	3.450E+05	3.210E+05	7.7	33.4	1.00
10-Ne	0.026	-0.809	3.260E-04	2.350E-04	13.4	74.6	1.16
	0.030	-0.778	1.500E-03	1.260E-03	13.4	44.8	1.12
	0.035	-0.745	6.410E-03	6.060E-03	9.7	31.6	1.10
	0.040	-0.716	1.960E-02	1.990E-02	9.7	30.1	1.08
	0.045	-0.690	4.770E-02	5.080E-02	9.7	29.2	1.07
	0.050	-0.667	9.900E-02	1.090E-01	9.7	28.6	1.07
	0.055	-0.646	1.830E-01	2.050E-01	3.0	26.6	1.07
	0.060	-0.627	3.110E-01	3.530E-01	3.0	26.3	1.07
	0.070	-0.594	7.460E-01	8.590E-01	3.0	18.4	1.06
	0.080	-0.565	1.520E+00	1.750E+00	3.0	18.4	1.07
	0.090	-0.539	2.750E+00	3.150E+00	1.6	15.9	1.07
	0.100	-0.517	4.610E+00	5.230E+00	1.6	16.0	1.07
	0.110	-0.496	7.270E+00	8.160E+00	1.6	8.9	1.07
	0.120	-0.477	1.100E+01	1.210E+01	1.6	9.0	1.07
	0.140	-0.443	2.240E+01	2.410E+01	1.4	8.8	1.07
	0.160	-0.414	4.130E+01	4.310E+01	1.4	8.8	1.08
	0.180	-0.389	7.050E+01	7.200E+01	1.4	9.0	1.08
	0.200	-0.366	1.130E+02	1.140E+02	1.4	8.8	1.08
	0.230	-0.336	2.120E+02	2.070E+02	1.4	9.0	1.08
	0.260	-0.309	3.630E+02	3.520E+02	1.4	9.1	1.09
	0.300	-0.278	6.780E+02	6.470E+02	1.4	3.0	1.09
	0.350	-0.245	1.310E+03	1.240E+03	1.8	3.2	1.09
	0.400	-0.216	2.280E+03	2.140E+03	1.8	3.2	1.09
	0.450	-0.190	3.680E+03	3.450E+03	1.8	3.2	1.09
	0.500	-0.167	5.550E+03	5.210E+03	2.7	4.0	1.10
	0.550	-0.146	7.960E+03	7.480E+03	2.7	4.0	1.10
	0.600	-0.127	1.090E+04	1.030E+04	2.7	4.0	1.10
	0.700	-0.094	1.870E+04	1.740E+04	2.7	4.1	1.10
	0.800	-0.065	2.870E+04	2.690E+04	2.5	3.7	1.11
	0.900	-0.039	4.080E+04	3.850E+04	2.5	3.7	1.11
	1.000	-0.017	5.460E+04	5.200E+04	2.5	3.7	1.11
	1.100	0.004	6.980E+04	6.680E+04	2.5	3.7	1.12
	1.200	0.023	8.600E+04	8.280E+04	2.9	4.2	1.12
	1.400	0.057	1.200E+05	1.170E+05	2.9	4.2	1.12
	1.600	0.086	1.540E+05	1.520E+05	2.9	4.2	1.12
	1.800	0.111	1.880E+05	1.860E+05	2.9	6.0	1.13
	2.000	0.134	2.200E+05	2.180E+05	2.8	5.5	1.13
	2.300	0.164	2.630E+05	2.620E+05	2.8	5.5	1.13
	2.600	0.191	3.000E+05	2.990E+05	2.8	5.5	1.12
	3.000	0.222	3.410E+05	3.410E+05	2.2	5.0	1.12
	3.500	0.255	3.790E+05	3.820E+05	2.2	5.0	1.11
	4.000	0.284	4.070E+05	4.100E+05	2.2	5.0	1.11
	4.500	0.310	4.260E+05	4.310E+05	2.2	7.7	1.10
	5.000	0.333	4.390E+05	4.450E+05	2.7	7.0	1.09
	5.500	0.354	4.470E+05	4.550E+05	2.7	7.0	1.08
	6.000	0.373	4.510E+05	4.610E+05	2.7	7.0	1.08
	7.000	0.406	4.520E+05	4.620E+05	2.7	7.0	1.06
	8.000	0.435	4.470E+05	4.560E+05	3.3	7.9	1.05
	9.000	0.461	4.380E+05	4.460E+05	3.3	7.9	1.05
	10.000	0.483	4.270E+05	4.340E+05	3.3	7.9	1.04
	11.000	0.504	4.160E+05	4.210E+05	3.3	7.9	1.03
	12.000	0.523	4.040E+05	4.080E+05	4.6	9.2	1.03
	14.000	0.557	3.810E+05	3.830E+05	4.6	9.3	1.02
	16.000	0.586	3.600E+05	3.580E+05	4.6	9.3	1.01
	18.000	0.611	3.400E+05	3.360E+05	4.6	28.1	1.01
	20.000	0.634	3.220E+05	3.160E+05	6.6	34.0	1.01
	23.000	0.664	2.980E+05	2.900E+05	6.6	34.2	1.01
	26.000	0.691	2.770E+05	2.670E+05	6.6	34.5	1.00
	30.000	0.722	2.540E+05	2.420E+05	7.7	30.2	1.00
	35.000	0.755	2.300E+05	2.160E+05	7.7	30.5	1.00
	40.000	0.784	2.100E+05	1.950E+05	7.7	30.8	1.00
11-Na	0.035	-0.792	9.080E-04	7.150E-04	13.4	45.0	1.04
	0.040	-0.763	3.250E-03	2.890E-03	13.4	41.4	1.03
	0.045	-0.738	8.940E-03	8.620E-03	9.7	29.7	1.03
	0.050	-0.715	2.050E-02	2.080E-02	9.7	28.6	1.02
	0.055	-0.694	4.090E-02	4.330E-02	9.7	27.9	1.02
	0.060	-0.675	7.380E-02	8.040E-02	9.7	27.4	1.02
	0.070	-0.642	1.940E-01	2.190E-01	3.0	25.2	1.02
	0.080	-0.613	4.190E-01	4.800E-01	3.0	24.9	1.02
	0.090	-0.587	7.940E-01	9.150E-01	3.0	17.5	1.02
	0.100	-0.564	1.370E+00	1.580E+00	3.0	17.5	1.02
	0.110	-0.544	2.210E+00	2.530E+00	1.6	15.0	1.02
	0.120	-0.525	3.380E+00	3.850E+00	1.6	15.1	1.03
	0.140	-0.491	7.030E+00	7.870E+00	1.6	8.5	1.03
	0.160	-0.462	1.310E+01	1.430E+01	1.4	8.0	1.03
	0.180	-0.437	2.240E+01	2.400E+01	1.4	8.2	1.03

TABLE I. Empirical Reference Cross Sections for *K*-Shell Ionization by α Particles

See page 84 for Explanation of Tables

TARGET	ENERGY [MeV]	LOG XI	SIGMA [barn] THEORY	SIGMA [barn] REFERENCE	ERROR [%] RND	ERROR [%] TOT	COR
11-Na	0.200	-0.414	3.600E+01	3.750E+01	1.4	8.4	1.03
	0.230	-0.383	6.710E+01	6.830E+01	1.4	8.5	1.03
	0.260	-0.357	1.150E+02	1.150E+02	1.4	8.4	1.04
	0.300	-0.326	2.140E+02	2.090E+02	1.4	8.5	1.04
	0.350	-0.292	4.130E+02	3.970E+02	1.4	2.8	1.04
	0.400	-0.263	7.230E+02	6.860E+02	1.8	3.1	1.04
	0.450	-0.238	1.170E+03	1.100E+03	1.8	3.1	1.05
	0.500	-0.215	1.780E+03	1.670E+03	1.8	3.1	1.05
	0.550	-0.194	2.580E+03	2.420E+03	1.8	3.1	1.05
	0.600	-0.175	3.600E+03	3.370E+03	1.8	3.1	1.05
	0.700	-0.142	6.310E+03	5.920E+03	2.7	3.9	1.06
	0.800	-0.113	9.980E+03	9.290E+03	2.7	4.0	1.06
	0.900	-0.087	1.460E+04	1.370E+04	2.7	3.9	1.06
	1.000	-0.064	2.020E+04	1.890E+04	2.5	3.6	1.07
	1.100	-0.044	2.650E+04	2.500E+04	2.5	3.6	1.07
	1.200	-0.025	3.360E+04	3.190E+04	2.5	3.6	1.07
	1.400	0.009	4.930E+04	4.720E+04	2.5	3.6	1.08
	1.600	0.038	6.630E+04	6.410E+04	2.9	4.1	1.08
	1.800	0.063	8.380E+04	8.160E+04	2.9	4.1	1.08
	2.000	0.086	1.010E+05	9.990E+04	2.9	4.1	1.08
	2.300	0.117	1.270E+05	1.250E+05	2.9	5.7	1.09
	2.600	0.143	1.500E+05	1.490E+05	2.8	5.3	1.09
	3.000	0.174	1.770E+05	1.760E+05	2.8	5.3	1.09
	3.500	0.208	2.060E+05	2.060E+05	2.8	5.3	1.09
	4.000	0.237	2.290E+05	2.300E+05	2.2	4.7	1.08
	4.500	0.262	2.470E+05	2.480E+05	2.2	4.7	1.08
	5.000	0.285	2.600E+05	2.620E+05	2.2	4.7	1.08
	5.500	0.306	2.700E+05	2.730E+05	2.2	7.2	1.07
	6.000	0.325	2.780E+05	2.820E+05	2.7	6.5	1.07
	7.000	0.358	2.870E+05	2.920E+05	2.7	6.5	1.06
	8.000	0.387	2.900E+05	2.970E+05	2.7	6.5	1.05
	9.000	0.413	2.900E+05	2.960E+05	2.7	6.5	1.05
	10.000	0.436	2.870E+05	2.930E+05	3.3	7.4	1.04
	11.000	0.456	2.830E+05	2.890E+05	3.3	7.4	1.03
	12.000	0.475	2.780E+05	2.830E+05	3.3	7.4	1.03
	14.000	0.509	2.670E+05	2.700E+05	3.3	7.4	1.02
	16.000	0.538	2.550E+05	2.570E+05	4.6	8.7	1.02
	18.000	0.563	2.440E+05	2.440E+05	4.6	8.8	1.01
	20.000	0.586	2.330E+05	2.320E+05	4.6	8.8	1.01
	23.000	0.617	2.180E+05	2.150E+05	6.6	31.3	1.01
	26.000	0.643	2.040E+05	2.000E+05	6.6	31.5	1.01
	30.000	0.674	1.880E+05	1.830E+05	6.6	31.8	1.00
	35.000	0.708	1.720E+05	1.640E+05	6.6	32.1	1.00
	40.000	0.737	1.580E+05	1.490E+05	7.7	27.7	1.00
	45.000	0.762	1.460E+05	1.370E+05	7.7	27.9	1.00
	50.000	0.785	1.360E+05	1.260E+05	7.7	28.1	1.00
12-Mg	0.040	-0.809	5.100E-04	3.660E-04	13.4	67.7	1.02
	0.045	-0.784	1.600E-03	1.310E-03	13.4	42.0	1.02
	0.050	-0.761	4.080E-03	3.660E-03	13.4	39.5	1.02
	0.055	-0.740	8.900E-03	8.520E-03	9.7	28.2	1.02
	0.060	-0.721	1.730E-02	1.730E-02	9.7	27.4	1.02
	0.070	-0.688	5.060E-02	5.410E-02	9.7	26.3	1.02
	0.080	-0.659	1.180E-01	1.310E-01	3.0	24.2	1.02
	0.090	-0.633	2.360E-01	2.670E-01	3.0	23.8	1.02
	0.100	-0.610	4.230E-01	4.850E-01	3.0	23.5	1.02
	0.110	-0.590	7.020E-01	8.090E-01	3.0	16.6	1.02
	0.120	-0.571	1.100E+00	1.270E+00	3.0	16.6	1.02
	0.140	-0.537	2.350E+00	2.690E+00	1.6	14.2	1.02
	0.160	-0.508	4.450E+00	5.020E+00	1.6	14.3	1.02
	0.180	-0.483	7.690E+00	8.550E+00	1.6	8.1	1.02
	0.200	-0.460	1.240E+01	1.360E+01	1.4	7.6	1.02
	0.230	-0.429	2.330E+01	2.480E+01	1.4	7.8	1.02
	0.260	-0.403	4.000E+01	4.130E+01	1.4	7.9	1.02
	0.300	-0.372	7.450E+01	7.510E+01	1.4	7.8	1.03
	0.350	-0.338	1.440E+02	1.410E+02	1.4	8.0	1.03
	0.400	-0.309	2.510E+02	2.430E+02	1.4	8.0	1.03
	0.450	-0.284	4.080E+02	3.900E+02	1.4	2.7	1.03
	0.500	-0.261	6.240E+02	5.920E+02	1.8	3.0	1.03
	0.550	-0.240	9.100E+02	8.570E+02	1.8	3.0	1.03
	0.600	-0.221	1.280E+03	1.200E+03	1.8	3.0	1.04
	0.700	-0.188	2.280E+03	2.130E+03	1.8	3.0	1.04
	0.800	-0.159	3.680E+03	3.450E+03	2.7	3.8	1.04
	0.900	-0.133	5.500E+03	5.180E+03	2.7	3.8	1.04
	1.000	-0.110	7.770E+03	7.230E+03	2.7	3.8	1.05
	1.100	-0.090	1.040E+04	9.750E+03	2.7	3.8	1.05
	1.200	-0.071	1.350E+04	1.270E+04	2.5	3.5	1.05
	1.400	-0.037	2.070E+04	1.950E+04	2.5	3.5	1.05
	1.600	-0.008	2.890E+04	2.760E+04	2.5	3.5	1.06
	1.800	0.017	3.780E+04	3.630E+04	2.5	3.5	1.06
	2.000	0.040	4.710E+04	4.560E+04	2.9	4.0	1.06

TARGET	ENERGY [MeV]	LOG XI	SIGMA [barn] THEORY	SIGMA [barn] REFERENCE	ERROR [%] RND	ERROR [%] TOT	COR
12-Mg	2.300	0.071	6.130E+04	5.980E+04	2.9	4.0	1.06
	2.600	0.097	7.510E+04	7.420E+04	2.9	4.0	1.07
	3.000	0.128	9.250E+04	9.190E+04	2.8	5.0	1.07
	3.500	0.162	1.120E+05	1.120E+05	2.8	5.0	1.07
	4.000	0.191	1.280E+05	1.280E+05	2.8	5.0	1.07
	4.500	0.216	1.420E+05	1.420E+05	2.8	5.0	1.07
	5.000	0.239	1.540E+05	1.540E+05	2.2	4.5	1.06
	5.500	0.260	1.630E+05	1.640E+05	2.2	4.5	1.06
	6.000	0.279	1.700E+05	1.710E+05	2.2	4.5	1.06
	7.000	0.312	1.810E+05	1.830E+05	2.2	6.7	1.05
	8.000	0.341	1.880E+05	1.910E+05	2.7	6.0	1.05
	9.000	0.367	1.910E+05	1.960E+05	2.7	6.0	1.04
	10.000	0.390	1.930E+05	1.970E+05	2.7	6.0	1.04
	11.000	0.410	1.920E+05	1.970E+05	2.7	6.0	1.03
	12.000	0.429	1.910E+05	1.950E+05	3.3	6.9	1.03
	14.000	0.463	1.870E+05	1.910E+05	3.3	6.9	1.02
	16.000	0.492	1.810E+05	1.840E+05	3.3	6.9	1.02
	18.000	0.517	1.750E+05	1.770E+05	3.3	6.9	1.02
	20.000	0.540	1.690E+05	1.700E+05	4.6	8.3	1.01
	23.000	0.571	1.600E+05	1.600E+05	4.6	8.3	1.01
	26.000	0.597	1.510E+05	1.500E+05	4.6	8.3	1.01
	30.000	0.628	1.410E+05	1.390E+05	6.6	28.8	1.00
	35.000	0.662	1.290E+05	1.260E+05	6.6	29.1	1.00
	40.000	0.691	1.200E+05	1.150E+05	6.6	29.3	1.00
	45.000	0.716	1.110E+05	1.060E+05	7.7	24.9	1.00
	50.000	0.739	1.040E+05	9.830E+04	7.7	25.0	1.00
	55.000	0.760	9.770E+04	9.150E+04	7.7	25.2	1.00
	60.000	0.779	9.220E+04	8.550E+04	7.7	25.4	1.00
13-Al	0.050	-0.803	8.340E-04	6.210E-04	13.4	62.4	1.02
	0.055	-0.782	1.990E-03	1.640E-03	13.4	39.9	1.02
	0.060	-0.763	4.150E-03	3.690E-03	13.4	38.0	1.02
	0.070	-0.730	1.360E-02	1.340E-02	9.7	26.2	1.02
	0.080	-0.701	3.440E-02	3.600E-02	9.7	25.2	1.01
	0.090	-0.675	7.300E-02	7.950E-02	9.7	24.6	1.01
	0.100	-0.652	1.370E-01	1.530E-01	3.0	22.7	1.01
	0.110	-0.632	2.360E-01	2.670E-01	3.0	22.4	1.01
	0.120	-0.613	3.780E-01	4.330E-01	3.0	22.2	1.01
	0.140	-0.579	8.410E-01	9.700E-01	3.0	15.7	1.01
	0.160	-0.550	1.630E+00	1.870E+00	1.6	13.3	1.02
	0.180	-0.525	2.860E+00	3.250E+00	1.6	13.4	1.02
	0.200	-0.502	4.670E+00	5.250E+00	1.6	13.5	1.02
	0.230	-0.472	8.820E+00	9.720E+00	1.6	7.7	1.02
	0.260	-0.445	1.520E+01	1.640E+01	1.4	7.2	1.02
	0.300	-0.414	2.850E+01	2.970E+01	1.4	7.4	1.02
	0.350	-0.380	5.510E+01	5.590E+01	1.4	7.6	1.02
	0.400	-0.351	9.650E+01	9.570E+01	1.4	7.4	1.02
	0.450	-0.326	1.570E+02	1.530E+02	1.4	7.5	1.02
	0.500	-0.303	2.400E+02	2.320E+02	1.4	7.5	1.02
	0.550	-0.282	3.520E+02	3.360E+02	1.4	2.6	1.02
	0.600	-0.263	4.950E+02	4.700E+02	1.8	2.9	1.03
	0.700	-0.230	8.930E+02	8.390E+02	1.8	2.9	1.03
	0.800	-0.201	1.460E+03	1.370E+03	1.8	2.9	1.03
	0.900	-0.175	2.220E+03	2.080E+03	1.8	2.9	1.03
	1.000	-0.152	3.190E+03	2.990E+03	2.7	3.7	1.03
	1.100	-0.132	4.360E+03	4.100E+03	2.7	3.7	1.03
	1.200	-0.113	5.740E+03	5.340E+03	2.7	3.7	1.04
	1.400	-0.079	9.070E+03	8.480E+03	2.7	3.7	1.04
	1.600	-0.050	1.310E+04	1.230E+04	2.5	3.4	1.04
	1.800	-0.025	1.760E+04	1.680E+04	2.5	3.4	1.04
	2.000	-0.002	2.260E+04	2.160E+04	2.5	3.4	1.04
	2.300	0.028	3.040E+04	2.930E+04	2.9	3.9	1.05
	2.600	0.055	3.850E+04	3.740E+04	2.9	3.9	1.05
	3.000	0.086	4.910E+04	4.840E+04	2.9	3.9	1.05
	3.500	0.120	6.160E+04	6.110E+04	2.8	4.8	1.05
	4.000	0.149	7.300E+04	7.270E+04	2.8	4.8	1.05
	4.500	0.174	8.300E+04	8.250E+04	2.8	4.8	1.05
	5.000	0.197	9.170E+04	9.150E+04	2.8	4.8	1.05
	5.500	0.218	9.920E+04	9.940E+04	2.2	4.2	1.05
	6.000	0.237	1.060E+05	1.060E+05	2.2	4.2	1.05
	7.000	0.270	1.160E+05	1.160E+05	2.2	4.2	1.05
	8.000	0.299	1.230E+05	1.240E+05	2.2	4.2	1.04
	9.000	0.325	1.270E+05	1.290E+05	2.7	5.5	1.04
	10.000	0.348	1.300E+05	1.320E+05	2.7	5.5	1.04
	11.000	0.368	1.320E+05	1.350E+05	2.7	5.5	1.03
	12.000	0.387	1.330E+05	1.360E+05	2.7	5.5	1.03
	14.000	0.421	1.320E+05	1.350E+05	3.3	6.4	1.03
	16.000	0.450	1.300E+05	1.330E+05	3.3	6.4	1.02
	18.000	0.475	1.270E+05	1.300E+05	3.3	6.4	1.02
	20.000	0.498	1.240E+05	1.260E+05	3.3	6.4	1.01
	23.000	0.528	1.190E+05	1.200E+05	4.6	7.7	1.01
	26.000	0.555	1.130E+05	1.140E+05	4.6	7.8	1.01

TABLE I. Empirical Reference Cross Sections for *K*-Shell Ionization by α Particles

See page 84 for Explanation of Tables

TARGET	ENERGY [MeV]	LOG XI	SIGMA [barn] THEORY	SIGMA [barn] REFERENCE	ERROR [%] RND	ERROR [%] TOT	COR
13-Al	30.000	0.586	1.070E+05	1.060E+05	4.6	7.8	1.01
	35.000	0.620	9.890E+04	9.770E+04	6.6	26.2	1.00
	40.000	0.649	9.210E+04	9.020E+04	6.6	26.4	1.00
	45.000	0.674	8.620E+04	8.360E+04	6.6	26.6	1.00
	50.000	0.697	8.090E+04	7.780E+04	6.6	26.8	1.00
	55.000	0.718	7.620E+04	7.270E+04	7.7	22.3	1.00
	60.000	0.737	7.210E+04	6.820E+04	7.7	22.4	1.00
	70.000	0.770	6.510E+04	6.060E+04	7.7	22.6	1.00
	80.000	0.799	5.930E+04	5.450E+04	7.7	22.8	1.00
14-Si	0.060	-0.802	9.600E-04	7.170E-04	13.4	58.9	1.02
	0.070	-0.769	3.580E-03	3.120E-03	13.4	36.8	1.01
	0.080	-0.740	9.940E-03	9.520E-03	9.7	25.1	1.01
	0.090	-0.714	2.260E-02	2.310E-02	9.7	24.2	1.01
	0.100	-0.691	4.490E-02	4.770E-02	9.7	23.6	1.01
	0.110	-0.671	8.030E-02	8.790E-02	9.7	23.2	1.01
	0.120	-0.652	1.330E-01	1.490E-01	3.0	21.4	1.01
	0.140	-0.618	3.110E-01	3.550E-01	3.0	21.0	1.01
	0.160	-0.589	6.200E-01	7.140E-01	3.0	14.9	1.01
	0.180	-0.564	1.110E+00	1.280E+00	3.0	14.9	1.01
	0.200	-0.541	1.840E+00	2.110E+00	1.6	12.4	1.01
	0.230	-0.511	3.530E+00	3.990E+00	1.6	12.6	1.01
	0.260	-0.484	6.150E+00	6.850E+00	1.6	7.2	1.01
	0.300	-0.453	1.160E+01	1.260E+01	1.4	6.7	1.01
	0.350	-0.419	2.260E+01	2.380E+01	1.4	6.9	1.02
	0.400	-0.390	3.970E+01	4.060E+01	1.4	7.0	1.02
	0.450	-0.365	6.470E+01	6.490E+01	1.4	6.8	1.02
	0.500	-0.342	9.960E+01	9.800E+01	1.4	6.9	1.02
	0.550	-0.321	1.460E+02	1.420E+02	1.4	7.0	1.02
	0.600	-0.302	2.060E+02	1.990E+02	1.4	7.0	1.02
	0.700	-0.269	3.740E+02	3.560E+02	1.8	2.8	1.02
	0.800	-0.240	6.190E+02	5.830E+02	1.8	2.8	1.02
	0.900	-0.214	9.500E+02	8.920E+02	1.8	2.8	1.02
	1.000	-0.191	1.380E+03	1.290E+03	1.8	2.8	1.02
	1.100	-0.171	1.910E+03	1.790E+03	2.7	3.6	1.02
	1.200	-0.152	2.540E+03	2.390E+03	2.7	3.6	1.03
	1.400	-0.118	4.130E+03	3.840E+03	2.7	3.6	1.03
	1.600	-0.089	6.120E+03	5.710E+03	2.7	3.6	1.03
	1.800	-0.064	8.450E+03	7.930E+03	2.5	3.3	1.03
	2.000	-0.041	1.110E+04	1.040E+04	2.5	3.3	1.03
	2.300	-0.011	1.540E+04	1.470E+04	2.5	3.3	1.03
	2.600	0.016	2.000E+04	1.930E+04	2.5	3.3	1.04
	3.000	0.047	2.650E+04	2.560E+04	2.9	3.8	1.04
	3.500	0.081	3.440E+04	3.390E+04	2.9	3.8	1.04
	4.000	0.110	4.190E+04	4.150E+04	2.9	5.0	1.04
	4.500	0.135	4.890E+04	4.860E+04	2.8	4.5	1.04
	5.000	0.158	5.520E+04	5.510E+04	2.8	4.5	1.04
	5.500	0.179	6.090E+04	6.060E+04	2.8	4.5	1.04
	6.000	0.198	6.590E+04	6.580E+04	2.8	4.5	1.04
	7.000	0.231	7.420E+04	7.440E+04	2.2	4.0	1.04
	8.000	0.260	8.050E+04	8.110E+04	2.2	4.0	1.04
	9.000	0.286	8.520E+04	8.580E+04	2.2	4.0	1.04
	10.000	0.309	8.860E+04	8.960E+04	2.2	5.7	1.03
	11.000	0.329	9.110E+04	9.230E+04	2.7	5.0	1.03
	12.000	0.348	9.270E+04	9.430E+04	2.7	5.0	1.03
	14.000	0.382	9.430E+04	9.640E+04	2.7	5.0	1.02
	16.000	0.411	9.430E+04	9.640E+04	2.7	5.0	1.02
	18.000	0.436	9.340E+04	9.530E+04	3.3	5.9	1.02
	20.000	0.459	9.190E+04	9.360E+04	3.3	5.9	1.02
	23.000	0.489	8.910E+04	9.040E+04	3.3	5.9	1.01
	26.000	0.516	8.590E+04	8.690E+04	3.3	6.0	1.01
	30.000	0.547	8.160E+04	8.210E+04	4.6	7.3	1.01
	35.000	0.581	7.650E+04	7.630E+04	4.6	7.3	1.00
	40.000	0.610	7.180E+04	7.110E+04	4.6	18.0	1.00
	45.000	0.635	6.750E+04	6.630E+04	6.6	23.8	1.00
	50.000	0.658	6.370E+04	6.210E+04	6.6	23.9	1.00
	55.000	0.679	6.020E+04	5.830E+04	6.6	24.0	1.00
	60.000	0.698	5.710E+04	5.490E+04	6.6	24.2	1.00
	70.000	0.731	5.180E+04	4.910E+04	7.7	19.7	1.00
	80.000	0.760	4.740E+04	4.440E+04	7.7	19.9	1.00
	90.000	0.786	4.380E+04	4.050E+04	7.7	20.0	1.00
15-P	0.070	-0.806	9.370E-04	6.860E-04	13.4	56.4	1.01
	0.080	-0.777	2.870E-03	2.420E-03	13.4	35.8	1.01
	0.090	-0.751	7.050E-03	6.530E-03	9.7	24.0	1.01
	0.100	-0.729	1.480E-02	1.460E-02	9.7	23.1	1.01
	0.110	-0.708	2.780E-02	2.870E-02	9.7	22.5	1.01
	0.120	-0.689	4.770E-02	5.100E-02	9.7	22.1	1.01
	0.140	-0.655	1.170E-01	1.310E-01	3.0	20.1	1.01
	0.160	-0.626	2.430E-01	2.760E-01	3.0	19.7	1.01
	0.180	-0.601	4.460E-01	5.130E-01	3.0	19.6	1.01
15-P	0.200	-0.578	7.530E-01	8.680E-01	3.0	14.0	1.01
	0.230	-0.548	1.470E+00	1.690E+00	1.6	11.5	1.01
	0.260	-0.521	2.590E+00	2.950E+00	1.6	11.6	1.01
	0.300	-0.490	4.940E+00	5.520E+00	1.6	6.7	1.01
	0.350	-0.456	9.690E+00	1.060E+01	1.4	6.2	1.01
	0.400	-0.427	1.710E+01	1.820E+01	1.4	6.4	1.01
	0.450	-0.402	2.810E+01	2.900E+01	1.4	6.5	1.01
	0.500	-0.379	4.330E+01	4.390E+01	1.4	6.6	1.01
	0.550	-0.358	6.370E+01	6.350E+01	1.4	6.3	1.01
	0.600	-0.339	9.020E+01	8.860E+01	1.4	6.4	1.01
	0.700	-0.306	1.650E+02	1.590E+02	1.4	6.5	1.02
	0.800	-0.277	2.740E+02	2.620E+02	1.8	2.7	1.02
	0.900	-0.251	4.250E+02	4.010E+02	1.8	2.7	1.02
	1.000	-0.229	6.210E+02	5.830E+02	1.8	2.7	1.02
	1.100	-0.208	8.690E+02	8.150E+02	1.8	2.7	1.02
	1.200	-0.189	1.170E+03	1.100E+03	1.8	2.7	1.02
	1.400	-0.155	1.940E+03	1.820E+03	2.7	3.5	1.02
	1.600	-0.126	2.930E+03	2.760E+03	2.7	3.5	1.02
	1.800	-0.101	4.130E+03	3.850E+03	2.7	3.5	1.02
	2.000	-0.078	5.520E+03	5.170E+03	2.5	3.2	1.02
	2.300	-0.048	7.910E+03	7.450E+03	2.5	3.2	1.03
	2.600	-0.021	1.060E+04	1.010E+04	2.5	3.2	1.03
	3.000	0.010	1.440E+04	1.380E+04	2.5	3.2	1.03
	3.500	0.044	1.930E+04	1.870E+04	2.9	3.7	1.03
	4.000	0.073	2.420E+04	2.380E+04	2.9	3.7	1.03
	4.500	0.098	2.900E+04	2.860E+04	2.9	3.6	1.03
	5.000	0.121	3.340E+04	3.310E+04	2.8	4.3	1.03
	5.500	0.142	3.750E+04	3.730E+04	2.8	4.3	1.03
	6.000	0.161	4.130E+04	4.120E+04	2.8	4.3	1.03
	7.000	0.194	4.780E+04	4.760E+04	2.8	4.3	1.03
	8.000	0.223	5.300E+04	5.310E+04	2.2	3.7	1.03
	9.000	0.249	5.720E+04	5.750E+04	2.2	3.7	1.03
	10.000	0.271	6.040E+04	6.070E+04	2.2	3.7	1.03
	11.000	0.292	6.300E+04	6.340E+04	2.2	3.7	1.03
	12.000	0.311	6.490E+04	6.560E+04	2.2	5.2	1.03
	14.000	0.345	6.730E+04	6.840E+04	2.7	4.5	1.02
	16.000	0.374	6.840E+04	7.000E+04	2.7	4.5	1.02
	18.000	0.399	6.870E+04	7.020E+04	2.7	4.5	1.02
	20.000	0.422	6.830E+04	6.980E+04	3.3	5.4	1.02
	23.000	0.452	6.710E+04	6.840E+04	3.3	5.4	1.01
	26.000	0.479	6.540E+04	6.650E+04	3.3	5.5	1.01
	30.000	0.510	6.280E+04	6.360E+04	3.3	5.5	1.01
	35.000	0.544	5.950E+04	5.990E+04	4.6	6.8	1.01
	40.000	0.573	5.630E+04	5.630E+04	4.6	6.8	1.00
	45.000	0.598	5.330E+04	5.290E+04	4.6	6.8	1.00
	50.000	0.621	5.050E+04	4.980E+04	6.6	21.2	1.00
	55.000	0.642	4.800E+04	4.700E+04	6.6	21.3	1.00
	60.000	0.661	4.570E+04	4.450E+04	6.6	21.4	1.00
	70.000	0.694	4.160E+04	4.010E+04	6.6	21.6	1.00
	80.000	0.723	3.830E+04	3.640E+04	7.7	17.0	1.00
	90.000	0.749	3.540E+04	3.330E+04	7.7	17.2	1.00
	100.000	0.771	3.300E+04	3.070E+04	7.7	17.3	1.00
	110.000	0.792	3.080E+04	2.840E+04	7.7	17.3	1.00
16-S	0.080	-0.809	8.460E-04	6.080E-04	13.4	53.8	1.01
	0.090	-0.784	2.260E-03	1.850E-03	13.4	34.6	1.01
	0.100	-0.761	5.060E-03	4.540E-03	13.4	32.8	1.01
	0.110	-0.740	9.980E-03	9.560E-03	9.7	22.0	1.01
	0.120	-0.721	1.790E-02	1.800E-02	9.7	21.4	1.01
	0.140	-0.688	4.670E-02	5.000E-02	9.7	20.7	1.01
	0.160	-0.659	1.010E-01	1.120E-01	3.0	18.8	1.01
	0.180	-0.633	1.910E-01	2.160E-01	3.0	18.5	1.01
	0.200	-0.610	3.300E-01	3.780E-01	3.0	18.3	1.01
	0.230	-0.580	6.590E-01	7.600E-01	3.0	13.1	1.01
	0.260	-0.553	1.180E+00	1.360E+00	1.6	10.6	1.01
	0.300	-0.522	2.280E+00	2.600E+00	1.6	10.7	1.01
	0.350	-0.489	4.530E+00	5.060E+00	1.6	6.3	1.01
	0.400	-0.460	8.070E+00	8.820E+00	1.4	5.8	1.01
	0.450	-0.434	1.330E+01	1.420E+01	1.4	5.9	1.01
	0.500	-0.411	2.060E+01	2.140E+01	1.4	6.0	1.01
	0.550	-0.391	3.040E+01	3.110E+01	1.4	6.1	1.01
	0.600	-0.372	4.310E+01	4.350E+01	1.4	5.8	1.01
	0.700	-0.338	7.930E+01	7.780E+01	1.4	5.9	1.01
	0.800	-0.309	1.330E+02	1.280E+02	1.4	6.0	1.01
	0.900	-0.284	2.060E+02	1.980E+02	1.4	2.3	1.01
	1.000	-0.261	3.040E+02	2.880E+02	1.8	2.6	1.01
	1.100	-0.240	4.280E+02	4.030E+02	1.8	2.6	1.01
	1.200	-0.221	5.800E+02	5.450E+02	1.8	2.6	1.01
	1.400	-0.188	9.770E+02	9.160E+02	1.8	2.6	1.02
	1.600	-0.159	1.500E+03	1.410E+03	2.7	3.4	1.02
	1.800	-0.133	2.150E+03	2.020E+03	2.7	3.4	1.02

TABLE I. Empirical Reference Cross Sections for *K*-Shell Ionization by α Particles

See page 84 for Explanation of Tables

TARGET	ENERGY [MeV]	LOG XI	SIGMA [barn] THEORY	SIGMA [barn] REFERENCE	ERROR [%] RND	ERROR [%] TOT	COR
16-S	2.000	-0.110	2.920E+03	2.720E+03	2.7	3.4	1.02
	2.300	-0.080	4.270E+03	4.000E+03	2.7	3.4	1.02
	2.600	-0.053	5.830E+03	5.490E+03	2.5	3.1	1.02
	3.000	-0.022	8.140E+03	7.750E+03	2.5	3.1	1.02
	3.500	0.011	1.130E+04	1.080E+04	2.5	3.1	1.02
	4.000	0.040	1.450E+04	1.400E+04	2.9	3.6	1.02
	4.500	0.066	1.770E+04	1.720E+04	2.9	3.6	1.03
	5.000	0.089	2.080E+04	2.050E+04	2.9	3.5	1.03
	5.500	0.109	2.370E+04	2.350E+04	2.9	4.5	1.03
	6.000	0.128	2.650E+04	2.630E+04	2.8	4.0	1.03
	7.000	0.162	3.140E+04	3.140E+04	2.8	4.0	1.03
	8.000	0.191	3.560E+04	3.550E+04	2.8	4.0	1.03
	9.000	0.216	3.910E+04	3.920E+04	2.8	4.0	1.03
	10.000	0.239	4.200E+04	4.220E+04	2.2	3.5	1.03
	11.000	0.260	4.430E+04	4.460E+04	2.2	3.5	1.03
	12.000	0.279	4.620E+04	4.640E+04	2.2	3.5	1.02
	14.000	0.312	4.890E+04	4.940E+04	2.2	4.7	1.02
	16.000	0.341	5.050E+04	5.120E+04	2.7	4.0	1.02
	18.000	0.367	5.130E+04	5.240E+04	2.7	4.0	1.02
	20.000	0.390	5.160E+04	5.270E+04	2.7	4.0	1.02
	23.000	0.420	5.130E+04	5.240E+04	3.3	5.0	1.01
	26.000	0.447	5.060E+04	5.160E+04	3.3	5.0	1.01
	30.000	0.478	4.910E+04	5.000E+04	3.3	5.0	1.01
	35.000	0.511	4.710E+04	4.760E+04	3.3	5.0	1.01
	40.000	0.540	4.490E+04	4.520E+04	4.6	6.3	1.00
	45.000	0.566	4.280E+04	4.280E+04	4.6	6.3	1.00
	50.000	0.589	4.080E+04	4.060E+04	4.6	6.3	1.00
	55.000	0.609	3.890E+04	3.850E+04	4.6	12.9	1.00
	60.000	0.628	3.720E+04	3.660E+04	6.6	18.7	1.00
	70.000	0.662	3.410E+04	3.320E+04	6.6	18.8	1.00
	80.000	0.691	3.150E+04	3.030E+04	6.6	18.9	1.00
	90.000	0.716	2.920E+04	2.790E+04	7.7	14.4	1.00
	100.000	0.739	2.730E+04	2.580E+04	7.7	14.5	1.00
	110.000	0.760	2.560E+04	2.390E+04	7.7	14.5	1.00
	120.000	0.779	2.410E+04	2.240E+04	7.7	14.6	1.00
17-Cl	0.090	-0.814	7.290E-04	5.070E-04	13.4	51.5	1.02
	0.100	-0.792	1.750E-03	1.380E-03	13.4	33.5	1.02
	0.110	-0.771	3.630E-03	3.140E-03	13.4	31.8	1.02
	0.120	-0.752	6.800E-03	6.290E-03	9.7	20.8	1.01
	0.140	-0.719	1.900E-02	1.910E-02	9.7	19.8	1.01
	0.160	-0.690	4.280E-02	4.560E-02	9.7	19.3	1.01
	0.180	-0.664	8.370E-02	9.240E-02	3.0	17.5	1.01
	0.200	-0.641	1.480E-01	1.670E-01	3.0	17.2	1.01
	0.230	-0.611	3.040E-01	3.480E-01	3.0	17.0	1.01
	0.260	-0.584	5.540E-01	6.390E-01	3.0	12.3	1.01
	0.300	-0.553	1.090E+00	1.250E+00	1.6	9.8	1.01
	0.350	-0.520	2.190E+00	2.490E+00	1.6	9.9	1.01
	0.400	-0.491	3.940E+00	4.400E+00	1.6	5.8	1.01
	0.450	-0.465	6.520E+00	7.150E+00	1.4	5.3	1.01
	0.500	-0.442	1.010E+01	1.090E+01	1.4	5.4	1.01
	0.550	-0.421	1.500E+01	1.590E+01	1.4	5.4	1.01
	0.600	-0.403	2.140E+01	2.210E+01	1.4	5.5	1.01
	0.700	-0.369	3.960E+01	3.980E+01	1.4	5.3	1.01
	0.800	-0.340	6.650E+01	6.540E+01	1.4	5.4	1.01
	0.900	-0.314	1.040E+02	1.010E+02	1.4	5.4	1.01
	1.000	-0.292	1.540E+02	1.480E+02	1.4	2.2	1.01
	1.100	-0.271	2.180E+02	2.070E+02	1.8	2.4	1.01
	1.200	-0.252	2.970E+02	2.810E+02	1.8	2.5	1.01
	1.400	-0.219	5.060E+02	4.750E+02	1.8	2.5	1.01
	1.600	-0.190	7.870E+02	7.380E+02	1.8	2.5	1.01
	1.800	-0.164	1.140E+03	1.070E+03	2.7	3.3	1.01
	2.000	-0.141	1.570E+03	1.480E+03	2.7	3.3	1.02
	2.300	-0.111	2.350E+03	2.190E+03	2.7	3.3	1.02
	2.600	-0.084	3.260E+03	3.050E+03	2.7	3.3	1.02
	3.000	-0.053	4.660E+03	4.390E+03	2.5	3.0	1.02
	3.500	-0.020	6.620E+03	6.300E+03	2.5	3.0	1.02
	4.000	0.009	8.710E+03	8.350E+03	2.5	3.0	1.02
	4.500	0.035	1.090E+04	1.050E+04	2.9	3.5	1.02
	5.000	0.058	1.300E+04	1.260E+04	2.9	3.5	1.02
	5.500	0.079	1.510E+04	1.480E+04	2.9	3.4	1.02
	6.000	0.097	1.710E+04	1.690E+04	2.9	3.4	1.02
	7.000	0.131	2.080E+04	2.060E+04	2.8	3.8	1.02
	8.000	0.160	2.400E+04	2.400E+04	2.8	3.8	1.02
	9.000	0.186	2.690E+04	2.670E+04	2.8	3.8	1.02
	10.000	0.208	2.920E+04	2.920E+04	2.8	3.8	1.02
	11.000	0.229	3.130E+04	3.140E+04	2.2	3.2	1.02
	12.000	0.248	3.300E+04	3.320E+04	2.2	3.2	1.02
	14.000	0.281	3.550E+04	3.580E+04	2.2	3.2	1.02
	16.000	0.310	3.730E+04	3.770E+04	2.2	4.2	1.02
	18.000	0.336	3.840E+04	3.890E+04	2.7	3.5	1.02
17-Cl	20.000	0.359	3.900E+04	3.970E+04	2.7	3.5	1.01
	23.000	0.389	3.930E+04	4.020E+04	2.7	3.5	1.01
	26.000	0.416	3.920E+04	4.010E+04	2.7	3.5	1.01
	30.000	0.447	3.850E+04	3.930E+04	3.3	4.5	1.01
	35.000	0.480	3.730E+04	3.800E+04	3.3	4.5	1.01
	40.000	0.509	3.590E+04	3.640E+04	3.3	4.5	1.01
	45.000	0.535	3.450E+04	3.480E+04	4.6	5.8	1.00
	50.000	0.558	3.310E+04	3.320E+04	4.6	5.8	1.00
	55.000	0.579	3.170E+04	3.160E+04	4.6	5.8	1.00
	60.000	0.597	3.040E+04	3.020E+04	4.6	5.8	1.00
	70.000	0.631	2.810E+04	2.760E+04	6.6	16.1	1.00
	80.000	0.660	2.600E+04	2.540E+04	6.6	16.2	1.00
	90.000	0.686	2.430E+04	2.340E+04	6.6	16.3	1.00
	100.000	0.708	2.270E+04	2.170E+04	6.6	16.4	1.00
	110.000	0.729	2.130E+04	2.030E+04	7.7	11.8	1.00
	120.000	0.748	2.010E+04	1.900E+04	7.7	11.8	1.00
	140.000	0.781	1.810E+04	1.680E+04	7.7	11.9	1.00
18-Ar	0.110	-0.801	1.290E-03	9.710E-04	13.4	45.3	1.01
	0.120	-0.782	2.530E-03	2.090E-03	13.4	30.8	1.01
	0.140	-0.749	7.580E-03	7.090E-03	9.7	19.1	1.01
	0.160	-0.720	1.800E-02	1.820E-02	9.7	18.4	1.01
	0.180	-0.694	3.660E-02	3.880E-02	9.7	17.9	1.01
	0.200	-0.671	6.660E-02	7.280E-02	9.7	17.7	1.01
	0.230	-0.641	1.410E-01	1.590E-01	3.0	15.9	1.01
	0.260	-0.614	2.620E-01	3.000E-01	3.0	15.7	1.01
	0.300	-0.583	5.250E-01	6.050E-01	3.0	11.4	1.01
	0.350	-0.550	1.070E+00	1.230E+00	1.6	8.9	1.01
	0.400	-0.521	1.950E+00	2.220E+00	1.6	9.0	1.01
	0.450	-0.495	3.250E+00	3.650E+00	1.6	5.4	1.01
	0.500	-0.472	5.090E+00	5.620E+00	1.6	5.4	1.01
	0.550	-0.451	7.580E+00	8.230E+00	1.4	4.9	1.01
	0.600	-0.433	1.080E+01	1.160E+01	1.4	4.9	1.01
	0.700	-0.399	2.020E+01	2.080E+01	1.4	5.1	1.01
	0.800	-0.370	3.410E+01	3.430E+01	1.4	4.8	1.01
	0.900	-0.344	5.360E+01	5.280E+01	1.4	4.9	1.01
	1.000	-0.322	7.960E+01	7.760E+01	1.4	4.9	1.01
	1.100	-0.301	1.130E+02	1.090E+02	1.4	5.0	1.01
	1.200	-0.282	1.550E+02	1.480E+02	1.4	2.1	1.01
	1.400	-0.249	2.670E+02	2.520E+02	1.8	2.3	1.01
	1.600	-0.220	4.200E+02	3.940E+02	1.8	2.3	1.01
	1.800	-0.194	6.160E+02	5.780E+02	1.8	2.4	1.01
	2.000	-0.171	8.580E+02	8.040E+02	2.7	3.2	1.01
	2.300	-0.141	1.300E+03	1.220E+03	2.7	3.2	1.01
	2.600	-0.114	1.840E+03	1.710E+03	2.7	3.2	1.01
	3.000	-0.083	2.690E+03	2.510E+03	2.7	3.2	1.01
	3.500	-0.050	3.910E+03	3.680E+03	2.5	2.9	1.01
	4.000	-0.021	5.250E+03	5.000E+03	2.5	2.9	1.02
	4.500	0.005	6.670E+03	6.390E+03	2.5	2.9	1.02
	5.000	0.028	8.130E+03	7.830E+03	2.9	3.4	1.02
	5.500	0.049	9.580E+03	9.290E+03	2.9	3.4	1.02
	6.000	0.067	1.100E+04	1.070E+04	2.9	3.4	1.02
	7.000	0.101	1.370E+04	1.360E+04	2.9	4.0	1.02
	8.000	0.130	1.620E+04	1.610E+04	2.8	3.5	1.02
	9.000	0.156	1.840E+04	1.840E+04	2.8	3.5	1.02
	10.000	0.178	2.040E+04	2.030E+04	2.8	3.5	1.02
	11.000	0.199	2.200E+04	2.200E+04	2.8	3.5	1.02
	12.000	0.218	2.350E+04	2.350E+04	2.2	3.0	1.02
	14.000	0.251	2.580E+04	2.600E+04	2.2	3.0	1.02
	16.000	0.280	2.750E+04	2.770E+04	2.2	3.0	1.02
	18.000	0.306	2.870E+04	2.900E+04	2.2	3.7	1.02
	20.000	0.329	2.950E+04	2.990E+04	2.7	3.0	1.01
	23.000	0.359	3.020E+04	3.070E+04	2.7	3.0	1.01
	26.000	0.386	3.040E+04	3.110E+04	2.7	3.0	1.01
	30.000	0.417	3.030E+04	3.090E+04	3.3	4.0	1.01
	35.000	0.450	2.970E+04	3.020E+04	3.3	4.0	1.01
	40.000	0.479	2.880E+04	2.930E+04	3.3	4.0	1.01
	45.000	0.505	2.790E+04	2.820E+04	3.3	4.0	1.00
	50.000	0.528	2.690E+04	2.710E+04	4.6	5.3	1.00
	55.000	0.549	2.590E+04	2.600E+04	4.6	5.3	1.00
	60.000	0.567	2.490E+04	2.500E+04	4.6	5.3	1.00
	70.000	0.601	2.320E+04	2.300E+04	4.6	7.9	1.00
	80.000	0.630	2.160E+04	2.130E+04	6.6	13.6	1.00
	90.000	0.656	2.020E+04	1.970E+04	6.6	13.7	1.00
	100.000	0.678	1.900E+04	1.840E+04	6.6	13.7	1.00
	110.000	0.699	1.790E+04	1.720E+04	6.6	13.8	1.00
	120.000	0.718	1.690E+04	1.610E+04	7.7	9.2	1.00
	140.000	0.751	1.530E+04	1.440E+04	7.7	9.2	1.00
	160.000	0.780	1.390E+04	1.290E+04	7.7	9.2	1.00
19-K	0.120	-0.809	9.140E-04	6.560E-04	13.4	43.4	1.01
	0.140	-0.776	2.990E-03	2.530E-03	13.4	28.6	1.01

TABLE I. Empirical Reference Cross Sections for *K*-Shell Ionization by α Particles

See page 84 for Explanation of Tables

TARGET	ENERGY [MeV]	LOG XI	SIGMA [barn] THEORY	SIGMA [barn] REFERENCE	ERROR [%] RND	ERROR [%] TOT	COR
19-K	0.160	-0.747	7.550E-03	7.090E-03	9.7	17.4	1.00
	0.180	-0.721	1.610E-02	1.610E-02	9.7	16.9	1.00
	0.200	-0.699	3.020E-02	3.180E-02	9.7	16.6	1.00
	0.230	-0.668	6.630E-02	7.290E-02	9.7	16.3	1.00
	0.260	-0.642	1.270E-01	1.430E-01	3.0	14.6	1.00
	0.300	-0.611	2.600E-01	2.990E-01	3.0	14.4	1.00
	0.350	-0.577	5.430E-01	6.270E-01	3.0	10.5	1.00
	0.400	-0.548	1.000E+00	1.150E+00	1.6	8.0	1.01
	0.450	-0.522	1.690E+00	1.920E+00	1.6	8.1	1.01
	0.500	-0.500	2.660E+00	2.990E+00	1.6	4.9	1.01
	0.550	-0.479	3.980E+00	4.420E+00	1.6	4.9	1.01
	0.600	-0.460	5.720E+00	6.250E+00	1.4	4.4	1.01
	0.700	-0.427	1.070E+01	1.140E+01	1.4	4.5	1.01
	0.800	-0.398	1.820E+01	1.870E+01	1.4	4.6	1.01
	0.900	-0.372	2.880E+01	2.900E+01	1.4	4.3	1.01
	1.000	-0.349	4.300E+01	4.260E+01	1.4	4.4	1.01
	1.100	-0.328	6.140E+01	6.000E+01	1.4	4.4	1.01
	1.200	-0.309	8.440E+01	8.180E+01	1.4	4.4	1.01
	1.400	-0.276	1.470E+02	1.400E+02	1.8	2.2	1.01
	1.600	-0.247	2.330E+02	2.200E+02	1.8	2.2	1.01
	1.800	-0.221	3.450E+02	3.240E+02	1.8	2.2	1.01
	2.000	-0.199	4.840E+02	4.540E+02	1.8	2.2	1.01
	2.300	-0.168	7.460E+02	7.000E+02	2.7	3.1	1.01
	2.600	-0.142	1.070E+03	1.000E+03	2.7	3.1	1.01
	3.000	-0.111	1.590E+03	1.480E+03	2.7	3.1	1.01
	3.500	-0.077	2.360E+03	2.210E+03	2.5	2.8	1.01
	4.000	-0.048	3.230E+03	3.050E+03	2.5	2.8	1.01
	4.500	-0.022	4.180E+03	3.980E+03	2.5	2.8	1.01
	5.000	0.000	5.170E+03	4.950E+03	2.5	2.8	1.01
	5.500	0.021	6.190E+03	5.950E+03	2.9	3.3	1.01
	6.000	0.040	7.200E+03	6.970E+03	2.9	3.3	1.01
	7.000	0.073	9.190E+03	9.040E+03	2.9	3.2	1.02
	8.000	0.102	1.110E+04	1.090E+04	2.9	3.7	1.02
	9.000	0.128	1.280E+04	1.270E+04	2.8	3.3	1.02
	10.000	0.151	1.430E+04	1.430E+04	2.8	3.3	1.02
	11.000	0.172	1.570E+04	1.560E+04	2.8	3.3	1.02
	12.000	0.191	1.690E+04	1.690E+04	2.8	3.3	1.02
	14.000	0.224	1.900E+04	1.900E+04	2.2	2.7	1.02
	16.000	0.253	2.050E+04	2.060E+04	2.2	2.7	1.01
	18.000	0.279	2.170E+04	2.180E+04	2.2	2.7	1.01
	20.000	0.301	2.250E+04	2.270E+04	2.2	3.2	1.01
	23.000	0.332	2.330E+04	2.370E+04	2.7	2.8	1.01
	26.000	0.358	2.380E+04	2.420E+04	2.7	2.8	1.01
	30.000	0.389	2.390E+04	2.450E+04	2.7	2.8	1.01
	35.000	0.423	2.370E+04	2.430E+04	3.3	3.5	1.01
	40.000	0.452	2.330E+04	2.370E+04	3.3	3.5	1.01
	45.000	0.478	2.270E+04	2.310E+04	3.3	3.5	1.00
	50.000	0.500	2.200E+04	2.230E+04	3.3	3.5	1.00
	55.000	0.521	2.130E+04	2.160E+04	3.3	3.5	1.00
	60.000	0.540	2.070E+04	2.080E+04	4.6	4.8	1.00
	70.000	0.573	1.930E+04	1.930E+04	4.6	4.8	1.00
	80.000	0.602	1.810E+04	1.800E+04	4.6	5.4	1.00
	90.000	0.628	1.700E+04	1.680E+04	6.6	11.1	1.00
	100.000	0.651	1.600E+04	1.570E+04	6.6	11.1	1.00
	110.000	0.672	1.520E+04	1.470E+04	6.6	11.1	1.00
	120.000	0.691	1.440E+04	1.380E+04	6.6	11.2	1.00
	140.000	0.724	1.300E+04	1.240E+04	7.7	8.9	1.00
	160.000	0.753	1.190E+04	1.120E+04	7.7	9.0	1.00
	180.000	0.779	1.100E+04	1.020E+04	7.7	9.0	1.00
20-Ca	0.140	-0.802	1.180E-03	8.840E-04	13.4	38.8	1.00
	0.160	-0.773	3.190E-03	2.730E-03	13.4	26.7	1.00
	0.180	-0.748	7.120E-03	6.670E-03	9.7	15.8	1.00
	0.200	-0.725	1.390E-02	1.380E-02	9.7	15.5	1.00
	0.230	-0.694	3.180E-02	3.360E-02	9.7	15.1	1.00
	0.260	-0.668	6.260E-02	6.880E-02	9.7	14.9	1.00
	0.300	-0.637	1.320E-01	1.490E-01	3.0	13.2	1.00
	0.350	-0.603	2.820E-01	3.240E-01	3.0	13.0	1.00
	0.400	-0.574	5.270E-01	6.080E-01	3.0	9.7	1.00
	0.450	-0.549	8.980E-01	1.030E+00	1.6	7.2	1.00
	0.500	-0.526	1.430E+00	1.630E+00	1.6	7.2	1.00
	0.550	-0.505	2.150E+00	2.430E+00	1.6	7.3	1.00
	0.600	-0.486	3.110E+00	3.460E+00	1.6	4.5	1.00
	0.700	-0.453	5.860E+00	6.370E+00	1.4	3.9	1.00
	0.800	-0.424	1.000E+01	1.060E+01	1.4	4.0	1.01
	0.900	-0.398	1.590E+01	1.640E+01	1.4	4.1	1.01
	1.000	-0.375	2.390E+01	2.410E+01	1.4	4.1	1.01
	1.100	-0.355	3.420E+01	3.400E+01	1.4	3.8	1.01
	1.200	-0.336	4.730E+01	4.630E+01	1.4	3.9	1.01
	1.400	-0.302	8.260E+01	7.970E+01	1.4	3.9	1.01
	1.600	-0.273	1.320E+02	1.260E+02	1.8	2.1	1.01
	1.800	-0.248	1.970E+02	1.860E+02	1.8	2.1	1.01

TARGET	ENERGY [MeV]	LOG XI	SIGMA [barn] THEORY	SIGMA [barn] REFERENCE	ERROR [%] RND	ERROR [%] TOT	COR
20-Ca	2.000	-0.225	2.790E+02	2.620E+02	1.8	2.1	1.01
	2.300	-0.194	4.360E+02	4.080E+02	1.8	2.1	1.01
	2.600	-0.168	6.320E+02	5.930E+02	2.7	3.0	1.01
	3.000	-0.137	9.530E+02	8.960E+02	2.7	3.0	1.01
	3.500	-0.103	1.440E+03	1.340E+03	2.7	3.0	1.01
	4.000	-0.074	2.010E+03	1.880E+03	2.5	2.7	1.01
	4.500	-0.049	2.640E+03	2.490E+03	2.5	2.7	1.01
	5.000	-0.026	3.320E+03	3.160E+03	2.5	2.7	1.01
	5.500	-0.005	4.020E+03	3.840E+03	2.5	2.7	1.01
	6.000	0.014	4.740E+03	4.550E+03	2.5	2.7	1.01
	7.000	0.047	6.180E+03	5.990E+03	2.9	3.1	1.01
	8.000	0.076	7.580E+03	7.460E+03	2.9	3.1	1.01
	9.000	0.102	8.890E+03	8.790E+03	2.9	3.5	1.01
	10.000	0.125	1.010E+04	1.000E+04	2.8	3.0	1.01
	11.000	0.145	1.120E+04	1.120E+04	2.8	3.0	1.01
	12.000	0.164	1.220E+04	1.220E+04	2.8	3.0	1.01
	14.000	0.198	1.390E+04	1.390E+04	2.8	3.0	1.01
	16.000	0.227	1.530E+04	1.530E+04	2.2	2.5	1.01
	18.000	0.252	1.640E+04	1.650E+04	2.2	2.5	1.01
	20.000	0.275	1.720E+04	1.730E+04	2.2	2.5	1.01
	23.000	0.306	1.810E+04	1.830E+04	2.2	2.7	1.01
	26.000	0.332	1.860E+04	1.890E+04	2.7	3.3	1.01
	30.000	0.363	1.900E+04	1.940E+04	2.7	3.3	1.01
	35.000	0.397	1.910E+04	1.950E+04	2.7	3.3	1.01
	40.000	0.426	1.890E+04	1.930E+04	3.3	3.6	1.01
	45.000	0.451	1.850E+04	1.890E+04	3.3	3.6	1.00
	50.000	0.474	1.810E+04	1.840E+04	3.3	3.6	1.00
	55.000	0.495	1.760E+04	1.790E+04	3.3	3.6	1.00
	60.000	0.514	1.710E+04	1.730E+04	3.3	3.6	1.00
	70.000	0.547	1.620E+04	1.630E+04	4.6	5.0	1.00
	80.000	0.576	1.520E+04	1.520E+04	4.6	5.0	1.00
	90.000	0.602	1.440E+04	1.430E+04	4.6	6.4	1.00
	100.000	0.625	1.360E+04	1.340E+04	6.6	8.5	1.00
	110.000	0.645	1.290E+04	1.260E+04	6.6	8.5	1.00
	120.000	0.664	1.230E+04	1.190E+04	6.6	8.5	1.00
	140.000	0.698	1.110E+04	1.070E+04	6.6	8.6	1.00
	160.000	0.727	1.020E+04	9.700E+03	7.7	11.6	1.00
	180.000	0.752	9.430E+03	8.860E+03	7.7	11.6	1.00
	200.000	0.775	8.760E+03	8.140E+03	7.7	11.7	1.00
21-Sc	0.160	-0.798	1.380E-03	1.060E-03	13.4	26.3	1.00
	0.180	-0.772	3.240E-03	2.790E-03	13.4	24.9	1.00
	0.200	-0.749	6.580E-03	6.130E-03	9.7	14.2	1.00
	0.230	-0.719	1.570E-02	1.580E-02	9.7	13.9	1.00
	0.260	-0.692	3.190E-02	3.380E-02	9.7	13.7	1.00
	0.300	-0.661	6.910E-02	7.640E-02	3.0	12.1	1.00
	0.350	-0.628	1.510E-01	1.720E-01	3.0	11.8	1.00
	0.400	-0.599	2.870E-01	3.300E-01	3.0	8.8	1.00
	0.450	-0.573	4.950E-01	5.710E-01	3.0	8.8	1.00
	0.500	-0.550	7.930E-01	9.120E-01	1.6	6.3	1.00
	0.550	-0.530	1.200E+00	1.370E+00	1.6	6.3	1.00
	0.600	-0.511	1.750E+00	1.980E+00	1.6	6.4	1.00
	0.700	-0.477	3.320E+00	3.680E+00	1.6	4.0	1.00
	0.800	-0.448	5.710E+00	6.180E+00	1.4	3.5	1.00
	0.900	-0.423	9.110E+00	9.630E+00	1.4	3.5	1.00
	1.000	-0.400	1.370E+01	1.410E+01	1.4	3.6	1.00
	1.100	-0.379	1.970E+01	2.000E+01	1.4	3.6	1.00
	1.200	-0.360	2.740E+01	2.730E+01	1.4	3.3	1.00
	1.400	-0.327	4.820E+01	4.710E+01	1.4	3.4	1.01
	1.600	-0.298	7.750E+01	7.460E+01	1.4	1.8	1.01
	1.800	-0.272	1.160E+02	1.110E+02	1.8	2.0	1.01
	2.000	-0.249	1.660E+02	1.570E+02	1.8	2.0	1.01
	2.300	-0.219	2.610E+02	2.450E+02	1.8	2.0	1.01
	2.600	-0.192	3.830E+02	3.590E+02	1.8	2.0	1.01
	3.000	-0.161	5.860E+02	5.490E+02	2.7	2.9	1.01
	3.500	-0.128	9.010E+02	8.470E+02	2.7	2.9	1.01
	4.000	-0.099	1.280E+03	1.190E+03	2.7	2.9	1.01
	4.500	-0.073	1.700E+03	1.590E+03	2.5	2.6	1.01
	5.000	-0.050	2.160E+03	2.040E+03	2.5	2.6	1.01
	5.500	-0.030	2.650E+03	2.520E+03	2.5	2.6	1.01
	6.000	-0.011	3.160E+03	3.020E+03	2.5	2.6	1.01
	7.000	0.023	4.210E+03	4.050E+03	2.9	3.0	1.01
	8.000	0.052	5.250E+03	5.090E+03	2.9	3.0	1.01
	9.000	0.077	6.250E+03	6.150E+03	2.9	3.0	1.01
	10.000	0.100	7.200E+03	7.120E+03	2.9	3.2	1.01
	11.000	0.121	8.090E+03	8.020E+03	2.8	2.9	1.01
	12.000	0.140	8.910E+03	8.860E+03	2.8	2.9	1.01
	14.000	0.173	1.030E+04	1.030E+04	2.8	2.9	1.01
	16.000	0.202	1.150E+04	1.150E+04	2.8	2.9	1.01
	18.000	0.228	1.250E+04	1.250E+04	2.2	2.2	1.01
	20.000	0.251	1.320E+04	1.330E+04	2.2	2.2	1.01

TABLE I. Empirical Reference Cross Sections for *K*-Shell Ionization by α Particles

See page 84 for Explanation of Tables

TARGET	ENERGY [MeV]	LOG XI	SIGMA [barn] THEORY	SIGMA [barn] REFERENCE	ERROR [%] RND	ERROR [%] TOT	COR
21-Sc	23.000	0.281	1.410E+04	1.420E+04	2.2	2.2	1.01
	26.000	0.308	1.470E+04	1.480E+04	2.2	2.3	1.01
	30.000	0.339	1.510E+04	1.540E+04	2.7	3.8	1.01
	35.000	0.372	1.540E+04	1.570E+04	2.7	3.8	1.01
	40.000	0.401	1.540E+04	1.570E+04	2.7	3.8	1.01
	45.000	0.427	1.520E+04	1.550E+04	3.3	4.1	1.00
	50.000	0.450	1.500E+04	1.530E+04	3.3	4.1	1.00
	55.000	0.470	1.470E+04	1.490E+04	3.3	4.1	1.00
	60.000	0.489	1.430E+04	1.450E+04	3.3	4.1	1.00
	70.000	0.523	1.360E+04	1.370E+04	4.6	5.5	1.00
	80.000	0.552	1.290E+04	1.300E+04	4.6	5.5	1.00
	90.000	0.577	1.220E+04	1.220E+04	4.6	5.5	1.00
	100.000	0.600	1.160E+04	1.150E+04	4.6	8.9	1.00
	110.000	0.621	1.100E+04	1.090E+04	6.6	7.3	1.00
	120.000	0.640	1.050E+04	1.030E+04	6.6	7.3	1.00
	140.000	0.673	9.600E+03	9.320E+03	6.6	7.3	1.00
	160.000	0.702	8.830E+03	8.470E+03	6.6	7.3	1.00
	180.000	0.728	8.170E+03	7.760E+03	7.7	14.2	1.00
	200.000	0.751	7.610E+03	7.150E+03	7.7	14.3	1.00
	230.000	0.781	6.900E+03	6.390E+03	7.7	14.4	1.00
22-Ti	0.180	-0.795	1.460E-03	1.130E-03	13.4	24.2	1.00
	0.200	-0.773	3.090E-03	2.660E-03	13.4	23.2	1.00
	0.230	-0.742	7.740E-03	7.370E-03	9.7	12.6	1.00
	0.260	-0.716	1.630E-02	1.660E-02	9.7	12.4	1.00
	0.300	-0.685	3.650E-02	3.920E-02	9.7	12.2	1.00
	0.350	-0.651	8.200E-02	9.160E-02	3.0	10.6	1.00
	0.400	-0.622	1.590E-01	1.810E-01	3.0	10.5	1.00
	0.450	-0.597	2.770E-01	3.190E-01	3.0	7.9	1.00
	0.500	-0.574	4.490E-01	5.180E-01	3.0	7.9	1.00
	0.550	-0.553	6.870E-01	7.900E-01	1.6	5.4	1.00
	0.600	-0.534	1.000E+00	1.150E+00	1.6	5.4	1.00
	0.700	-0.501	1.920E+00	2.160E+00	1.6	5.5	1.00
	0.800	-0.472	3.330E+00	3.670E+00	1.6	3.6	1.00
	0.900	-0.446	5.340E+00	5.760E+00	1.4	3.0	1.00
	1.000	-0.423	8.070E+00	8.540E+00	1.4	3.1	1.00
	1.100	-0.402	1.170E+01	1.200E+01	1.4	3.1	1.00
	1.200	-0.384	1.620E+01	1.650E+01	1.4	3.1	1.00
	1.400	-0.350	2.870E+01	2.840E+01	1.4	2.8	1.00
	1.600	-0.321	4.650E+01	4.530E+01	1.4	2.9	1.00
	1.800	-0.295	7.020E+01	6.750E+01	1.4	1.7	1.00
	2.000	-0.273	1.010E+02	9.590E+01	1.8	1.9	1.00
	2.300	-0.242	1.600E+02	1.510E+02	1.8	1.9	1.01
	2.600	-0.216	2.360E+02	2.220E+02	1.8	1.9	1.01
	3.000	-0.185	3.660E+02	3.430E+02	1.8	1.9	1.01
	3.500	-0.151	5.700E+02	5.350E+02	2.7	2.8	1.01
	4.000	-0.122	8.190E+02	7.610E+02	2.7	2.8	1.01
	4.500	-0.097	1.110E+03	1.030E+03	2.7	2.8	1.01
	5.000	-0.074	1.420E+03	1.330E+03	2.5	2.6	1.01
	5.500	-0.053	1.770E+03	1.660E+03	2.5	2.6	1.01
	6.000	-0.034	2.130E+03	2.010E+03	2.5	2.6	1.01
	7.000	-0.001	2.890E+03	2.760E+03	2.5	2.6	1.01
	8.000	0.028	3.660E+03	3.530E+03	2.9	2.9	1.01
	9.000	0.054	4.420E+03	4.290E+03	2.9	2.9	1.01
	10.000	0.077	5.160E+03	5.080E+03	2.9	2.9	1.01
	11.000	0.098	5.860E+03	5.790E+03	2.9	2.9	1.01
	12.000	0.116	6.510E+03	6.450E+03	2.9	3.0	1.01
	14.000	0.150	7.690E+03	7.660E+03	2.8	3.2	1.01
	16.000	0.179	8.690E+03	8.640E+03	2.8	3.2	1.01
	18.000	0.205	9.520E+03	9.510E+03	2.8	3.2	1.01
	20.000	0.227	1.020E+04	1.020E+04	2.2	2.5	1.01
	23.000	0.258	1.100E+04	1.110E+04	2.2	2.5	1.01
	26.000	0.284	1.160E+04	1.170E+04	2.2	2.5	1.01
	30.000	0.315	1.210E+04	1.230E+04	2.7	4.3	1.01
	35.000	0.349	1.250E+04	1.270E+04	2.7	4.3	1.01
	40.000	0.378	1.260E+04	1.290E+04	2.7	4.2	1.01
	45.000	0.403	1.250E+04	1.280E+04	2.7	4.2	1.00
	50.000	0.426	1.240E+04	1.270E+04	3.3	4.6	1.00
	55.000	0.447	1.220E+04	1.250E+04	3.3	4.6	1.00
	60.000	0.466	1.200E+04	1.220E+04	3.3	4.6	1.00
	70.000	0.499	1.150E+04	1.170E+04	3.3	4.6	1.00
	80.000	0.528	1.100E+04	1.110E+04	4.6	6.0	1.00
	90.000	0.554	1.050E+04	1.050E+04	4.6	6.0	1.00
	100.000	0.577	9.960E+03	9.950E+03	4.6	6.0	1.00
	110.000	0.598	9.500E+03	9.440E+03	4.6	6.0	1.00
	120.000	0.616	9.080E+03	8.970E+03	6.6	9.8	1.00
	140.000	0.650	8.320E+03	8.140E+03	6.6	9.8	1.00
	160.000	0.679	7.670E+03	7.430E+03	6.6	9.9	1.00
	180.000	0.705	7.120E+03	6.830E+03	6.6	9.9	1.00
	200.000	0.727	6.640E+03	6.310E+03	7.7	16.8	1.00
	230.000	0.758	6.040E+03	5.660E+03	7.7	17.0	1.00

TARGET	ENERGY [MeV]	LOG XI	SIGMA [barn] THEORY	SIGMA [barn] REFERENCE	ERROR [%] RND	ERROR [%] TOT	COR
22-Ti	260.000	0.784	5.540E+03	5.120E+03	7.7	17.1	1.00
23-V	0.180	-0.818	6.570E-04	4.480E-04	13.4	30.3	1.00
	0.200	-0.795	1.460E-03	1.130E-03	13.4	22.3	1.00
	0.230	-0.764	3.850E-03	3.410E-03	13.4	21.2	1.00
	0.260	-0.738	8.400E-03	8.090E-03	9.7	11.0	1.00
	0.300	-0.707	1.950E-02	2.020E-02	9.7	10.9	1.00
	0.350	-0.673	4.520E-02	4.930E-02	9.7	10.8	1.00
	0.400	-0.644	8.940E-02	1.010E-01	3.0	9.3	1.00
	0.450	-0.619	1.590E-01	1.810E-01	3.0	9.2	1.00
	0.500	-0.596	2.600E-01	2.990E-01	3.0	7.1	1.00
	0.550	-0.575	4.000E-01	4.610E-01	3.0	7.1	1.00
	0.600	-0.556	5.820E-01	6.770E-01	1.6	4.6	1.00
	0.700	-0.523	1.140E+00	1.300E+00	1.6	4.6	1.00
	0.800	-0.494	1.990E+00	2.220E+00	1.6	3.1	1.00
	0.900	-0.468	3.200E+00	3.520E+00	1.6	3.2	1.00
	1.000	-0.445	4.860E+00	5.250E+00	1.4	2.6	1.00
	1.100	-0.425	7.050E+00	7.470E+00	1.4	2.6	1.00
	1.200	-0.406	9.840E+00	1.020E+01	1.4	2.6	1.00
	1.400	-0.372	1.750E+01	1.770E+01	1.4	2.3	1.00
	1.600	-0.343	2.850E+01	2.810E+01	1.4	2.3	1.00
	1.800	-0.318	4.330E+01	4.210E+01	1.4	2.4	1.00
	2.000	-0.295	6.240E+01	6.000E+01	1.4	1.6	1.00
	2.300	-0.264	9.980E+01	9.480E+01	1.8	1.8	1.00
	2.600	-0.238	1.490E+02	1.400E+02	1.8	1.8	1.00
	3.000	-0.207	2.320E+02	2.180E+02	1.8	1.8	1.00
	3.500	-0.173	3.670E+02	3.440E+02	2.7	2.7	1.01
	4.000	-0.144	5.330E+02	5.010E+02	2.7	2.7	1.01
	4.500	-0.119	7.280E+02	6.770E+02	2.7	2.7	1.01
	5.000	-0.096	9.480E+02	8.840E+02	2.7	2.7	1.01
	5.500	-0.075	1.190E+03	1.110E+03	2.5	2.7	1.01
	6.000	-0.056	1.450E+03	1.360E+03	2.5	2.7	1.01
	7.000	-0.023	1.990E+03	1.900E+03	2.5	2.7	1.01
	8.000	0.006	2.570E+03	2.460E+03	2.5	2.7	1.01
	9.000	0.032	3.150E+03	3.040E+03	2.9	3.0	1.01
	10.000	0.055	3.710E+03	3.610E+03	2.9	3.0	1.01
	11.000	0.075	4.260E+03	4.200E+03	2.9	3.0	1.01
	12.000	0.094	4.790E+03	4.730E+03	2.9	3.0	1.01
	14.000	0.128	5.750E+03	5.710E+03	2.8	3.4	1.01
	16.000	0.157	6.580E+03	6.570E+03	2.8	3.4	1.01
	18.000	0.182	7.300E+03	7.270E+03	2.8	3.4	1.01
	20.000	0.205	7.910E+03	7.900E+03	2.8	3.4	1.01
	23.000	0.236	8.640E+03	8.670E+03	2.2	2.7	1.01
	26.000	0.262	9.190E+03	9.260E+03	2.2	2.7	1.01
	30.000	0.293	9.720E+03	9.790E+03	2.2	2.7	1.01
	35.000	0.327	1.010E+04	1.020E+04	2.7	4.8	1.01
	40.000	0.356	1.030E+04	1.050E+04	2.7	4.7	1.01
	45.000	0.381	1.040E+04	1.060E+04	2.7	4.7	1.00
	50.000	0.404	1.030E+04	1.060E+04	2.7	4.7	1.00
	55.000	0.425	1.020E+04	1.050E+04	3.3	5.1	1.00
	60.000	0.444	1.010E+04	1.030E+04	3.3	5.1	1.00
	70.000	0.477	9.750E+03	9.920E+03	3.3	5.1	1.00
	80.000	0.506	9.360E+03	9.480E+03	3.3	5.1	1.00
	90.000	0.532	8.970E+03	9.040E+03	4.6	6.5	1.00
	100.000	0.555	8.580E+03	8.610E+03	4.6	6.5	1.00
	110.000	0.575	8.210E+03	8.200E+03	4.6	6.5	1.00
	120.000	0.594	7.870E+03	7.820E+03	4.6	6.5	1.00
	140.000	0.628	7.240E+03	7.130E+03	6.6	12.4	1.00
	160.000	0.657	6.700E+03	6.540E+03	6.6	12.4	1.00
	180.000	0.682	6.230E+03	6.030E+03	6.6	12.5	1.00
	200.000	0.705	5.830E+03	5.590E+03	6.6	12.5	1.00
	230.000	0.736	5.310E+03	5.030E+03	7.7	19.5	1.00
	260.000	0.762	4.880E+03	4.560E+03	7.7	19.7	1.00
	300.000	0.793	4.410E+03	4.060E+03	7.7	19.8	1.00
24-Cr	0.200	-0.816	6.790E-04	4.680E-04	13.4	26.5	1.00
	0.230	-0.785	1.900E-03	1.540E-03	13.4	20.0	1.00
	0.260	-0.759	4.320E-03	3.910E-03	9.7	9.9	1.00
	0.300	-0.728	1.050E-02	1.030E-02	9.7	9.9	1.00
	0.350	-0.694	2.510E-02	2.660E-02	9.7	9.9	1.00
	0.400	-0.665	5.090E-02	5.600E-02	3.0	8.0	1.00
	0.450	-0.640	9.180E-02	1.040E-01	3.0	7.9	1.00
	0.500	-0.617	1.520E-01	1.740E-01	3.0	7.8	1.00
	0.550	-0.596	2.370E-01	2.730E-01	3.0	6.2	1.00
	0.600	-0.577	3.510E-01	4.040E-01	3.0	6.2	1.00
	0.700	-0.544	6.870E-01	7.880E-01	1.6	3.7	1.00
	0.800	-0.515	1.210E+00	1.370E+00	1.6	3.7	1.00
	0.900	-0.489	1.960E+00	2.180E+00	1.6	2.7	1.00
	1.000	-0.466	2.980E+00	3.280E+00	1.6	2.7	1.00
	1.100	-0.446	4.340E+00	4.690E+00	1.4	2.1	1.00
	1.200	-0.427	6.080E+00	6.460E+00	1.4	2.1	1.00

TABLE I. Empirical Reference Cross Sections for *K*-Shell Ionization by α Particles

See page 84 for Explanation of Tables

TARGET	ENERGY [MeV]	LOG XI	SIGMA [barn] THEORY	SIGMA [barn] REFERENCE	ERROR [%] RND	ERROR [%] TOT	COR
24-Cr	1.400	-0.393	1.090E+01	1.120E+01	1.4	2.1	1.00
	1.600	-0.364	1.780E+01	1.780E+01	1.4	1.8	1.00
	1.800	-0.339	2.720E+01	2.670E+01	1.4	1.8	1.00
	2.000	-0.316	3.940E+01	3.820E+01	1.4	1.8	1.00
	2.300	-0.285	6.340E+01	6.070E+01	1.4	1.5	1.00
	2.600	-0.259	9.510E+01	9.010E+01	1.8	1.8	1.00
	3.000	-0.228	1.500E+02	1.410E+02	1.8	1.8	1.00
	3.500	-0.194	2.390E+02	2.240E+02	1.8	1.8	1.00
	4.000	-0.165	3.520E+02	3.300E+02	2.7	2.8	1.00
	4.500	-0.140	4.850E+02	4.560E+02	2.7	2.8	1.00
	5.000	-0.117	6.380E+02	5.930E+02	2.7	2.8	1.00
	5.500	-0.096	8.070E+02	7.530E+02	2.7	2.8	1.01
	6.000	-0.077	9.910E+02	9.270E+02	2.5	2.8	1.01
	7.000	-0.044	1.390E+03	1.310E+03	2.5	2.8	1.01
	8.000	-0.015	1.820E+03	1.730E+03	2.5	2.8	1.01
	9.000	0.011	2.250E+03	2.160E+03	2.5	2.8	1.01
	10.000	0.034	2.690E+03	2.600E+03	2.9	3.1	1.01
	11.000	0.054	3.120E+03	3.030E+03	2.9	3.1	1.01
	12.000	0.073	3.540E+03	3.480E+03	2.9	3.1	1.01
	14.000	0.107	4.320E+03	4.270E+03	2.9	3.4	1.01
	16.000	0.136	5.010E+03	4.980E+03	2.8	3.7	1.01
	18.000	0.161	5.620E+03	5.610E+03	2.8	3.7	1.01
	20.000	0.184	6.150E+03	6.120E+03	2.8	3.7	1.01
	23.000	0.215	6.800E+03	6.800E+03	2.8	3.7	1.01
	26.000	0.241	7.310E+03	7.350E+03	2.2	3.0	1.01
	30.000	0.272	7.820E+03	7.850E+03	2.2	3.0	1.01
	35.000	0.306	8.230E+03	8.320E+03	2.2	3.7	1.01
	40.000	0.335	8.470E+03	8.600E+03	2.7	5.2	1.01
	45.000	0.360	8.590E+03	8.780E+03	2.7	5.2	1.00
	50.000	0.383	8.620E+03	8.820E+03	2.7	5.2	1.00
	55.000	0.404	8.590E+03	8.780E+03	2.7	5.2	1.00
	60.000	0.423	8.520E+03	8.700E+03	3.3	5.6	1.00
	70.000	0.456	8.290E+03	8.450E+03	3.3	5.6	1.00
	80.000	0.485	8.010E+03	8.140E+03	3.3	5.6	1.00
	90.000	0.511	7.710E+03	7.810E+03	3.3	5.6	1.00
	100.000	0.534	7.410E+03	7.470E+03	4.6	7.0	1.00
	110.000	0.554	7.120E+03	7.150E+03	4.6	7.0	1.00
	120.000	0.573	6.840E+03	6.840E+03	4.6	7.0	1.00
	140.000	0.607	6.330E+03	6.270E+03	4.6	16.5	1.00
	160.000	0.636	5.860E+03	5.770E+03	6.6	14.9	1.00
	180.000	0.661	5.480E+03	5.340E+03	6.6	15.0	1.00
	200.000	0.684	5.130E+03	4.960E+03	6.6	15.0	1.00
	230.000	0.715	4.690E+03	4.480E+03	7.7	22.0	1.00
	260.000	0.741	4.320E+03	4.080E+03	7.7	22.2	1.00
	300.000	0.772	3.910E+03	3.640E+03	7.7	22.4	1.00
25-Mn	0.230	-0.805	9.430E-04	6.930E-04	13.4	22.3	1.00
	0.260	-0.779	2.250E-03	1.880E-03	13.4	18.1	1.00
	0.300	-0.748	5.680E-03	5.320E-03	9.7	11.5	1.00
	0.350	-0.714	1.420E-02	1.440E-02	9.7	11.4	1.00
	0.400	-0.685	2.940E-02	3.160E-02	9.7	11.3	1.00
	0.450	-0.660	5.400E-02	5.990E-02	3.0	6.6	1.00
	0.500	-0.637	9.080E-02	1.030E-01	3.0	6.6	1.00
	0.550	-0.616	1.430E-01	1.630E-01	3.0	6.5	1.00
	0.600	-0.597	2.130E-01	2.450E-01	3.0	5.3	1.00
	0.700	-0.564	4.230E-01	4.870E-01	3.0	5.3	1.00
	0.800	-0.535	7.490E-01	8.570E-01	1.6	2.8	1.00
	0.900	-0.509	1.220E+00	1.380E+00	1.6	2.8	1.00
	1.000	-0.486	1.870E+00	2.090E+00	1.6	2.2	1.00
	1.100	-0.466	2.740E+00	3.000E+00	1.4	1.6	1.00
	1.200	-0.447	3.840E+00	4.150E+00	1.4	1.6	1.00
	1.400	-0.413	6.920E+00	7.220E+00	1.4	1.7	1.00
	1.600	-0.384	1.140E+01	1.160E+01	1.4	1.7	1.00
	1.800	-0.359	1.740E+01	1.740E+01	1.4	1.5	1.00
	2.000	-0.336	2.540E+01	2.480E+01	1.4	1.5	1.00
	2.300	-0.305	4.110E+01	3.970E+01	1.4	1.5	1.00
	2.600	-0.279	6.190E+01	5.920E+01	1.4	1.4	1.00
	3.000	-0.248	9.840E+01	9.290E+01	1.8	1.9	1.00
	3.500	-0.214	1.590E+02	1.490E+02	1.8	1.9	1.00
	4.000	-0.185	2.350E+02	2.210E+02	1.8	1.9	1.00
	4.500	-0.160	3.280E+02	3.070E+02	2.7	2.9	1.00
	5.000	-0.137	4.350E+02	4.080E+02	2.7	2.9	1.00
	5.500	-0.116	5.550E+02	5.160E+02	2.7	2.9	1.00
	6.000	-0.097	6.860E+02	6.400E+02	2.7	2.9	1.00
	7.000	-0.064	9.770E+02	9.170E+02	2.5	2.9	1.00
	8.000	-0.035	1.290E+03	1.220E+03	2.5	2.9	1.01
	9.000	-0.009	1.620E+03	1.550E+03	2.5	2.9	1.01
	10.000	0.014	1.960E+03	1.880E+03	2.5	2.9	1.01
	11.000	0.034	2.300E+03	2.220E+03	2.9	3.2	1.01
	12.000	0.053	2.630E+03	2.550E+03	2.9	3.2	1.01
	14.000	0.087	3.260E+03	3.210E+03	2.9	3.2	1.01
	16.000	0.116	3.830E+03	3.800E+03	2.9	3.6	1.01

TARGET	ENERGY [MeV]	LOG XI	SIGMA [barn] THEORY	SIGMA [barn] REFERENCE	ERROR [%] RND	ERROR [%] TOT	COR
25-Mn	18.000	0.141	4.340E+03	4.320E+03	2.8	3.9	1.01
	20.000	0.164	4.800E+03	4.790E+03	2.8	3.9	1.01
	23.000	0.195	5.370E+03	5.360E+03	2.8	3.9	1.01
	26.000	0.221	5.840E+03	5.850E+03	2.2	3.2	1.01
	30.000	0.252	6.310E+03	6.350E+03	2.2	3.2	1.01
	35.000	0.286	6.730E+03	6.770E+03	2.2	3.2	1.01
	40.000	0.315	6.990E+03	7.070E+03	2.7	5.7	1.01
	45.000	0.340	7.140E+03	7.250E+03	2.7	5.7	1.00
	50.000	0.363	7.210E+03	7.380E+03	2.7	5.7	1.00
	55.000	0.384	7.230E+03	7.390E+03	2.7	5.7	1.00
	60.000	0.403	7.200E+03	7.370E+03	2.7	5.7	1.00
	70.000	0.436	7.070E+03	7.220E+03	3.3	6.1	1.00
	80.000	0.465	6.880E+03	7.010E+03	3.3	6.1	1.00
	90.000	0.491	6.660E+03	6.760E+03	3.3	6.1	1.00
	100.000	0.514	6.430E+03	6.500E+03	3.3	6.1	1.00
	110.000	0.534	6.200E+03	6.250E+03	4.6	7.5	1.00
	120.000	0.553	5.970E+03	6.000E+03	4.6	7.5	1.00
	140.000	0.587	5.550E+03	5.530E+03	4.6	7.5	1.00
	160.000	0.616	5.170E+03	5.110E+03	6.6	17.4	1.00
	180.000	0.641	4.840E+03	4.750E+03	6.6	17.5	1.00
	200.000	0.664	4.540E+03	4.420E+03	6.6	17.6	1.00
	230.000	0.695	4.160E+03	4.010E+03	6.6	17.7	1.00
	260.000	0.721	3.840E+03	3.660E+03	7.7	24.7	1.00
	300.000	0.752	3.480E+03	3.270E+03	7.7	24.9	1.00
	350.000	0.786	3.120E+03	2.890E+03	7.7	25.2	1.00
26-Fe	0.260	-0.798	1.160E-03	8.060E-04	13.4	16.9	1.00
	0.300	-0.767	3.070E-03	2.470E-03	13.4	16.4	1.00
	0.350	-0.734	7.980E-03	7.160E-03	8.0	20.3	1.00
	0.400	-0.705	1.710E-02	1.640E-02	8.0	19.5	1.00
	0.450	-0.679	3.210E-02	3.230E-02	8.0	19.0	1.00
	0.500	-0.656	5.470E-02	5.680E-02	2.9	10.2	1.00
	0.550	-0.635	8.710E-02	9.230E-02	2.9	10.1	1.00
	0.600	-0.617	1.310E-01	1.410E-01	2.9	10.0	1.00
	0.700	-0.583	2.640E-01	2.880E-01	2.9	7.5	1.00
	0.800	-0.554	4.710E-01	5.160E-01	1.7	4.2	1.00
	0.900	-0.529	7.730E-01	8.440E-01	1.7	4.2	1.00
	1.000	-0.506	1.190E+00	1.290E+00	1.7	4.2	1.00
	1.100	-0.485	1.750E+00	1.880E+00	1.7	3.0	1.00
	1.200	-0.466	2.470E+00	2.620E+00	1.2	2.0	1.00
	1.400	-0.433	4.470E+00	4.660E+00	1.2	2.0	1.00
	1.600	-0.404	7.370E+00	7.540E+00	1.2	2.1	1.00
	1.800	-0.378	1.140E+01	1.140E+01	1.2	2.1	1.00
	2.000	-0.355	1.660E+01	1.640E+01	1.3	1.7	1.00
	2.300	-0.325	2.700E+01	2.630E+01	1.3	1.7	1.00
	2.600	-0.298	4.090E+01	3.950E+01	1.3	1.4	1.00
	3.000	-0.267	6.550E+01	6.250E+01	1.5	1.6	1.00
	3.500	-0.234	1.060E+02	1.010E+02	1.5	1.6	1.00
	4.000	-0.205	1.590E+02	1.500E+02	1.5	1.6	1.00
	4.500	-0.179	2.240E+02	2.110E+02	1.5	1.6	1.00
	5.000	-0.156	2.990E+02	2.820E+02	2.0	2.1	1.00
	5.500	-0.135	3.850E+02	3.640E+02	2.0	2.1	1.00
	6.000	-0.117	4.790E+02	4.510E+02	2.0	2.1	1.00
	7.000	-0.083	6.920E+02	6.540E+02	2.0	2.1	1.00
	8.000	-0.054	9.280E+02	8.820E+02	2.3	2.7	1.00
	9.000	-0.029	1.180E+03	1.130E+03	2.3	2.7	1.00
	10.000	-0.006	1.440E+03	1.390E+03	2.3	2.7	1.00
	11.000	0.015	1.700E+03	1.650E+03	2.3	2.7	1.01
	12.000	0.034	1.960E+03	1.910E+03	2.2	2.8	1.01
	14.000	0.067	2.470E+03	2.420E+03	2.2	2.8	1.01
	16.000	0.096	2.940E+03	2.920E+03	2.2	2.8	1.01
	18.000	0.122	3.370E+03	3.360E+03	2.1	3.4	1.01
	20.000	0.145	3.750E+03	3.770E+03	2.1	3.3	1.01
	23.000	0.175	4.260E+03	4.280E+03	2.1	3.3	1.01
	26.000	0.202	4.670E+03	4.730E+03	2.1	3.3	1.01
	30.000	0.233	5.110E+03	5.200E+03	2.6	2.8	1.01
	35.000	0.266	5.510E+03	5.590E+03	2.6	2.8	1.01
	40.000	0.295	5.780E+03	5.880E+03	2.6	2.8	1.00
	45.000	0.321	5.950E+03	6.060E+03	3.9	4.1	1.00
	50.000	0.344	6.050E+03	6.170E+03	3.9	4.1	1.00
	55.000	0.365	6.100E+03	6.210E+03	3.9	4.1	1.00
	60.000	0.383	6.110E+03	6.200E+03	3.9	4.1	1.00
	70.000	0.417	6.050E+03	6.100E+03	3.9	4.1	1.00
	80.000	0.446	5.920E+03	5.920E+03	5.9	6.3	1.00
	90.000	0.471	5.760E+03	5.700E+03	5.9	6.3	1.00
	100.000	0.494	5.580E+03	5.450E+03	5.9	6.3	1.00
	110.000	0.515	5.400E+03	5.210E+03	5.9	6.3	1.00
	120.000	0.534	5.220E+03	4.960E+03	7.3	7.8	1.00
	140.000	0.567	4.880E+03	4.490E+03	7.3	7.8	1.00
	160.000	0.596	4.570E+03	4.060E+03	7.3	7.8	1.00
	180.000	0.622	4.290E+03	3.670E+03	5.8	9.3	1.00

TABLE I. Empirical Reference Cross Sections for K-Shell Ionization by α Particles

See page 84 for Explanation of Tables

TARGET	ENERGY [MeV]	LOG XI	SIGMA [barn] THEORY	SIGMA [barn] REFERENCE	ERROR [%] RND	ERROR [%] TOT	COR
26-Fe	200.000	0.645	4.030E+03	3.330E+03	5.8	9.4	1.00
27-Co	0.260	-0.817	5.980E-04	3.710E-04	13.4	15.8	1.00
	0.300	-0.786	1.670E-03	1.240E-03	13.4	14.6	1.00
	0.350	-0.752	4.530E-03	3.840E-03	8.0	19.3	1.00
	0.400	-0.723	1.000E-02	9.220E-03	8.0	18.4	1.00
	0.450	-0.698	1.920E-02	1.870E-02	8.0	17.8	1.00
	0.500	-0.675	3.330E-02	3.370E-02	8.0	17.4	1.00
	0.550	-0.654	5.370E-02	5.580E-02	2.9	8.8	1.00
	0.600	-0.635	8.160E-02	8.660E-02	2.9	8.7	1.00
	0.700	-0.602	1.660E-01	1.800E-01	2.9	8.6	1.00
	0.800	-0.573	3.000E-01	3.280E-01	2.9	6.6	1.00
	0.900	-0.547	4.970E-01	5.430E-01	1.7	3.3	1.00
	1.000	-0.524	7.690E-01	8.390E-01	1.7	3.3	1.00
	1.100	-0.504	1.130E+00	1.230E+00	1.7	3.3	1.00
	1.200	-0.485	1.600E+00	1.720E+00	1.7	2.5	1.00
	1.400	-0.451	2.920E+00	3.080E+00	1.2	1.6	1.00
	1.600	-0.422	4.850E+00	5.020E+00	1.2	1.6	1.00
	1.800	-0.397	7.490E+00	7.620E+00	1.2	1.6	1.00
	2.000	-0.374	1.100E+01	1.100E+01	1.3	1.4	1.00
	2.300	-0.343	1.800E+01	1.770E+01	1.3	1.4	1.00
	2.600	-0.317	2.740E+01	2.660E+01	1.3	1.4	1.00
	3.000	-0.286	4.410E+01	4.230E+01	1.3	1.3	1.00
	3.500	-0.252	7.220E+01	6.870E+01	1.5	1.6	1.00
	4.000	-0.223	1.090E+02	1.030E+02	1.5	1.6	1.00
	4.500	-0.198	1.540E+02	1.450E+02	1.5	1.6	1.00
	5.000	-0.175	2.070E+02	1.960E+02	2.0	2.0	1.00
	5.500	-0.154	2.690E+02	2.540E+02	2.0	2.0	1.00
	6.000	-0.135	3.370E+02	3.190E+02	2.0	2.0	1.00
	7.000	-0.102	4.930E+02	4.650E+02	2.0	2.0	1.00
	8.000	-0.073	6.690E+02	6.340E+02	2.3	2.6	1.00
	9.000	-0.047	8.590E+02	8.180E+02	2.3	2.6	1.00
	10.000	-0.024	1.060E+03	1.020E+03	2.3	2.6	1.00
	11.000	-0.004	1.260E+03	1.220E+03	2.3	2.6	1.00
	12.000	0.015	1.470E+03	1.420E+03	2.3	2.6	1.00
	14.000	0.049	1.870E+03	1.830E+03	2.2	2.7	1.00
	16.000	0.078	2.260E+03	2.240E+03	2.2	2.7	1.01
	18.000	0.103	2.620E+03	2.610E+03	2.2	3.4	1.01
	20.000	0.126	2.940E+03	2.950E+03	2.1	3.1	1.01
	23.000	0.157	3.380E+03	3.390E+03	2.1	3.1	1.01
	26.000	0.183	3.740E+03	3.780E+03	2.1	3.1	1.01
	30.000	0.214	4.140E+03	4.200E+03	2.6	2.6	1.01
	35.000	0.248	4.520E+03	4.600E+03	2.6	2.6	1.00
	40.000	0.277	4.780E+03	4.860E+03	2.6	2.6	1.00
	45.000	0.302	4.960E+03	5.050E+03	2.6	2.6	1.00
	50.000	0.325	5.080E+03	5.180E+03	3.9	4.6	1.00
	55.000	0.346	5.150E+03	5.250E+03	3.9	4.6	1.00
	60.000	0.365	5.190E+03	5.280E+03	3.9	4.6	1.00
	70.000	0.398	5.180E+03	5.240E+03	3.9	4.6	1.00
	80.000	0.427	5.100E+03	5.130E+03	5.9	5.9	1.00
	90.000	0.453	4.990E+03	4.980E+03	5.9	5.9	1.00
	100.000	0.476	4.860E+03	4.800E+03	5.9	5.9	1.00
	110.000	0.496	4.720E+03	4.610E+03	5.9	5.9	1.00
	120.000	0.515	4.580E+03	4.410E+03	5.9	5.9	1.00
	140.000	0.549	4.300E+03	4.030E+03	7.3	7.4	1.00
	160.000	0.578	4.040E+03	3.670E+03	7.3	7.4	1.00
	180.000	0.603	3.800E+03	3.350E+03	7.3	7.9	1.00
	200.000	0.626	3.590E+03	3.060E+03	5.8	6.4	1.00
	230.000	0.657	3.310E+03	2.670E+03	5.8	6.4	1.00
28-Ni	0.300	-0.803	9.020E-04	6.080E-04	13.4	14.9	
	0.350	-0.770	2.570E-03	2.040E-03	13.4	14.2	
	0.400	-0.741	5.880E-03	5.160E-03	8.0	17.2	
	0.450	-0.715	1.160E-02	1.090E-02	8.0	16.6	
	0.500	-0.692	2.040E-02	2.010E-02	8.0	16.2	
	0.550	-0.672	3.340E-02	3.400E-02	8.0	15.9	
	0.600	-0.653	5.140E-02	5.360E-02	2.9	7.3	
	0.700	-0.619	1.070E-01	1.140E-01	2.9	7.2	
	0.800	-0.590	1.940E-01	2.120E-01	2.9	5.7	
	0.900	-0.565	3.240E-01	3.540E-01	2.9	5.7	
	1.000	-0.542	5.050E-01	5.520E-01	1.7	2.4	
	1.100	-0.521	7.470E-01	8.140E-01	1.7	2.4	
	1.200	-0.502	1.060E+00	1.150E+00	1.7	2.4	
	1.400	-0.469	1.950E+00	2.070E+00	1.7	2.1	
	1.600	-0.440	3.240E+00	3.400E+00	1.2	1.4	
	1.800	-0.414	5.030E+00	5.170E+00	1.2	1.4	
	2.000	-0.391	7.390E+00	7.500E+00	1.2	1.4	
	2.300	-0.361	1.220E+01	1.210E+01	1.3	1.9	
	2.600	-0.334	1.860E+01	1.820E+01	1.3	1.9	
	3.000	-0.303	3.010E+01	2.910E+01	1.3	1.9	
	3.500	-0.270	4.970E+01	4.750E+01	1.5	1.7	
28-Ni	4.000	-0.241	7.540E+01	7.160E+01	1.5	1.7	
	4.500	-0.215	1.070E+02	1.010E+02	1.5	1.7	
	5.000	-0.192	1.460E+02	1.370E+02	1.5	1.7	
	5.500	-0.172	1.900E+02	1.790E+02	2.0	2.1	
	6.000	-0.153	2.400E+02	2.270E+02	2.0	2.1	
	7.000	-0.119	3.550E+02	3.340E+02	2.0	2.1	
	8.000	-0.090	4.870E+02	4.600E+02	2.0	2.1	
	9.000	-0.065	6.320E+02	5.990E+02	2.3	2.5	
	10.000	-0.042	7.850E+02	7.490E+02	2.3	2.5	
	11.000	-0.021	9.450E+02	9.080E+02	2.3	2.5	
	12.000	-0.002	1.110E+03	1.070E+03	2.3	2.5	
	14.000	0.031	1.430E+03	1.390E+03	2.2	2.6	
	16.000	0.060	1.750E+03	1.710E+03	2.2	2.6	
	18.000	0.086	2.040E+03	2.030E+03	2.2	2.6	
	20.000	0.109	2.320E+03	2.310E+03	2.2	3.2	
	23.000	0.139	2.690E+03	2.700E+03	2.1	2.9	
	26.000	0.166	3.010E+03	3.030E+03	2.1	2.9	
	30.000	0.197	3.370E+03	3.410E+03	2.1	2.8	
	35.000	0.230	3.710E+03	3.780E+03	2.6	2.8	
	40.000	0.259	3.970E+03	4.050E+03	2.6	2.8	
	45.000	0.285	4.150E+03	4.220E+03	2.6	2.8	
	50.000	0.308	4.280E+03	4.360E+03	3.9	5.1	
	55.000	0.328	4.360E+03	4.450E+03	3.9	5.1	
	60.000	0.347	4.410E+03	4.500E+03	3.9	5.1	
	70.000	0.381	4.440E+03	4.520E+03	3.9	5.1	
	80.000	0.410	4.410E+03	4.460E+03	3.9	5.1	
	90.000	0.435	4.340E+03	4.350E+03	5.9	6.4	
	100.000	0.458	4.250E+03	4.220E+03	5.9	6.4	
	110.000	0.479	4.140E+03	4.080E+03	5.9	6.4	
	120.000	0.498	4.030E+03	3.930E+03	5.9	6.4	
	140.000	0.531	3.800E+03	3.620E+03	7.3	8.0	
	160.000	0.560	3.590E+03	3.320E+03	7.3	8.0	
	180.000	0.586	3.390E+03	3.050E+03	7.3	8.0	
	200.000	0.609	3.210E+03	2.800E+03	7.3	10.8	
	230.000	0.639	2.960E+03	2.470E+03	5.8	8.3	
	260.000	0.666	2.750E+03	2.180E+03	5.8	8.4	
29-Cu	0.350	-0.787	1.470E-03	1.080E-03	13.4	16.3	
	0.400	-0.758	3.480E-03	2.890E-03	8.0	15.9	
	0.450	-0.732	7.030E-03	6.330E-03	8.0	15.3	
	0.500	-0.710	1.270E-02	1.210E-02	8.0	14.9	
	0.550	-0.689	2.100E-02	2.080E-02	8.0	14.6	
	0.600	-0.670	3.280E-02	3.340E-02	8.0	14.4	
	0.700	-0.636	6.900E-02	7.310E-02	2.9	5.9	
	0.800	-0.607	1.270E-01	1.380E-01	2.9	5.8	
	0.900	-0.582	2.140E-01	2.330E-01	2.9	4.8	
	1.000	-0.559	3.350E-01	3.670E-01	1.7	2.0	
	1.100	-0.538	4.990E-01	5.450E-01	1.7	2.0	
	1.200	-0.519	7.110E-01	7.740E-01	1.7	2.0	
	1.400	-0.486	1.310E+00	1.410E+00	1.7	1.9	
	1.600	-0.457	2.190E+00	2.320E+00	1.2	1.8	
	1.800	-0.431	3.420E+00	3.560E+00	1.2	1.8	
	2.000	-0.409	5.040E+00	5.170E+00	1.2	1.9	
	2.300	-0.378	8.330E+00	8.380E+00	1.2	1.9	
	2.600	-0.352	1.280E+01	1.270E+01	1.3	2.4	
	3.000	-0.320	2.080E+01	2.030E+01	1.3	2.5	
	3.500	-0.287	3.460E+01	3.320E+01	1.3	1.6	
	4.000	-0.258	5.280E+01	5.030E+01	1.5	1.8	
	4.500	-0.232	7.570E+01	7.170E+01	1.5	1.8	
	5.000	-0.210	1.030E+02	9.730E+01	1.5	1.8	
	5.500	-0.189	1.350E+02	1.280E+02	1.5	1.8	
	6.000	-0.170	1.720E+02	1.620E+02	2.0	2.2	
	7.000	-0.136	2.570E+02	2.430E+02	2.0	2.2	
	8.000	-0.107	3.560E+02	3.350E+02	2.0	2.2	
	9.000	-0.082	4.660E+02	4.410E+02	2.0	2.2	
	10.000	-0.059	5.850E+02	5.560E+02	2.3	2.4	
	11.000	-0.038	7.090E+02	6.770E+02	2.3	2.4	
	12.000	-0.019	8.370E+02	8.060E+02	2.3	2.4	
	14.000	0.014	1.100E+03	1.060E+03	2.3	2.4	
	16.000	0.043	1.350E+03	1.320E+03	2.2	2.5	
	18.000	0.069	1.600E+03	1.570E+03	2.2	2.5	
	20.000	0.091	1.830E+03	1.820E+03	2.2	2.5	
	23.000	0.122	2.150E+03	2.150E+03	2.1	2.6	
	26.000	0.148	2.430E+03	2.440E+03	2.1	2.6	
	30.000	0.180	2.750E+03	2.770E+03	2.1	2.6	
	35.000	0.213	3.060E+03	3.100E+03	2.6	3.1	
	40.000	0.242	3.300E+03	3.360E+03	2.6	3.1	
	45.000	0.268	3.480E+03	3.530E+03	2.6	3.1	
	50.000	0.290	3.610E+03	3.670E+03	2.6	3.1	
	55.000	0.311	3.700E+03	3.770E+03	3.9	5.6	
	60.000	0.330	3.760E+03	3.840E+03	3.9	5.6	

TABLE I. Empirical Reference Cross Sections for *K*-Shell Ionization by α Particles

See page 84 for Explanation of Tables

TARGET	ENERGY [MeV]	LOG XI	SIGMA [barn] THEORY	SIGMA [barn] REFERENCE	ERROR [%] RND	ERROR [%] TOT
29-Cu	70.000	0.364	3.820E+03	3.890E+03	3.9	5.6
	80.000	0.393	3.820E+03	3.870E+03	3.9	5.6
	90.000	0.418	3.780E+03	3.810E+03	3.9	5.6
	100.000	0.441	3.710E+03	3.720E+03	5.9	6.9
	110.000	0.462	3.630E+03	3.610E+03	5.9	6.9
	120.000	0.481	3.550E+03	3.490E+03	5.9	6.9
	140.000	0.514	3.370E+03	3.250E+03	5.9	7.0
	160.000	0.543	3.190E+03	3.010E+03	7.3	8.5
	180.000	0.569	3.020E+03	2.780E+03	7.3	8.5
	200.000	0.591	2.870E+03	2.570E+03	7.3	8.6
	230.000	0.622	2.660E+03	2.280E+03	5.8	11.1
	260.000	0.648	2.480E+03	2.030E+03	5.8	11.4
30-Zn	0.350	-0.804	8.300E-04	5.590E-04	13.4	22.3
	0.400	-0.775	2.050E-03	1.590E-03	13.4	18.1
	0.450	-0.749	4.250E-03	3.640E-03	8.0	13.9
	0.500	-0.726	7.840E-03	7.170E-03	8.0	13.5
	0.550	-0.706	1.320E-02	1.270E-02	8.0	13.3
	0.600	-0.687	2.090E-02	2.070E-02	8.0	13.1
	0.700	-0.653	4.480E-02	4.670E-02	2.9	4.5
	0.800	-0.624	8.370E-02	8.950E-02	2.9	4.4
	0.900	-0.599	1.420E-01	1.540E-01	2.9	3.9
	1.000	-0.576	2.240E-01	2.450E-01	2.9	3.9
	1.100	-0.555	3.350E-01	3.670E-01	1.7	2.9
	1.200	-0.536	4.790E-01	5.240E-01	1.7	2.9
	1.400	-0.503	8.900E-01	9.640E-01	1.7	2.9
	1.600	-0.474	1.500E+00	1.600E+00	1.7	2.3
	1.800	-0.448	2.340E+00	2.460E+00	1.2	2.3
	2.000	-0.425	3.460E+00	3.590E+00	1.2	2.3
	2.300	-0.395	5.740E+00	5.840E+00	1.2	2.4
	2.600	-0.368	8.860E+00	8.860E+00	1.3	2.9
	3.000	-0.337	1.450E+01	1.420E+01	1.3	3.0
	3.500	-0.304	2.420E+01	2.340E+01	1.3	3.0
	4.000	-0.275	3.720E+01	3.560E+01	1.5	1.9
	4.500	-0.249	5.360E+01	5.090E+01	1.5	1.9
	5.000	-0.226	7.350E+01	6.950E+01	1.5	1.9
	5.500	-0.206	9.680E+01	9.130E+01	1.5	1.9
	6.000	-0.187	1.240E+02	1.170E+02	1.5	1.9
	7.000	-0.153	1.870E+02	1.760E+02	2.0	2.3
	8.000	-0.124	2.610E+02	2.460E+02	2.0	2.3
	9.000	-0.099	3.450E+02	3.260E+02	2.0	2.3
	10.000	-0.076	4.370E+02	4.130E+02	2.3	2.3
	11.000	-0.055	5.340E+02	5.070E+02	2.3	2.3
	12.000	-0.036	6.340E+02	6.060E+02	2.3	2.3
	14.000	-0.003	8.420E+02	8.130E+02	2.3	2.3
	16.000	0.026	1.050E+03	1.020E+03	2.2	2.4
	18.000	0.052	1.250E+03	1.230E+03	2.2	2.4
	20.000	0.075	1.450E+03	1.430E+03	2.2	2.4
	23.000	0.105	1.720E+03	1.710E+03	2.2	2.7
	26.000	0.132	1.960E+03	1.960E+03	2.1	2.4
	30.000	0.163	2.240E+03	2.250E+03	2.1	2.4
	35.000	0.196	2.520E+03	2.550E+03	2.1	2.4
	40.000	0.225	2.750E+03	2.790E+03	2.6	3.3
	45.000	0.251	2.920E+03	2.970E+03	2.6	3.3
	50.000	0.274	3.050E+03	3.090E+03	2.6	3.3
	55.000	0.294	3.140E+03	3.200E+03	2.6	3.3
	60.000	0.313	3.210E+03	3.270E+03	3.9	6.1
	70.000	0.347	3.290E+03	3.350E+03	3.9	6.1
	80.000	0.376	3.310E+03	3.360E+03	3.9	6.1
	90.000	0.401	3.290E+03	3.330E+03	3.9	6.1
	100.000	0.424	3.250E+03	3.270E+03	5.9	7.4
	110.000	0.445	3.190E+03	3.190E+03	5.9	7.4
	120.000	0.464	3.130E+03	3.100E+03	5.9	7.4
	140.000	0.497	2.990E+03	2.910E+03	5.9	7.5
	160.000	0.526	2.840E+03	2.710E+03	7.3	9.0
	180.000	0.552	2.700E+03	2.520E+03	7.3	9.0
	200.000	0.575	2.570E+03	2.340E+03	7.3	9.1
	230.000	0.605	2.390E+03	2.100E+03	7.3	16.4
	260.000	0.632	2.230E+03	1.880E+03	5.8	14.2
	300.000	0.663	2.050E+03	1.630E+03	5.8	14.6
31-Ga	0.400	-0.791	1.210E-03	8.690E-04	13.4	20.5
	0.450	-0.766	2.580E-03	2.090E-03	13.4	19.7
	0.500	-0.743	4.870E-03	4.260E-03	8.0	12.1
	0.550	-0.722	8.370E-03	7.750E-03	8.0	11.9
	0.600	-0.703	1.340E-02	1.290E-02	8.0	11.7
	0.700	-0.670	2.940E-02	3.000E-02	8.0	11.5
	0.800	-0.641	5.560E-02	5.870E-02	2.9	3.0
	0.900	-0.615	9.520E-02	1.030E-01	2.9	3.0
	1.000	-0.592	1.510E-01	1.650E-01	2.9	3.0
31-Ga	1.100	-0.571	2.280E-01	2.490E-01	2.9	3.0
	1.200	-0.553	3.270E-01	3.580E-01	1.7	3.8
	1.400	-0.519	6.110E-01	6.660E-01	1.7	3.9
	1.600	-0.490	1.030E+00	1.110E+00	1.7	2.8
	1.800	-0.464	1.620E+00	1.720E+00	1.2	2.8
	2.000	-0.442	2.400E+00	2.520E+00	1.2	2.8
	2.300	-0.411	4.010E+00	4.120E+00	1.2	2.8
	2.600	-0.385	6.210E+00	6.270E+00	1.2	2.9
	3.000	-0.354	1.020E+01	1.010E+01	1.3	3.4
	3.500	-0.320	1.710E+01	1.670E+01	1.3	3.5
	4.000	-0.291	2.640E+01	2.540E+01	1.3	1.8
	4.500	-0.266	3.830E+01	3.650E+01	1.5	2.0
	5.000	-0.243	5.280E+01	5.010E+01	1.5	2.0
	5.500	-0.222	6.990E+01	6.600E+01	1.5	2.0
	6.000	-0.203	8.970E+01	8.460E+01	1.5	2.0
	7.000	-0.170	1.370E+02	1.290E+02	2.0	2.4
	8.000	-0.141	1.930E+02	1.830E+02	2.0	2.4
	9.000	-0.115	2.570E+02	2.420E+02	2.0	2.4
	10.000	-0.092	3.280E+02	3.090E+02	2.0	2.4
	11.000	-0.071	4.030E+02	3.820E+02	2.3	2.4
	12.000	-0.053	4.830E+02	4.590E+02	2.3	2.4
	14.000	-0.019	6.490E+02	6.240E+02	2.3	2.4
	16.000	0.010	8.180E+02	7.920E+02	2.3	2.4
	18.000	0.036	9.850E+02	9.600E+02	2.2	2.3
	20.000	0.058	1.150E+03	1.120E+03	2.2	2.3
	23.000	0.089	1.380E+03	1.370E+03	2.2	2.3
	26.000	0.115	1.580E+03	1.580E+03	2.1	2.2
	30.000	0.146	1.830E+03	1.840E+03	2.1	2.2
	35.000	0.180	2.080E+03	2.100E+03	2.1	2.2
	40.000	0.209	2.290E+03	2.320E+03	2.6	3.6
	45.000	0.234	2.450E+03	2.490E+03	2.6	3.6
	50.000	0.257	2.570E+03	2.620E+03	2.6	3.6
	55.000	0.278	2.670E+03	2.710E+03	2.6	3.6
	60.000	0.297	2.740E+03	2.790E+03	2.6	3.6
	70.000	0.330	2.830E+03	2.880E+03	3.9	6.6
	80.000	0.359	2.870E+03	2.920E+03	3.9	6.6
	90.000	0.385	2.870E+03	2.910E+03	3.9	6.6
	100.000	0.408	2.850E+03	2.880E+03	3.9	6.6
	110.000	0.429	2.810E+03	2.820E+03	5.9	7.9
	120.000	0.447	2.760E+03	2.760E+03	5.9	7.9
	140.000	0.481	2.650E+03	2.610E+03	5.9	8.0
	160.000	0.510	2.530E+03	2.450E+03	5.9	8.0
	180.000	0.536	2.420E+03	2.290E+03	7.3	9.5
	200.000	0.558	2.310E+03	2.140E+03	7.3	9.6
	230.000	0.589	2.150E+03	1.930E+03	7.3	9.7
	260.000	0.615	2.010E+03	1.740E+03	5.8	16.8
	300.000	0.646	1.850E+03	1.520E+03	5.8	17.4
32-Ge	0.400	-0.807	7.050E-04	4.660E-04	13.4	30.1
	0.450	-0.781	1.560E-03	1.180E-03	13.4	22.2
	0.500	-0.759	3.020E-03	2.510E-03	8.0	10.5
	0.550	-0.738	5.300E-03	4.700E-03	8.0	10.3
	0.600	-0.719	8.610E-03	8.030E-03	8.0	10.2
	0.700	-0.685	1.930E-02	1.920E-02	8.0	10.1
	0.800	-0.656	3.710E-02	3.850E-02	2.9	4.1
	0.900	-0.631	6.430E-02	6.840E-02	2.9	4.1
	1.000	-0.608	1.030E-01	1.110E-01	2.9	4.1
	1.100	-0.587	1.560E-01	1.700E-01	2.9	3.7
	1.200	-0.568	2.250E-01	2.460E-01	2.9	3.7
	1.400	-0.535	4.230E-01	4.620E-01	1.7	4.8
	1.600	-0.506	7.180E-01	7.790E-01	1.7	4.8
	1.800	-0.480	1.130E+00	1.210E+00	1.7	3.3
	2.000	-0.458	1.680E+00	1.780E+00	1.2	3.3
	2.300	-0.427	2.820E+00	2.930E+00	1.2	3.3
	2.600	-0.401	4.380E+00	4.470E+00	1.2	3.3
	3.000	-0.369	7.230E+00	7.240E+00	1.3	3.9
	3.500	-0.336	1.220E+01	1.200E+01	1.3	4.0
	4.000	-0.307	1.890E+01	1.830E+01	1.3	4.0
	4.500	-0.281	2.750E+01	2.640E+01	1.3	1.9
	5.000	-0.259	3.810E+01	3.630E+01	1.5	2.1
	5.500	-0.238	5.080E+01	4.810E+01	1.5	2.1
	6.000	-0.219	6.540E+01	6.170E+01	1.5	2.1
	7.000	-0.185	1.010E+02	9.490E+01	1.5	2.1
	8.000	-0.156	1.430E+02	1.350E+02	2.0	2.5
	9.000	-0.131	1.920E+02	1.820E+02	2.0	2.5
	10.000	-0.108	2.470E+02	2.320E+02	2.0	2.5
	11.000	-0.087	3.060E+02	2.890E+02	2.0	2.5
	12.000	-0.068	3.680E+02	3.490E+02	2.3	2.5
	14.000	-0.035	5.010E+02	4.780E+02	2.3	2.5
	16.000	-0.006	6.380E+02	6.160E+02	2.3	2.5
	18.000	0.020	7.760E+02	7.530E+02	2.2	2.2

TABLE I. Empirical Reference Cross Sections for *K*-Shell Ionization by α Particles

See page 84 for Explanation of Tables

TARGET	ENERGY [MeV]	LOG XI	SIGMA [barn] THEORY	SIGMA [barn] REFERENCE	ERROR [%] RND	ERROR [%] TOT
32-Ge	20.000	0.042	9.110E+02	8.890E+02	2.2	2.2
	23.000	0.073	1.100E+03	1.090E+03	2.2	2.2
	26.000	0.099	1.280E+03	1.280E+03	2.2	2.2
	30.000	0.131	1.500E+03	1.500E+03	2.1	2.4
	35.000	0.164	1.720E+03	1.730E+03	2.1	2.4
	40.000	0.193	1.910E+03	1.930E+03	2.1	2.4
	45.000	0.219	2.060E+03	2.090E+03	2.6	3.8
	50.000	0.241	2.180E+03	2.220E+03	2.6	3.8
	55.000	0.262	2.270E+03	2.300E+03	2.6	3.8
	60.000	0.281	2.340E+03	2.380E+03	2.6	3.8
	70.000	0.315	2.440E+03	2.490E+03	3.9	7.1
	80.000	0.344	2.490E+03	2.540E+03	3.9	7.1
	90.000	0.369	2.500E+03	2.550E+03	3.9	7.1
	100.000	0.392	2.490E+03	2.530E+03	3.9	7.1
	110.000	0.413	2.470E+03	2.500E+03	3.9	7.1
	120.000	0.432	2.440E+03	2.450E+03	5.9	8.4
	140.000	0.465	2.360E+03	2.340E+03	5.9	8.5
	160.000	0.494	2.260E+03	2.210E+03	5.9	8.5
	180.000	0.520	2.170E+03	2.080E+03	5.9	8.5
	200.000	0.542	2.070E+03	1.950E+03	7.3	10.1
	230.000	0.573	1.940E+03	1.770E+03	7.3	10.2
	260.000	0.599	1.820E+03	1.610E+03	7.3	10.3
	300.000	0.631	1.680E+03	1.420E+03	5.8	20.1
	350.000	0.664	1.530E+03	1.220E+03	5.8	20.9
33-As	0.450	-0.797	9.450E-04	6.610E-04	13.4	25.0
	0.500	-0.774	1.880E-03	1.470E-03	13.4	23.8
	0.550	-0.753	3.360E-03	2.840E-03	8.0	8.7
	0.600	-0.734	5.560E-03	4.980E-03	8.0	8.6
	0.700	-0.701	1.280E-02	1.240E-02	8.0	8.6
	0.800	-0.672	2.500E-02	2.540E-02	8.0	8.6
	0.900	-0.646	4.370E-02	4.590E-02	2.9	5.6
	1.000	-0.623	7.070E-02	7.560E-02	2.9	5.5
	1.100	-0.603	1.070E-01	1.160E-01	2.9	5.5
	1.200	-0.584	1.550E-01	1.700E-01	2.9	4.6
	1.400	-0.550	2.960E-01	3.240E-01	1.7	5.7
	1.600	-0.521	5.050E-01	5.500E-01	1.7	5.7
	1.800	-0.496	7.990E-01	8.630E-01	1.7	3.7
	2.000	-0.473	1.190E+00	1.280E+00	1.7	3.7
	2.300	-0.443	2.000E+00	2.100E+00	1.2	3.8
	2.600	-0.416	3.130E+00	3.220E+00	1.2	3.8
	3.000	-0.385	5.180E+00	5.240E+00	1.2	3.8
	3.500	-0.351	8.780E+00	8.690E+00	1.3	4.5
	4.000	-0.322	1.370E+01	1.330E+01	1.3	4.5
	4.500	-0.297	2.000E+01	1.930E+01	1.3	2.0
	5.000	-0.274	2.780E+01	2.660E+01	1.5	2.2
	5.500	-0.253	3.720E+01	3.540E+01	1.5	2.2
	6.000	-0.234	4.810E+01	4.560E+01	1.5	2.2
	7.000	-0.201	7.450E+01	7.030E+01	1.5	2.2
	8.000	-0.172	1.070E+02	1.010E+02	2.0	2.6
	9.000	-0.146	1.450E+02	1.370E+02	2.0	2.6
	10.000	-0.123	1.870E+02	1.760E+02	2.0	2.6
	11.000	-0.103	2.330E+02	2.190E+02	2.0	2.6
	12.000	-0.084	2.820E+02	2.670E+02	2.0	2.6
	14.000	-0.050	3.880E+02	3.690E+02	2.3	2.6
	16.000	-0.021	4.990E+02	4.800E+02	2.3	2.6
	18.000	0.004	6.130E+02	5.920E+02	2.3	2.6
	20.000	0.027	7.250E+02	7.050E+02	2.2	2.3
	23.000	0.057	8.880E+02	8.690E+02	2.2	2.3
	26.000	0.084	1.040E+03	1.030E+03	2.2	2.3
	30.000	0.115	1.230E+03	1.220E+03	2.1	2.7
	35.000	0.149	1.430E+03	1.430E+03	2.1	2.7
	40.000	0.178	1.590E+03	1.600E+03	2.1	2.7
	45.000	0.203	1.730E+03	1.750E+03	2.1	2.7
	50.000	0.226	1.840E+03	1.870E+03	2.6	4.1
	55.000	0.247	1.930E+03	1.970E+03	2.6	4.1
	60.000	0.266	2.000E+03	2.040E+03	2.6	4.1
	70.000	0.299	2.100E+03	2.140E+03	2.6	4.1
	80.000	0.328	2.160E+03	2.200E+03	3.9	7.6
	90.000	0.354	2.190E+03	2.230E+03	3.9	7.6
	100.000	0.377	2.190E+03	2.230E+03	3.9	7.6
	110.000	0.397	2.180E+03	2.210E+03	3.9	7.6
	120.000	0.416	2.160E+03	2.180E+03	3.9	7.6
	140.000	0.450	2.100E+03	2.090E+03	5.9	8.9
	160.000	0.479	2.020E+03	1.990E+03	5.9	9.0
	180.000	0.504	1.940E+03	1.890E+03	5.9	9.0
	200.000	0.527	1.870E+03	1.780E+03	7.3	10.6
	230.000	0.557	1.750E+03	1.630E+03	7.3	10.7
	260.000	0.584	1.650E+03	1.490E+03	7.3	10.8
	300.000	0.615	1.530E+03	1.320E+03	5.8	22.6
	350.000	0.649	1.390E+03	1.140E+03	5.8	23.6
34-Se	0.450	-0.812	5.710E-04	3.660E-04	13.4	38.4
	0.500	-0.789	1.170E-03	8.530E-04	13.4	26.6
	0.550	-0.768	2.140E-03	1.710E-03	13.4	25.4
	0.600	-0.749	3.600E-03	3.080E-03	8.0	9.1
	0.700	-0.716	8.500E-03	7.980E-03	8.0	9.0
	0.800	-0.687	1.690E-02	1.680E-02	8.0	9.0
	0.900	-0.661	3.000E-02	3.090E-02	2.9	7.1
	1.000	-0.638	4.890E-02	5.170E-02	2.9	7.0
	1.100	-0.618	7.490E-02	8.050E-02	2.9	6.9
	1.200	-0.599	1.090E-01	1.190E-01	2.9	5.5
	1.400	-0.565	2.090E-01	2.280E-01	2.9	5.5
	1.600	-0.536	3.580E-01	3.920E-01	1.7	6.6
	1.800	-0.511	5.690E-01	6.180E-01	1.7	6.6
	2.000	-0.488	8.530E-01	9.180E-01	1.7	4.2
	2.300	-0.457	1.440E+00	1.520E+00	1.2	4.2
	2.600	-0.431	2.250E+00	2.340E+00	1.2	4.3
	3.000	-0.400	3.750E+00	3.820E+00	1.2	4.3
	3.500	-0.366	6.380E+00	6.370E+00	1.3	4.9
	4.000	-0.337	9.970E+00	9.780E+00	1.3	5.0
	4.500	-0.312	1.460E+01	1.420E+01	1.3	5.0
	5.000	-0.289	2.040E+01	1.960E+01	1.3	2.1
	5.500	-0.268	2.740E+01	2.620E+01	1.5	2.3
	6.000	-0.249	3.560E+01	3.380E+01	1.5	2.3
	7.000	-0.216	5.560E+01	5.240E+01	1.5	2.3
	8.000	-0.187	8.020E+01	7.570E+01	1.5	2.3
	9.000	-0.161	1.090E+02	1.030E+02	2.0	2.7
	10.000	-0.138	1.420E+02	1.340E+02	2.0	2.7
	11.000	-0.118	1.780E+02	1.680E+02	2.0	2.7
	12.000	-0.099	2.170E+02	2.050E+02	2.0	2.7
	14.000	-0.065	3.020E+02	2.860E+02	2.3	2.7
	16.000	-0.036	3.920E+02	3.750E+02	2.3	2.7
	18.000	-0.011	4.850E+02	4.680E+02	2.3	2.7
	20.000	0.012	5.780E+02	5.600E+02	2.3	2.7
	23.000	0.043	7.150E+02	6.980E+02	2.2	2.4
	26.000	0.069	8.460E+02	8.300E+02	2.2	2.4
	30.000	0.100	1.010E+03	1.000E+03	2.2	2.7
	35.000	0.134	1.180E+03	1.180E+03	2.1	2.9
	40.000	0.163	1.330E+03	1.340E+03	2.1	2.9
	45.000	0.188	1.460E+03	1.470E+03	2.1	2.9
	50.000	0.211	1.560E+03	1.580E+03	2.6	4.3
	55.000	0.232	1.650E+03	1.680E+03	2.6	4.3
	60.000	0.251	1.720E+03	1.750E+03	2.6	4.3
	70.000	0.284	1.820E+03	1.850E+03	2.6	4.3
	80.000	0.313	1.880E+03	1.920E+03	3.9	8.1
	90.000	0.339	1.910E+03	1.950E+03	3.9	8.1
	100.000	0.362	1.930E+03	1.960E+03	3.9	8.1
	110.000	0.382	1.920E+03	1.950E+03	3.9	8.1
	120.000	0.401	1.910E+03	1.940E+03	3.9	8.1
	140.000	0.435	1.870E+03	1.880E+03	5.9	9.4
	160.000	0.464	1.810E+03	1.800E+03	5.9	9.5
	180.000	0.489	1.750E+03	1.710E+03	5.9	9.5
	200.000	0.512	1.680E+03	1.620E+03	5.9	9.6
	230.000	0.543	1.590E+03	1.490E+03	7.3	11.2
	260.000	0.569	1.500E+03	1.370E+03	7.3	11.3
	300.000	0.600	1.390E+03	1.230E+03	7.3	27.7
	350.000	0.634	1.270E+03	1.070E+03	5.8	26.1
	400.000	0.663	1.170E+03	9.350E+02	5.8	27.2
35-Br	0.500	-0.803	7.250E-04	4.890E-04	13.4	40.8
	0.550	-0.783	1.360E-03	1.020E-03	13.4	28.2
	0.600	-0.764	2.340E-03	1.900E-03	13.4	27.1
	0.700	-0.730	5.670E-03	5.140E-03	8.0	10.7
	0.800	-0.701	1.150E-02	1.110E-02	8.0	10.5
	0.900	-0.676	2.070E-02	2.090E-02	8.0	10.4
	1.000	-0.653	3.410E-02	3.550E-02	2.9	8.5
	1.100	-0.632	5.260E-02	5.590E-02	2.9	8.3
	1.200	-0.613	7.720E-02	8.320E-02	2.9	8.3
	1.400	-0.580	1.490E-01	1.620E-01	2.9	6.4
	1.600	-0.551	2.570E-01	2.810E-01	1.7	7.5
	1.800	-0.525	4.090E-01	4.470E-01	1.7	7.5
	2.000	-0.502	6.160E-01	6.670E-01	1.7	7.6
	2.300	-0.472	1.040E+00	1.110E+00	1.7	4.7
	2.600	-0.445	1.640E+00	1.720E+00	1.2	4.7
	3.000	-0.414	2.730E+00	2.810E+00	1.2	4.8
	3.500	-0.381	4.670E+00	4.710E+00	1.2	4.8
	4.000	-0.352	7.340E+00	7.260E+00	1.3	5.5
	4.500	-0.326	1.080E+01	1.050E+01	1.3	5.5
	5.000	-0.303	1.510E+01	1.460E+01	1.3	5.6
	5.500	-0.283	2.040E+01	1.950E+01	1.3	2.2
	6.000	-0.264	2.650E+01	2.530E+01	1.5	2.4
	7.000	-0.230	4.170E+01	3.950E+01	1.5	2.4
	8.000	-0.201	6.070E+01	5.720E+01	1.5	2.4

TABLE I. Empirical Reference Cross Sections for *K*-Shell Ionization by α Particles

See page 84 for Explanation of Tables

TARGET	ENERGY [MeV]	LOG XI	SIGMA [barn] THEORY	SIGMA [barn] REFERENCE	ERROR [%] RND	ERROR [%] TOT
35-Br	9.000	-0.176	8.310E+01	7.840E+01	2.0	2.8
	10.000	-0.153	1.090E+02	1.030E+02	2.0	2.8
	11.000	-0.132	1.370E+02	1.300E+02	2.0	2.8
	12.000	-0.113	1.680E+02	1.580E+02	2.0	2.8
	14.000	-0.080	2.360E+02	2.230E+02	2.3	2.8
	16.000	-0.051	3.090E+02	2.940E+02	2.3	2.8
	18.000	-0.025	3.850E+02	3.700E+02	2.3	2.8
	20.000	-0.002	4.630E+02	4.470E+02	2.3	2.8
	23.000	0.028	5.780E+02	5.620E+02	2.2	2.5
	26.000	0.055	6.890E+02	6.750E+02	2.2	2.5
	30.000	0.086	8.280E+02	8.210E+02	2.2	2.5
	35.000	0.119	9.830E+02	9.820E+02	2.1	3.2
	40.000	0.148	1.120E+03	1.120E+03	2.1	3.2
	45.000	0.174	1.230E+03	1.240E+03	2.1	3.2
	50.000	0.197	1.330E+03	1.340E+03	2.1	3.2
	55.000	0.217	1.410E+03	1.430E+03	2.6	4.6
	60.000	0.236	1.470E+03	1.500E+03	2.6	4.6
	70.000	0.270	1.570E+03	1.600E+03	2.6	4.6
	80.000	0.299	1.640E+03	1.670E+03	2.6	4.6
	90.000	0.324	1.680E+03	1.710E+03	3.9	8.5
	100.000	0.347	1.700E+03	1.730E+03	3.9	8.5
	110.000	0.368	1.700E+03	1.730E+03	3.9	8.6
	120.000	0.387	1.700E+03	1.720E+03	3.9	8.6
	140.000	0.420	1.670E+03	1.680E+03	5.9	9.9
	160.000	0.449	1.620E+03	1.620E+03	5.9	9.9
	180.000	0.475	1.570E+03	1.550E+03	5.9	10.0
	200.000	0.498	1.520E+03	1.480E+03	5.9	10.0
	230.000	0.528	1.440E+03	1.370E+03	7.3	11.6
	260.000	0.555	1.360E+03	1.270E+03	7.3	11.7
	300.000	0.586	1.270E+03	1.140E+03	7.3	11.9
	350.000	0.619	1.160E+03	1.000E+03	5.8	28.5
	400.000	0.648	1.070E+03	8.810E+02	5.8	29.6
36-Kr	0.500	-0.818	4.480E-04	4.280E-04	9.4	40.4
	0.550	-0.797	8.620E-04	8.350E-04	9.4	27.8
	0.600	-0.778	1.510E-03	1.480E-03	9.4	27.6
	0.700	-0.745	3.780E-03	3.750E-03	6.5	19.9
	0.800	-0.716	7.840E-03	7.830E-03	6.5	19.8
	0.900	-0.690	1.430E-02	1.440E-02	6.5	19.7
	1.000	-0.667	2.380E-02	2.390E-02	6.5	19.7
	1.100	-0.646	3.710E-02	3.730E-02	3.1	8.0
	1.200	-0.627	5.480E-02	5.510E-02	3.1	8.0
	1.400	-0.594	1.060E-01	1.070E-01	3.1	5.4
	1.600	-0.565	1.850E-01	1.840E-01	2.3	5.8
	1.800	-0.539	2.960E-01	2.940E-01	2.3	5.8
	2.000	-0.517	4.470E-01	4.410E-01	2.3	5.9
	2.300	-0.486	7.600E-01	7.440E-01	2.3	4.1
	2.600	-0.460	1.200E+00	1.160E+00	1.6	4.3
	3.000	-0.428	2.000E+00	1.940E+00	1.6	4.3
	3.500	-0.395	3.440E+00	3.300E+00	1.6	4.3
	4.000	-0.366	5.420E+00	5.170E+00	1.7	5.6
	4.500	-0.340	8.010E+00	7.610E+00	1.7	5.6
	5.000	-0.318	1.130E+01	1.060E+01	1.7	5.6
	5.500	-0.297	1.520E+01	1.430E+01	1.7	2.5
	6.000	-0.278	1.990E+01	1.870E+01	1.3	2.2
	7.000	-0.245	3.150E+01	2.970E+01	1.3	2.2
	8.000	-0.216	4.600E+01	4.340E+01	1.3	2.2
	9.000	-0.190	6.330E+01	6.000E+01	1.3	2.2
	10.000	-0.167	8.340E+01	7.920E+01	1.3	2.1
	11.000	-0.146	1.060E+02	1.010E+02	1.3	2.1
	12.000	-0.127	1.300E+02	1.250E+02	1.3	2.1
	14.000	-0.094	1.850E+02	1.770E+02	1.3	2.1
	16.000	-0.065	2.440E+02	2.350E+02	1.4	2.1
	18.000	-0.039	3.070E+02	2.970E+02	1.4	2.1
	20.000	-0.017	3.710E+02	3.610E+02	1.4	2.1
	23.000	0.014	4.670E+02	4.580E+02	1.4	2.1
	26.000	0.040	5.620E+02	5.540E+02	1.5	2.2
	30.000	0.072	6.810E+02	6.780E+02	1.5	2.2
	35.000	0.105	8.170E+02	8.190E+02	2.3	3.0
	40.000	0.134	9.360E+02	9.440E+02	2.3	3.0
	45.000	0.160	1.040E+03	1.060E+03	2.3	3.0
	50.000	0.182	1.130E+03	1.160E+03	2.3	3.0
	55.000	0.203	1.200E+03	1.240E+03	2.3	3.0
	60.000	0.222	1.270E+03	1.320E+03	3.9	5.6
	70.000	0.255	1.360E+03	1.430E+03	3.9	5.6
	80.000	0.284	1.430E+03	1.510E+03	3.9	5.6
	90.000	0.310	1.470E+03	1.560E+03	3.9	7.3
	100.000	0.333	1.490E+03	1.590E+03	4.9	8.3
	110.000	0.354	1.500E+03	1.610E+03	4.9	8.3
	120.000	0.373	1.500E+03	1.610E+03	4.9	8.3
	140.000	0.406	1.490E+03	1.600E+03	4.9	8.3
	160.000	0.435	1.460E+03	1.560E+03	7.7	11.1
36-Kr	180.000	0.461	1.410E+03	1.510E+03	7.7	11.1
	200.000	0.483	1.370E+03	1.460E+03	7.7	11.2
	230.000	0.514	1.300E+03	1.370E+03	7.7	11.2
	260.000	0.540	1.240E+03	1.290E+03	7.9	11.5
	300.000	0.572	1.160E+03	1.190E+03	7.9	11.5
37-Rb	0.550	-0.811	5.460E-04	5.240E-04	9.4	37.7
	0.600	-0.792	9.790E-04	9.500E-04	9.4	26.2
	0.700	-0.758	2.530E-03	2.500E-03	6.5	18.4
	0.800	-0.729	5.380E-03	5.350E-03	6.5	18.3
	0.900	-0.704	9.970E-03	9.980E-03	6.5	18.3
	1.000	-0.681	1.680E-02	1.690E-02	6.5	18.2
	1.100	-0.660	2.640E-02	2.650E-02	3.1	6.5
	1.200	-0.641	3.920E-02	3.940E-02	3.1	6.5
	1.400	-0.608	7.700E-02	7.720E-02	3.1	6.6
	1.600	-0.579	1.340E-01	1.340E-01	3.1	5.4
	1.800	-0.553	2.160E-01	2.150E-01	2.3	4.8
	2.000	-0.530	3.280E-01	3.250E-01	2.3	4.8
	2.300	-0.500	5.600E-01	5.500E-01	2.3	3.6
	2.600	-0.473	8.840E-01	8.630E-01	1.6	3.8
	3.000	-0.442	1.490E+00	1.440E+00	1.6	3.8
	3.500	-0.409	2.560E+00	2.460E+00	1.6	3.8
	4.000	-0.380	4.050E+00	3.870E+00	1.7	5.1
	4.500	-0.354	6.000E+00	5.710E+00	1.7	5.1
	5.000	-0.331	8.460E+00	7.970E+00	1.7	5.1
	5.500	-0.311	1.150E+01	1.080E+01	1.7	5.1
	6.000	-0.292	1.500E+01	1.410E+01	1.7	2.4
	7.000	-0.258	2.390E+01	2.250E+01	1.3	2.1
	8.000	-0.229	3.510E+01	3.320E+01	1.3	2.1
	9.000	-0.204	4.870E+01	4.600E+01	1.3	2.1
	10.000	-0.181	6.440E+01	6.100E+01	1.3	2.1
	11.000	-0.160	8.210E+01	7.810E+01	1.3	2.0
	12.000	-0.141	1.020E+02	9.700E+01	1.3	2.0
	14.000	-0.108	1.450E+02	1.390E+02	1.3	2.0
	16.000	-0.079	1.930E+02	1.860E+02	1.3	2.0
	18.000	-0.053	2.450E+02	2.370E+02	1.4	2.0
	20.000	-0.030	2.980E+02	2.900E+02	1.4	2.0
	23.000	0.000	3.790E+02	3.710E+02	1.4	2.0
	26.000	0.027	4.600E+02	4.520E+02	1.5	2.1
	30.000	0.058	5.620E+02	5.560E+02	1.5	2.1
	35.000	0.091	6.810E+02	6.800E+02	1.5	2.1
	40.000	0.120	7.870E+02	7.910E+02	2.3	2.7
	45.000	0.146	8.800E+02	8.950E+02	2.3	2.7
	50.000	0.169	9.600E+02	9.840E+02	2.3	2.7
	55.000	0.189	1.030E+03	1.060E+03	2.3	2.7
	60.000	0.208	1.090E+03	1.130E+03	2.3	2.7
	70.000	0.242	1.180E+03	1.240E+03	3.9	5.4
	80.000	0.271	1.250E+03	1.310E+03	3.9	5.4
	90.000	0.296	1.290E+03	1.370E+03	3.9	5.4
	100.000	0.319	1.320E+03	1.400E+03	3.9	6.9
	110.000	0.340	1.330E+03	1.420E+03	4.9	7.9
	120.000	0.359	1.340E+03	1.430E+03	4.9	7.9
	140.000	0.392	1.330E+03	1.430E+03	4.9	7.8
	160.000	0.421	1.310E+03	1.400E+03	4.9	7.8
	180.000	0.447	1.280E+03	1.370E+03	7.7	10.7
	200.000	0.470	1.240E+03	1.320E+03	7.7	10.7
	230.000	0.500	1.180E+03	1.250E+03	7.7	10.7
	260.000	0.527	1.130E+03	1.180E+03	7.9	11.0
	300.000	0.558	1.060E+03	1.090E+03	7.9	11.0
38-Sr	0.550	-0.824	3.450E-04	3.280E-04	9.4	35.3
	0.600	-0.805	6.330E-04	6.090E-04	9.4	35.0
	0.700	-0.772	1.700E-03	1.660E-03	9.4	24.5
	0.800	-0.743	3.690E-03	3.660E-03	6.5	16.9
	0.900	-0.717	6.960E-03	6.950E-03	6.5	16.8
	1.000	-0.694	1.190E-02	1.190E-02	6.5	16.7
	1.100	-0.674	1.880E-02	1.890E-02	6.5	16.7
	1.200	-0.655	2.820E-02	2.840E-02	3.1	5.1
	1.400	-0.621	5.600E-02	5.620E-02	3.1	5.1
	1.600	-0.592	9.850E-02	9.860E-02	3.1	4.4
	1.800	-0.567	1.590E-01	1.590E-01	2.3	3.8
	2.000	-0.544	2.420E-01	2.400E-01	2.3	3.8
	2.300	-0.513	4.150E-01	4.090E-01	2.3	3.8
	2.600	-0.487	6.570E-01	6.440E-01	2.3	3.1
	3.000	-0.456	1.110E+00	1.080E+00	1.6	3.2
	3.500	-0.422	1.920E+00	1.850E+00	1.6	3.3
	4.000	-0.393	3.040E+00	2.920E+00	1.6	3.3
	4.500	-0.368	4.520E+00	4.310E+00	1.7	4.5
	5.000	-0.345	6.390E+00	6.070E+00	1.7	4.6
	5.500	-0.324	8.680E+00	8.180E+00	1.7	4.6
	6.000	-0.305	1.140E+01	1.070E+01	1.7	4.6

TABLE I. Empirical Reference Cross Sections for *K*-Shell Ionization by α Particles

See page 84 for Explanation of Tables

TARGET	ENERGY [MeV]	LOG XI	SIGMA [barn] THEORY	SIGMA [barn] REFERENCE	ERROR [%] RND	ERROR [%] TOT
38-Sr	7.000	-0.272	1.820E+01	1.720E+01	1.3	2.0
	8.000	-0.243	2.700E+01	2.540E+01	1.3	2.0
	9.000	-0.217	3.750E+01	3.540E+01	1.3	2.0
	10.000	-0.194	4.990E+01	4.720E+01	1.3	2.0
	11.000	-0.174	6.390E+01	6.070E+01	1.3	1.9
	12.000	-0.155	7.950E+01	7.570E+01	1.3	1.9
	14.000	-0.121	1.150E+02	1.090E+02	1.3	1.9
	16.000	-0.092	1.540E+02	1.480E+02	1.3	1.9
	18.000	-0.067	1.960E+02	1.890E+02	1.4	1.9
	20.000	-0.044	2.400E+02	2.330E+02	1.4	1.9
	23.000	-0.013	3.080E+02	3.010E+02	1.4	1.9
	26.000	0.013	3.760E+02	3.690E+02	1.4	1.9
	30.000	0.044	4.650E+02	4.590E+02	1.5	2.0
	35.000	0.078	5.680E+02	5.660E+02	1.5	2.0
	40.000	0.107	6.620E+02	6.630E+02	2.3	2.5
	45.000	0.132	7.450E+02	7.500E+02	2.3	2.5
	50.000	0.155	8.180E+02	8.350E+02	2.3	2.5
	55.000	0.176	8.810E+02	9.050E+02	2.3	2.5
	60.000	0.195	9.360E+02	9.670E+02	2.3	2.5
	70.000	0.228	1.020E+03	1.070E+03	3.9	5.2
	80.000	0.257	1.090E+03	1.140E+03	3.9	5.1
	90.000	0.283	1.130E+03	1.200E+03	3.9	5.1
	100.000	0.306	1.160E+03	1.230E+03	3.9	6.4
	110.000	0.326	1.180E+03	1.260E+03	3.9	6.4
	120.000	0.345	1.190E+03	1.270E+03	4.9	7.4
	140.000	0.379	1.190E+03	1.280E+03	4.9	7.4
	160.000	0.408	1.170E+03	1.260E+03	4.9	7.4
	180.000	0.433	1.150E+03	1.230E+03	7.7	10.2
	200.000	0.456	1.120E+03	1.200E+03	7.7	10.2
	230.000	0.487	1.080E+03	1.140E+03	7.7	10.2
	260.000	0.513	1.030E+03	1.080E+03	7.7	10.2
	300.000	0.544	9.660E+02	1.010E+03	7.9	10.5
	350.000	0.578	8.950E+02	9.180E+02	7.9	10.6
39-Y	0.600	-0.818	4.080E-04	3.890E-04	9.4	32.6
	0.700	-0.785	1.130E-03	1.110E-03	9.4	23.0
	0.800	-0.756	2.530E-03	2.500E-03	6.5	15.4
	0.900	-0.730	4.870E-03	4.850E-03	6.5	15.3
	1.000	-0.707	8.440E-03	8.440E-03	6.5	15.3
	1.100	-0.687	1.350E-02	1.360E-02	6.5	15.2
	1.200	-0.668	2.040E-02	2.050E-02	6.5	15.2
	1.400	-0.634	4.090E-02	4.120E-02	3.1	3.6
	1.600	-0.605	7.260E-02	7.280E-02	3.1	3.6
	1.800	-0.580	1.180E-01	1.180E-01	3.1	3.4
	2.000	-0.557	1.800E-01	1.790E-01	2.3	2.8
	2.300	-0.526	3.100E-01	3.070E-01	2.3	2.8
	2.600	-0.500	4.920E-01	4.840E-01	2.3	2.5
	3.000	-0.469	8.340E-01	8.130E-01	1.6	2.7
	3.500	-0.435	1.450E+00	1.400E+00	1.6	2.7
	4.000	-0.406	2.300E+00	2.210E+00	1.6	2.7
	4.500	-0.381	3.430E+00	3.280E+00	1.7	4.0
	5.000	-0.358	4.860E+00	4.630E+00	1.7	4.0
	5.500	-0.337	6.620E+00	6.280E+00	1.7	4.0
	6.000	-0.318	8.720E+00	8.220E+00	1.7	4.0
	7.000	-0.285	1.400E+01	1.320E+01	1.7	2.2
	8.000	-0.256	2.080E+01	1.960E+01	1.3	1.9
	9.000	-0.230	2.910E+01	2.750E+01	1.3	1.9
	10.000	-0.207	3.880E+01	3.670E+01	1.3	1.9
	11.000	-0.187	5.000E+01	4.730E+01	1.3	1.9
	12.000	-0.168	6.240E+01	5.930E+01	1.3	1.8
	14.000	-0.134	9.060E+01	8.670E+01	1.3	1.8
	16.000	-0.105	1.230E+02	1.170E+02	1.3	1.8
	18.000	-0.080	1.570E+02	1.510E+02	1.3	1.8
	20.000	-0.057	1.940E+02	1.880E+02	1.4	1.8
	23.000	-0.026	2.510E+02	2.440E+02	1.4	1.8
	26.000	0.000	3.090E+02	3.020E+02	1.4	1.8
	30.000	0.031	3.850E+02	3.790E+02	1.5	1.9
	35.000	0.065	4.750E+02	4.720E+02	1.5	1.9
	40.000	0.094	5.570E+02	5.570E+02	1.5	1.9
	45.000	0.119	6.320E+02	6.350E+02	2.3	2.4
	50.000	0.142	6.980E+02	7.040E+02	2.3	2.4
	55.000	0.163	7.560E+02	7.730E+02	2.3	2.4
	60.000	0.182	8.070E+02	8.300E+02	2.3	2.4
	70.000	0.215	8.890E+02	9.240E+02	3.9	4.9
	80.000	0.244	9.500E+02	9.950E+02	3.9	4.9
	90.000	0.270	9.940E+02	1.050E+03	3.9	4.9
	100.000	0.293	1.020E+03	1.090E+03	3.9	4.9
	110.000	0.313	1.050E+03	1.110E+03	3.9	5.9
	120.000	0.332	1.060E+03	1.130E+03	4.9	6.9
	140.000	0.366	1.070E+03	1.140E+03	4.9	6.9
	160.000	0.395	1.060E+03	1.130E+03	4.9	6.9
	180.000	0.420	1.040E+03	1.120E+03	4.9	6.9
39-Y	200.000	0.443	1.020E+03	1.090E+03	7.7	9.7
	230.000	0.474	9.790E+02	1.040E+03	7.7	9.7
	260.000	0.500	9.380E+02	9.940E+02	7.7	9.8
	300.000	0.531	8.850E+02	9.280E+02	7.9	10.0
	350.000	0.565	8.230E+02	8.490E+02	7.9	10.1
40-Zr	0.700	-0.798	7.600E-04	7.350E-04	9.4	21.6
	0.800	-0.769	1.750E-03	1.710E-03	9.4	21.4
	0.900	-0.743	3.430E-03	3.400E-03	6.5	13.8
	1.000	-0.720	6.020E-03	6.010E-03	6.5	13.8
	1.100	-0.699	9.760E-03	9.770E-03	6.5	13.8
	1.200	-0.680	1.490E-02	1.490E-02	6.5	13.7
	1.400	-0.647	3.020E-02	3.030E-02	3.1	4.0
	1.600	-0.618	5.390E-02	5.410E-02	3.1	4.0
	1.800	-0.592	8.810E-02	8.820E-02	3.1	3.7
	2.000	-0.570	1.350E-01	1.350E-01	3.1	3.7
	2.300	-0.539	2.330E-01	2.310E-01	2.3	2.7
	2.600	-0.513	3.720E-01	3.670E-01	2.3	2.7
	3.000	-0.482	6.320E-01	6.180E-01	2.3	2.5
	3.500	-0.448	1.100E+00	1.070E+00	1.6	2.2
	4.000	-0.419	1.750E+00	1.690E+00	1.6	2.2
	4.500	-0.393	2.620E+00	2.510E+00	1.6	2.2
	5.000	-0.371	3.720E+00	3.550E+00	1.7	3.5
	5.500	-0.350	5.080E+00	4.830E+00	1.7	3.5
	6.000	-0.331	6.710E+00	6.330E+00	1.7	3.5
	7.000	-0.298	1.080E+01	1.020E+01	1.7	2.1
	8.000	-0.269	1.620E+01	1.520E+01	1.3	1.8
	9.000	-0.243	2.270E+01	2.140E+01	1.3	1.8
	10.000	-0.220	3.040E+01	2.870E+01	1.3	1.8
	11.000	-0.199	3.930E+01	3.710E+01	1.3	1.8
	12.000	-0.180	4.920E+01	4.670E+01	1.3	1.8
	14.000	-0.147	7.210E+01	6.870E+01	1.3	1.7
	16.000	-0.118	9.820E+01	9.370E+01	1.3	1.7
	18.000	-0.092	1.270E+02	1.220E+02	1.3	1.7
	20.000	-0.070	1.570E+02	1.520E+02	1.4	1.7
	23.000	-0.039	2.050E+02	1.990E+02	1.4	1.7
	26.000	-0.013	2.540E+02	2.480E+02	1.4	1.7
	30.000	0.018	3.190E+02	3.130E+02	1.5	1.8
	35.000	0.052	3.970E+02	3.930E+02	1.5	1.8
	40.000	0.081	4.700E+02	4.690E+02	1.5	1.8
	45.000	0.107	5.360E+02	5.370E+02	2.3	2.6
	50.000	0.129	5.960E+02	6.000E+02	2.3	2.6
	55.000	0.150	6.490E+02	6.610E+02	2.3	2.6
	60.000	0.169	6.960E+02	7.130E+02	2.3	2.6
	70.000	0.202	7.730E+02	8.000E+02	2.3	2.6
	80.000	0.231	8.310E+02	8.680E+02	3.9	4.7
	90.000	0.257	8.740E+02	9.190E+02	3.9	4.7
	100.000	0.280	9.060E+02	9.570E+02	3.9	4.7
	110.000	0.301	9.280E+02	9.850E+02	3.9	5.5
	120.000	0.320	9.420E+02	1.000E+03	3.9	5.5
	140.000	0.353	9.540E+02	1.020E+03	4.9	6.5
	160.000	0.382	9.520E+02	1.020E+03	4.9	6.4
	180.000	0.408	9.400E+02	1.010E+03	4.9	6.4
	200.000	0.430	9.230E+02	9.890E+02	7.7	9.3
	230.000	0.461	8.920E+02	9.520E+02	7.7	9.3
	260.000	0.487	8.570E+02	9.110E+02	7.7	9.3
	300.000	0.518	8.110E+02	8.550E+02	7.7	9.3
	350.000	0.552	7.570E+02	7.860E+02	7.9	9.6
41-Nb	0.700	-0.810	5.100E-04	4.900E-04	9.4	27.2
	0.800	-0.781	1.210E-03	1.180E-03	9.4	19.9
	0.900	-0.755	2.420E-03	2.390E-03	6.5	12.3
	1.000	-0.732	4.320E-03	4.300E-03	6.5	12.3
	1.100	-0.712	7.090E-03	7.080E-03	6.5	12.3
	1.200	-0.693	1.090E-02	1.090E-02	6.5	12.3
	1.400	-0.659	2.240E-02	2.250E-02	3.1	5.5
	1.600	-0.630	4.040E-02	4.060E-02	3.1	5.5
	1.800	-0.605	6.640E-02	6.660E-02	3.1	5.5
	2.000	-0.582	1.020E-01	1.020E-01	3.1	4.7
	2.300	-0.552	1.770E-01	1.760E-01	2.3	3.7
	2.600	-0.525	2.840E-01	2.810E-01	2.3	3.7
	3.000	-0.494	4.830E-01	4.740E-01	2.3	3.0
	3.500	-0.460	8.430E-01	8.200E-01	1.6	1.7
	4.000	-0.431	1.350E+00	1.300E+00	1.6	1.7
	4.500	-0.406	2.020E+00	1.940E+00	1.6	1.7
	5.000	-0.383	2.880E+00	2.750E+00	1.6	1.7
	5.500	-0.362	3.930E+00	3.750E+00	1.7	3.0
	6.000	-0.343	5.210E+00	4.950E+00	1.7	3.0
	7.000	-0.310	8.430E+00	7.950E+00	1.7	3.0
	8.000	-0.281	1.260E+01	1.190E+01	1.7	1.9
	9.000	-0.255	1.780E+01	1.680E+01	1.3	1.7

TABLE I. Empirical Reference Cross Sections for *K*-Shell Ionization by α Particles

See page 84 for Explanation of Tables

TARGET	ENERGY [MeV]	LOG XI	SIGMA [barn] THEORY	SIGMA [barn] REFERENCE	ERROR [%] RND	ERROR [%] TOT
41-Nb	10.000	-0.232	2.400E+01	2.260E+01	1.3	1.7
	11.000	-0.212	3.110E+01	2.930E+01	1.3	1.7
	12.000	-0.193	3.910E+01	3.700E+01	1.3	1.7
	14.000	-0.159	5.760E+01	5.480E+01	1.3	1.6
	16.000	-0.130	7.900E+01	7.550E+01	1.3	1.6
	18.000	-0.105	1.030E+02	9.820E+01	1.3	1.6
	20.000	-0.082	1.280E+02	1.230E+02	1.3	1.6
	23.000	-0.052	1.680E+02	1.630E+02	1.4	1.6
	26.000	-0.025	2.100E+02	2.040E+02	1.4	1.6
	30.000	0.006	2.660E+02	2.600E+02	1.4	1.6
	35.000	0.040	3.340E+02	3.290E+02	1.5	1.7
	40.000	0.069	3.980E+02	3.950E+02	1.5	1.7
	45.000	0.094	4.570E+02	4.560E+02	1.5	1.7
	50.000	0.117	5.100E+02	5.120E+02	2.3	2.9
	55.000	0.138	5.580E+02	5.630E+02	2.3	2.9
	60.000	0.157	6.010E+02	6.140E+02	2.3	2.9
	70.000	0.190	6.730E+02	6.940E+02	2.3	2.9
	80.000	0.219	7.280E+02	7.580E+02	3.9	4.4
	90.000	0.245	7.710E+02	8.070E+02	3.9	4.4
	100.000	0.268	8.020E+02	8.450E+02	3.9	4.4
	110.000	0.288	8.250E+02	8.730E+02	3.9	4.4
	120.000	0.307	8.400E+02	8.930E+02	3.9	5.0
	140.000	0.341	8.570E+02	9.150E+02	4.9	6.0
	160.000	0.370	8.580E+02	9.200E+02	4.9	6.0
	180.000	0.395	8.510E+02	9.130E+02	4.9	6.0
	200.000	0.418	8.380E+02	8.990E+02	4.9	6.0
	230.000	0.448	8.130E+02	8.700E+02	7.7	8.8
	260.000	0.475	7.850E+02	8.360E+02	7.7	8.8
	300.000	0.506	7.450E+02	7.880E+02	7.7	8.8
	350.000	0.540	6.970E+02	7.290E+02	7.9	9.1
	400.000	0.569	6.530E+02	6.730E+02	7.9	9.1
42-Mo	0.700	-0.822	3.410E-04	3.250E-04	9.4	24.8
	0.800	-0.793	8.320E-04	8.070E-04	9.4	18.5
	0.900	-0.767	1.710E-03	1.680E-03	9.4	18.3
	1.000	-0.744	3.100E-03	3.070E-03	6.5	10.8
	1.100	-0.724	5.160E-03	5.140E-03	6.5	10.8
	1.200	-0.705	8.010E-03	8.010E-03	6.5	10.8
	1.400	-0.671	1.670E-02	1.680E-02	6.5	10.7
	1.600	-0.642	3.040E-02	3.050E-02	3.1	7.0
	1.800	-0.617	5.020E-02	5.040E-02	3.1	7.0
	2.000	-0.594	7.750E-02	7.770E-02	3.1	5.7
	2.300	-0.564	1.350E-01	1.350E-01	2.3	4.7
	2.600	-0.537	2.170E-01	2.150E-01	2.3	4.7
	3.000	-0.506	3.710E-01	3.660E-01	2.3	4.8
	3.500	-0.472	6.500E-01	6.340E-01	1.6	2.1
	4.000	-0.443	1.040E+00	1.010E+00	1.6	2.1
	4.500	-0.418	1.560E+00	1.510E+00	1.6	2.1
	5.000	-0.395	2.230E+00	2.140E+00	1.6	2.1
	5.500	-0.374	3.060E+00	2.920E+00	1.7	2.4
	6.000	-0.355	4.050E+00	3.860E+00	1.7	2.4
	7.000	-0.322	6.600E+00	6.220E+00	1.7	2.5
	8.000	-0.293	9.910E+00	9.330E+00	1.7	1.8
	9.000	-0.267	1.400E+01	1.320E+01	1.3	1.6
	10.000	-0.244	1.890E+01	1.790E+01	1.3	1.6
	11.000	-0.224	2.460E+01	2.320E+01	1.3	1.6
	12.000	-0.205	3.110E+01	2.940E+01	1.3	1.6
	14.000	-0.171	4.610E+01	4.380E+01	1.3	1.5
	16.000	-0.142	6.360E+01	6.070E+01	1.3	1.5
	18.000	-0.117	8.320E+01	7.940E+01	1.3	1.5
	20.000	-0.094	1.040E+02	1.000E+02	1.3	1.5
	23.000	-0.064	1.380E+02	1.330E+02	1.4	1.5
	26.000	-0.037	1.730E+02	1.680E+02	1.4	1.5
	30.000	-0.006	2.210E+02	2.160E+02	1.4	1.5
	35.000	0.028	2.800E+02	2.760E+02	1.5	1.6
	40.000	0.057	3.360E+02	3.330E+02	1.5	1.6
	45.000	0.082	3.890E+02	3.880E+02	1.5	1.6
	50.000	0.105	4.370E+02	4.380E+02	2.3	3.1
	55.000	0.126	4.800E+02	4.830E+02	2.3	3.1
	60.000	0.145	5.200E+02	5.280E+02	2.3	3.1
	70.000	0.178	5.860E+02	6.020E+02	2.3	3.1
	80.000	0.207	6.390E+02	6.620E+02	2.3	3.1
	90.000	0.233	6.790E+02	7.090E+02	3.9	4.2
	100.000	0.256	7.100E+02	7.460E+02	3.9	4.2
	110.000	0.276	7.330E+02	7.740E+02	3.9	4.2
	120.000	0.295	7.500E+02	7.950E+02	3.9	4.2
	140.000	0.329	7.690E+02	8.200E+02	4.9	5.5
	160.000	0.358	7.740E+02	8.290E+02	4.9	5.5
	180.000	0.383	7.710E+02	8.270E+02	4.9	5.5
	200.000	0.406	7.620E+02	8.170E+02	4.9	5.5
	230.000	0.436	7.420E+02	7.950E+02	7.7	8.3
	260.000	0.463	7.180E+02	7.670E+02	7.7	8.3
42-Mo	300.000	0.494	6.850E+02	7.260E+02	7.7	8.3
	350.000	0.528	6.430E+02	6.750E+02	7.9	8.6
	400.000	0.557	6.040E+02	6.260E+02	7.9	8.6
43-Tc	0.800	-0.805	5.740E-04	5.530E-04	9.4	22.0
	0.900	-0.779	1.210E-03	1.180E-03	9.4	16.9
	1.000	-0.756	2.230E-03	2.200E-03	6.5	9.3
	1.100	-0.736	3.760E-03	3.740E-03	6.5	9.3
	1.200	-0.717	5.910E-03	5.900E-03	6.5	9.3
	1.400	-0.683	1.250E-02	1.260E-02	6.5	9.3
	1.600	-0.654	2.300E-02	2.310E-02	3.1	8.5
	1.800	-0.629	3.820E-02	3.840E-02	3.1	8.5
	2.000	-0.606	5.930E-02	5.950E-02	3.1	8.5
	2.300	-0.575	1.040E-01	1.040E-01	3.1	6.7
	2.600	-0.549	1.680E-01	1.670E-01	2.3	5.7
	3.000	-0.518	2.870E-01	2.840E-01	2.3	5.8
	3.500	-0.484	5.040E-01	4.940E-01	2.3	4.0
	4.000	-0.455	8.110E-01	7.870E-01	1.6	2.6
	4.500	-0.430	1.220E+00	1.180E+00	1.6	2.6
	5.000	-0.407	1.740E+00	1.680E+00	1.6	2.6
	5.500	-0.386	2.390E+00	2.290E+00	1.6	2.6
	6.000	-0.367	3.180E+00	3.030E+00	1.7	1.9
	7.000	-0.334	5.190E+00	4.920E+00	1.7	1.9
	8.000	-0.305	7.820E+00	7.370E+00	1.7	1.9
	9.000	-0.279	1.110E+01	1.050E+01	1.3	1.5
	10.000	-0.256	1.500E+01	1.420E+01	1.3	1.5
	11.000	-0.236	1.960E+01	1.850E+01	1.3	1.5
	12.000	-0.217	2.490E+01	2.350E+01	1.3	1.5
	14.000	-0.183	3.710E+01	3.520E+01	1.3	1.5
	16.000	-0.154	5.150E+01	4.900E+01	1.3	1.4
	18.000	-0.129	6.760E+01	6.470E+01	1.3	1.4
	20.000	-0.106	8.530E+01	8.160E+01	1.3	1.4
	23.000	-0.075	1.140E+02	1.090E+02	1.3	1.4
	26.000	-0.049	1.440E+02	1.390E+02	1.4	1.4
	30.000	-0.018	1.850E+02	1.800E+02	1.4	1.4
	35.000	0.016	2.360E+02	2.310E+02	1.4	1.4
	40.000	0.045	2.850E+02	2.810E+02	1.5	1.5
	45.000	0.070	3.320E+02	3.300E+02	1.5	1.5
	50.000	0.093	3.750E+02	3.740E+02	1.5	1.5
	55.000	0.114	4.140E+02	4.160E+02	2.3	3.4
	60.000	0.133	4.500E+02	4.530E+02	2.3	3.4
	70.000	0.166	5.110E+02	5.230E+02	2.3	3.4
	80.000	0.195	5.600E+02	5.790E+02	2.3	3.3
	90.000	0.221	5.990E+02	6.240E+02	3.9	4.0
	100.000	0.244	6.290E+02	6.590E+02	3.9	4.0
	110.000	0.264	6.520E+02	6.870E+02	3.9	4.0
	120.000	0.283	6.700E+02	7.080E+02	3.9	4.0
	140.000	0.317	6.910E+02	7.350E+02	3.9	4.0
	160.000	0.346	6.990E+02	7.470E+02	4.9	5.0
	180.000	0.371	6.990E+02	7.490E+02	4.9	5.0
	200.000	0.394	6.930E+02	7.430E+02	4.9	5.0
	230.000	0.425	6.780E+02	7.270E+02	4.9	5.0
	260.000	0.451	6.580E+02	7.040E+02	7.7	7.9
	300.000	0.482	6.300E+02	6.700E+02	7.7	7.9
	350.000	0.516	5.930E+02	6.250E+02	7.7	7.9
	400.000	0.545	5.590E+02	5.820E+02	7.9	8.1
	450.000	0.570	5.270E+02	5.420E+02	7.9	8.1
44-Ru	0.800	-0.816	3.950E-04	3.770E-04	9.4	19.5
	0.900	-0.791	8.520E-04	8.270E-04	9.4	15.4
	1.000	-0.768	1.610E-03	1.580E-03	9.4	15.3
	1.100	-0.747	2.750E-03	2.720E-03	6.5	7.8
	1.200	-0.728	4.370E-03	4.350E-03	6.5	7.8
	1.400	-0.695	9.410E-03	9.430E-03	6.5	7.8
	1.600	-0.666	1.750E-02	1.750E-02	3.1	10.0
	1.800	-0.640	2.930E-02	2.940E-02	3.1	10.0
	2.000	-0.617	4.560E-02	4.580E-02	3.1	10.0
	2.300	-0.587	8.050E-02	8.060E-02	3.1	7.7
	2.600	-0.560	1.300E-01	1.300E-01	2.3	6.7
	3.000	-0.529	2.240E-01	2.220E-01	2.3	6.8
	3.500	-0.496	3.940E-01	3.870E-01	2.3	4.5
	4.000	-0.467	6.340E-01	6.180E-01	1.6	3.1
	4.500	-0.441	9.560E-01	9.250E-01	1.6	3.1
	5.000	-0.418	1.370E+00	1.320E+00	1.6	3.1
	5.500	-0.398	1.880E+00	1.810E+00	1.6	3.2
	6.000	-0.379	2.510E+00	2.400E+00	1.7	2.0
	7.000	-0.345	4.100E+00	3.900E+00	1.7	2.0
	8.000	-0.316	6.200E+00	5.840E+00	1.7	2.0
	9.000	-0.291	8.830E+00	8.320E+00	1.7	1.7
	10.000	-0.268	1.200E+01	1.130E+01	1.3	1.4
	11.000	-0.247	1.570E+01	1.480E+01	1.3	1.4

TABLE I. Empirical Reference Cross Sections for *K*-Shell Ionization by α Particles

See page 84 for Explanation of Tables

TARGET	ENERGY [MeV]	LOG XI	SIGMA [barn] THEORY	SIGMA [barn] REFERENCE	ERROR [%] RND	ERROR [%] TOT
44-Ru	12.000	-0.228	2.000E+01	1.880E+01	1.3	1.4
	14.000	-0.195	2.990E+01	2.830E+01	1.3	1.4
	16.000	-0.166	4.180E+01	3.970E+01	1.3	1.3
	18.000	-0.140	5.520E+01	5.270E+01	1.3	1.3
	20.000	-0.117	6.990E+01	6.670E+01	1.3	1.3
	23.000	-0.087	9.380E+01	9.010E+01	1.3	1.3
	26.000	-0.060	1.190E+02	1.150E+02	1.4	1.4
	30.000	-0.029	1.550E+02	1.500E+02	1.4	1.4
	35.000	0.004	1.990E+02	1.950E+02	1.4	1.4
	40.000	0.033	2.420E+02	2.380E+02	1.5	1.6
	45.000	0.059	2.830E+02	2.800E+02	1.5	1.6
	50.000	0.082	3.220E+02	3.210E+02	1.5	1.6
	55.000	0.102	3.570E+02	3.580E+02	2.3	3.6
	60.000	0.121	3.900E+02	3.920E+02	2.3	3.6
	70.000	0.155	4.460E+02	4.550E+02	2.3	3.6
	80.000	0.184	4.920E+02	5.070E+02	2.3	3.6
	90.000	0.209	5.290E+02	5.490E+02	2.3	3.6
	100.000	0.232	5.580E+02	5.830E+02	3.9	4.0
	110.000	0.253	5.810E+02	6.100E+02	3.9	4.0
	120.000	0.272	5.980E+02	6.310E+02	3.9	4.0
	140.000	0.305	6.210E+02	6.590E+02	3.9	4.2
	160.000	0.334	6.310E+02	6.740E+02	4.9	5.2
	180.000	0.360	6.340E+02	6.780E+02	4.9	5.2
	200.000	0.383	6.310E+02	6.760E+02	4.9	5.2
	230.000	0.413	6.190E+02	6.640E+02	4.9	5.2
	260.000	0.440	6.040E+02	6.460E+02	7.7	8.1
	300.000	0.471	5.790E+02	6.180E+02	7.7	8.1
	350.000	0.504	5.480E+02	5.800E+02	7.7	8.1
	400.000	0.533	5.170E+02	5.420E+02	7.9	8.3
	450.000	0.559	4.890E+02	5.060E+02	7.9	8.3
45-Rh	0.800	-0.828	2.710E-04	2.570E-04	9.4	16.9
	0.900	-0.802	6.010E-04	5.800E-04	9.4	16.8
	1.000	-0.779	1.160E-03	1.130E-03	9.4	13.8
	1.100	-0.758	2.010E-03	1.980E-03	6.5	6.7
	1.200	-0.740	3.240E-03	3.210E-03	6.5	6.7
	1.400	-0.706	7.100E-03	7.100E-03	6.5	6.7
	1.600	-0.677	1.330E-02	1.340E-02	6.5	6.7
	1.800	-0.652	2.250E-02	2.260E-02	3.1	11.5
	2.000	-0.629	3.530E-02	3.550E-02	3.1	11.5
	2.300	-0.598	6.260E-02	6.270E-02	3.1	8.7
	2.600	-0.572	1.020E-01	1.010E-01	3.1	8.7
	3.000	-0.541	1.750E-01	1.740E-01	2.3	7.8
	3.500	-0.507	3.090E-01	3.050E-01	2.3	7.8
	4.000	-0.478	4.990E-01	4.880E-01	2.3	5.1
	4.500	-0.453	7.540E-01	7.320E-01	1.6	3.7
	5.000	-0.430	1.080E+00	1.050E+00	1.6	3.7
	5.500	-0.409	1.490E+00	1.430E+00	1.6	3.7
	6.000	-0.390	1.990E+00	1.900E+00	1.6	3.7
	7.000	-0.357	3.260E+00	3.110E+00	1.7	2.5
	8.000	-0.328	4.940E+00	4.660E+00	1.7	2.5
	9.000	-0.302	7.060E+00	6.650E+00	1.7	2.5
	10.000	-0.279	9.620E+00	9.060E+00	1.3	1.5
	11.000	-0.258	1.260E+01	1.190E+01	1.3	1.5
	12.000	-0.240	1.610E+01	1.520E+01	1.3	1.5
	14.000	-0.206	2.430E+01	2.290E+01	1.3	1.5
	16.000	-0.177	3.400E+01	3.230E+01	1.3	1.5
	18.000	-0.152	4.510E+01	4.300E+01	1.3	1.4
	20.000	-0.129	5.740E+01	5.500E+01	1.3	1.4
	23.000	-0.098	7.760E+01	7.440E+01	1.3	1.4
	26.000	-0.072	9.930E+01	9.560E+01	1.4	1.5
	30.000	-0.041	1.290E+02	1.250E+02	1.4	1.5
	35.000	-0.007	1.680E+02	1.640E+02	1.4	1.5
	40.000	0.022	2.060E+02	2.020E+02	1.5	1.7
	45.000	0.047	2.420E+02	2.390E+02	1.5	1.7
	50.000	0.070	2.770E+02	2.750E+02	1.5	1.7
	55.000	0.091	3.090E+02	3.080E+02	1.5	1.7
	60.000	0.110	3.380E+02	3.390E+02	2.3	3.9
	70.000	0.143	3.900E+02	3.940E+02	2.3	3.9
	80.000	0.172	4.330E+02	4.440E+02	2.3	3.8
	90.000	0.198	4.670E+02	4.830E+02	2.3	3.8
	100.000	0.221	4.950E+02	5.160E+02	3.9	4.3
	110.000	0.242	5.180E+02	5.420E+02	3.9	4.3
	120.000	0.260	5.350E+02	5.630E+02	3.9	4.3
	140.000	0.294	5.580E+02	5.920E+02	3.9	4.3
	160.000	0.323	5.710E+02	6.080E+02	3.9	4.6
	180.000	0.348	5.750E+02	6.150E+02	4.9	5.7
	200.000	0.371	5.740E+02	6.150E+02	4.9	5.7
	230.000	0.402	5.660E+02	6.070E+02	4.9	5.7
	260.000	0.428	5.540E+02	5.940E+02	4.9	5.7
	300.000	0.459	5.340E+02	5.700E+02	7.7	8.6
	350.000	0.493	5.070E+02	5.380E+02	7.7	8.6
45-Rh	400.000	0.522	4.800E+02	5.040E+02	7.9	8.7
	450.000	0.547	4.540E+02	4.730E+02	7.9	8.7
	500.000	0.570	4.300E+02	4.430E+02	7.9	8.7
46-Pd	0.900	-0.813	4.240E-04	3.990E-04	5.2	21.0
	1.000	-0.790	8.360E-04	7.960E-04	5.2	14.6
	1.100	-0.769	1.480E-03	1.420E-03	5.2	14.5
	1.200	-0.751	2.410E-03	2.340E-03	3.6	9.1
	1.400	-0.717	5.380E-03	5.280E-03	3.6	9.0
	1.600	-0.688	1.020E-02	1.010E-02	3.6	9.0
	1.800	-0.663	1.740E-02	1.730E-02	2.5	7.5
	2.000	-0.640	2.750E-02	2.730E-02	2.5	7.5
	2.300	-0.609	4.900E-02	4.880E-02	2.5	7.4
	2.600	-0.583	7.970E-02	7.950E-02	2.5	5.8
	3.000	-0.552	1.380E-01	1.380E-01	1.6	8.3
	3.500	-0.518	2.450E-01	2.430E-01	1.6	8.3
	4.000	-0.489	3.960E-01	3.920E-01	1.6	5.0
	4.500	-0.464	5.990E-01	5.910E-01	1.2	4.8
	5.000	-0.441	8.600E-01	8.460E-01	1.2	4.8
	5.500	-0.420	1.190E+00	1.160E+00	1.2	4.8
	6.000	-0.401	1.580E+00	1.550E+00	1.2	4.8
	7.000	-0.368	2.610E+00	2.540E+00	1.2	4.7
	8.000	-0.339	3.960E+00	3.850E+00	1.2	4.7
	9.000	-0.313	5.670E+00	5.490E+00	1.2	4.7
	10.000	-0.290	7.750E+00	7.490E+00	1.2	1.9
	11.000	-0.269	1.020E+01	9.850E+00	0.9	1.5
	12.000	-0.251	1.300E+01	1.250E+01	0.9	1.5
	14.000	-0.217	1.970E+01	1.900E+01	0.9	1.5
	16.000	-0.188	2.780E+01	2.680E+01	0.9	1.5
	18.000	-0.163	3.710E+01	3.590E+01	0.8	1.5
	20.000	-0.140	4.740E+01	4.600E+01	0.8	1.5
	23.000	-0.109	6.440E+01	6.280E+01	0.8	1.5
	26.000	-0.083	8.280E+01	8.080E+01	0.8	1.5
	30.000	-0.052	1.090E+02	1.060E+02	0.9	1.5
	35.000	-0.018	1.420E+02	1.390E+02	0.9	1.5
	40.000	0.011	1.750E+02	1.730E+02	0.9	1.5
	45.000	0.036	2.070E+02	2.070E+02	1.3	1.6
	50.000	0.059	2.380E+02	2.390E+02	1.3	1.6
	55.000	0.080	2.670E+02	2.690E+02	1.3	1.6
	60.000	0.099	2.940E+02	2.970E+02	1.3	1.6
	70.000	0.132	3.410E+02	3.490E+02	1.7	3.5
	80.000	0.161	3.810E+02	3.920E+02	1.7	3.5
	90.000	0.187	4.130E+02	4.320E+02	1.7	3.5
	100.000	0.210	4.400E+02	4.630E+02	1.7	3.5
	110.000	0.231	4.620E+02	4.890E+02	1.8	3.7
	120.000	0.249	4.790E+02	5.100E+02	1.8	3.7
	140.000	0.283	5.030E+02	5.410E+02	1.8	3.7
	160.000	0.312	5.160E+02	5.590E+02	1.8	5.5
	180.000	0.337	5.220E+02	5.680E+02	3.0	5.3
	200.000	0.360	5.230E+02	5.710E+02	3.0	5.3
	230.000	0.391	5.180E+02	5.670E+02	3.0	5.3
	260.000	0.417	5.090E+02	5.570E+02	4.7	7.6
	300.000	0.448	4.920E+02	5.370E+02	4.7	7.6
	350.000	0.482	4.690E+02	5.090E+02	4.7	7.6
	400.000	0.511	4.450E+02	4.790E+02	7.0	8.3
	450.000	0.536	4.220E+02	4.500E+02	7.0	8.4
	500.000	0.559	4.010E+02	4.220E+02	7.0	8.4
47-Ag	0.900	-0.824	2.980E-04	2.790E-04	5.2	6.6
	1.000	-0.801	6.010E-04	5.690E-04	5.2	6.5
	1.100	-0.780	1.080E-03	1.040E-03	5.2	6.5
	1.200	-0.761	1.790E-03	1.730E-03	3.6	4.2
	1.400	-0.728	4.080E-03	3.990E-03	3.6	4.2
	1.600	-0.699	7.850E-03	7.740E-03	3.6	4.2
	1.800	-0.673	1.350E-02	1.340E-02	3.6	4.2
	2.000	-0.651	2.140E-02	2.130E-02	2.5	3.1
	2.300	-0.620	3.850E-02	3.830E-02	2.5	3.1
	2.600	-0.594	6.290E-02	6.270E-02	2.5	3.1
	3.000	-0.563	1.090E-01	1.090E-01	1.6	3.1
	3.500	-0.529	1.940E-01	1.930E-01	1.6	3.1
	4.000	-0.500	3.150E-01	3.120E-01	1.6	3.1
	4.500	-0.474	4.770E-01	4.710E-01	1.6	3.1
	5.000	-0.452	6.860E-01	6.760E-01	1.2	2.7
	5.500	-0.431	9.480E-01	9.310E-01	1.2	2.7
	6.000	-0.412	1.270E+00	1.240E+00	1.2	2.7
	7.000	-0.379	2.090E+00	2.040E+00	1.2	2.7
	8.000	-0.350	3.190E+00	3.100E+00	1.2	2.7
	9.000	-0.324	4.570E+00	4.430E+00	1.2	2.7
	10.000	-0.301	6.260E+00	6.050E+00	1.2	2.7

TABLE I. Empirical Reference Cross Sections for *K*-Shell Ionization by α Particles

See page 84 for Explanation of Tables

TARGET	ENERGY [MeV]	LOG XI	SIGMA [barn] THEORY	SIGMA [barn] REFERENCE	ERROR [%] RND	ERROR [%] TOT
47-Ag	11.000	-0.280	8.260E+00	7.980E+00	1.2	2.7
	12.000	-0.261	1.060E+01	1.020E+01	0.9	2.3
	14.000	-0.228	1.610E+01	1.550E+01	0.9	2.3
	16.000	-0.199	2.280E+01	2.190E+01	0.9	2.3
	18.000	-0.173	3.050E+01	2.950E+01	0.9	2.3
	20.000	-0.151	3.910E+01	3.790E+01	0.8	2.2
	23.000	-0.120	5.350E+01	5.210E+01	0.8	2.2
	26.000	-0.094	6.920E+01	6.750E+01	0.8	2.2
	30.000	-0.063	9.140E+01	8.930E+01	0.9	2.3
	35.000	-0.029	1.200E+02	1.180E+02	0.9	2.3
	40.000	0.000	1.490E+02	1.470E+02	0.9	2.3
	45.000	0.026	1.780E+02	1.770E+02	1.3	1.9
	50.000	0.048	2.050E+02	2.050E+02	1.3	1.9
	55.000	0.069	2.310E+02	2.320E+02	1.3	1.9
	60.000	0.088	2.550E+02	2.580E+02	1.3	1.9
	70.000	0.121	2.980E+02	3.040E+02	1.7	4.9
	80.000	0.150	3.350E+02	3.440E+02	1.7	4.9
	90.000	0.176	3.660E+02	3.810E+02	1.7	4.8
	100.000	0.199	3.910E+02	4.100E+02	1.7	4.8
	110.000	0.220	4.120E+02	4.350E+02	1.7	7.9
	120.000	0.239	4.290E+02	4.550E+02	1.8	8.3
	140.000	0.272	4.530E+02	4.850E+02	1.8	8.2
	160.000	0.301	4.670E+02	5.040E+02	1.8	8.2
	180.000	0.327	4.740E+02	5.150E+02	3.0	6.6
	200.000	0.349	4.770E+02	5.200E+02	3.0	6.6
	230.000	0.380	4.740E+02	5.190E+02	3.0	6.6
	260.000	0.406	4.670E+02	5.110E+02	3.0	6.6
	300.000	0.437	4.540E+02	4.960E+02	4.7	9.5
	350.000	0.471	4.340E+02	4.720E+02	4.7	9.5
	400.000	0.500	4.130E+02	4.460E+02	4.7	9.6
	450.000	0.526	3.930E+02	4.200E+02	7.0	8.7
	500.000	0.548	3.740E+02	3.960E+02	7.0	8.7
	550.000	0.569	3.560E+02	3.730E+02	7.0	8.8
48-Cd	1.000	-0.812	4.320E-04	4.070E-04	5.2	6.3
	1.100	-0.791	7.930E-04	7.550E-04	5.2	6.3
	1.200	-0.772	1.330E-03	1.260E-03	5.2	6.3
	1.400	-0.739	3.100E-03	3.020E-03	3.6	4.0
	1.600	-0.710	6.060E-03	5.950E-03	3.6	4.0
	1.800	-0.684	1.050E-02	1.040E-02	3.6	4.0
	2.000	-0.661	1.680E-02	1.670E-02	2.5	2.8
	2.300	-0.631	3.040E-02	3.020E-02	2.5	2.8
	2.600	-0.604	4.980E-02	4.970E-02	2.5	2.8
	3.000	-0.573	8.700E-02	8.670E-02	2.5	2.8
	3.500	-0.540	1.550E-01	1.540E-01	1.6	2.8
	4.000	-0.511	2.520E-01	2.500E-01	1.6	2.8
	4.500	-0.485	3.820E-01	3.780E-01	1.6	2.8
	5.000	-0.462	5.510E-01	5.430E-01	1.2	2.5
	5.500	-0.442	7.610E-01	7.490E-01	1.2	2.5
	6.000	-0.423	1.020E+00	9.990E-01	1.2	2.5
	7.000	-0.389	1.680E+00	1.650E+00	1.2	2.5
	8.000	-0.360	2.570E+00	2.510E+00	1.2	2.4
	9.000	-0.335	3.700E+00	3.600E+00	1.2	2.4
	10.000	-0.312	5.080E+00	4.910E+00	1.2	2.4
	11.000	-0.291	6.710E+00	6.490E+00	1.2	2.4
	12.000	-0.272	8.600E+00	8.320E+00	0.9	2.0
	14.000	-0.239	1.320E+01	1.270E+01	0.9	2.0
	16.000	-0.210	1.870E+01	1.800E+01	0.9	2.0
	18.000	-0.184	2.520E+01	2.430E+01	0.9	2.0
	20.000	-0.161	3.240E+01	3.140E+01	0.8	2.0
	23.000	-0.131	4.450E+01	4.330E+01	0.8	2.0
	26.000	-0.104	5.790E+01	5.640E+01	0.8	2.0
	30.000	-0.073	7.700E+01	7.520E+01	0.9	2.0
	35.000	-0.040	1.020E+02	9.980E+01	0.9	2.0
	40.000	-0.011	1.270E+02	1.250E+02	0.9	2.0
	45.000	0.015	1.530E+02	1.510E+02	1.3	1.6
	50.000	0.038	1.770E+02	1.760E+02	1.3	1.6
	55.000	0.058	2.000E+02	2.000E+02	1.3	1.6
	60.000	0.077	2.220E+02	2.230E+02	1.3	1.6
	70.000	0.111	2.610E+02	2.660E+02	1.3	1.9
	80.000	0.140	2.950E+02	3.020E+02	1.7	4.4
	90.000	0.165	3.240E+02	3.340E+02	1.7	4.4
	100.000	0.188	3.480E+02	3.630E+02	1.7	4.3
	110.000	0.209	3.680E+02	3.870E+02	1.7	6.9
	120.000	0.228	3.840E+02	4.060E+02	1.8	7.4
	140.000	0.261	4.080E+02	4.360E+02	1.8	7.3
	160.000	0.290	4.230E+02	4.560E+02	1.8	7.3
	180.000	0.316	4.310E+02	4.670E+02	1.8	7.2
	200.000	0.339	4.350E+02	4.730E+02	3.0	5.7
	230.000	0.369	4.350E+02	4.750E+02	3.0	5.7
	260.000	0.396	4.290E+02	4.700E+02	3.0	5.7
	300.000	0.427	4.180E+02	4.580E+02	4.7	8.6
48-Cd	350.000	0.460	4.010E+02	4.370E+02	4.7	8.6
	400.000	0.489	3.830E+02	4.150E+02	4.7	8.6
	450.000	0.515	3.650E+02	3.930E+02	7.0	7.8
	500.000	0.538	3.480E+02	3.710E+02	7.0	7.8
	550.000	0.558	3.320E+02	3.500E+02	7.0	7.8
49-In	1.000	-0.822	3.100E-04	2.890E-04	5.2	6.0
	1.100	-0.802	5.790E-04	5.490E-04	5.2	6.0
	1.200	-0.783	9.890E-04	9.460E-04	5.2	6.0
	1.400	-0.749	2.360E-03	2.290E-03	3.6	3.7
	1.600	-0.720	4.680E-03	4.580E-03	3.6	3.7
	1.800	-0.695	8.210E-03	8.100E-03	3.6	3.7
	2.000	-0.672	1.320E-02	1.310E-02	3.6	3.7
	2.300	-0.641	2.400E-02	2.390E-02	2.5	2.6
	2.600	-0.615	3.970E-02	3.950E-02	2.5	2.6
	3.000	-0.584	6.950E-02	6.930E-02	2.5	2.6
	3.500	-0.550	1.240E-01	1.240E-01	1.6	2.6
	4.000	-0.521	2.020E-01	2.010E-01	1.6	2.6
	4.500	-0.496	3.080E-01	3.050E-01	1.6	2.6
	5.000	-0.473	4.440E-01	4.380E-01	1.6	2.6
	5.500	-0.452	6.140E-01	6.050E-01	1.2	2.2
	6.000	-0.433	8.220E-01	8.080E-01	1.2	2.2
	7.000	-0.400	1.360E+00	1.330E+00	1.2	2.2
	8.000	-0.371	2.090E+00	2.040E+00	1.2	2.2
	9.000	-0.345	3.000E+00	2.920E+00	1.2	2.2
	10.000	-0.322	4.130E+00	4.000E+00	1.2	2.2
	11.000	-0.302	5.470E+00	5.290E+00	1.2	2.2
	12.000	-0.283	7.030E+00	6.800E+00	1.2	2.2
	14.000	-0.249	1.080E+01	1.040E+01	0.9	1.7
	16.000	-0.220	1.540E+01	1.480E+01	0.9	1.7
	18.000	-0.195	2.080E+01	2.000E+01	0.9	1.7
	20.000	-0.172	2.690E+01	2.600E+01	0.9	1.7
	23.000	-0.141	3.720E+01	3.610E+01	0.8	1.7
	26.000	-0.115	4.850E+01	4.730E+01	0.8	1.7
	30.000	-0.084	6.500E+01	6.340E+01	0.8	1.7
	35.000	-0.050	8.670E+01	8.470E+01	0.9	1.8
	40.000	-0.021	1.090E+02	1.070E+02	0.9	1.8
	45.000	0.004	1.310E+02	1.290E+02	0.9	1.8
	50.000	0.027	1.530E+02	1.520E+02	1.3	1.4
	55.000	0.048	1.730E+02	1.730E+02	1.3	1.4
	60.000	0.067	1.930E+02	1.940E+02	1.3	1.4
	70.000	0.100	2.290E+02	2.320E+02	1.3	1.4
	80.000	0.129	2.600E+02	2.660E+02	1.7	3.9
	90.000	0.155	2.870E+02	2.950E+02	1.7	3.9
	100.000	0.178	3.090E+02	3.220E+02	1.7	3.9
	110.000	0.198	3.280E+02	3.440E+02	1.7	3.8
	120.000	0.217	3.440E+02	3.630E+02	1.7	6.0
	140.000	0.251	3.670E+02	3.920E+02	1.8	6.4
	160.000	0.280	3.830E+02	4.110E+02	1.8	6.4
	180.000	0.305	3.920E+02	4.240E+02	1.8	6.3
	200.000	0.328	3.970E+02	4.310E+02	3.0	4.8
	230.000	0.359	3.980E+02	4.340E+02	3.0	4.8
	260.000	0.385	3.950E+02	4.320E+02	3.0	4.8
	300.000	0.416	3.860E+02	4.220E+02	4.7	7.7
	350.000	0.450	3.720E+02	4.060E+02	4.7	7.7
	400.000	0.479	3.560E+02	3.870E+02	4.7	7.7
	450.000	0.504	3.400E+02	3.670E+02	4.7	7.7
	500.000	0.527	3.250E+02	3.480E+02	7.0	7.2
	550.000	0.548	3.100E+02	3.290E+02	7.0	7.2
	600.000	0.567	2.970E+02	3.110E+02	7.0	7.2
50-Sn	1.100	-0.812	4.230E-04	3.990E-04	5.2	5.7
	1.200	-0.793	7.350E-04	6.990E-04	5.2	5.7
	1.400	-0.760	1.800E-03	1.740E-03	3.6	3.7
	1.600	-0.731	3.620E-03	3.540E-03	3.6	3.7
	1.800	-0.705	6.430E-03	6.330E-03	3.6	3.7
	2.000	-0.682	1.040E-02	1.030E-02	3.6	3.7
	2.300	-0.652	1.910E-02	1.900E-02	2.5	2.7
	2.600	-0.625	3.170E-02	3.160E-02	2.5	2.7
	3.000	-0.594	5.580E-02	5.560E-02	2.5	2.7
	3.500	-0.561	1.000E-01	9.980E-02	1.6	2.3
	4.000	-0.532	1.630E-01	1.620E-01	1.6	2.3
	4.500	-0.506	2.490E-01	2.470E-01	1.6	2.3
	5.000	-0.483	3.590E-01	3.550E-01	1.6	2.3
	5.500	-0.462	4.980E-01	4.910E-01	1.2	2.0
	6.000	-0.444	6.670E-01	6.570E-01	1.2	2.0
	7.000	-0.410	1.110E+00	1.090E+00	1.2	2.0
	8.000	-0.381	1.700E+00	1.660E+00	1.2	2.0
	9.000	-0.356	2.450E+00	2.390E+00	1.2	1.9
	10.000	-0.333	3.380E+00	3.280E+00	1.2	1.9
	11.000	-0.312	4.480E+00	4.340E+00	1.2	1.9

TABLE I. Empirical Reference Cross Sections for *K*-Shell Ionization by α Particles

See page 84 for Explanation of Tables

TARGET	ENERGY [MeV]	LOG XI	SIGMA [barn] THEORY	SIGMA [barn] REFERENCE	ERROR [%] RND	ERROR [%] TOT
50-Sn	12.000	-0.293	5.770E+00	5.580E+00	1.2	1.9
	14.000	-0.260	8.890E+00	8.580E+00	0.9	1.5
	16.000	-0.231	1.270E+01	1.220E+01	0.9	1.5
	18.000	-0.205	1.720E+01	1.660E+01	0.9	1.5
	20.000	-0.182	2.240E+01	2.160E+01	0.9	1.5
	23.000	-0.152	3.110E+01	3.010E+01	0.8	1.5
	26.000	-0.125	4.080E+01	3.970E+01	0.8	1.5
	30.000	-0.094	5.490E+01	5.350E+01	0.8	1.5
	35.000	-0.061	7.370E+01	7.200E+01	0.9	1.5
	40.000	-0.032	9.320E+01	9.120E+01	0.9	1.5
	45.000	-0.006	1.130E+02	1.110E+02	0.9	1.5
	50.000	0.017	1.320E+02	1.310E+02	1.3	1.5
	55.000	0.038	1.500E+02	1.500E+02	1.3	1.5
	60.000	0.056	1.680E+02	1.680E+02	1.3	1.5
	70.000	0.090	2.010E+02	2.030E+02	1.3	1.5
	80.000	0.119	2.290E+02	2.340E+02	1.7	3.4
	90.000	0.144	2.540E+02	2.610E+02	1.7	3.4
	100.000	0.167	2.750E+02	2.840E+02	1.7	3.4
	110.000	0.188	2.930E+02	3.060E+02	1.7	3.4
	120.000	0.207	3.080E+02	3.240E+02	1.7	5.0
	140.000	0.240	3.310E+02	3.520E+02	1.8	5.5
	160.000	0.269	3.470E+02	3.720E+02	1.8	5.4
	180.000	0.295	3.570E+02	3.850E+02	1.8	5.4
	200.000	0.318	3.620E+02	3.930E+02	3.0	3.9
	230.000	0.348	3.650E+02	3.980E+02	3.0	3.9
	260.000	0.375	3.630E+02	3.970E+02	3.0	3.9
	300.000	0.406	3.560E+02	3.900E+02	3.0	3.9
	350.000	0.439	3.450E+02	3.760E+02	4.7	6.8
	400.000	0.468	3.310E+02	3.600E+02	4.7	6.8
	450.000	0.494	3.170E+02	3.430E+02	4.7	6.8
	500.000	0.517	3.030E+02	3.260E+02	7.0	8.1
	550.000	0.538	2.900E+02	3.090E+02	7.0	8.2
	600.000	0.556	2.780E+02	2.930E+02	7.0	8.2
51-Sb	1.100	-0.822	3.090E-04	2.890E-04	5.2	5.5
	1.200	-0.803	5.450E-04	5.160E-04	5.2	5.5
	1.400	-0.770	1.370E-03	1.320E-03	5.2	5.5
	1.600	-0.741	2.810E-03	2.740E-03	3.6	3.9
	1.800	-0.715	5.050E-03	4.960E-03	3.6	3.9
	2.000	-0.692	8.260E-03	8.150E-03	3.6	3.9
	2.300	-0.662	1.530E-02	1.520E-02	2.5	2.9
	2.600	-0.635	2.550E-02	2.540E-02	2.5	2.9
	3.000	-0.604	4.510E-02	4.490E-02	2.5	2.9
	3.500	-0.571	8.110E-02	8.080E-02	1.6	2.1
	4.000	-0.542	1.330E-01	1.320E-01	1.6	2.1
	4.500	-0.516	2.020E-01	2.010E-01	1.6	2.1
	5.000	-0.493	2.920E-01	2.900E-01	1.6	2.1
	5.500	-0.473	4.060E-01	4.010E-01	1.6	2.1
	6.000	-0.454	5.440E-01	5.360E-01	1.2	1.7
	7.000	-0.420	9.050E-01	8.870E-01	1.2	1.7
	8.000	-0.391	1.390E+00	1.360E+00	1.2	1.7
	9.000	-0.366	2.010E+00	1.960E+00	1.2	1.7
	10.000	-0.343	2.770E+00	2.700E+00	1.2	1.7
	11.000	-0.322	3.680E+00	3.570E+00	1.2	1.7
	12.000	-0.303	4.750E+00	4.590E+00	1.2	1.7
	14.000	-0.270	7.350E+00	7.100E+00	0.9	1.2
	16.000	-0.241	1.060E+01	1.020E+01	0.9	1.2
	18.000	-0.215	1.430E+01	1.380E+01	0.9	1.2
	20.000	-0.192	1.870E+01	1.800E+01	0.9	1.2
	23.000	-0.162	2.600E+01	2.520E+01	0.8	1.2
	26.000	-0.135	3.430E+01	3.340E+01	0.8	1.2
	30.000	-0.104	4.650E+01	4.530E+01	0.8	1.2
	35.000	-0.071	6.280E+01	6.130E+01	0.9	1.3
	40.000	-0.042	7.990E+01	7.810E+01	0.9	1.3
	45.000	-0.016	9.710E+01	9.520E+01	0.9	1.3
	50.000	0.007	1.140E+02	1.120E+02	0.9	1.3
	55.000	0.027	1.310E+02	1.300E+02	1.3	1.7
	60.000	0.046	1.470E+02	1.460E+02	1.3	1.7
	70.000	0.080	1.760E+02	1.780E+02	1.3	1.7
	80.000	0.109	2.030E+02	2.060E+02	1.3	2.1
	90.000	0.134	2.250E+02	2.310E+02	1.7	2.9
	100.000	0.157	2.450E+02	2.520E+02	1.7	2.9
	110.000	0.178	2.620E+02	2.730E+02	1.7	2.9
	120.000	0.197	2.770E+02	2.900E+02	1.7	2.9
	140.000	0.230	2.990E+02	3.170E+02	1.8	4.5
	160.000	0.259	3.140E+02	3.360E+02	1.8	4.5
	180.000	0.285	3.240E+02	3.490E+02	1.8	4.5
	200.000	0.308	3.310E+02	3.580E+02	1.8	4.5
	230.000	0.338	3.350E+02	3.640E+02	3.0	3.0
	260.000	0.365	3.340E+02	3.650E+02	3.0	3.0
	300.000	0.396	3.290E+02	3.600E+02	3.0	3.0
	350.000	0.429	3.190E+02	3.490E+02	4.7	5.8
51-Sb	400.000	0.458	3.080E+02	3.360E+02	4.7	5.8
	450.000	0.484	2.960E+02	3.210E+02	4.7	5.9
	500.000	0.507	2.830E+02	3.050E+02	7.0	9.1
	550.000	0.527	2.720E+02	2.900E+02	7.0	9.1
	600.000	0.546	2.600E+02	2.760E+02	7.0	9.1
52-Te	1.200	-0.813	4.050E-04	3.810E-04	5.2	5.2
	1.400	-0.780	1.040E-03	1.000E-03	5.2	5.2
	1.600	-0.751	2.190E-03	2.120E-03	3.6	4.2
	1.800	-0.725	3.980E-03	3.890E-03	3.6	4.2
	2.000	-0.702	6.570E-03	6.470E-03	3.6	4.2
	2.300	-0.672	1.230E-02	1.220E-02	3.6	4.2
	2.600	-0.645	2.060E-02	2.040E-02	2.5	3.2
	3.000	-0.614	3.650E-02	3.640E-02	2.5	3.2
	3.500	-0.581	6.600E-02	6.580E-02	2.5	3.2
	4.000	-0.552	1.080E-01	1.080E-01	1.6	1.8
	4.500	-0.526	1.650E-01	1.640E-01	1.6	1.8
	5.000	-0.503	2.390E-01	2.370E-01	1.6	1.8
	5.500	-0.483	3.320E-01	3.280E-01	1.6	1.8
	6.000	-0.464	4.460E-01	4.400E-01	1.2	1.5
	7.000	-0.430	7.420E-01	7.290E-01	1.2	1.5
	8.000	-0.401	1.140E+00	1.120E+00	1.2	1.5
	9.000	-0.376	1.650E+00	1.610E+00	1.2	1.5
	10.000	-0.353	2.280E+00	2.220E+00	1.2	1.4
	11.000	-0.332	3.040E+00	2.960E+00	1.2	1.4
	12.000	-0.313	3.920E+00	3.800E+00	1.2	1.4
	14.000	-0.280	6.090E+00	5.880E+00	1.2	1.4
	16.000	-0.251	8.780E+00	8.460E+00	0.9	1.0
	18.000	-0.225	1.200E+01	1.150E+01	0.9	1.0
	20.000	-0.202	1.560E+01	1.500E+01	0.9	1.0
	23.000	-0.172	2.190E+01	2.120E+01	0.9	1.0
	26.000	-0.145	2.900E+01	2.810E+01	0.8	1.0
	30.000	-0.114	3.940E+01	3.840E+01	0.8	1.0
	35.000	-0.081	5.360E+01	5.230E+01	0.8	1.0
	40.000	-0.052	6.850E+01	6.700E+01	0.9	1.0
	45.000	-0.026	8.370E+01	8.200E+01	0.9	1.0
	50.000	-0.003	9.880E+01	9.720E+01	0.9	1.0
	55.000	0.017	1.140E+02	1.130E+02	1.3	2.0
	60.000	0.036	1.280E+02	1.270E+02	1.3	2.0
	70.000	0.070	1.550E+02	1.560E+02	1.3	2.0
	80.000	0.099	1.790E+02	1.810E+02	1.3	2.0
	90.000	0.124	2.000E+02	2.040E+02	1.7	2.4
	100.000	0.147	2.190E+02	2.240E+02	1.7	2.4
	110.000	0.168	2.350E+02	2.420E+02	1.7	2.4
	120.000	0.187	2.480E+02	2.590E+02	1.7	2.4
	140.000	0.220	2.700E+02	2.850E+02	1.7	3.1
	160.000	0.249	2.850E+02	3.040E+02	1.8	3.6
	180.000	0.275	2.950E+02	3.170E+02	1.8	3.6
	200.000	0.298	3.020E+02	3.260E+02	1.8	3.6
	230.000	0.328	3.070E+02	3.330E+02	3.0	4.0
	260.000	0.355	3.080E+02	3.360E+02	3.0	4.0
	300.000	0.386	3.040E+02	3.330E+02	3.0	4.0
	350.000	0.419	2.960E+02	3.240E+02	4.7	4.9
	400.000	0.448	2.860E+02	3.130E+02	4.7	4.9
	450.000	0.474	2.760E+02	3.000E+02	4.7	4.9
	500.000	0.497	2.650E+02	2.860E+02	4.7	4.9
	550.000	0.517	2.540E+02	2.730E+02	7.0	10.0
	600.000	0.536	2.440E+02	2.600E+02	7.0	10.0
	700.000	0.570	2.260E+02	2.360E+02	7.0	10.1
53-I	1.200	-0.823	2.990E-04	2.790E-04	5.2	5.5
	1.400	-0.789	7.940E-04	7.570E-04	5.2	5.5
	1.600	-0.760	1.700E-03	1.640E-03	3.6	4.5
	1.800	-0.735	3.130E-03	3.060E-03	3.6	4.5
	2.000	-0.712	5.230E-03	5.130E-03	3.6	4.4
	2.300	-0.682	9.860E-03	9.750E-03	3.6	4.4
	2.600	-0.655	1.660E-02	1.650E-02	2.5	3.5
	3.000	-0.624	2.970E-02	2.960E-02	2.5	3.4
	3.500	-0.591	5.390E-02	5.370E-02	2.5	3.4
	4.000	-0.562	8.860E-02	8.820E-02	1.6	1.7
	4.500	-0.536	1.350E-01	1.350E-01	1.6	1.7
	5.000	-0.513	1.960E-01	1.950E-01	1.6	1.7
	5.500	-0.492	2.730E-01	2.700E-01	1.6	1.7
	6.000	-0.473	3.670E-01	3.620E-01	1.6	1.7
	7.000	-0.440	6.110E-01	6.010E-01	1.2	1.2
	8.000	-0.411	9.410E-01	9.220E-01	1.2	1.2
	9.000	-0.385	1.360E+00	1.330E+00	1.2	1.2
	10.000	-0.363	1.890E+00	1.840E+00	1.2	1.3
	11.000	-0.342	2.520E+00	2.450E+00	1.2	1.3
	12.000	-0.323	3.250E+00	3.150E+00	1.2	1.3
	14.000	-0.289	5.060E+00	4.900E+00	1.2	1.3

TABLE I. Empirical Reference Cross Sections for *K*-Shell Ionization by α Particles

See page 84 for Explanation of Tables

TARGET	ENERGY [MeV]	LOG XI	SIGMA [barn] THEORY	SIGMA [barn] REFERENCE	ERROR [%] RND	ERROR [%] TOT
53-I	16.000	-0.260	7.320E+00	7.060E+00	0.9	1.0
	18.000	-0.235	1.000E+01	9.630E+00	0.9	1.0
	20.000	-0.212	1.310E+01	1.260E+01	0.9	1.0
	23.000	-0.182	1.840E+01	1.780E+01	0.9	1.0
	26.000	-0.155	2.450E+01	2.380E+01	0.8	0.8
	30.000	-0.124	3.350E+01	3.260E+01	0.8	0.8
	35.000	-0.091	4.580E+01	4.470E+01	0.8	0.8
	40.000	-0.062	5.890E+01	5.750E+01	0.9	1.0
	45.000	-0.036	7.230E+01	7.070E+01	0.9	1.0
	50.000	-0.013	8.570E+01	8.410E+01	0.9	1.0
	55.000	0.008	9.890E+01	9.750E+01	0.9	1.0
	60.000	0.027	1.120E+02	1.110E+02	1.3	2.2
	70.000	0.060	1.360E+02	1.360E+02	1.3	2.2
	80.000	0.089	1.580E+02	1.600E+02	1.3	2.2
	90.000	0.115	1.780E+02	1.810E+02	1.7	1.9
	100.000	0.137	1.950E+02	2.000E+02	1.7	1.9
	110.000	0.158	2.100E+02	2.160E+02	1.7	1.9
	120.000	0.177	2.230E+02	2.320E+02	1.7	1.9
	140.000	0.211	2.440E+02	2.560E+02	1.7	2.2
	160.000	0.240	2.590E+02	2.750E+02	1.8	2.7
	180.000	0.265	2.690E+02	2.880E+02	1.8	2.6
	200.000	0.288	2.760E+02	2.970E+02	1.8	2.6
	230.000	0.318	2.820E+02	3.050E+02	3.0	4.9
	260.000	0.345	2.830E+02	3.090E+02	3.0	4.9
	300.000	0.376	2.810E+02	3.070E+02	3.0	4.9
	350.000	0.409	2.750E+02	3.010E+02	3.0	4.9
	400.000	0.438	2.670E+02	2.910E+02	4.7	5.5
	450.000	0.464	2.570E+02	2.800E+02	4.7	5.5
	500.000	0.487	2.480E+02	2.680E+02	4.7	5.5
	550.000	0.508	2.380E+02	2.570E+02	7.0	10.9
	600.000	0.527	2.290E+02	2.450E+02	7.0	10.9
	700.000	0.560	2.120E+02	2.230E+02	7.0	11.0
54-Xe	1.400	-0.799	6.040E-04	5.730E-04	5.2	5.8
	1.600	-0.770	1.320E-03	1.270E-03	5.2	5.8
	1.800	-0.745	2.470E-03	2.400E-03	3.6	4.7
	2.000	-0.722	4.170E-03	4.080E-03	3.6	4.7
	2.300	-0.691	7.950E-03	7.840E-03	3.6	4.7
	2.600	-0.665	1.350E-02	1.340E-02	2.5	3.7
	3.000	-0.634	2.430E-02	2.410E-02	2.5	3.7
	3.500	-0.600	4.420E-02	4.400E-02	2.5	3.7
	4.000	-0.571	7.270E-02	7.250E-02	1.6	2.0
	4.500	-0.546	1.110E-01	1.110E-01	1.6	2.0
	5.000	-0.523	1.620E-01	1.610E-01	1.6	2.0
	5.500	-0.502	2.250E-01	2.230E-01	1.6	2.0
	6.000	-0.483	3.020E-01	2.990E-01	1.6	2.0
	7.000	-0.450	5.050E-01	4.970E-01	1.2	1.5
	8.000	-0.421	7.780E-01	7.630E-01	1.2	1.5
	9.000	-0.395	1.130E+00	1.110E+00	1.2	1.5
	10.000	-0.372	1.570E+00	1.530E+00	1.2	1.6
	11.000	-0.352	2.090E+00	2.040E+00	1.2	1.6
	12.000	-0.333	2.710E+00	2.630E+00	1.2	1.6
	14.000	-0.299	4.220E+00	4.080E+00	1.2	1.6
	16.000	-0.270	6.110E+00	5.910E+00	0.9	1.3
	18.000	-0.245	8.380E+00	8.070E+00	0.9	1.3
	20.000	-0.222	1.100E+01	1.060E+01	0.9	1.3
	23.000	-0.191	1.560E+01	1.500E+01	0.9	1.3
	26.000	-0.165	2.070E+01	2.010E+01	0.8	1.1
	30.000	-0.134	2.850E+01	2.770E+01	0.8	1.1
	35.000	-0.100	3.920E+01	3.820E+01	0.8	1.1
	40.000	-0.071	5.060E+01	4.940E+01	0.9	1.3
	45.000	-0.046	6.240E+01	6.100E+01	0.9	1.3
	50.000	-0.023	7.430E+01	7.280E+01	0.9	1.3
	55.000	-0.002	8.610E+01	8.470E+01	0.9	1.3
	60.000	0.017	9.770E+01	9.690E+01	1.3	2.5
	70.000	0.050	1.200E+02	1.200E+02	1.3	2.5
	80.000	0.079	1.400E+02	1.410E+02	1.3	2.5
	90.000	0.105	1.580E+02	1.600E+02	1.3	3.6
	100.000	0.128	1.740E+02	1.780E+02	1.7	2.0
	110.000	0.148	1.880E+02	1.930E+02	1.7	2.0
	120.000	0.167	2.000E+02	2.070E+02	1.7	2.0
	140.000	0.201	2.200E+02	2.310E+02	1.7	2.2
	160.000	0.230	2.350E+02	2.490E+02	1.8	1.9
	180.000	0.255	2.450E+02	2.620E+02	1.8	1.9
	200.000	0.278	2.520E+02	2.710E+02	1.8	1.9
	230.000	0.309	2.590E+02	2.800E+02	1.8	1.9
	260.000	0.335	2.610E+02	2.840E+02	3.0	5.8
	300.000	0.366	2.600E+02	2.840E+02	3.0	5.8
	350.000	0.400	2.550E+02	2.790E+02	3.0	5.8
	400.000	0.429	2.480E+02	2.710E+02	4.7	6.4
	450.000	0.454	2.400E+02	2.620E+02	4.7	6.4
54-Xe	500.000	0.477	2.320E+02	2.520E+02	4.7	6.4
	550.000	0.498	2.230E+02	2.410E+02	4.7	6.4
	600.000	0.517	2.150E+02	2.310E+02	7.0	11.9
	700.000	0.550	1.990E+02	2.110E+02	7.0	11.9
55-Cs	1.400	-0.809	4.590E-04	4.330E-04	5.2	6.0
	1.600	-0.780	1.030E-03	9.820E-04	5.2	6.0
	1.800	-0.754	1.950E-03	1.890E-03	3.6	5.0
	2.000	-0.731	3.330E-03	3.260E-03	3.6	5.0
	2.300	-0.701	6.430E-03	6.330E-03	3.6	4.9
	2.600	-0.674	1.100E-02	1.090E-02	3.6	4.9
	3.000	-0.643	1.990E-02	1.980E-02	2.5	4.0
	3.500	-0.610	3.640E-02	3.630E-02	2.5	3.9
	4.000	-0.581	6.010E-02	5.990E-02	2.5	3.9
	4.500	-0.555	9.220E-02	9.180E-02	1.6	2.2
	5.000	-0.532	1.340E-01	1.330E-01	1.6	2.2
	5.500	-0.512	1.870E-01	1.850E-01	1.6	2.2
	6.000	-0.493	2.510E-01	2.490E-01	1.6	2.2
	7.000	-0.459	4.200E-01	4.140E-01	1.2	1.7
	8.000	-0.430	6.470E-01	6.360E-01	1.2	1.7
	9.000	-0.405	9.410E-01	9.210E-01	1.2	1.7
	10.000	-0.382	1.300E+00	1.270E+00	1.2	1.7
	11.000	-0.361	1.740E+00	1.700E+00	1.2	1.8
	12.000	-0.342	2.260E+00	2.200E+00	1.2	1.8
	14.000	-0.309	3.530E+00	3.420E+00	1.2	1.8
	16.000	-0.280	5.130E+00	4.960E+00	1.2	1.8
	18.000	-0.254	7.050E+00	6.800E+00	0.9	1.5
	20.000	-0.231	9.280E+00	8.930E+00	0.9	1.5
	23.000	-0.201	1.320E+01	1.270E+01	0.9	1.5
	26.000	-0.174	1.760E+01	1.700E+01	0.9	1.5
	30.000	-0.143	2.430E+01	2.360E+01	0.8	1.4
	35.000	-0.110	3.360E+01	3.280E+01	0.8	1.4
	40.000	-0.081	4.360E+01	4.260E+01	0.8	1.4
	45.000	-0.055	5.400E+01	5.280E+01	0.9	1.5
	50.000	-0.032	6.460E+01	6.320E+01	0.9	1.5
	55.000	-0.012	7.520E+01	7.380E+01	0.9	1.5
	60.000	0.007	8.550E+01	8.430E+01	0.9	1.5
	70.000	0.041	1.050E+02	1.050E+02	1.3	2.7
	80.000	0.070	1.240E+02	1.240E+02	1.3	2.7
	90.000	0.095	1.400E+02	1.420E+02	1.3	2.7
	100.000	0.118	1.550E+02	1.580E+02	1.7	2.5
	110.000	0.139	1.680E+02	1.720E+02	1.7	2.5
	120.000	0.158	1.800E+02	1.850E+02	1.7	2.4
	140.000	0.191	1.990E+02	2.080E+02	1.7	2.4
	160.000	0.220	2.130E+02	2.250E+02	1.7	3.1
	180.000	0.246	2.230E+02	2.380E+02	1.8	2.9
	200.000	0.269	2.310E+02	2.470E+02	1.8	2.9
	230.000	0.299	2.370E+02	2.560E+02	1.8	2.8
	260.000	0.326	2.400E+02	2.610E+02	3.0	6.7
	300.000	0.357	2.410E+02	2.630E+02	3.0	6.7
	350.000	0.390	2.370E+02	2.590E+02	3.0	6.7
	400.000	0.419	2.310E+02	2.530E+02	4.7	7.3
	450.000	0.445	2.240E+02	2.450E+02	4.7	7.3
	500.000	0.468	2.170E+02	2.360E+02	4.7	7.3
	550.000	0.488	2.090E+02	2.270E+02	4.7	7.3
	600.000	0.507	2.020E+02	2.170E+02	7.0	12.8
	700.000	0.541	1.880E+02	1.990E+02	7.0	12.9
	800.000	0.570	1.750E+02	1.830E+02	7.0	12.9
56-Ba	1.400	-0.818	3.490E-04	3.030E-04	6.1	6.2
	1.600	-0.789	7.980E-04	7.200E-04	6.1	6.2
	1.800	-0.763	1.550E-03	1.440E-03	4.7	4.9
	2.000	-0.741	2.670E-03	2.540E-03	4.7	4.9
	2.300	-0.710	5.220E-03	5.080E-03	4.7	4.9
	2.600	-0.684	9.010E-03	8.910E-03	4.7	4.9
	3.000	-0.653	1.640E-02	1.640E-02	3.0	4.1
	3.500	-0.619	3.010E-02	3.050E-02	3.0	4.1
	4.000	-0.590	4.990E-02	5.080E-02	3.0	4.1
	4.500	-0.564	7.660E-02	7.820E-02	1.9	2.6
	5.000	-0.542	1.120E-01	1.140E-01	1.9	2.6
	5.500	-0.521	1.550E-01	1.580E-01	1.9	2.6
	6.000	-0.502	2.090E-01	2.130E-01	1.9	2.6
	7.000	-0.469	3.500E-01	3.540E-01	1.3	2.1
	8.000	-0.440	5.400E-01	5.440E-01	1.3	2.1
	9.000	-0.414	7.860E-01	7.860E-01	1.3	2.1
	10.000	-0.391	1.090E+00	1.090E+00	1.3	2.1
	11.000	-0.370	1.460E+00	1.450E+00	1.2	2.1
	12.000	-0.352	1.890E+00	1.870E+00	1.2	2.1
	14.000	-0.318	2.970E+00	2.900E+00	1.2	2.1
	16.000	-0.289	4.320E+00	4.210E+00	1.2	2.1
	18.000	-0.263	5.950E+00	5.790E+00	0.9	1.6

TABLE I. Empirical Reference Cross Sections for *K*-Shell Ionization by α Particles

See page 84 for Explanation of Tables

TARGET	ENERGY [MeV]	LOG XI	SIGMA [barn] THEORY	SIGMA [barn] REFERENCE	ERROR [%] RND	ERROR [%] TOT
56-Ba	20.000	-0.241	7.840E+00	7.610E+00	0.9	1.6
	23.000	-0.210	1.120E+01	1.080E+01	0.9	1.6
	26.000	-0.184	1.500E+01	1.450E+01	0.9	1.6
	30.000	-0.153	2.080E+01	2.020E+01	0.9	1.5
	35.000	-0.119	2.890E+01	2.820E+01	0.9	1.5
	40.000	-0.090	3.770E+01	3.670E+01	0.9	1.5
	45.000	-0.064	4.680E+01	4.580E+01	1.0	1.2
	50.000	-0.042	5.620E+01	5.510E+01	1.0	1.2
	55.000	-0.021	6.560E+01	6.460E+01	1.0	1.2
	60.000	-0.002	7.500E+01	7.400E+01	1.0	1.2
	70.000	0.031	9.290E+01	9.250E+01	1.9	2.3
	80.000	0.060	1.100E+02	1.100E+02	1.9	2.3
	90.000	0.086	1.250E+02	1.260E+02	1.9	2.2
	100.000	0.109	1.390E+02	1.410E+02	1.9	2.6
	110.000	0.130	1.510E+02	1.540E+02	1.9	1.9
	120.000	0.148	1.620E+02	1.660E+02	1.9	1.9
	140.000	0.182	1.800E+02	1.860E+02	1.9	1.9
	160.000	0.211	1.930E+02	2.020E+02	1.9	2.0
	180.000	0.237	2.040E+02	2.150E+02	2.0	2.0
	200.000	0.259	2.110E+02	2.240E+02	2.0	2.0
	230.000	0.290	2.180E+02	2.340E+02	2.0	2.0
	260.000	0.316	2.220E+02	2.390E+02	3.4	4.1
	300.000	0.347	2.230E+02	2.430E+02	3.4	4.1
	350.000	0.381	2.200E+02	2.420E+02	3.4	4.1
	400.000	0.410	2.150E+02	2.390E+02	5.7	5.8
	450.000	0.436	2.090E+02	2.340E+02	5.7	5.8
	500.000	0.458	2.030E+02	2.280E+02	5.7	5.8
57-La	1.400	-0.827	2.650E-04	2.260E-04	6.1	6.5
	1.600	-0.798	6.200E-04	5.530E-04	6.1	6.5
	1.800	-0.773	1.220E-03	1.130E-03	6.1	6.5
	2.000	-0.750	2.140E-03	2.020E-03	4.7	4.7
	2.300	-0.719	4.240E-03	4.100E-03	4.7	4.7
	2.600	-0.693	7.390E-03	7.270E-03	4.7	4.7
	3.000	-0.662	1.360E-02	1.350E-02	3.0	3.9
	3.500	-0.628	2.500E-02	2.530E-02	3.0	3.9
	4.000	-0.599	4.160E-02	4.230E-02	3.0	3.9
	4.500	-0.574	6.400E-02	6.520E-02	3.0	3.9
	5.000	-0.551	9.320E-02	9.500E-02	1.9	2.4
	5.500	-0.530	1.300E-01	1.330E-01	1.9	2.4
	6.000	-0.511	1.750E-01	1.780E-01	1.9	2.4
	7.000	-0.478	2.930E-01	2.970E-01	1.9	2.4
	8.000	-0.449	4.530E-01	4.570E-01	1.3	1.8
	9.000	-0.423	6.590E-01	6.610E-01	1.3	1.8
	10.000	-0.400	9.160E-01	9.140E-01	1.3	1.8
	11.000	-0.380	1.230E+00	1.220E+00	1.3	1.8
	12.000	-0.361	1.590E+00	1.580E+00	1.2	1.8
	14.000	-0.327	2.500E+00	2.470E+00	1.2	1.8
	16.000	-0.298	3.650E+00	3.560E+00	1.2	1.8
	18.000	-0.273	5.030E+00	4.900E+00	1.2	1.8
	20.000	-0.250	6.650E+00	6.460E+00	0.9	1.4
	23.000	-0.219	9.500E+00	9.190E+00	0.9	1.4
	26.000	-0.193	1.280E+01	1.240E+01	0.9	1.4
	30.000	-0.162	1.780E+01	1.730E+01	0.9	1.2
	35.000	-0.128	2.490E+01	2.430E+01	0.9	1.2
	40.000	-0.099	3.260E+01	3.170E+01	0.9	1.2
	45.000	-0.074	4.070E+01	3.970E+01	1.0	1.2
	50.000	-0.051	4.900E+01	4.800E+01	1.0	1.2
	55.000	-0.030	5.740E+01	5.640E+01	1.0	1.2
	60.000	-0.011	6.580E+01	6.490E+01	1.0	1.2
	70.000	0.022	8.200E+01	8.150E+01	1.9	2.0
	80.000	0.051	9.730E+01	9.730E+01	1.9	2.0
	90.000	0.077	1.110E+02	1.120E+02	1.9	2.0
	100.000	0.100	1.240E+02	1.260E+02	1.9	2.0
	110.000	0.120	1.350E+02	1.380E+02	1.9	2.3
	120.000	0.139	1.460E+02	1.490E+02	1.9	2.3
	140.000	0.173	1.630E+02	1.680E+02	1.9	2.3
	160.000	0.202	1.760E+02	1.830E+02	1.9	2.7
	180.000	0.227	1.860E+02	1.950E+02	2.0	2.9
	200.000	0.250	1.930E+02	2.040E+02	2.0	2.9
	230.000	0.281	2.010E+02	2.140E+02	2.0	2.9
	260.000	0.307	2.050E+02	2.200E+02	2.0	2.9
	300.000	0.338	2.060E+02	2.240E+02	3.4	5.0
	350.000	0.372	2.050E+02	2.250E+02	3.4	5.0
	400.000	0.401	2.010E+02	2.220E+02	5.7	6.8
	450.000	0.426	1.960E+02	2.180E+02	5.7	6.7
	500.000	0.449	1.900E+02	2.130E+02	5.7	6.7
	550.000	0.470	1.840E+02	2.070E+02	5.7	6.7
58-Ce	1.400	-0.836	2.000E-04	1.690E-04	6.1	6.8
	1.600	-0.807	4.820E-04	4.240E-04	6.1	6.8
58-Ce	1.800	-0.782	9.690E-04	8.830E-04	6.1	6.8
	2.000	-0.759	1.720E-03	1.610E-03	4.7	4.9
	2.300	-0.728	3.460E-03	3.320E-03	4.7	4.9
	2.600	-0.702	6.080E-03	5.950E-03	4.7	4.9
	3.000	-0.671	1.120E-02	1.120E-02	3.0	3.6
	3.500	-0.637	2.090E-02	2.100E-02	3.0	3.6
	4.000	-0.608	3.480E-02	3.530E-02	3.0	3.6
	4.500	-0.583	5.360E-02	5.460E-02	3.0	3.6
	5.000	-0.560	7.820E-02	7.970E-02	1.9	2.1
	5.500	-0.539	1.090E-01	1.110E-01	1.9	2.1
	6.000	-0.520	1.470E-01	1.500E-01	1.9	2.2
	7.000	-0.487	2.470E-01	2.500E-01	1.9	2.2
	8.000	-0.458	3.810E-01	3.850E-01	1.3	1.6
	9.000	-0.432	5.550E-01	5.580E-01	1.3	1.6
	10.000	-0.409	7.720E-01	7.720E-01	1.3	1.6
	11.000	-0.389	1.030E+00	1.030E+00	1.3	1.6
	12.000	-0.370	1.340E+00	1.340E+00	1.2	1.6
	14.000	-0.336	2.110E+00	2.090E+00	1.2	1.6
	16.000	-0.307	3.090E+00	3.020E+00	1.2	1.6
	18.000	-0.282	4.270E+00	4.160E+00	1.2	1.6
	20.000	-0.259	5.650E+00	5.500E+00	0.9	1.1
	23.000	-0.228	8.100E+00	7.840E+00	0.9	1.1
	26.000	-0.202	1.090E+01	1.060E+01	0.9	1.1
	30.000	-0.171	1.530E+01	1.480E+01	0.9	1.1
	35.000	-0.137	2.140E+01	2.090E+01	0.9	1.0
	40.000	-0.108	2.820E+01	2.750E+01	0.9	1.0
	45.000	-0.083	3.540E+01	3.440E+01	0.9	1.0
	50.000	-0.060	4.280E+01	4.180E+01	1.0	1.4
	55.000	-0.039	5.020E+01	4.930E+01	1.0	1.4
	60.000	-0.020	5.770E+01	5.680E+01	1.0	1.4
	70.000	0.013	7.240E+01	7.180E+01	1.9	2.1
	80.000	0.042	8.630E+01	8.610E+01	1.9	2.1
	90.000	0.068	9.920E+01	9.960E+01	1.9	2.1
	100.000	0.091	1.110E+02	1.120E+02	1.9	2.1
	110.000	0.111	1.220E+02	1.240E+02	1.9	2.2
	120.000	0.130	1.310E+02	1.340E+02	1.9	2.8
	140.000	0.164	1.470E+02	1.520E+02	1.9	2.8
	160.000	0.193	1.600E+02	1.660E+02	1.9	2.8
	180.000	0.218	1.700E+02	1.780E+02	1.9	3.7
	200.000	0.241	1.770E+02	1.870E+02	2.0	3.8
	230.000	0.272	1.840E+02	1.960E+02	2.0	3.8
	260.000	0.298	1.890E+02	2.030E+02	2.0	3.8
	300.000	0.329	1.910E+02	2.070E+02	3.4	5.9
	350.000	0.363	1.900E+02	2.080E+02	3.4	5.9
	400.000	0.392	1.870E+02	2.070E+02	3.4	5.9
	450.000	0.417	1.830E+02	2.040E+02	5.7	7.6
	500.000	0.440	1.780E+02	1.990E+02	5.7	7.6
	550.000	0.461	1.730E+02	1.940E+02	5.7	7.6
59-Pr	1.600	-0.816	3.740E-04	3.250E-04	6.1	7.1
	1.800	-0.790	7.670E-04	6.910E-04	6.1	7.0
	2.000	-0.768	1.380E-03	1.280E-03	4.7	5.2
	2.300	-0.737	2.820E-03	2.690E-03	4.7	5.2
	2.600	-0.711	5.020E-03	4.880E-03	4.7	5.2
	3.000	-0.680	9.350E-03	9.260E-03	4.7	5.2
	3.500	-0.646	1.750E-02	1.760E-02	3.0	3.4
	4.000	-0.617	2.920E-02	2.960E-02	3.0	3.4
	4.500	-0.592	4.520E-02	4.600E-02	3.0	3.4
	5.000	-0.569	6.590E-02	6.720E-02	1.9	1.9
	5.500	-0.548	9.210E-02	9.390E-02	1.9	1.9
	6.000	-0.529	1.240E-01	1.270E-01	1.9	1.9
	7.000	-0.496	2.080E-01	2.120E-01	1.9	1.9
	8.000	-0.467	3.220E-01	3.260E-01	1.3	1.3
	9.000	-0.441	4.700E-01	4.730E-01	1.3	1.3
	10.000	-0.418	6.530E-01	6.540E-01	1.3	1.3
	11.000	-0.397	8.760E-01	8.740E-01	1.3	1.3
	12.000	-0.379	1.140E+00	1.130E+00	1.3	1.3
	14.000	-0.345	1.790E+00	1.770E+00	1.2	1.3
	16.000	-0.316	2.620E+00	2.570E+00	1.2	1.3
	18.000	-0.290	3.630E+00	3.540E+00	1.2	1.3
	20.000	-0.268	4.820E+00	4.700E+00	0.9	1.0
	23.000	-0.237	6.920E+00	6.710E+00	0.9	1.0
	26.000	-0.211	9.380E+00	9.080E+00	0.9	1.0
	30.000	-0.180	1.320E+01	1.280E+01	0.9	1.0
	35.000	-0.146	1.850E+01	1.800E+01	0.9	1.0
	40.000	-0.117	2.450E+01	2.390E+01	0.9	1.0
	45.000	-0.092	3.080E+01	3.000E+01	0.9	1.0
	50.000	-0.069	3.740E+01	3.650E+01	1.0	1.7
	55.000	-0.048	4.410E+01	4.320E+01	1.0	1.7
	60.000	-0.029	5.080E+01	4.990E+01	1.0	1.7
	70.000	0.004	6.400E+01	6.340E+01	1.0	1.7

TABLE I. Empirical Reference Cross Sections for *K*-Shell Ionization by α Particles

See page 84 for Explanation of Tables

TARGET	ENERGY [MeV]	LOG XI	SIGMA [barn] THEORY	SIGMA [barn] REFERENCE	ERROR [%] RND	ERROR [%] TOT
59-Pr	80.000	0.033	7.670E+01	7.640E+01	1.9	2.3
	90.000	0.059	8.850E+01	8.870E+01	1.9	2.3
	100.000	0.082	9.940E+01	1.000E+02	1.9	2.3
	110.000	0.103	1.090E+02	1.110E+02	1.9	2.7
	120.000	0.121	1.180E+02	1.200E+02	1.9	3.3
	140.000	0.155	1.340E+02	1.370E+02	1.9	3.3
	160.000	0.184	1.460E+02	1.510E+02	1.9	3.3
	180.000	0.210	1.550E+02	1.620E+02	1.9	4.6
	200.000	0.232	1.620E+02	1.710E+02	2.0	4.8
	230.000	0.263	1.700E+02	1.800E+02	2.0	4.7
	260.000	0.289	1.740E+02	1.870E+02	2.0	4.7
	300.000	0.320	1.770E+02	1.910E+02	3.4	6.9
	350.000	0.354	1.770E+02	1.930E+02	3.4	6.8
	400.000	0.383	1.750E+02	1.930E+02	3.4	6.8
	450.000	0.408	1.710E+02	1.900E+02	5.7	8.6
	500.000	0.431	1.670E+02	1.860E+02	5.7	8.5
	550.000	0.452	1.620E+02	1.820E+02	5.7	8.5
60-Nd	1.600	-0.825	2.900E-04	2.490E-04	6.1	7.4
	1.800	-0.799	6.080E-04	5.420E-04	6.1	7.3
	2.000	-0.776	1.110E-03	1.020E-03	6.1	7.3
	2.300	-0.746	2.310E-03	2.190E-03	4.7	5.5
	2.600	-0.719	4.150E-03	4.010E-03	4.7	5.5
	3.000	-0.688	7.810E-03	7.700E-03	4.7	5.4
	3.500	-0.655	1.470E-02	1.470E-02	3.0	3.1
	4.000	-0.626	2.470E-02	2.490E-02	3.0	3.1
	4.500	-0.600	3.820E-02	3.880E-02	3.0	3.1
	5.000	-0.577	5.580E-02	5.690E-02	3.0	3.1
	5.500	-0.557	7.800E-02	7.960E-02	1.9	2.1
	6.000	-0.538	1.050E-01	1.070E-01	1.9	2.1
	7.000	-0.504	1.770E-01	1.800E-01	1.9	2.1
	8.000	-0.475	2.740E-01	2.770E-01	1.9	2.1
	9.000	-0.450	3.990E-01	4.020E-01	1.3	1.6
	10.000	-0.427	5.550E-01	5.570E-01	1.3	1.6
	11.000	-0.406	7.440E-01	7.440E-01	1.3	1.6
	12.000	-0.387	9.680E-01	9.650E-01	1.3	1.6
	14.000	-0.354	1.530E+00	1.510E+00	1.2	1.4
	16.000	-0.325	2.240E+00	2.190E+00	1.2	1.4
	18.000	-0.299	3.100E+00	3.030E+00	1.2	1.4
	20.000	-0.276	4.130E+00	4.020E+00	1.2	1.4
	23.000	-0.246	5.940E+00	5.760E+00	0.9	1.3
	26.000	-0.219	8.070E+00	7.800E+00	0.9	1.3
	30.000	-0.188	1.130E+01	1.100E+01	0.9	1.3
	35.000	-0.155	1.600E+01	1.560E+01	0.9	1.3
	40.000	-0.126	2.130E+01	2.070E+01	0.9	1.3
	45.000	-0.100	2.690E+01	2.620E+01	0.9	1.3
	50.000	-0.077	3.270E+01	3.190E+01	1.0	1.9
	55.000	-0.057	3.870E+01	3.790E+01	1.0	1.9
	60.000	-0.038	4.470E+01	4.390E+01	1.0	1.9
	70.000	-0.004	5.670E+01	5.600E+01	1.0	1.9
	80.000	0.025	6.820E+01	6.780E+01	1.9	2.6
	90.000	0.050	7.900E+01	7.900E+01	1.9	2.6
	100.000	0.073	8.910E+01	8.960E+01	1.9	2.6
	110.000	0.094	9.830E+01	9.940E+01	1.9	2.6
	120.000	0.113	1.070E+02	1.080E+02	1.9	3.2
	140.000	0.146	1.210E+02	1.240E+02	1.9	3.8
	160.000	0.175	1.330E+02	1.370E+02	1.9	3.8
	180.000	0.201	1.420E+02	1.480E+02	1.9	5.6
	200.000	0.224	1.490E+02	1.560E+02	1.9	5.6
	230.000	0.254	1.560E+02	1.660E+02	2.0	5.7
	260.000	0.281	1.610E+02	1.720E+02	2.0	5.7
	300.000	0.312	1.640E+02	1.770E+02	3.4	7.8
	350.000	0.345	1.650E+02	1.800E+02	3.4	7.8
	400.000	0.374	1.630E+02	1.790E+02	3.4	7.7
	450.000	0.400	1.600E+02	1.770E+02	5.7	9.5
	500.000	0.423	1.570E+02	1.740E+02	5.7	9.4
	550.000	0.443	1.520E+02	1.710E+02	5.7	9.4
	600.000	0.462	1.480E+02	1.670E+02	5.7	9.4
61-Pm	1.600	-0.833	2.250E-04	1.900E-04	6.1	7.7
	1.800	-0.808	4.810E-04	4.240E-04	6.1	7.6
	2.000	-0.785	8.960E-04	8.130E-04	6.1	7.6
	2.300	-0.755	1.890E-03	1.780E-03	4.7	5.7
	2.600	-0.728	3.440E-03	3.300E-03	4.7	5.7
	3.000	-0.697	6.540E-03	6.410E-03	4.7	5.7
	3.500	-0.663	1.240E-02	1.240E-02	3.0	3.1
	4.000	-0.634	2.090E-02	2.110E-02	3.0	3.1
	4.500	-0.609	3.240E-02	3.290E-02	3.0	3.1
	5.000	-0.586	4.740E-02	4.830E-02	3.0	3.1
	5.500	-0.565	6.640E-02	6.770E-02	1.9	2.4

TARGET	ENERGY [MeV]	LOG XI	SIGMA [barn] THEORY	SIGMA [barn] REFERENCE	ERROR [%] RND	ERROR [%] TOT
61-Pm	6.000	-0.546	8.960E-02	9.140E-02	1.9	2.4
	7.000	-0.513	1.510E-01	1.530E-01	1.9	2.4
	8.000	-0.484	2.330E-01	2.370E-01	1.9	2.4
	9.000	-0.458	3.400E-01	3.430E-01	1.3	1.8
	10.000	-0.435	4.730E-01	4.760E-01	1.3	1.8
	11.000	-0.415	6.350E-01	6.350E-01	1.3	1.8
	12.000	-0.396	8.260E-01	8.240E-01	1.3	1.8
	14.000	-0.362	1.300E+00	1.290E+00	1.2	1.6
	16.000	-0.333	1.910E+00	1.890E+00	1.2	1.6
	18.000	-0.308	2.660E+00	2.600E+00	1.2	1.6
	20.000	-0.285	3.540E+00	3.450E+00	1.2	1.6
	23.000	-0.255	5.100E+00	4.960E+00	0.9	1.5
	26.000	-0.228	6.950E+00	6.730E+00	0.9	1.5
	30.000	-0.197	9.800E+00	9.490E+00	0.9	1.5
	35.000	-0.163	1.390E+01	1.350E+01	0.9	1.5
	40.000	-0.134	1.850E+01	1.800E+01	0.9	1.5
	45.000	-0.109	2.350E+01	2.290E+01	0.9	1.5
	50.000	-0.086	2.870E+01	2.790E+01	0.9	1.5
	55.000	-0.065	3.400E+01	3.320E+01	1.0	2.2
	60.000	-0.046	3.940E+01	3.860E+01	1.0	2.2
	70.000	-0.013	5.020E+01	4.950E+01	1.0	2.2
	80.000	0.016	6.070E+01	6.020E+01	1.9	2.8
	90.000	0.042	7.060E+01	7.050E+01	1.9	2.8
	100.000	0.065	7.990E+01	8.020E+01	1.9	2.8
	110.000	0.085	8.840E+01	8.920E+01	1.9	2.8
	120.000	0.104	9.620E+01	9.760E+01	1.9	3.7
	140.000	0.138	1.100E+02	1.120E+02	1.9	4.3
	160.000	0.167	1.210E+02	1.250E+02	1.9	4.2
	180.000	0.192	1.290E+02	1.350E+02	1.9	4.2
	200.000	0.215	1.360E+02	1.430E+02	1.9	6.5
	230.000	0.245	1.440E+02	1.520E+02	2.0	6.7
	260.000	0.272	1.490E+02	1.580E+02	2.0	6.6
	300.000	0.303	1.520E+02	1.640E+02	2.0	6.6
	350.000	0.337	1.540E+02	1.670E+02	3.4	8.7
	400.000	0.366	1.530E+02	1.670E+02	3.4	8.7
	450.000	0.391	1.500E+02	1.660E+02	3.4	8.6
	500.000	0.414	1.470E+02	1.630E+02	5.7	10.4
	550.000	0.435	1.430E+02	1.600E+02	5.7	10.3
	600.000	0.454	1.390E+02	1.570E+02	5.7	10.3
62-Sm	1.800	-0.816	3.800E-04	3.310E-04	6.1	7.9
	2.000	-0.793	7.200E-04	6.460E-04	6.1	7.9
	2.300	-0.763	1.550E-03	1.440E-03	4.7	6.0
	2.600	-0.736	2.850E-03	2.720E-03	4.7	6.0
	3.000	-0.705	5.480E-03	5.350E-03	4.7	6.0
	3.500	-0.672	1.050E-02	1.040E-02	3.0	3.3
	4.000	-0.643	1.770E-02	1.780E-02	3.0	3.3
	4.500	-0.617	2.760E-02	2.790E-02	3.0	3.3
	5.000	-0.594	4.040E-02	4.110E-02	3.0	3.3
	5.500	-0.574	5.660E-02	5.770E-02	3.0	3.3
	6.000	-0.555	7.650E-02	7.800E-02	1.9	2.6
	7.000	-0.521	1.290E-01	1.310E-01	1.9	2.6
	8.000	-0.492	1.990E-01	2.020E-01	1.9	2.6
	9.000	-0.467	2.910E-01	2.940E-01	1.3	2.1
	10.000	-0.444	4.050E-01	4.080E-01	1.3	2.1
	11.000	-0.423	5.430E-01	5.440E-01	1.3	2.1
	12.000	-0.404	7.070E-01	7.060E-01	1.3	2.1
	14.000	-0.371	1.120E+00	1.110E+00	1.2	1.9
	16.000	-0.342	1.640E+00	1.620E+00	1.2	1.9
	18.000	-0.316	2.280E+00	2.230E+00	1.2	1.9
	20.000	-0.293	3.040E+00	2.970E+00	1.2	1.9
	23.000	-0.263	4.390E+00	4.280E+00	0.9	1.8
	26.000	-0.236	5.990E+00	5.810E+00	0.9	1.8
	30.000	-0.205	8.480E+00	8.210E+00	0.9	1.8
	35.000	-0.172	1.210E+01	1.170E+01	0.9	1.8
	40.000	-0.143	1.610E+01	1.570E+01	0.9	1.8
	45.000	-0.117	2.050E+01	2.000E+01	0.9	1.8
	50.000	-0.094	2.510E+01	2.450E+01	0.9	1.8
	55.000	-0.074	2.990E+01	2.920E+01	1.0	2.4
	60.000	-0.055	3.480E+01	3.410E+01	1.0	2.4
	70.000	-0.021	4.450E+01	4.380E+01	1.0	2.4
	80.000	0.008	5.410E+01	5.350E+01	1.0	2.4
	90.000	0.033	6.320E+01	6.290E+01	1.9	3.1
	100.000	0.056	7.170E+01	7.180E+01	1.9	3.1
	110.000	0.077	7.960E+01	8.010E+01	1.9	3.1
	120.000	0.096	8.690E+01	8.790E+01	1.9	3.1
	140.000	0.129	9.960E+01	1.020E+02	1.9	4.8
	160.000	0.158	1.100E+02	1.130E+02	1.9	4.7
	180.000	0.184	1.180E+02	1.230E+02	1.9	4.7
	200.000	0.207	1.250E+02	1.310E+02	1.9	7.5
	230.000	0.237	1.320E+02	1.400E+02	2.0	7.6

TABLE I. Empirical Reference Cross Sections for *K*-Shell Ionization by α Particles

See page 84 for Explanation of Tables

TARGET	ENERGY [MeV]	LOG XI	SIGMA [barn] THEORY	SIGMA [barn] REFERENCE	ERROR [%] RND	ERROR [%] TOT
62-Sm	260.000	0.264	1.370E+02	1.460E+02	2.0	7.6
	300.000	0.295	1.410E+02	1.520E+02	2.0	7.5
	350.000	0.328	1.430E+02	1.550E+02	3.4	9.6
	400.000	0.357	1.420E+02	1.560E+02	3.4	9.6
	450.000	0.383	1.410E+02	1.550E+02	3.4	9.5
	500.000	0.406	1.380E+02	1.530E+02	5.7	11.3
	550.000	0.426	1.350E+02	1.500E+02	5.7	11.2
	600.000	0.445	1.310E+02	1.470E+02	5.7	11.2
63-Eu	1.800	-0.825	3.000E-04	2.570E-04	6.1	8.3
	2.000	-0.802	5.780E-04	5.130E-04	6.1	8.2
	2.300	-0.771	1.270E-03	1.170E-03	6.1	8.1
	2.600	-0.745	2.370E-03	2.240E-03	4.7	6.3
	3.000	-0.714	4.610E-03	4.470E-03	4.7	6.2
	3.500	-0.680	8.880E-03	8.790E-03	4.7	6.2
	4.000	-0.651	1.510E-02	1.510E-02	3.0	3.6
	4.500	-0.626	2.360E-02	2.380E-02	3.0	3.6
	5.000	-0.603	3.460E-02	3.510E-02	3.0	3.6
	5.500	-0.582	4.850E-02	4.940E-02	3.0	3.6
	6.000	-0.563	6.560E-02	6.690E-02	1.9	2.8
	7.000	-0.530	1.100E-01	1.120E-01	1.9	2.8
	8.000	-0.501	1.710E-01	1.740E-01	1.9	2.8
	9.000	-0.475	2.490E-01	2.530E-01	1.9	2.8
	10.000	-0.452	3.470E-01	3.500E-01	1.3	2.3
	11.000	-0.432	4.660E-01	4.680E-01	1.3	2.3
	12.000	-0.413	6.070E-01	6.070E-01	1.3	2.3
	14.000	-0.379	9.590E-01	9.540E-01	1.3	2.3
	16.000	-0.350	1.410E+00	1.400E+00	1.2	2.1
	18.000	-0.325	1.960E+00	1.920E+00	1.2	2.2
	20.000	-0.302	2.620E+00	2.560E+00	1.2	2.2
	23.000	-0.271	3.790E+00	3.700E+00	0.9	2.0
	26.000	-0.245	5.190E+00	5.030E+00	0.9	2.0
	30.000	-0.214	7.360E+00	7.120E+00	0.9	2.0
	35.000	-0.180	1.050E+01	1.020E+01	0.9	2.0
	40.000	-0.151	1.410E+01	1.370E+01	0.9	2.1
	45.000	-0.126	1.800E+01	1.750E+01	0.9	2.1
	50.000	-0.103	2.210E+01	2.150E+01	0.9	2.1
	55.000	-0.082	2.640E+01	2.560E+01	0.9	2.1
	60.000	-0.063	3.070E+01	3.000E+01	1.0	2.7
	70.000	-0.030	3.950E+01	3.880E+01	1.0	2.7
	80.000	-0.001	4.820E+01	4.760E+01	1.0	2.7
	90.000	0.025	5.650E+01	5.620E+01	1.9	3.3
	100.000	0.048	6.440E+01	6.430E+01	1.9	3.3
	110.000	0.068	7.170E+01	7.200E+01	1.9	3.3
	120.000	0.087	7.850E+01	7.920E+01	1.9	3.3
	140.000	0.121	9.040E+01	9.210E+01	1.9	5.3
	160.000	0.150	1.000E+02	1.030E+02	1.9	5.2
	180.000	0.175	1.080E+02	1.120E+02	1.9	5.2
	200.000	0.198	1.150E+02	1.200E+02	1.9	5.2
	230.000	0.229	1.220E+02	1.280E+02	2.0	8.6
	260.000	0.255	1.270E+02	1.350E+02	2.0	8.5
	300.000	0.286	1.310E+02	1.400E+02	2.0	8.5
	350.000	0.320	1.330E+02	1.440E+02	3.4	10.6
	400.000	0.349	1.330E+02	1.450E+02	3.4	10.5
	450.000	0.374	1.320E+02	1.450E+02	3.4	10.5
	500.000	0.397	1.290E+02	1.430E+02	3.4	10.4
	550.000	0.418	1.270E+02	1.410E+02	5.7	12.1
	600.000	0.437	1.240E+02	1.380E+02	5.7	12.1
64-Gd	1.800	-0.833	2.360E-04	2.000E-04	6.1	8.6
	2.000	-0.810	4.650E-04	4.080E-04	6.1	8.5
	2.300	-0.780	1.040E-03	9.510E-04	6.1	8.4
	2.600	-0.753	1.970E-03	1.850E-03	4.7	6.5
	3.000	-0.722	3.880E-03	3.740E-03	4.7	6.5
	3.500	-0.688	7.550E-03	7.440E-03	4.7	6.5
	4.000	-0.659	1.290E-02	1.290E-02	3.0	3.8
	4.500	-0.634	2.020E-02	2.040E-02	3.0	3.8
	5.000	-0.611	2.970E-02	3.010E-02	3.0	3.8
	5.500	-0.590	4.170E-02	4.240E-02	3.0	3.8
	6.000	-0.571	5.640E-02	5.750E-02	1.9	3.1
	7.000	-0.538	9.490E-02	9.680E-02	1.9	3.1
	8.000	-0.509	1.470E-01	1.500E-01	1.9	3.1
	9.000	-0.483	2.150E-01	2.180E-01	1.9	3.1
	10.000	-0.461	2.990E-01	3.020E-01	1.3	2.6
	11.000	-0.440	4.010E-01	4.040E-01	1.3	2.6
	12.000	-0.421	5.230E-01	5.240E-01	1.3	2.6
	14.000	-0.387	8.270E-01	8.230E-01	1.3	2.6
	16.000	-0.358	1.220E+00	1.210E+00	1.2	2.4
	18.000	-0.333	1.690E+00	1.670E+00	1.2	2.4
	20.000	-0.310	2.260E+00	2.210E+00	1.2	2.4
64-Gd	23.000	-0.280	3.280E+00	3.200E+00	1.2	2.4
	26.000	-0.253	4.500E+00	4.370E+00	0.9	2.3
	30.000	-0.222	6.400E+00	6.190E+00	0.9	2.3
	35.000	-0.188	9.170E+00	8.890E+00	0.9	2.3
	40.000	-0.159	1.230E+01	1.200E+01	0.9	2.3
	45.000	-0.134	1.580E+01	1.540E+01	0.9	2.3
	50.000	-0.111	1.940E+01	1.890E+01	0.9	2.3
	55.000	-0.090	2.320E+01	2.260E+01	0.9	2.3
	60.000	-0.071	2.720E+01	2.650E+01	1.0	2.9
	70.000	-0.038	3.510E+01	3.450E+01	1.0	2.9
	80.000	-0.009	4.300E+01	4.240E+01	1.0	2.9
	90.000	0.017	5.060E+01	5.020E+01	1.9	3.6
	100.000	0.039	5.780E+01	5.770E+01	1.9	3.6
	110.000	0.060	6.460E+01	6.480E+01	1.9	3.6
	120.000	0.079	7.090E+01	7.140E+01	1.9	3.6
	140.000	0.113	8.210E+01	8.340E+01	1.9	5.2
	160.000	0.142	9.140E+01	9.370E+01	1.9	5.7
	180.000	0.167	9.910E+01	1.020E+02	1.9	5.7
	200.000	0.190	1.050E+02	1.090E+02	1.9	5.7
	230.000	0.220	1.130E+02	1.180E+02	1.9	9.4
	260.000	0.247	1.180E+02	1.240E+02	2.0	9.5
	300.000	0.278	1.220E+02	1.300E+02	2.0	9.4
	350.000	0.312	1.240E+02	1.340E+02	3.4	11.5
	400.000	0.341	1.240E+02	1.350E+02	3.4	11.4
	450.000	0.366	1.230E+02	1.350E+02	3.4	11.4
	500.000	0.389	1.220E+02	1.340E+02	3.4	11.3
	550.000	0.410	1.190E+02	1.320E+02	5.7	13.1
	600.000	0.429	1.170E+02	1.300E+02	5.7	13.0
	700.000	0.462	1.110E+02	1.250E+02	5.7	13.0
65-Tb	2.000	-0.818	3.720E-04	3.230E-04	6.1	8.8
	2.300	-0.788	8.530E-04	7.710E-04	6.1	8.7
	2.600	-0.761	1.640E-03	1.530E-03	4.7	6.8
	3.000	-0.730	3.270E-03	3.140E-03	4.7	6.8
	3.500	-0.697	6.420E-03	6.310E-03	4.7	6.7
	4.000	-0.668	1.100E-02	1.100E-02	3.0	4.1
	4.500	-0.642	1.740E-02	1.750E-02	3.0	4.1
	5.000	-0.619	2.560E-02	2.590E-02	3.0	4.0
	5.500	-0.598	3.590E-02	3.650E-02	3.0	4.0
	6.000	-0.580	4.860E-02	4.960E-02	3.0	4.0
	7.000	-0.546	8.190E-02	8.360E-02	1.9	3.3
	8.000	-0.517	1.270E-01	1.290E-01	1.9	3.3
	9.000	-0.492	1.850E-01	1.880E-01	1.9	3.3
	10.000	-0.469	2.580E-01	2.610E-01	1.3	2.8
	11.000	-0.448	3.460E-01	3.490E-01	1.3	2.8
	12.000	-0.429	4.510E-01	4.530E-01	1.3	2.8
	14.000	-0.396	7.140E-01	7.120E-01	1.3	2.8
	16.000	-0.367	1.050E+00	1.040E+00	1.2	2.6
	18.000	-0.341	1.470E+00	1.450E+00	1.2	2.7
	20.000	-0.318	1.960E+00	1.920E+00	1.2	2.7
	23.000	-0.288	2.850E+00	2.780E+00	1.2	2.7
	26.000	-0.261	3.910E+00	3.800E+00	0.9	2.6
	30.000	-0.230	5.570E+00	5.390E+00	0.9	2.6
	35.000	-0.197	8.010E+00	7.760E+00	0.9	2.6
	40.000	-0.168	1.080E+01	1.050E+01	0.9	2.6
	45.000	-0.142	1.380E+01	1.350E+01	0.9	2.6
	50.000	-0.119	1.710E+01	1.670E+01	0.9	2.6
	55.000	-0.098	2.050E+01	2.000E+01	0.9	2.6
	60.000	-0.080	2.400E+01	2.350E+01	1.0	3.2
	70.000	-0.046	3.120E+01	3.060E+01	1.0	3.2
	80.000	-0.017	3.840E+01	3.780E+01	1.0	3.2
	90.000	0.008	4.530E+01	4.490E+01	1.0	3.2
	100.000	0.031	5.200E+01	5.180E+01	1.9	3.8
	110.000	0.052	5.830E+01	5.830E+01	1.9	3.8
	120.000	0.071	6.410E+01	6.450E+01	1.9	3.8
	140.000	0.104	7.460E+01	7.560E+01	1.9	5.7
	160.000	0.133	8.340E+01	8.520E+01	1.9	6.2
	180.000	0.159	9.070E+01	9.340E+01	1.9	6.2
	200.000	0.182	9.670E+01	1.000E+02	1.9	6.2
	230.000	0.212	1.040E+02	1.090E+02	1.9	10.4
	260.000	0.239	1.090E+02	1.150E+02	2.0	10.5
	300.000	0.270	1.130E+02	1.200E+02	2.0	10.4
	350.000	0.303	1.160E+02	1.240E+02	2.0	10.3
	400.000	0.332	1.160E+02	1.260E+02	3.4	12.4
	450.000	0.358	1.160E+02	1.260E+02	3.4	12.3
	500.000	0.381	1.140E+02	1.260E+02	3.4	12.3
	550.000	0.402	1.120E+02	1.240E+02	5.7	14.0
	600.000	0.420	1.100E+02	1.220E+02	5.7	13.9
	700.000	0.454	1.050E+02	1.180E+02	5.7	13.9
66-Dy	2.000	-0.826	2.980E-04	1.990E-04	9.6	11.0
	2.300	-0.796	6.990E-04	5.330E-04	9.6	10.9

TABLE I. Empirical Reference Cross Sections for *K*-Shell Ionization by α Particles

See page 84 for Explanation of Tables

TARGET	ENERGY [MeV]	LOG XI	SIGMA [barn] THEORY	SIGMA [barn] REFERENCE	ERROR [%] RND	ERROR [%] TOT
66-Dy	2.600	-0.769	1.360E-03	1.140E-03	7.1	8.3
	3.000	-0.738	2.760E-03	2.490E-03	7.1	8.2
	3.500	-0.705	5.480E-03	5.280E-03	7.1	8.2
	4.000	-0.676	9.480E-03	9.500E-03	2.8	4.4
	4.500	-0.650	1.500E-02	1.540E-02	2.8	4.3
	5.000	-0.627	2.210E-02	2.310E-02	2.8	4.3
	5.500	-0.607	3.110E-02	3.290E-02	2.8	4.3
	6.000	-0.588	4.210E-02	4.490E-02	2.8	4.3
	7.000	-0.554	7.100E-02	7.620E-02	2.8	4.0
	8.000	-0.525	1.100E-01	1.180E-01	2.8	4.0
	9.000	-0.500	1.610E-01	1.710E-01	2.8	4.0
	10.000	-0.477	2.240E-01	2.370E-01	2.1	3.6
	11.000	-0.456	3.000E-01	3.160E-01	2.1	3.7
	12.000	-0.437	3.910E-01	4.080E-01	2.1	3.7
	14.000	-0.404	6.190E-01	6.370E-01	2.1	3.7
	16.000	-0.375	9.120E-01	9.250E-01	1.8	2.3
	18.000	-0.349	1.270E+00	1.270E+00	1.8	2.3
	20.000	-0.326	1.700E+00	1.680E+00	1.8	2.3
	23.000	-0.296	2.480E+00	2.410E+00	1.8	2.3
	26.000	-0.269	3.400E+00	3.280E+00	1.5	2.8
	30.000	-0.238	4.860E+00	4.660E+00	1.5	2.8
	35.000	-0.205	7.010E+00	6.710E+00	1.5	2.8
	40.000	-0.176	9.460E+00	9.060E+00	2.1	2.7
	45.000	-0.150	1.220E+01	1.170E+01	2.1	2.7
	50.000	-0.127	1.510E+01	1.460E+01	2.1	2.7
	55.000	-0.107	1.810E+01	1.770E+01	2.1	2.7
	60.000	-0.088	2.130E+01	2.090E+01	2.9	4.8
	70.000	-0.054	2.780E+01	2.770E+01	2.9	4.8
	80.000	-0.025	3.430E+01	3.470E+01	2.9	4.8
	90.000	0.000	4.070E+01	4.170E+01	2.9	4.7
	100.000	0.023	4.680E+01	4.860E+01	3.6	5.1
	110.000	0.044	5.260E+01	5.520E+01	3.6	5.1
	120.000	0.063	5.800E+01	6.170E+01	3.6	5.1
	140.000	0.096	6.780E+01	7.360E+01	3.6	5.1
	160.000	0.125	7.610E+01	8.410E+01	8.4	12.0
	180.000	0.151	8.310E+01	9.340E+01	8.4	11.9
	200.000	0.174	8.890E+01	1.010E+02	8.4	11.9
	230.000	0.204	9.560E+01	1.110E+02	8.4	15.2
	260.000	0.231	1.010E+02	1.190E+02	7.1	13.0
	300.000	0.262	1.050E+02	1.270E+02	7.1	12.9
	350.000	0.295	1.080E+02	1.330E+02	7.1	12.7
67-Ho	2.300	-0.804	5.710E-04	4.230E-04	9.6	10.6
	2.600	-0.777	1.140E-03	9.230E-04	9.6	10.5
	3.000	-0.746	2.330E-03	2.070E-03	7.1	8.0
	3.500	-0.713	4.690E-03	4.450E-03	7.1	7.9
	4.000	-0.684	8.160E-03	8.100E-03	7.1	7.9
	4.500	-0.658	1.290E-02	1.320E-02	2.8	4.1
	5.000	-0.635	1.910E-02	1.990E-02	2.8	4.1
	5.500	-0.615	2.700E-02	2.850E-02	2.8	4.0
	6.000	-0.596	3.660E-02	3.890E-02	2.8	4.0
	7.000	-0.562	6.170E-02	6.620E-02	2.8	3.8
	8.000	-0.533	9.570E-02	1.030E-01	2.8	3.8
	9.000	-0.508	1.400E-01	1.490E-01	2.8	3.8
	10.000	-0.485	1.940E-01	2.070E-01	2.8	3.8
	11.000	-0.464	2.610E-01	2.750E-01	2.1	3.4
	12.000	-0.445	3.400E-01	3.560E-01	2.1	3.4
	14.000	-0.412	5.380E-01	5.550E-01	2.1	3.5
	16.000	-0.383	7.930E-01	8.070E-01	2.1	3.5
	18.000	-0.357	1.110E+00	1.110E+00	1.8	2.0
	20.000	-0.334	1.480E+00	1.470E+00	1.8	2.0
	23.000	-0.304	2.160E+00	2.110E+00	1.8	2.1
	26.000	-0.277	2.970E+00	2.870E+00	1.5	2.6
	30.000	-0.246	4.250E+00	4.080E+00	1.5	2.6
	35.000	-0.213	6.140E+00	5.880E+00	1.5	2.6
	40.000	-0.184	8.310E+00	7.960E+00	2.1	2.5
	45.000	-0.158	1.070E+01	1.030E+01	2.1	2.5
	50.000	-0.135	1.330E+01	1.290E+01	2.1	2.5
	55.000	-0.115	1.610E+01	1.560E+01	2.1	2.5
	60.000	-0.096	1.890E+01	1.850E+01	2.1	2.5
	70.000	-0.062	2.480E+01	2.460E+01	2.9	4.5
	80.000	-0.033	3.070E+01	3.090E+01	2.9	4.5
	90.000	-0.008	3.650E+01	3.720E+01	2.9	4.5
	100.000	0.015	4.210E+01	4.350E+01	3.6	4.9
	110.000	0.036	4.750E+01	4.960E+01	3.6	4.9
	120.000	0.055	5.250E+01	5.550E+01	3.6	4.9
	140.000	0.088	6.160E+01	6.660E+01	3.6	4.9
	160.000	0.117	6.950E+01	7.640E+01	8.4	11.6
	180.000	0.143	7.610E+01	8.510E+01	8.4	11.5
	200.000	0.166	8.160E+01	9.270E+01	8.4	11.5
	230.000	0.196	8.820E+01	1.020E+02	8.4	11.4
	260.000	0.223	9.310E+01	1.100E+02	8.4	14.3
67-Ho	300.000	0.254	9.750E+01	1.170E+02	7.1	12.1
	350.000	0.287	1.010E+02	1.230E+02	7.1	12.0
68-Er	2.300	-0.812	4.670E-04	3.340E-04	9.6	10.2
	2.600	-0.785	9.440E-04	7.480E-04	9.6	10.2
	3.000	-0.754	1.970E-03	1.710E-03	7.1	7.7
	3.500	-0.721	4.010E-03	3.750E-03	7.1	7.7
	4.000	-0.692	7.040E-03	6.910E-03	7.1	7.6
	4.500	-0.666	1.120E-02	1.130E-02	2.8	3.8
	5.000	-0.643	1.660E-02	1.720E-02	2.8	3.8
	5.500	-0.622	2.350E-02	2.460E-02	2.8	3.8
	6.000	-0.603	3.180E-02	3.380E-02	2.8	3.8
	7.000	-0.570	5.380E-02	5.760E-02	2.8	3.6
	8.000	-0.541	8.340E-02	8.950E-02	2.8	3.6
	9.000	-0.515	1.220E-01	1.300E-01	2.8	3.6
	10.000	-0.493	1.690E-01	1.810E-01	2.8	3.6
	11.000	-0.472	2.270E-01	2.410E-01	2.1	3.2
	12.000	-0.453	2.960E-01	3.110E-01	2.1	3.2
	14.000	-0.420	4.690E-01	4.860E-01	2.1	3.2
	16.000	-0.391	6.910E-01	7.060E-01	2.1	3.2
	18.000	-0.365	9.650E-01	9.740E-01	1.8	1.8
	20.000	-0.342	1.290E+00	1.290E+00	1.8	1.8
	23.000	-0.312	1.680E+00	1.840E+00	1.8	1.8
	26.000	-0.285	2.590E+00	2.520E+00	1.8	1.8
	30.000	-0.254	3.720E+00	3.580E+00	1.5	2.3
	35.000	-0.221	5.390E+00	5.160E+00	1.5	2.3
	40.000	-0.192	7.320E+00	7.000E+00	1.5	2.3
	45.000	-0.166	9.460E+00	9.060E+00	2.1	2.2
	50.000	-0.143	1.180E+01	1.140E+01	2.1	2.2
	55.000	-0.122	1.420E+01	1.380E+01	2.1	2.2
	60.000	-0.103	1.680E+01	1.640E+01	2.1	2.2
	70.000	-0.070	2.210E+01	2.180E+01	2.9	4.3
	80.000	-0.041	2.740E+01	2.750E+01	2.9	4.3
	90.000	-0.015	3.280E+01	3.330E+01	2.9	4.3
	100.000	0.007	3.790E+01	3.900E+01	3.6	4.7
	110.000	0.028	4.290E+01	4.460E+01	3.6	4.7
	120.000	0.047	4.760E+01	5.010E+01	3.6	4.7
	140.000	0.080	5.610E+01	6.020E+01	3.6	4.6
	160.000	0.109	6.340E+01	6.940E+01	8.4	11.1
	180.000	0.135	6.970E+01	7.760E+01	8.4	11.1
	200.000	0.158	7.500E+01	8.480E+01	8.4	11.0
	230.000	0.188	8.140E+01	9.380E+01	8.4	11.0
	260.000	0.215	8.610E+01	1.010E+02	8.4	13.5
	300.000	0.246	9.050E+01	1.080E+02	7.1	11.3
	350.000	0.279	9.370E+01	1.140E+02	7.1	11.2
69-Tm	2.300	-0.819	3.810E-04	2.630E-04	9.6	9.9
	2.600	-0.793	7.840E-04	6.050E-04	9.6	9.9
	3.000	-0.762	1.670E-03	1.420E-03	7.1	7.4
	3.500	-0.728	3.440E-03	3.170E-03	7.1	7.4
	4.000	-0.699	6.080E-03	5.900E-03	7.1	7.4
	4.500	-0.674	9.720E-03	9.760E-03	2.8	3.6
	5.000	-0.651	1.450E-02	1.490E-02	2.8	3.6
	5.500	-0.630	2.050E-02	2.140E-02	2.8	3.6
	6.000	-0.611	2.780E-02	2.940E-02	2.8	3.6
	7.000	-0.578	4.700E-02	5.030E-02	2.8	3.6
	8.000	-0.549	7.300E-02	7.830E-02	2.8	3.3
	9.000	-0.523	1.060E-01	1.140E-01	2.8	3.3
	10.000	-0.500	1.480E-01	1.580E-01	2.8	3.3
	11.000	-0.480	1.990E-01	2.110E-01	2.1	2.9
	12.000	-0.461	2.590E-01	2.730E-01	2.1	2.9
	14.000	-0.427	4.100E-01	4.260E-01	2.1	3.0
	16.000	-0.398	6.040E-01	6.200E-01	2.1	3.0
	18.000	-0.373	8.430E-01	8.550E-01	1.8	2.0
	20.000	-0.350	1.130E+00	1.130E+00	1.8	2.1
	23.000	-0.319	1.650E+00	1.620E+00	1.8	2.1
	26.000	-0.293	2.270E+00	2.210E+00	1.8	2.1
	30.000	-0.262	3.260E+00	3.150E+00	1.5	2.0
	35.000	-0.228	4.740E+00	4.540E+00	1.5	2.0
	40.000	-0.199	6.450E+00	6.170E+00	1.5	2.0
	45.000	-0.174	8.360E+00	8.000E+00	2.1	2.2
	50.000	-0.151	1.040E+01	1.000E+01	2.1	2.2
	55.000	-0.130	1.260E+01	1.220E+01	2.1	2.2
	60.000	-0.111	1.490E+01	1.450E+01	2.1	2.2
	70.000	-0.078	1.970E+01	1.940E+01	2.9	4.0
	80.000	-0.049	2.460E+01	2.460E+01	2.9	4.0
	90.000	-0.023	2.940E+01	2.980E+01	2.9	4.0
	100.000	0.000	3.420E+01	3.500E+01	2.9	4.0
	110.000	0.020	3.870E+01	4.010E+01	3.6	4.4
	120.000	0.039	4.310E+01	4.510E+01	3.6	4.4
	140.000	0.073	5.100E+01	5.460E+01	3.6	4.4
	160.000	0.102	5.800E+01	6.310E+01	3.6	5.2

TABLE I. Empirical Reference Cross Sections for *K*-Shell Ionization by α Particles

See page 84 for Explanation of Tables

TARGET	ENERGY [MeV]	LOG XI	SIGMA [barn] THEORY	SIGMA [barn] REFERENCE	ERROR [%] RND	ERROR [%] TOT
69-Tm	180.000	0.127	6.390E+01	7.080E+01	8.4	10.6
	200.000	0.150	6.900E+01	7.750E+01	8.4	10.6
	230.000	0.181	7.510E+01	8.610E+01	8.4	10.6
	260.000	0.207	7.980E+01	9.300E+01	8.4	12.6
	300.000	0.238	8.410E+01	1.000E+02	7.1	10.4
	350.000	0.272	8.740E+01	1.060E+02	7.1	10.4
70-Yb	2.300	-0.827	3.100E-04	2.060E-04	9.6	9.7
	2.600	-0.801	6.520E-04	4.880E-04	9.6	9.7
	3.000	-0.769	1.410E-03	1.180E-03	7.1	7.2
	3.500	-0.736	2.950E-03	2.680E-03	7.1	7.2
	4.000	-0.707	5.260E-03	5.050E-03	7.1	7.2
	4.500	-0.681	8.460E-03	8.420E-03	7.1	7.2
	5.000	-0.659	1.260E-02	1.290E-02	2.8	3.3
	5.500	-0.638	1.790E-02	1.860E-02	2.8	3.3
	6.000	-0.619	2.440E-02	2.560E-02	2.8	3.3
	7.000	-0.586	4.120E-02	4.400E-02	2.8	3.3
	8.000	-0.557	6.400E-02	6.870E-02	2.8	3.1
	9.000	-0.531	9.340E-02	1.000E-01	2.8	3.1
	10.000	-0.508	1.300E-01	1.390E-01	2.8	3.1
	11.000	-0.487	1.740E-01	1.860E-01	2.8	3.1
	12.000	-0.468	2.270E-01	2.400E-01	2.1	2.7
	14.000	-0.435	3.590E-01	3.750E-01	2.1	2.7
	16.000	-0.406	5.290E-01	5.450E-01	2.1	2.7
	18.000	-0.380	7.390E-01	7.520E-01	2.1	2.7
	20.000	-0.358	9.910E-01	9.960E-01	1.8	2.3
	23.000	-0.327	1.450E+00	1.430E+00	1.8	2.3
	26.000	-0.301	2.000E+00	1.950E+00	1.8	2.3
	30.000	-0.269	2.870E+00	2.770E+00	1.5	1.8
	35.000	-0.236	4.180E+00	4.000E+00	1.5	1.8
	40.000	-0.207	5.690E+00	5.450E+00	1.5	1.8
	45.000	-0.181	7.390E+00	7.070E+00	2.1	2.5
	50.000	-0.159	9.240E+00	8.860E+00	2.1	2.5
	55.000	-0.138	1.120E+01	1.080E+01	2.1	2.5
	60.000	-0.119	1.330E+01	1.290E+01	2.1	2.5
	70.000	-0.086	1.760E+01	1.730E+01	2.9	3.8
	80.000	-0.057	2.200E+01	2.200E+01	2.9	3.8
	90.000	-0.031	2.650E+01	2.670E+01	2.9	3.8
	100.000	-0.008	3.080E+01	3.150E+01	2.9	3.8
	110.000	0.013	3.500E+01	3.620E+01	3.6	4.2
	120.000	0.032	3.910E+01	4.070E+01	3.6	4.2
	140.000	0.065	4.650E+01	4.950E+01	3.6	4.2
	160.000	0.094	5.300E+01	5.740E+01	3.6	4.2
	180.000	0.120	5.860E+01	6.460E+01	8.4	10.2
	200.000	0.142	6.350E+01	7.100E+01	8.4	10.2
	230.000	0.173	6.940E+01	7.910E+01	8.4	10.1
	260.000	0.199	7.390E+01	8.570E+01	8.4	10.1
	300.000	0.231	7.820E+01	9.260E+01	7.1	9.6
	350.000	0.264	8.160E+01	9.870E+01	7.1	9.6
	400.000	0.293	8.330E+01	1.030E+02	7.1	9.5
71-Lu	2.600	-0.808	5.410E-04	3.930E-04	9.6	10.0
	3.000	-0.777	1.190E-03	9.710E-04	9.6	10.0
	3.500	-0.744	2.530E-03	2.260E-03	7.1	7.4
	4.000	-0.715	4.560E-03	4.320E-03	7.1	7.4
	4.500	-0.689	7.370E-03	7.260E-03	7.1	7.4
	5.000	-0.666	1.110E-02	1.120E-02	2.8	3.1
	5.500	-0.645	1.570E-02	1.620E-02	2.8	3.1
	6.000	-0.627	2.140E-02	2.240E-02	2.8	3.1
	7.000	-0.593	3.630E-02	3.860E-02	2.8	3.1
	8.000	-0.564	5.630E-02	6.040E-02	2.8	2.9
	9.000	-0.539	8.210E-02	8.820E-02	2.8	2.9
	10.000	-0.516	1.140E-01	1.220E-01	2.8	2.9
	11.000	-0.495	1.530E-01	1.640E-01	2.8	2.9
	12.000	-0.476	2.000E-01	2.120E-01	2.1	2.5
	14.000	-0.443	3.160E-01	3.300E-01	2.1	2.5
	16.000	-0.414	4.650E-01	4.810E-01	2.1	2.5
	18.000	-0.388	6.500E-01	6.630E-01	2.1	2.5
	20.000	-0.365	8.700E-01	8.790E-01	1.8	2.5
	23.000	-0.335	1.270E+00	1.260E+00	1.8	2.6
	26.000	-0.308	1.760E+00	1.720E+00	1.8	2.6
	30.000	-0.277	2.530E+00	2.450E+00	1.5	1.5
	35.000	-0.244	3.690E+00	3.540E+00	1.5	1.5
	40.000	-0.215	5.030E+00	4.820E+00	1.5	1.5
	45.000	-0.189	6.550E+00	6.270E+00	1.5	1.5
	50.000	-0.166	8.200E+00	7.860E+00	2.1	2.7
	55.000	-0.145	9.970E+00	9.580E+00	2.1	2.7
	60.000	-0.127	1.180E+01	1.150E+01	2.1	2.7
	70.000	-0.093	1.570E+01	1.540E+01	2.1	2.7
	80.000	-0.064	1.980E+01	1.960E+01	2.9	3.5
	90.000	-0.039	2.380E+01	2.390E+01	2.9	3.5
71-Lu	100.000	-0.016	2.780E+01	2.830E+01	2.9	3.5
	110.000	0.005	3.170E+01	3.260E+01	3.6	3.9
	120.000	0.024	3.540E+01	3.680E+01	3.6	3.9
	140.000	0.057	4.230E+01	4.480E+01	3.6	3.9
	160.000	0.086	4.850E+01	5.230E+01	3.6	3.9
	180.000	0.112	5.380E+01	5.900E+01	8.4	9.8
	200.000	0.135	5.840E+01	6.500E+01	8.4	9.7
	230.000	0.165	6.400E+01	7.270E+01	8.4	9.7
	260.000	0.192	6.840E+01	7.900E+01	8.4	9.7
	300.000	0.223	7.270E+01	8.570E+01	8.4	10.9
	350.000	0.256	7.610E+01	9.160E+01	7.1	8.7
	400.000	0.285	7.800E+01	9.560E+01	7.1	8.7
72-Hf	2.600	-0.816	4.470E-04	3.140E-04	9.6	10.4
	3.000	-0.785	1.010E-03	7.990E-04	9.6	10.3
	3.500	-0.751	2.170E-03	1.900E-03	7.1	7.7
	4.000	-0.722	3.950E-03	3.690E-03	7.1	7.7
	4.500	-0.697	6.430E-03	6.270E-03	7.1	7.7
	5.000	-0.674	9.680E-03	9.730E-03	2.8	2.9
	5.500	-0.653	1.380E-02	1.420E-02	2.8	2.9
	6.000	-0.634	1.880E-02	1.960E-02	2.8	2.9
	7.000	-0.601	3.190E-02	3.390E-02	2.8	2.9
	8.000	-0.572	4.960E-02	5.310E-02	2.8	2.9
	9.000	-0.546	7.240E-02	7.770E-02	2.8	2.9
	10.000	-0.523	1.010E-01	1.080E-01	2.8	2.9
	11.000	-0.503	1.350E-01	1.440E-01	2.8	2.9
	12.000	-0.484	1.760E-01	1.870E-01	2.8	2.9
	14.000	-0.450	2.780E-01	2.920E-01	2.1	2.2
	16.000	-0.421	4.090E-01	4.240E-01	2.1	2.2
	18.000	-0.396	5.710E-01	5.860E-01	2.1	2.2
	20.000	-0.373	7.660E-01	7.760E-01	1.8	2.8
	23.000	-0.342	1.120E+00	1.120E+00	1.8	2.8
	26.000	-0.316	1.550E+00	1.520E+00	1.8	2.8
	30.000	-0.285	2.230E+00	2.160E+00	1.8	2.8
	35.000	-0.251	3.250E+00	3.130E+00	1.5	1.7
	40.000	-0.222	4.450E+00	4.270E+00	1.5	1.7
	45.000	-0.197	5.800E+00	5.550E+00	1.5	1.7
	50.000	-0.174	7.280E+00	6.970E+00	2.1	3.0
	55.000	-0.153	8.870E+00	8.510E+00	2.1	3.0
	60.000	-0.134	1.060E+01	1.020E+01	2.1	3.0
	70.000	-0.101	1.410E+01	1.380E+01	2.1	3.0
	80.000	-0.072	1.780E+01	1.760E+01	2.9	3.3
	90.000	-0.046	2.150E+01	2.150E+01	2.9	3.3
	100.000	-0.023	2.510E+01	2.540E+01	2.9	3.3
	110.000	-0.003	2.870E+01	2.940E+01	2.9	3.3
	120.000	0.016	3.210E+01	3.320E+01	3.6	3.7
	140.000	0.050	3.860E+01	4.070E+01	3.6	3.7
	160.000	0.079	4.430E+01	4.750E+01	3.6	3.7
	180.000	0.104	4.930E+01	5.380E+01	3.6	3.8
	200.000	0.127	5.370E+01	5.950E+01	8.4	9.3
	230.000	0.158	5.910E+01	6.680E+01	8.4	9.3
	260.000	0.184	6.340E+01	7.280E+01	8.4	9.3
	300.000	0.215	6.760E+01	7.920E+01	8.4	10.1
	350.000	0.249	7.100E+01	8.510E+01	7.1	7.9
	400.000	0.278	7.300E+01	8.900E+01	7.1	7.9
73-Ta	2.600	-0.823	3.700E-04	2.510E-04	9.6	10.8
	3.000	-0.792	8.500E-04	6.570E-04	9.6	10.7
	3.500	-0.759	1.870E-03	1.610E-03	7.1	8.0
	4.000	-0.730	3.440E-03	3.160E-03	7.1	8.0
	4.500	-0.704	5.630E-03	5.420E-03	7.1	7.9
	5.000	-0.681	8.510E-03	8.470E-03	7.1	7.9
	5.500	-0.661	1.220E-02	1.240E-02	2.8	2.9
	6.000	-0.642	1.660E-02	1.720E-02	2.8	2.9
	7.000	-0.608	2.820E-02	2.990E-02	2.8	2.9
	8.000	-0.579	4.390E-02	4.690E-02	2.8	2.9
	9.000	-0.554	6.400E-02	6.870E-02	2.8	3.1
	10.000	-0.531	8.910E-02	9.560E-02	2.8	3.1
	11.000	-0.510	1.190E-01	1.280E-01	2.8	3.1
	12.000	-0.491	1.550E-01	1.660E-01	2.8	3.1
	14.000	-0.458	2.450E-01	2.580E-01	2.1	2.1
	16.000	-0.429	3.610E-01	3.760E-01	2.1	2.1
	18.000	-0.403	5.050E-01	5.190E-01	2.1	2.1
	20.000	-0.380	6.760E-01	6.880E-01	2.1	2.1
	23.000	-0.350	9.880E-01	9.900E-01	1.8	3.1
	26.000	-0.323	1.370E+00	1.340E+00	1.8	3.1
	30.000	-0.292	1.970E+00	1.920E+00	1.8	3.1
	35.000	-0.259	2.880E+00	2.780E+00	1.5	1.9
	40.000	-0.230	3.950E+00	3.790E+00	1.5	1.9
	45.000	-0.204	5.160E+00	4.940E+00	1.5	1.9
	50.000	-0.181	6.480E+00	6.210E+00	2.1	3.3
	55.000	-0.161	7.910E+00	7.580E+00	2.1	3.3

TABLE I. Empirical Reference Cross Sections for *K*-Shell Ionization by α Particles

See page 84 for Explanation of Tables

TARGET	ENERGY [MeV]	LOG XI	SIGMA [barn] THEORY	SIGMA [barn] REFERENCE	ERROR [%] RND	ERROR [%] TOT
73-Ta	60.000	-0.142	9.420E+00	9.100E+00	2.1	3.2
	70.000	-0.108	1.260E+01	1.230E+01	2.1	3.2
	80.000	-0.079	1.600E+01	1.570E+01	2.9	3.0
	90.000	-0.054	1.930E+01	1.930E+01	2.9	3.0
	100.000	-0.031	2.270E+01	2.290E+01	2.9	3.0
	110.000	-0.010	2.600E+01	2.650E+01	2.9	3.0
	120.000	0.009	2.920E+01	3.010E+01	3.6	3.7
	140.000	0.042	3.520E+01	3.690E+01	3.6	3.7
	160.000	0.071	4.050E+01	4.330E+01	3.6	3.7
	180.000	0.097	4.530E+01	4.920E+01	3.6	3.7
	200.000	0.120	4.940E+01	5.450E+01	8.4	8.8
	230.000	0.150	5.460E+01	6.140E+01	8.4	8.8
	260.000	0.177	5.880E+01	6.720E+01	8.4	8.8
	300.000	0.208	6.290E+01	7.340E+01	8.4	9.2
	350.000	0.241	6.630E+01	7.900E+01	7.1	7.1
	400.000	0.270	6.830E+01	8.290E+01	7.1	7.1
	450.000	0.296	6.930E+01	8.550E+01	7.1	7.1
74-W	3.000	-0.800	7.170E-04	5.390E-04	9.6	11.0
	3.500	-0.766	1.610E-03	1.350E-03	7.1	8.4
	4.000	-0.737	2.990E-03	2.700E-03	7.1	8.3
	4.500	-0.712	4.930E-03	4.690E-03	7.1	8.2
	5.000	-0.689	7.490E-03	7.380E-03	7.1	8.2
	5.500	-0.668	1.070E-02	1.080E-02	2.8	3.2
	6.000	-0.649	1.470E-02	1.510E-02	2.8	3.2
	7.000	-0.616	2.500E-02	2.640E-02	2.8	3.2
	8.000	-0.587	3.890E-02	4.150E-02	2.8	3.2
	9.000	-0.561	5.680E-02	6.090E-02	2.8	3.3
	10.000	-0.538	7.890E-02	8.470E-02	2.8	3.3
	11.000	-0.518	1.060E-01	1.130E-01	2.8	3.3
	12.000	-0.499	1.380E-01	1.470E-01	2.8	3.3
	14.000	-0.465	2.170E-01	2.300E-01	2.1	2.4
	16.000	-0.436	3.200E-01	3.340E-01	2.1	2.4
	18.000	-0.411	4.460E-01	4.610E-01	2.1	2.4
	20.000	-0.388	5.980E-01	6.110E-01	2.1	2.4
	23.000	-0.357	8.740E-01	8.790E-01	1.8	3.3
	26.000	-0.331	1.210E+00	1.200E+00	1.8	3.3
	30.000	-0.300	1.750E+00	1.700E+00	1.8	3.3
	35.000	-0.266	2.560E+00	2.470E+00	1.5	2.2
	40.000	-0.237	3.510E+00	3.360E+00	1.5	2.2
	45.000	-0.212	4.590E+00	4.390E+00	1.5	2.2
	50.000	-0.189	5.780E+00	5.530E+00	1.5	2.2
	55.000	-0.168	7.060E+00	6.760E+00	2.1	3.5
	60.000	-0.149	8.420E+00	8.080E+00	2.1	3.5
	70.000	-0.116	1.130E+01	1.100E+01	2.1	3.5
	80.000	-0.087	1.430E+01	1.410E+01	2.9	3.1
	90.000	-0.061	1.740E+01	1.730E+01	2.9	3.1
	100.000	-0.038	2.050E+01	2.060E+01	2.9	3.1
	110.000	-0.018	2.360E+01	2.390E+01	2.9	3.1
	120.000	0.001	2.650E+01	2.720E+01	2.9	3.1
	140.000	0.035	3.210E+01	3.350E+01	3.6	3.9
	160.000	0.064	3.710E+01	3.950E+01	3.6	3.9
	180.000	0.089	4.160E+01	4.490E+01	3.6	3.9
	200.000	0.112	4.550E+01	4.990E+01	8.4	8.5
	230.000	0.143	5.050E+01	5.640E+01	8.4	8.5
	260.000	0.169	5.450E+01	6.200E+01	8.4	8.5
	300.000	0.200	5.850E+01	6.790E+01	8.4	8.5
	350.000	0.234	6.190E+01	7.340E+01	7.1	7.9
	400.000	0.263	6.400E+01	7.730E+01	7.1	7.9
	450.000	0.288	6.510E+01	7.990E+01	7.1	7.9
75-Re	3.000	-0.807	6.040E-04	4.410E-04	9.6	11.4
	3.500	-0.774	1.380E-03	1.140E-03	7.1	8.7
	4.000	-0.745	2.600E-03	2.310E-03	7.1	8.6
	4.500	-0.719	4.320E-03	4.050E-03	7.1	8.5
	5.000	-0.696	6.590E-03	6.430E-03	7.1	8.4
	5.500	-0.675	9.480E-03	9.500E-03	2.8	3.4
	6.000	-0.657	1.300E-02	1.330E-02	2.8	3.4
	7.000	-0.623	2.220E-02	2.330E-02	2.8	3.4
	8.000	-0.594	3.460E-02	3.680E-02	2.8	3.4
	9.000	-0.569	5.050E-02	5.410E-02	2.8	3.6
	10.000	-0.546	7.020E-02	7.530E-02	2.8	3.6
	11.000	-0.525	9.400E-02	1.010E-01	2.8	3.6
	12.000	-0.506	1.220E-01	1.310E-01	2.8	3.6
	14.000	-0.473	1.930E-01	2.040E-01	2.1	2.6
	16.000	-0.444	2.840E-01	2.970E-01	2.1	2.6
	18.000	-0.418	3.960E-01	4.100E-01	2.1	2.6
	20.000	-0.395	5.300E-01	5.430E-01	2.1	2.6
	23.000	-0.365	7.750E-01	7.820E-01	1.8	3.5
	26.000	-0.338	1.070E+00	1.070E+00	1.8	3.6
	30.000	-0.307	1.550E+00	1.510E+00	1.8	3.6
	35.000	-0.274	2.270E+00	2.200E+00	1.5	2.4

TARGET	ENERGY [MeV]	LOG XI	SIGMA [barn] THEORY	SIGMA [barn] REFERENCE	ERROR [%] RND	ERROR [%] TOT
75-Re	40.000	-0.245	3.120E+00	2.990E+00	1.5	2.5
	45.000	-0.219	4.080E+00	3.910E+00	1.5	2.5
	50.000	-0.196	5.150E+00	4.930E+00	1.5	2.5
	55.000	-0.175	6.310E+00	6.040E+00	2.1	3.8
	60.000	-0.157	7.530E+00	7.220E+00	2.1	3.8
	70.000	-0.123	1.020E+01	9.850E+00	2.1	3.8
	80.000	-0.094	1.290E+01	1.270E+01	2.1	3.7
	90.000	-0.069	1.570E+01	1.560E+01	2.9	3.3
	100.000	-0.046	1.860E+01	1.860E+01	2.9	3.3
	110.000	-0.025	2.140E+01	2.160E+01	2.9	3.3
	120.000	-0.006	2.410E+01	2.460E+01	2.9	3.3
	140.000	0.027	2.930E+01	3.050E+01	3.6	4.2
	160.000	0.056	3.400E+01	3.600E+01	3.6	4.2
	180.000	0.082	3.820E+01	4.110E+01	3.6	4.1
	200.000	0.105	4.190E+01	4.570E+01	3.6	4.7
	230.000	0.135	4.670E+01	5.190E+01	8.4	8.9
	260.000	0.162	5.050E+01	5.720E+01	8.4	8.9
	300.000	0.193	5.440E+01	6.290E+01	8.4	8.9
	350.000	0.226	5.780E+01	6.820E+01	7.1	8.8
	400.000	0.255	5.990E+01	7.210E+01	7.1	8.7
	450.000	0.281	6.110E+01	7.470E+01	7.1	8.7
76-Os	3.000	-0.814	5.090E-04	4.500E-04	6.8	9.8
	3.500	-0.781	1.180E-03	1.110E-03	6.8	9.6
	4.000	-0.752	2.260E-03	2.210E-03	5.6	7.0
	4.500	-0.726	3.790E-03	3.810E-03	5.6	6.9
	5.000	-0.703	5.820E-03	5.970E-03	5.6	6.9
	5.500	-0.683	8.390E-03	8.720E-03	5.6	6.9
	6.000	-0.664	1.160E-02	1.210E-02	2.9	3.6
	7.000	-0.630	1.980E-02	2.100E-02	2.9	3.6
	8.000	-0.601	3.080E-02	3.280E-02	2.9	3.6
	9.000	-0.576	4.500E-02	4.790E-02	2.8	3.2
	10.000	-0.553	6.250E-02	6.640E-02	2.8	3.2
	11.000	-0.532	8.370E-02	8.850E-02	2.8	3.2
	12.000	-0.513	1.090E-01	1.140E-01	2.8	3.2
	14.000	-0.480	1.720E-01	1.780E-01	2.1	3.0
	16.000	-0.451	2.520E-01	2.590E-01	2.1	3.0
	18.000	-0.425	3.520E-01	3.560E-01	2.1	3.1
	20.000	-0.402	4.710E-01	4.750E-01	2.1	3.1
	23.000	-0.372	6.890E-01	6.840E-01	2.1	3.1
	26.000	-0.345	9.530E-01	9.320E-01	2.5	3.7
	30.000	-0.314	1.380E+00	1.320E+00	2.5	3.7
	35.000	-0.281	2.020E+00	1.910E+00	2.5	3.7
	40.000	-0.252	2.780E+00	2.610E+00	2.0	3.0
	45.000	-0.226	3.640E+00	3.420E+00	2.0	3.0
	50.000	-0.203	4.600E+00	4.310E+00	2.0	3.0
	55.000	-0.183	5.640E+00	5.270E+00	3.0	3.5
	60.000	-0.164	6.750E+00	6.310E+00	3.0	3.5
	70.000	-0.130	9.120E+00	8.660E+00	3.0	3.5
	80.000	-0.101	1.160E+01	1.120E+01	3.0	3.5
	90.000	-0.076	1.420E+01	1.390E+01	3.3	4.0
	100.000	-0.053	1.680E+01	1.660E+01	3.3	4.0
	110.000	-0.032	1.940E+01	1.940E+01	3.3	4.0
	120.000	-0.013	2.190E+01	2.220E+01	3.3	4.0
	140.000	0.020	2.670E+01	2.770E+01	4.7	5.1
	160.000	0.049	3.110E+01	3.290E+01	4.7	5.1
	180.000	0.075	3.510E+01	3.780E+01	4.7	5.1
	200.000	0.098	3.860E+01	4.230E+01	4.7	5.1
	230.000	0.128	4.310E+01	4.830E+01	15.6	16.5
	260.000	0.155	4.680E+01	5.340E+01	15.6	16.5
	300.000	0.186	5.060E+01	5.910E+01	15.6	16.4
77-Ir	3.000	-0.822	4.270E-04	3.720E-04	6.8	9.6
	3.500	-0.788	1.020E-03	9.440E-04	6.8	9.4
	4.000	-0.759	1.970E-03	1.910E-03	5.6	6.7
	4.500	-0.734	3.330E-03	3.320E-03	5.6	6.7
	5.000	-0.711	5.140E-03	5.240E-03	5.6	6.7
	5.500	-0.690	7.440E-03	7.700E-03	5.6	6.7
	6.000	-0.671	1.030E-02	1.070E-02	5.6	6.7
	7.000	-0.638	1.760E-02	1.870E-02	2.9	3.3
	8.000	-0.609	2.750E-02	2.930E-02	2.9	3.3
	9.000	-0.583	4.020E-02	4.280E-02	2.9	3.3
	10.000	-0.560	5.580E-02	5.930E-02	2.8	3.0
	11.000	-0.540	7.480E-02	7.910E-02	2.8	3.0
	12.000	-0.521	9.710E-02	1.020E-01	2.8	3.0
	14.000	-0.487	1.530E-01	1.590E-01	2.8	3.0
	16.000	-0.458	2.250E-01	2.310E-01	2.1	2.8
	18.000	-0.433	3.130E-01	3.190E-01	2.1	2.8
	20.000	-0.410	4.200E-01	4.250E-01	2.1	2.8
	23.000	-0.379	6.130E-01	6.110E-01	2.1	2.8
	26.000	-0.353	8.480E-01	8.330E-01	2.5	3.4

TABLE I. Empirical Reference Cross Sections for *K*-Shell Ionization by α Particles

See page 84 for Explanation of Tables

TARGET	ENERGY [MeV]	LOG XI	SIGMA [barn] THEORY	SIGMA [barn] REFERENCE	ERROR [%] RND	ERROR [%] TOT
77-Ir	30.000	-0.322	1.230E+00	1.180E+00	2.5	3.5
	35.000	-0.288	1.800E+00	1.710E+00	2.5	3.5
	40.000	-0.259	2.480E+00	2.330E+00	2.0	2.8
	45.000	-0.234	3.250E+00	3.060E+00	2.0	2.8
	50.000	-0.211	4.110E+00	3.850E+00	2.0	2.8
	55.000	-0.190	5.050E+00	4.720E+00	2.0	2.8
	60.000	-0.171	6.050E+00	5.660E+00	3.0	3.3
	70.000	-0.138	8.200E+00	7.690E+00	3.0	3.3
	80.000	-0.109	1.050E+01	1.010E+01	3.0	3.3
	90.000	-0.083	1.280E+01	1.250E+01	3.3	3.7
	100.000	-0.060	1.520E+01	1.500E+01	3.3	3.7
	110.000	-0.040	1.760E+01	1.750E+01	3.3	3.7
	120.000	-0.021	1.990E+01	2.010E+01	3.3	3.7
	140.000	0.013	2.440E+01	2.510E+01	4.7	4.9
	160.000	0.042	2.850E+01	3.000E+01	4.7	4.9
	180.000	0.067	3.220E+01	3.450E+01	4.7	4.9
	200.000	0.090	3.560E+01	3.870E+01	4.7	4.9
	230.000	0.121	3.990E+01	4.440E+01	15.6	16.0
	260.000	0.147	4.340E+01	4.930E+01	15.6	16.0
	300.000	0.178	4.710E+01	5.470E+01	15.6	16.0
78-Pt	3.000	-0.829	3.580E-04	3.070E-04	6.8	9.3
	3.500	-0.795	8.710E-04	7.990E-04	6.8	9.2
	4.000	-0.766	1.710E-03	1.640E-03	5.6	6.5
	4.500	-0.741	2.920E-03	2.900E-03	5.6	6.5
	5.000	-0.718	4.540E-03	4.610E-03	5.6	6.4
	5.500	-0.697	6.610E-03	6.810E-03	5.6	6.4
	6.000	-0.678	9.150E-03	9.530E-03	5.6	6.4
	7.000	-0.645	1.580E-02	1.670E-02	2.9	3.1
	8.000	-0.616	2.460E-02	2.620E-02	2.9	3.1
	9.000	-0.590	3.600E-02	3.830E-02	2.9	3.1
	10.000	-0.567	5.000E-02	5.320E-02	2.8	2.9
	11.000	-0.547	6.690E-02	7.090E-02	2.8	2.9
	12.000	-0.528	8.690E-02	9.170E-02	2.8	2.9
	14.000	-0.494	1.370E-01	1.430E-01	2.8	2.9
	16.000	-0.465	2.010E-01	2.070E-01	2.1	2.5
	18.000	-0.440	2.800E-01	2.860E-01	2.1	2.6
	20.000	-0.417	3.740E-01	3.810E-01	2.1	2.6
	23.000	-0.387	5.470E-01	5.470E-01	2.1	2.6
	26.000	-0.360	7.560E-01	7.460E-01	2.5	3.2
	30.000	-0.329	1.090E+00	1.060E+00	2.5	3.2
	35.000	-0.295	1.610E+00	1.530E+00	2.5	3.2
	40.000	-0.266	2.210E+00	2.090E+00	2.0	2.5
	45.000	-0.241	2.910E+00	2.730E+00	2.0	2.5
	50.000	-0.218	3.680E+00	3.450E+00	2.0	2.5
	55.000	-0.197	4.530E+00	4.240E+00	2.0	2.5
	60.000	-0.178	5.430E+00	5.080E+00	3.0	3.1
	70.000	-0.145	7.380E+00	6.920E+00	3.0	3.1
	80.000	-0.116	9.460E+00	9.040E+00	3.0	3.0
	90.000	-0.090	1.160E+01	1.120E+01	3.0	3.0
	100.000	-0.067	1.380E+01	1.350E+01	3.3	3.5
	110.000	-0.047	1.600E+01	1.590E+01	3.3	3.5
	120.000	-0.028	1.820E+01	1.820E+01	3.3	3.5
	140.000	0.006	2.230E+01	2.280E+01	4.7	4.8
	160.000	0.035	2.610E+01	2.730E+01	4.7	4.8
	180.000	0.060	2.960E+01	3.160E+01	4.7	4.8
	200.000	0.083	3.280E+01	3.550E+01	4.7	4.8
	230.000	0.113	3.690E+01	4.090E+01	15.6	15.6
	260.000	0.140	4.030E+01	4.550E+01	15.6	15.6
	300.000	0.171	4.380E+01	5.070E+01	15.6	15.6
79-Au	3.000	-0.836	3.000E-04	2.530E-04	6.8	9.1
	3.500	-0.803	7.460E-04	6.760E-04	6.8	8.9
	4.000	-0.774	1.490E-03	1.410E-03	6.8	8.8
	4.500	-0.748	2.570E-03	2.530E-03	5.6	6.2
	5.000	-0.725	4.020E-03	4.050E-03	5.6	6.2
	5.500	-0.704	5.880E-03	6.020E-03	5.6	6.2
	6.000	-0.686	8.160E-03	8.470E-03	5.6	6.2
	7.000	-0.652	1.410E-02	1.490E-02	2.9	3.0
	8.000	-0.623	2.210E-02	2.350E-02	2.9	3.0
	9.000	-0.597	3.230E-02	3.440E-02	2.9	3.0
	10.000	-0.575	4.490E-02	4.780E-02	2.8	3.1
	11.000	-0.554	6.000E-02	6.370E-02	2.8	3.1
	12.000	-0.535	7.800E-02	8.240E-02	2.8	3.1
	14.000	-0.502	1.230E-01	1.280E-01	2.8	3.1
	16.000	-0.473	1.800E-01	1.860E-01	2.1	2.3
	18.000	-0.447	2.500E-01	2.560E-01	2.1	2.3
	20.000	-0.424	3.350E-01	3.390E-01	2.1	2.3
	23.000	-0.394	4.890E-01	4.910E-01	2.1	2.3
	26.000	-0.367	6.760E-01	6.690E-01	2.5	2.9
	30.000	-0.336	9.780E-01	9.520E-01	2.5	2.9
79-Au	35.000	-0.303	1.440E+00	1.370E+00	2.5	2.9
	40.000	-0.274	1.980E+00	1.870E+00	2.0	2.2
	45.000	-0.248	2.610E+00	2.450E+00	2.0	2.2
	50.000	-0.225	3.300E+00	3.100E+00	2.0	2.2
	55.000	-0.204	4.070E+00	3.800E+00	2.0	2.2
	60.000	-0.186	4.880E+00	4.560E+00	3.0	3.3
	70.000	-0.152	6.650E+00	6.230E+00	3.0	3.3
	80.000	-0.123	8.540E+00	8.140E+00	3.0	3.3
	90.000	-0.097	1.050E+01	1.010E+01	3.0	3.3
	100.000	-0.075	1.250E+01	1.220E+01	3.3	3.3
	110.000	-0.054	1.450E+01	1.430E+01	3.3	3.3
	120.000	-0.035	1.650E+01	1.650E+01	3.3	3.3
	140.000	-0.002	2.040E+01	2.080E+01	3.3	3.3
	160.000	0.027	2.400E+01	2.490E+01	4.7	5.1
	180.000	0.053	2.730E+01	2.890E+01	4.7	5.1
	200.000	0.076	3.020E+01	3.260E+01	4.7	5.1
	230.000	0.106	3.410E+01	3.760E+01	4.7	5.4
	260.000	0.133	3.740E+01	4.200E+01	15.6	16.0
	300.000	0.164	4.080E+01	4.690E+01	15.6	16.0
	350.000	0.197	4.400E+01	5.180E+01	15.6	16.0
80-Hg	3.500	-0.810	6.370E-04	5.700E-04	6.8	8.7
	4.000	-0.781	1.290E-03	1.220E-03	6.8	8.6
	4.500	-0.755	2.260E-03	2.200E-03	5.6	6.0
	5.000	-0.732	3.560E-03	3.560E-03	5.6	6.0
	5.500	-0.712	5.230E-03	5.330E-03	5.6	6.0
	6.000	-0.693	7.290E-03	7.530E-03	5.6	5.9
	7.000	-0.659	1.270E-02	1.330E-02	2.9	3.2
	8.000	-0.630	1.980E-02	2.110E-02	2.9	3.2
	9.000	-0.605	2.900E-02	3.090E-02	2.9	3.2
	10.000	-0.582	4.030E-02	4.300E-02	2.9	3.2
	11.000	-0.561	5.400E-02	5.740E-02	2.8	3.4
	12.000	-0.542	7.010E-02	7.420E-02	2.8	3.4
	14.000	-0.509	1.100E-01	1.160E-01	2.8	3.4
	16.000	-0.480	1.610E-01	1.680E-01	2.1	2.1
	18.000	-0.454	2.240E-01	2.310E-01	2.1	2.1
	20.000	-0.431	3.000E-01	3.050E-01	2.1	2.1
	23.000	-0.401	4.380E-01	4.410E-01	2.1	2.1
	26.000	-0.374	6.050E-01	6.020E-01	2.1	2.1
	30.000	-0.343	8.750E-01	8.550E-01	2.5	2.7
	35.000	-0.310	1.290E+00	1.230E+00	2.5	2.7
	40.000	-0.281	1.780E+00	1.680E+00	2.5	2.7
	45.000	-0.255	2.340E+00	2.200E+00	2.0	2.1
	50.000	-0.232	2.970E+00	2.790E+00	2.0	2.1
	55.000	-0.212	3.650E+00	3.420E+00	2.0	2.1
	60.000	-0.193	4.390E+00	4.110E+00	2.0	2.1
	70.000	-0.159	5.990E+00	5.610E+00	3.0	3.6
	80.000	-0.130	7.720E+00	7.340E+00	3.0	3.6
	90.000	-0.105	9.520E+00	9.150E+00	3.0	3.6
	100.000	-0.082	1.140E+01	1.100E+01	3.3	3.6
	110.000	-0.061	1.320E+01	1.300E+01	3.3	3.6
	120.000	-0.042	1.510E+01	1.500E+01	3.3	3.6
	140.000	-0.009	1.860E+01	1.890E+01	3.3	3.6
	160.000	0.020	2.200E+01	2.280E+01	4.7	5.3
	180.000	0.046	2.510E+01	2.640E+01	4.7	5.3
	200.000	0.069	2.790E+01	2.990E+01	4.7	5.3
	230.000	0.099	3.160E+01	3.460E+01	4.7	5.3
	260.000	0.126	3.470E+01	3.880E+01	15.6	16.5
	300.000	0.157	3.800E+01	4.350E+01	15.6	16.5
	350.000	0.190	4.110E+01	4.810E+01	15.6	16.4
81-Tl	3.500	-0.817	5.440E-04	4.800E-04	6.8	8.4
	4.000	-0.788	1.120E-03	1.040E-03	6.8	8.3
	4.500	-0.762	1.980E-03	1.920E-03	5.6	5.7
	5.000	-0.739	3.160E-03	3.130E-03	5.6	5.7
	5.500	-0.719	4.660E-03	4.720E-03	5.6	5.7
	6.000	-0.700	6.520E-03	6.700E-03	5.6	5.7
	7.000	-0.666	1.140E-02	1.190E-02	2.9	3.4
	8.000	-0.637	1.790E-02	1.890E-02	2.9	3.4
	9.000	-0.612	2.620E-02	2.780E-02	2.9	3.4
	10.000	-0.589	3.640E-02	3.870E-02	2.9	3.4
	11.000	-0.568	4.860E-02	5.170E-02	2.8	3.6
	12.000	-0.549	6.310E-02	6.690E-02	2.8	3.6
	14.000	-0.516	9.910E-02	1.040E-01	2.8	3.6
	16.000	-0.487	1.450E-01	1.510E-01	2.8	3.5
	18.000	-0.461	2.020E-01	2.080E-01	2.1	2.3
	20.000	-0.438	2.690E-01	2.750E-01	2.1	2.3
	23.000	-0.408	3.930E-01	3.980E-01	2.1	2.3
	26.000	-0.381	5.430E-01	5.420E-01	2.1	2.3
	30.000	-0.350	7.850E-01	7.700E-01	2.5	2.6
	35.000	-0.317	1.150E+00	1.110E+00	2.5	2.6

TABLE I. Empirical Reference Cross Sections for *K*-Shell Ionization by α Particles

See page 84 for Explanation of Tables

TARGET	ENERGY [MeV]	LOG XI	SIGMA [barn] THEORY	SIGMA [barn] REFERENCE	ERROR [%] RND	ERROR [%] TOT
81-Tl	40.000	-0.288	1.590E+00	1.510E+00	2.5	2.6
	45.000	-0.262	2.100E+00	1.980E+00	2.0	2.3
	50.000	-0.239	2.670E+00	2.510E+00	2.0	2.3
	55.000	-0.219	3.290E+00	3.080E+00	2.0	2.3
	60.000	-0.200	3.950E+00	3.700E+00	2.0	2.3
	70.000	-0.166	5.410E+00	5.060E+00	3.0	3.9
	80.000	-0.137	6.980E+00	6.550E+00	3.0	3.8
	90.000	-0.112	8.630E+00	8.270E+00	3.0	3.8
	100.000	-0.089	1.030E+01	9.990E+00	3.0	3.8
	110.000	-0.068	1.200E+01	1.180E+01	3.3	3.8
	120.000	-0.049	1.370E+01	1.360E+01	3.3	3.8
	140.000	-0.016	1.700E+01	1.720E+01	3.3	3.8
	160.000	0.013	2.020E+01	2.080E+01	4.7	5.6
	180.000	0.039	2.310E+01	2.420E+01	4.7	5.5
	200.000	0.062	2.570E+01	2.740E+01	4.7	5.5
	230.000	0.092	2.920E+01	3.190E+01	4.7	5.5
	260.000	0.119	3.220E+01	3.580E+01	15.6	16.9
	300.000	0.150	3.540E+01	4.030E+01	15.6	16.9
	350.000	0.183	3.840E+01	4.470E+01	15.6	16.9
82-Pb	3.500	-0.824	4.640E-04	4.020E-04	6.8	8.1
	4.000	-0.795	9.730E-04	8.950E-04	6.8	8.1
	4.500	-0.769	1.740E-03	1.670E-03	5.6	5.8
	5.000	-0.746	2.800E-03	2.750E-03	5.6	5.8
	5.500	-0.726	4.150E-03	4.180E-03	5.6	5.8
	6.000	-0.707	5.830E-03	5.970E-03	5.6	5.8
	7.000	-0.673	1.020E-02	1.070E-02	5.6	5.8
	8.000	-0.644	1.610E-02	1.700E-02	2.9	3.7
	9.000	-0.619	2.360E-02	2.510E-02	2.9	3.7
	10.000	-0.596	3.280E-02	3.500E-02	2.9	3.7
	11.000	-0.575	4.390E-02	4.670E-02	2.8	3.8
	12.000	-0.556	5.700E-02	6.050E-02	2.8	3.8
	14.000	-0.523	8.930E-02	9.420E-02	2.8	3.8
	16.000	-0.494	1.310E-01	1.370E-01	2.8	3.8
	18.000	-0.468	1.820E-01	1.880E-01	2.1	2.5
	20.000	-0.445	2.430E-01	2.480E-01	2.1	2.6
	23.000	-0.415	3.530E-01	3.590E-01	2.1	2.6
	26.000	-0.388	4.880E-01	4.890E-01	2.1	2.6
	30.000	-0.357	7.050E-01	6.940E-01	2.5	2.8
	35.000	-0.324	1.040E+00	1.000E+00	2.5	2.8
	40.000	-0.295	1.430E+00	1.360E+00	2.5	2.8
	45.000	-0.269	1.890E+00	1.780E+00	2.0	2.6
	50.000	-0.246	2.400E+00	2.250E+00	2.0	2.6
	55.000	-0.226	2.960E+00	2.780E+00	2.0	2.6
	60.000	-0.207	3.570E+00	3.340E+00	2.0	2.6
	70.000	-0.173	4.890E+00	4.570E+00	3.0	4.1
	80.000	-0.144	6.320E+00	5.930E+00	3.0	4.1
	90.000	-0.119	7.830E+00	7.480E+00	3.0	4.1
	100.000	-0.096	9.380E+00	9.050E+00	3.0	4.1
	110.000	-0.075	1.100E+01	1.070E+01	3.3	4.1
	120.000	-0.056	1.250E+01	1.230E+01	3.3	4.1
	140.000	-0.023	1.560E+01	1.570E+01	3.3	4.1
	160.000	0.006	1.850E+01	1.900E+01	4.7	5.8
	180.000	0.032	2.120E+01	2.220E+01	4.7	5.8
	200.000	0.055	2.370E+01	2.520E+01	4.7	5.8
	230.000	0.085	2.700E+01	2.930E+01	4.7	5.7
	260.000	0.112	2.990E+01	3.300E+01	15.6	17.4
	300.000	0.143	3.290E+01	3.730E+01	15.6	17.3
	350.000	0.176	3.590E+01	4.160E+01	15.6	17.3
83-Bi	3.500	-0.831	3.940E-04	3.370E-04	6.8	7.9
	4.000	-0.802	8.430E-04	7.660E-04	6.8	7.8
	4.500	-0.776	1.530E-03	1.450E-03	6.8	7.8
	5.000	-0.753	2.480E-03	2.420E-03	5.6	6.1
	5.500	-0.733	3.700E-03	3.700E-03	5.6	6.1
	6.000	-0.714	5.230E-03	5.320E-03	5.6	6.1
	7.000	-0.680	9.220E-03	9.590E-03	5.6	6.0
	8.000	-0.651	1.460E-02	1.540E-02	2.9	3.9
	9.000	-0.626	2.140E-02	2.270E-02	2.9	3.9
	10.000	-0.603	2.970E-02	3.160E-02	2.9	3.9
	11.000	-0.582	3.970E-02	4.230E-02	2.9	3.9
	12.000	-0.563	5.150E-02	5.480E-02	2.8	4.1
	14.000	-0.530	8.080E-02	8.530E-02	2.8	4.1
	16.000	-0.501	1.180E-01	1.240E-01	2.8	4.1
	18.000	-0.475	1.640E-01	1.700E-01	2.1	2.8
	20.000	-0.452	2.190E-01	2.250E-01	2.1	2.8
	23.000	-0.422	3.180E-01	3.250E-01	2.1	2.8
	26.000	-0.395	4.390E-01	4.420E-01	2.1	2.8
	30.000	-0.364	6.350E-01	6.280E-01	2.5	3.1
	35.000	-0.331	9.330E-01	9.060E-01	2.5	3.1
	40.000	-0.302	1.290E+00	1.230E+00	2.5	3.1
83-Bi	45.000	-0.276	1.700E+00	1.610E+00	2.0	2.9
	50.000	-0.253	2.160E+00	2.030E+00	2.0	2.9
	55.000	-0.233	2.670E+00	2.510E+00	2.0	2.9
	60.000	-0.214	3.220E+00	3.020E+00	2.0	2.9
	70.000	-0.180	4.420E+00	4.130E+00	3.0	4.4
	80.000	-0.151	5.730E+00	5.360E+00	3.0	4.4
	90.000	-0.126	7.110E+00	6.770E+00	3.0	4.4
	100.000	-0.103	8.530E+00	8.210E+00	3.0	4.3
	110.000	-0.082	9.980E+00	9.700E+00	3.3	4.4
	120.000	-0.063	1.140E+01	1.120E+01	3.3	4.3
	140.000	-0.030	1.430E+01	1.430E+01	3.3	4.3
	160.000	-0.001	1.700E+01	1.740E+01	3.3	4.3
	180.000	0.025	1.960E+01	2.030E+01	4.7	6.0
	200.000	0.048	2.190E+01	2.310E+01	4.7	6.0
	230.000	0.078	2.500E+01	2.700E+01	4.7	6.0
	260.000	0.105	2.770E+01	3.050E+01	4.7	7.2
	300.000	0.136	3.070E+01	3.460E+01	15.6	17.8
	350.000	0.169	3.350E+01	3.870E+01	15.6	17.7
	400.000	0.198	3.560E+01	4.190E+01	15.6	17.7
84-Po	3.500	-0.838	3.340E-04	2.810E-04	6.8	7.6
	4.000	-0.809	7.290E-04	6.530E-04	6.8	7.5
	4.500	-0.783	1.340E-03	1.260E-03	6.8	7.5
	5.000	-0.760	2.190E-03	2.120E-03	5.6	6.3
	5.500	-0.740	3.300E-03	3.280E-03	5.6	6.3
	6.000	-0.721	4.680E-03	4.740E-03	5.6	6.3
	7.000	-0.687	8.310E-03	8.610E-03	5.6	6.3
	8.000	-0.658	1.320E-02	1.380E-02	2.9	4.2
	9.000	-0.633	1.930E-02	2.050E-02	2.9	4.2
	10.000	-0.610	2.690E-02	2.870E-02	2.9	4.1
	11.000	-0.589	3.600E-02	3.840E-02	2.9	4.1
	12.000	-0.570	4.670E-02	4.970E-02	2.8	4.3
	14.000	-0.537	7.310E-02	7.740E-02	2.8	4.3
	16.000	-0.508	1.070E-01	1.120E-01	2.8	4.3
	18.000	-0.482	1.480E-01	1.540E-01	2.1	3.0
	20.000	-0.459	1.980E-01	2.030E-01	2.1	3.0
	23.000	-0.429	2.870E-01	2.920E-01	2.1	3.0
	26.000	-0.402	3.960E-01	4.000E-01	2.1	3.1
	30.000	-0.371	5.720E-01	5.680E-01	2.1	3.1
	35.000	-0.338	8.410E-01	8.190E-01	2.5	3.3
	40.000	-0.309	1.160E+00	1.110E+00	2.5	3.3
	45.000	-0.283	1.530E+00	1.450E+00	2.5	3.3
	50.000	-0.260	1.950E+00	1.840E+00	2.0	3.1
	55.000	-0.240	2.410E+00	2.270E+00	2.0	3.1
	60.000	-0.221	2.910E+00	2.730E+00	2.0	3.1
	70.000	-0.187	4.000E+00	3.740E+00	3.0	4.7
	80.000	-0.158	5.190E+00	4.860E+00	3.0	4.7
	90.000	-0.133	6.460E+00	6.130E+00	3.0	4.6
	100.000	-0.110	7.770E+00	7.440E+00	3.0	4.6
	110.000	-0.089	9.100E+00	8.810E+00	3.0	4.6
	120.000	-0.070	1.040E+01	1.020E+01	3.3	4.6
	140.000	-0.037	1.310E+01	1.300E+01	3.3	4.6
	160.000	-0.008	1.560E+01	1.590E+01	3.3	4.6
	180.000	0.018	1.800E+01	1.860E+01	4.7	6.3
	200.000	0.041	2.020E+01	2.120E+01	4.7	6.3
	230.000	0.071	2.320E+01	2.490E+01	4.7	6.2
	260.000	0.098	2.570E+01	2.820E+01	4.7	6.2
	300.000	0.129	2.860E+01	3.200E+01	15.6	18.3
	350.000	0.162	3.130E+01	3.600E+01	15.6	18.2
	400.000	0.191	3.330E+01	3.910E+01	15.6	18.1
85-At	4.000	-0.816	6.300E-04	5.560E-04	6.8	7.3
	4.500	-0.790	1.180E-03	1.090E-03	6.8	7.2
	5.000	-0.767	1.940E-03	1.860E-03	5.6	6.6
	5.500	-0.747	2.950E-03	2.900E-03	5.6	6.6
	6.000	-0.728	4.200E-03	4.220E-03	5.6	6.6
	7.000	-0.694	7.500E-03	7.740E-03	5.6	6.5
	8.000	-0.665	1.190E-02	1.250E-02	2.9	4.4
	9.000	-0.640	1.760E-02	1.860E-02	2.9	4.4
	10.000	-0.617	2.450E-02	2.600E-02	2.9	4.4
	11.000	-0.596	3.270E-02	3.480E-02	2.9	4.4
	12.000	-0.577	4.240E-02	4.520E-02	2.8	4.5
	14.000	-0.544	6.640E-02	7.030E-02	2.8	4.5
	16.000	-0.515	9.690E-02	1.020E-01	2.8	4.6
	18.000	-0.489	1.340E-01	1.400E-01	2.8	4.6
	20.000	-0.466	1.790E-01	1.850E-01	2.1	3.3
	23.000	-0.436	2.600E-01	2.650E-01	2.1	3.3
	26.000	-0.409	3.580E-01	3.630E-01	2.1	3.3
	30.000	-0.378	5.170E-01	5.150E-01	2.1	3.3
	35.000	-0.345	7.590E-01	7.430E-01	2.5	3.6
	40.000	-0.316	1.050E+00	1.010E+00	2.5	3.6

TABLE I. Empirical Reference Cross Sections for *K*-Shell Ionization by α Particles

See page 84 for Explanation of Tables

TARGET	ENERGY [MeV]	LOG XI	SIGMA [barn] THEORY	SIGMA [barn] REFERENCE	ERROR [%] RND	ERROR [%] TOT
85-At	45.000	-0.290	1.390E+00	1.320E+00	2.5	3.6
	50.000	-0.267	1.760E+00	1.660E+00	2.0	3.4
	55.000	-0.247	2.180E+00	2.040E+00	2.0	3.4
	60.000	-0.228	2.630E+00	2.470E+00	2.0	3.4
	70.000	-0.194	3.620E+00	3.390E+00	2.0	3.4
	80.000	-0.165	4.710E+00	4.410E+00	3.0	4.9
	90.000	-0.140	5.870E+00	5.510E+00	3.0	4.9
	100.000	-0.117	7.070E+00	6.760E+00	3.0	4.9
	110.000	-0.096	8.300E+00	8.010E+00	3.0	4.9
	120.000	-0.077	9.540E+00	9.290E+00	3.3	4.9
	140.000	-0.044	1.200E+01	1.190E+01	3.3	4.8
	160.000	-0.015	1.440E+01	1.450E+01	3.3	4.8
	180.000	0.011	1.660E+01	1.710E+01	4.7	6.5
	200.000	0.034	1.870E+01	1.950E+01	4.7	6.5
	230.000	0.064	2.150E+01	2.290E+01	4.7	6.5
	260.000	0.091	2.390E+01	2.610E+01	4.7	6.4
	300.000	0.122	2.660E+01	2.970E+01	15.6	18.7
	350.000	0.155	2.930E+01	3.350E+01	15.6	18.6
	400.000	0.184	3.120E+01	3.650E+01	15.6	18.6
86-Rn	4.000	-0.823	5.450E-04	4.670E-04	6.0	7.4
	4.500	-0.797	1.030E-03	9.160E-04	6.0	7.4
	5.000	-0.774	1.720E-03	1.570E-03	6.0	7.4
	5.500	-0.753	2.630E-03	2.460E-03	3.7	5.7
	6.000	-0.735	3.780E-03	3.580E-03	3.7	5.7
	7.000	-0.701	6.790E-03	6.600E-03	3.7	5.6
	8.000	-0.672	1.080E-02	1.070E-02	3.7	5.6
	9.000	-0.646	1.600E-02	1.600E-02	2.9	4.9
	10.000	-0.624	2.230E-02	2.240E-02	2.9	4.9
	11.000	-0.603	2.980E-02	3.020E-02	2.9	4.9
	12.000	-0.584	3.860E-02	3.920E-02	2.9	4.9
	14.000	-0.551	6.040E-02	6.150E-02	1.7	3.6
	16.000	-0.522	8.810E-02	8.960E-02	1.7	3.6
	18.000	-0.496	1.220E-01	1.240E-01	1.7	3.6
	20.000	-0.473	1.620E-01	1.640E-01	1.7	3.6
	23.000	-0.443	2.360E-01	2.370E-01	1.5	3.0
	26.000	-0.416	3.250E-01	3.250E-01	1.5	3.0
	30.000	-0.385	4.680E-01	4.640E-01	1.5	3.0
	35.000	-0.352	6.870E-01	6.730E-01	1.6	3.1
	40.000	-0.323	9.500E-01	9.190E-01	1.6	3.1
	45.000	-0.297	1.250E+00	1.200E+00	1.6	3.1
	50.000	-0.274	1.600E+00	1.510E+00	2.0	3.1
	55.000	-0.253	1.970E+00	1.860E+00	2.0	3.1
	60.000	-0.235	2.380E+00	2.230E+00	2.0	3.1
	70.000	-0.201	3.290E+00	3.030E+00	2.0	3.1
	80.000	-0.172	4.280E+00	3.910E+00	2.6	4.4
	90.000	-0.148	5.340E+00	4.840E+00	2.6	4.4
	100.000	-0.124	6.450E+00	5.810E+00	2.6	4.4
	110.000	-0.103	7.580E+00	6.800E+00	2.6	4.4
	120.000	-0.084	8.720E+00	7.800E+00	2.6	4.4
	140.000	-0.051	1.100E+01	9.820E+00	2.8	4.3
	160.000	-0.022	1.320E+01	1.180E+01	2.8	4.3
	180.000	0.004	1.530E+01	1.370E+01	2.8	4.3
	200.000	0.027	1.730E+01	1.560E+01	3.1	5.2
	230.000	0.057	1.990E+01	1.820E+01	3.1	5.2
87-Fr	4.000	-0.829	4.680E-04	3.980E-04	6.0	7.2
	4.500	-0.804	9.010E-04	7.940E-04	6.0	7.1
	5.000	-0.781	1.520E-03	1.380E-03	6.0	7.1
	5.500	-0.760	2.350E-03	2.180E-03	3.7	5.5
	6.000	-0.741	3.390E-03	3.200E-03	3.7	5.4
	7.000	-0.708	6.140E-03	5.940E-03	3.7	5.4
	8.000	-0.679	9.830E-03	9.690E-03	3.7	5.4
	9.000	-0.653	1.450E-02	1.450E-02	2.9	4.7
	10.000	-0.631	2.030E-02	2.040E-02	2.9	4.6
	11.000	-0.610	2.720E-02	2.740E-02	2.9	4.6
	12.000	-0.591	3.520E-02	3.570E-02	2.9	4.6
	14.000	-0.557	5.500E-02	5.600E-02	1.7	3.3
	16.000	-0.528	8.010E-02	8.150E-02	1.7	3.3
	18.000	-0.503	1.110E-01	1.130E-01	1.7	3.3
	20.000	-0.480	1.470E-01	1.490E-01	1.7	3.3
	23.000	-0.450	2.140E-01	2.150E-01	1.5	2.7
	26.000	-0.423	2.940E-01	2.940E-01	1.5	2.7
	30.000	-0.392	4.240E-01	4.210E-01	1.5	2.8
	35.000	-0.358	6.220E-01	6.110E-01	1.6	2.8
	40.000	-0.329	8.600E-01	8.340E-01	1.6	2.9
	45.000	-0.304	1.140E+00	1.090E+00	1.6	2.9
	50.000	-0.281	1.450E+00	1.380E+00	1.6	2.9
	55.000	-0.260	1.790E+00	1.690E+00	2.0	2.8
	60.000	-0.241	2.160E+00	2.020E+00	2.0	2.8
87-Fr	70.000	-0.208	2.990E+00	2.760E+00	2.0	2.8
	80.000	-0.179	3.890E+00	3.560E+00	2.0	2.9
	90.000	-0.153	4.860E+00	4.410E+00	2.6	4.1
	100.000	-0.131	5.880E+00	5.300E+00	2.6	4.1
	110.000	-0.110	6.920E+00	6.210E+00	2.6	4.1
	120.000	-0.091	7.980E+00	7.140E+00	2.6	4.1
	140.000	-0.057	1.010E+01	9.000E+00	2.8	4.0
	160.000	-0.028	1.210E+01	1.080E+01	2.8	4.0
	180.000	-0.003	1.410E+01	1.260E+01	2.8	4.0
	200.000	0.020	1.590E+01	1.440E+01	3.1	4.9
	230.000	0.050	1.840E+01	1.680E+01	3.1	4.9
88-Ra	4.000	-0.836	4.030E-04	3.380E-04	6.0	6.9
	4.500	-0.811	7.870E-04	6.870E-04	6.0	6.9
	5.000	-0.788	1.350E-03	1.210E-03	6.0	6.8
	5.500	-0.767	2.090E-03	1.930E-03	6.0	6.8
	6.000	-0.748	3.040E-03	2.850E-03	3.7	5.2
	7.000	-0.715	5.560E-03	5.360E-03	3.7	5.1
	8.000	-0.686	8.940E-03	8.780E-03	3.7	5.1
	9.000	-0.660	1.320E-02	1.320E-02	2.9	4.4
	10.000	-0.637	1.850E-02	1.860E-02	2.9	4.4
	11.000	-0.617	2.480E-02	2.500E-02	2.9	4.4
	12.000	-0.598	3.220E-02	3.260E-02	2.9	4.4
	14.000	-0.564	5.020E-02	5.110E-02	1.7	3.1
	16.000	-0.535	7.310E-02	7.440E-02	1.7	3.1
	18.000	-0.510	1.010E-01	1.030E-01	1.7	3.1
	20.000	-0.487	1.340E-01	1.360E-01	1.7	3.1
	23.000	-0.456	1.950E-01	1.960E-01	1.5	2.5
	26.000	-0.430	2.680E-01	2.680E-01	1.5	2.5
	30.000	-0.399	3.850E-01	3.830E-01	1.5	2.5
	35.000	-0.365	5.650E-01	5.560E-01	1.6	2.6
	40.000	-0.336	7.800E-01	7.590E-01	1.6	2.6
	45.000	-0.311	1.030E+00	9.920E-01	1.6	2.6
	50.000	-0.288	1.310E+00	1.250E+00	1.6	2.6
	55.000	-0.267	1.620E+00	1.540E+00	2.0	2.6
	60.000	-0.248	1.960E+00	1.840E+00	2.0	2.6
	70.000	-0.215	2.710E+00	2.510E+00	2.0	2.6
	80.000	-0.186	3.540E+00	3.250E+00	2.0	2.6
	90.000	-0.160	4.430E+00	4.030E+00	2.6	3.8
	100.000	-0.137	5.370E+00	4.850E+00	2.6	3.8
	110.000	-0.117	6.330E+00	5.690E+00	2.6	3.8
	120.000	-0.098	7.300E+00	6.540E+00	2.6	3.8
	140.000	-0.064	9.260E+00	8.270E+00	2.8	3.7
	160.000	-0.035	1.120E+01	9.980E+00	2.8	3.7
	180.000	-0.010	1.300E+01	1.160E+01	2.8	3.7
	200.000	0.013	1.470E+01	1.330E+01	3.1	4.6
	230.000	0.044	1.710E+01	1.550E+01	3.1	4.6
89-Ac	4.500	-0.817	6.860E-04	5.930E-04	6.0	6.6
	5.000	-0.795	1.190E-03	1.060E-03	6.0	6.6
	5.500	-0.774	1.870E-03	1.710E-03	6.0	6.5
	6.000	-0.755	2.730E-03	2.540E-03	3.7	4.9
	7.000	-0.722	5.030E-03	4.830E-03	3.7	4.9
	8.000	-0.693	8.140E-03	7.960E-03	3.7	4.9
	9.000	-0.667	1.210E-02	1.200E-02	3.7	4.8
	10.000	-0.644	1.690E-02	1.690E-02	2.9	4.1
	11.000	-0.623	2.270E-02	2.280E-02	2.9	4.1
	12.000	-0.604	2.940E-02	2.980E-02	2.9	4.1
	14.000	-0.571	4.590E-02	4.670E-02	2.9	4.1
	16.000	-0.542	6.680E-02	6.800E-02	1.7	2.8
	18.000	-0.516	9.230E-02	9.380E-02	1.7	2.8
	20.000	-0.494	1.230E-01	1.240E-01	1.7	2.8
	23.000	-0.463	1.770E-01	1.790E-01	1.5	2.2
	26.000	-0.437	2.440E-01	2.440E-01	1.5	2.2
	30.000	-0.406	3.500E-01	3.490E-01	1.5	2.2
	35.000	-0.372	5.130E-01	5.060E-01	1.5	2.3
	40.000	-0.343	7.090E-01	6.910E-01	1.6	2.3
	45.000	-0.317	9.350E-01	9.030E-01	1.6	2.4
	50.000	-0.295	1.190E+00	1.140E+00	1.6	2.4
	55.000	-0.274	1.480E+00	1.400E+00	2.0	2.3
	60.000	-0.255	1.780E+00	1.680E+00	2.0	2.3
	70.000	-0.222	2.470E+00	2.290E+00	2.0	2.3
	80.000	-0.193	3.230E+00	2.970E+00	2.0	2.3
	90.000	-0.167	4.040E+00	3.690E+00	2.6	3.5
	100.000	-0.144	4.900E+00	4.440E+00	2.6	3.6
	110.000	-0.123	5.790E+00	5.210E+00	2.6	3.6
	120.000	-0.104	6.690E+00	6.000E+00	2.6	3.6
	140.000	-0.071	8.500E+00	7.600E+00	2.8	3.5
	160.000	-0.042	1.030E+01	9.180E+00	2.8	3.5
	180.000	-0.016	1.200E+01	1.070E+01	2.8	3.5

TABLE I. Empirical Reference Cross Sections for *K*-Shell Ionization by α Particles

See page 84 for Explanation of Tables

TARGET	ENERGY [MeV]	LOG XI	SIGMA [barn] THEORY	SIGMA [barn] REFERENCE	ERROR [%] RND	ERROR [%] TOT
89-Ac	200.000	0.006	1.360E+01	1.220E+01	2.8	3.5
	230.000	0.037	1.590E+01	1.440E+01	3.1	4.3
90-Th	4.500	-0.824	5.980E-04	5.120E-04	6.0	6.3
	5.000	-0.801	1.050E-03	9.270E-04	6.0	6.3
	5.500	-0.781	1.660E-03	1.510E-03	6.0	6.3
	6.000	-0.762	2.450E-03	2.270E-03	3.7	4.7
	7.000	-0.728	4.560E-03	4.350E-03	3.7	4.6
	8.000	-0.699	7.410E-03	7.220E-03	3.7	4.6
	9.000	-0.674	1.110E-02	1.090E-02	3.7	4.6
	10.000	-0.651	1.550E-02	1.550E-02	2.9	3.9
	11.000	-0.630	2.080E-02	2.090E-02	2.9	3.9
	12.000	-0.611	2.700E-02	2.720E-02	2.9	3.9
	14.000	-0.578	4.210E-02	4.280E-02	2.9	3.9
	16.000	-0.549	6.120E-02	6.230E-02	1.7	2.6
	18.000	-0.523	8.440E-02	8.590E-02	1.7	2.6
	20.000	-0.500	1.120E-01	1.140E-01	1.7	2.6
	23.000	-0.470	1.620E-01	1.640E-01	1.7	2.6
	26.000	-0.443	2.220E-01	2.230E-01	1.5	2.0
	30.000	-0.412	3.190E-01	3.190E-01	1.5	2.0
	35.000	-0.379	4.670E-01	4.620E-01	1.5	2.0
	40.000	-0.350	6.450E-01	6.310E-01	1.6	2.1
	45.000	-0.324	8.510E-01	8.240E-01	1.6	2.1
	50.000	-0.301	1.080E+00	1.040E+00	1.6	2.1
	55.000	-0.281	1.340E+00	1.280E+00	1.6	2.1
	60.000	-0.262	1.620E+00	1.530E+00	2.0	2.0
	70.000	-0.228	2.250E+00	2.100E+00	2.0	2.0
	80.000	-0.199	2.940E+00	2.710E+00	2.0	2.0
	90.000	-0.174	3.690E+00	3.370E+00	2.6	3.3
	100.000	-0.151	4.480E+00	4.060E+00	2.6	3.3
	110.000	-0.130	5.300E+00	4.780E+00	2.6	3.3
	120.000	-0.111	6.130E+00	5.510E+00	2.6	3.3
	140.000	-0.078	7.810E+00	6.980E+00	2.6	3.3
	160.000	-0.049	9.470E+00	8.460E+00	2.8	3.2
	180.000	-0.023	1.110E+01	9.900E+00	2.8	3.2
	200.000	0.000	1.260E+01	1.130E+01	2.8	3.2
	230.000	0.030	1.470E+01	1.330E+01	3.1	4.1
	260.000	0.057	1.660E+01	1.520E+01	3.1	4.1
91-Pa	4.500	-0.831	5.190E-04	4.400E-04	6.0	6.0
	5.000	-0.808	9.240E-04	8.090E-04	6.0	6.0
	5.500	-0.787	1.480E-03	1.330E-03	6.0	6.0
	6.000	-0.768	2.200E-03	2.020E-03	6.0	6.0
	7.000	-0.735	4.130E-03	3.920E-03	3.7	4.4
	8.000	-0.706	6.760E-03	6.550E-03	3.7	4.4
	9.000	-0.680	1.010E-02	9.950E-03	3.7	4.3
	10.000	-0.658	1.420E-02	1.410E-02	2.9	3.7
	11.000	-0.637	1.910E-02	1.910E-02	2.9	3.6
	12.000	-0.618	2.470E-02	2.500E-02	2.9	3.6
	14.000	-0.584	3.860E-02	3.920E-02	2.9	3.6
	16.000	-0.555	5.610E-02	5.710E-02	1.7	2.3
	18.000	-0.530	7.740E-02	7.870E-02	1.7	2.3
	20.000	-0.507	1.030E-01	1.040E-01	1.7	2.3
	23.000	-0.477	1.480E-01	1.500E-01	1.7	2.4
	26.000	-0.450	2.030E-01	2.040E-01	1.5	1.7
	30.000	-0.419	2.910E-01	2.910E-01	1.5	1.7
	35.000	-0.385	4.260E-01	4.220E-01	1.5	1.7
	40.000	-0.357	5.880E-01	5.760E-01	1.6	1.8
	45.000	-0.331	7.750E-01	7.530E-01	1.6	1.8
	50.000	-0.308	9.870E-01	9.500E-01	1.6	1.8
	55.000	-0.287	1.220E+00	1.170E+00	1.6	1.8
	60.000	-0.268	1.480E+00	1.400E+00	2.0	2.2
	70.000	-0.235	2.050E+00	1.920E+00	2.0	2.2
	80.000	-0.206	2.690E+00	2.480E+00	2.0	2.2
	90.000	-0.180	3.380E+00	3.090E+00	2.0	2.2
	100.000	-0.158	4.100E+00	3.730E+00	2.6	3.0
	110.000	-0.137	4.850E+00	4.380E+00	2.6	3.0
	120.000	-0.118	5.620E+00	5.060E+00	2.6	3.0
	140.000	-0.084	7.180E+00	6.420E+00	2.6	3.0
	160.000	-0.055	8.730E+00	7.790E+00	2.8	2.9
	180.000	-0.030	1.020E+01	9.130E+00	2.8	2.9
	200.000	-0.007	1.170E+01	1.040E+01	2.8	2.9
	230.000	0.023	1.360E+01	1.230E+01	3.1	3.8
	260.000	0.050	1.540E+01	1.410E+01	3.1	3.8
92-U	4.500	-0.838	4.510E-04	3.770E-04	6.0	6.2
	5.000	-0.815	8.130E-04	7.060E-04	6.0	6.2
	5.500	-0.794	1.320E-03	1.180E-03	6.0	6.2
	6.000	-0.775	1.970E-03	1.800E-03	6.0	6.2
92-U	7.000	-0.742	3.750E-03	3.530E-03	3.7	4.1
	8.000	-0.713	6.170E-03	5.950E-03	3.7	4.1
	9.000	-0.687	9.250E-03	9.080E-03	3.7	4.1
	10.000	-0.664	1.300E-02	1.290E-02	3.7	4.1
	11.000	-0.644	1.750E-02	1.750E-02	2.9	3.4
	12.000	-0.625	2.270E-02	2.290E-02	2.9	3.4
	14.000	-0.591	3.550E-02	3.600E-02	2.9	3.4
	16.000	-0.562	5.160E-02	5.250E-02	1.7	2.1
	18.000	-0.537	7.100E-02	7.230E-02	1.7	2.1
	20.000	-0.514	9.410E-02	9.570E-02	1.7	2.1
	23.000	-0.483	1.360E-01	1.370E-01	1.7	2.1
	26.000	-0.457	1.860E-01	1.870E-01	1.5	1.6
	30.000	-0.426	2.660E-01	2.670E-01	1.5	1.6
	35.000	-0.392	3.890E-01	3.860E-01	1.5	1.6
	40.000	-0.363	5.360E-01	5.270E-01	1.6	1.7
	45.000	-0.338	7.070E-01	6.890E-01	1.6	1.7
	50.000	-0.315	9.010E-01	8.690E-01	1.6	1.7
	55.000	-0.294	1.120E+00	1.070E+00	1.6	1.7
	60.000	-0.275	1.350E+00	1.280E+00	2.0	2.4
	70.000	-0.242	1.870E+00	1.750E+00	2.0	2.4
	80.000	-0.213	2.460E+00	2.270E+00	2.0	2.4
	90.000	-0.187	3.090E+00	2.830E+00	2.0	2.4
	100.000	-0.164	3.750E+00	3.420E+00	2.6	2.7
	110.000	-0.144	4.450E+00	4.030E+00	2.6	2.7
	120.000	-0.125	5.160E+00	4.650E+00	2.6	2.7
	140.000	-0.091	6.610E+00	5.910E+00	2.6	2.7
	160.000	-0.062	8.050E+00	7.180E+00	2.8	3.1
	180.000	-0.037	9.450E+00	8.430E+00	2.8	3.1
	200.000	-0.014	1.080E+01	9.650E+00	2.8	3.1
	230.000	0.017	1.270E+01	1.140E+01	3.1	3.5
	260.000	0.043	1.430E+01	1.300E+01	3.1	3.5

TABLE IIa. Experimental Cross Sections for *K*-Shell X-Ray Production, Auger Electron Production, and Ionization by ^{1}H Ions

See page 84 for Explanation of Tables

TARGET	ENERGY [MeV]	SIGMA [barn]	ERROR [%]	REFERENCE	TARGET	ENERGY [MeV]	SIGMA [barn]	ERROR [%]	REFERENCE	TARGET	ENERGY [MeV]	SIGMA [barn]	ERROR [%]	REFERENCE
12-Mg	0.700	1.298E+04	8.0	INOR89	27-Co	0.100	3.370E-04	17.0	XKEI90					
12-Mg	0.750	7.180E+02	15.0	XNYU91B	27-Co	0.150	4.510E-03	17.0	XKEI90					
12-Mg	0.900	2.210E+04	8.0	INOR89	27-Co	0.200	2.040E-02	17.0	XKEI90					
12-Mg	1.000	9.620E+02	15.0	XNYU91B	27-Co	0.250	6.790E-02	17.0	XKEI90					
12-Mg	1.100	2.637E+04	8.0	INOR89	27-Co	0.300	1.690E-01	17.0	XKEI90					
12-Mg	1.300	3.614E+04	8.0	INOR89	27-Co	0.350	3.550E-01	17.0	XKEI90					
12-Mg	1.500	1.040E+03	15.0	XNYU91B	27-Co	0.400	5.820E-01	17.0	XKEI90					
12-Mg	2.000	1.210E+03	15.0	XNYU91B	27-Co	0.450	9.100E-01	17.0	XKEI90					
12-Mg	2.500	1.320E+03	15.0	XNYU91B	27-Co	0.500	4.200E+00	8.0	INOR89					
12-Mg	3.000	1.320E+03	15.0	XNYU91B	27-Co	0.500	1.361E+00	17.0	XKEI90					
12-Mg	3.500	1.500E+03	15.0	XNYU91B	27-Co	0.550	2.030E+00	17.0	XKEI90					
12-Mg	4.000	1.430E+03	15.0	XNYU91B	27-Co	0.700	1.170E+01	8.0	INOR89					
12-Mg	4.500	1.440E+03	15.0	XNYU91B	27-Co	1.100	4.210E+01	8.0	INOR89					
13-Al	0.500	4.605E+03	7.0	INOR89	27-Co	1.300	6.520E+01	8.0	INOR89					
13-Al	0.700	9.880E+03	3.0	INOR89	28-Ni	0.500	2.420E+00	7.0	INOR89					
13-Al	0.750	4.330E+02	15.0	XNYU91B	28-Ni	0.700	7.250E+00	7.0	INOR89					
13-Al	0.900	1.489E+04	7.0	INOR89	28-Ni	0.900	1.630E+01	7.0	INOR89					
13-Al	1.000	5.440E+02	15.0	XNYU91B	28-Ni	1.100	2.820E+01	7.0	INOR89					
13-Al	1.100	1.714E+04	7.0	INOR89	28-Ni	1.300	4.580E+01	7.0	INOR89					
13-Al	1.300	2.261E+04	6.0	INOR89	28-Ni	1.500	7.560E+01	7.0	INOR89					
13-Al	1.500	2.436E+04	8.0	INOR89	29-Cu	0.100	9.020E-05	17.0	XKEI90					
13-Al	1.500	8.490E+02	15.0	XNYU91B	29-Cu	0.150	1.580E-03	17.0	XKEI90					
13-Al	2.000	1.000E+03	15.0	XNYU91B	29-Cu	0.200	8.100E-03	17.0	XKEI90					
13-Al	2.500	1.080E+03	15.0	XNYU91B	29-Cu	0.250	3.190E-02	17.0	XKEI90					
13-Al	3.000	1.190E+03	15.0	XNYU91B	29-Cu	0.300	8.930E-02	17.0	XKEI90					
13-Al	3.500	1.200E+03	15.0	XNYU91B	29-Cu	0.350	2.020E-01	17.0	XKEI90					
13-Al	4.000	1.200E+03	15.0	XNYU91B	29-Cu	0.400	3.350E-01	17.0	XKEI90					
13-Al	4.500	1.200E+03	15.0	XNYU91B	29-Cu	0.450	5.150E-01	17.0	XKEI90					
14-Si	0.700	5.530E+03	8.0	INOR89	29-Cu	0.500	1.570E+00	7.0	INOR89					
14-Si	0.900	8.310E+03	8.0	INOR89	29-Cu	0.500	7.520E-01	17.0	XKEI90					
14-Si	1.000	5.460E+02	7.0	XNGE90	29-Cu	0.550	1.096E+00	17.0	XKEI90					
14-Si	1.100	1.038E+04	8.0	INOR89	29-Cu	0.700	4.890E+00	7.0	INOR89					
14-Si	1.300	1.230E+04	8.0	INOR89	29-Cu	0.900	1.220E+01	7.0	INOR89					
14-Si	1.500	1.993E+04	8.0	INOR89	29-Cu	1.100	1.940E+01	7.0	INOR89					
14-Si	2.000	9.170E+02	7.0	XNGE90	29-Cu	1.300	3.410E+01	6.0	INOR89					
15-P	0.500	1.370E+03	8.0	INOR89	30-Zn	1.000	5.480E+00	5.0	XNGE90					
15-P	0.700	3.350E+03	8.0	INOR89	30-Zn	2.000	3.370E+01	5.0	XNGE90					
15-P	0.900	5.400E+03	8.0	INOR89	31-Ga	0.500	6.000E-01	10.0	INOR89					
15-P	1.000	3.810E+02	6.0	XNGE90	31-Ga	0.700	2.460E+00	10.0	INOR89					
15-P	1.100	7.780E+03	8.0	INOR89	31-Ga	0.900	5.740E+00	10.0	INOR89					
15-P	1.300	9.600E+03	8.0	INOR89	31-Ga	1.000	4.120E+00	5.0	XNGE90					
15-P	1.500	1.100E+04	8.0	INOR89	31-Ga	1.100	9.740E+00	9.0	INOR89					
15-P	2.000	7.940E+02	6.0	XNGE90	31-Ga	1.300	1.780E+01	8.0	INOR89					
16-S	1.000	2.980E+02	6.0	XNGE90	31-Ga	1.500	2.910E+01	8.0	INOR89					
16-S	2.000	6.480E+02	6.0	XNGE90	31-Ga	2.000	2.600E+01	5.0	XNGE90					
17-Cl	0.500	4.450E+02	8.0	INOR89	35-Br	0.300	1.231E-02	8.2	XKSI91					
17-Cl	0.900	1.480E+03	8.0	INOR89	35-Br	0.320	1.681E-02	7.9	XKSI91					
17-Cl	1.100	2.090E+03	8.0	INOR89	35-Br	0.340	2.209E-02	7.7	XKSI91					
19-K	1.000	1.300E+02	6.0	XNGE90	35-Br	0.360	2.855E-02	7.5	XKSI91					
19-K	2.000	3.570E+02	6.0	XNGE90	35-Br	0.380	3.613E-02	7.3	XKSI91					
20-Ca	0.700	2.460E+02	7.0	INOR89	35-Br	0.400	4.509E-02	7.1	XKSI91					
20-Ca	0.900	4.760E+02	7.0	INOR89	37-Rb	0.300	6.010E-03	7.2	XKSI91					
20-Ca	1.000	9.540E+01	7.0	XNGE90	37-Rb	0.320	8.250E-03	6.6	XKSI91					
20-Ca	1.100	7.140E+02	7.0	INOR89	37-Rb	0.340	1.106E-02	6.7	XKSI91					
20-Ca	1.300	1.020E+03	7.0	INOR89	37-Rb	0.360	1.469E-02	6.0	XKSI91					
20-Ca	1.500	1.390E+03	7.0	INOR89	37-Rb	0.380	1.886E-02	5.9	XKSI91					
20-Ca	2.000	2.810E+02	7.0	XNGE90	37-Rb	0.400	2.373E-02	6.3	XKSI91					
22-Ti	0.500	3.690E+01	7.0	INOR89	39-Y	0.300	2.950E-03	7.0	XKSI91					
22-Ti	0.700	9.820E+01	7.0	INOR89	39-Y	0.320	3.890E-03	6.6	XKSI91					
22-Ti	0.900	1.890E+02	7.0	INOR89	39-Y	0.340	5.690E-03	5.7	XKSI91					
22-Ti	1.100	3.180E+02	7.0	INOR89	39-Y	0.360	7.600E-03	6.4	XKSI91					
22-Ti	1.300	4.580E+02	7.0	INOR89	39-Y	0.380	9.750E-03	6.0	XKSI91					
22-Ti	1.500	5.650E+02	7.0	INOR89	39-Y	0.400	1.212E-02	5.9	XKSI91					
23-V	0.075	3.500E-04	17.0	XKEI90	42-Mo	0.250	3.050E-04	17.0	XKEI90					
23-V	0.100	2.350E-03	17.0	XKEI90	42-Mo	0.300	1.140E-03	17.0	XKEI90					
23-V	0.150	2.330E-02	17.0	XKEI90	42-Mo	0.350	2.220E-03	17.0	XKEI90					
23-V	0.200	1.250E-01	17.0	XKEI90	42-Mo	0.400	4.350E-03	17.0	XKEI90					
23-V	0.250	4.400E-01	17.0	XKEI90	42-Mo	0.450	8.110E-03	17.0	XKEI90					
23-V	0.300	1.072E+00	17.0	XKEI90	42-Mo	0.500	1.290E-02	17.0	XKEI90					
23-V	0.350	2.051E+00	17.0	XKEI90	42-Mo	0.550	1.880E-02	17.0	XKEI90					
23-V	0.400	2.989E+00	17.0	XKEI90	47-Ag	0.200	5.200E-06	17.0	XKEI90					
23-V	0.450	4.058E+00	17.0	XKEI90	47-Ag	0.250	4.430E-05	17.0	XKEI90					
23-V	0.500	2.120E+01	7.0	INOR89	47-Ag	0.300	1.640E-04	17.0	XKEI90					
23-V	0.500	5.248E+00	17.0	XKEI90	47-Ag	0.350	5.010E-04	17.0	XKEI90					
23-V	0.550	6.830E+00	17.0	XKEI90	47-Ag	0.400	1.230E-03	17.0	XKEI90					
23-V	0.700	6.230E+01	7.0	INOR89	47-Ag	0.450	2.310E-03	17.0	XKEI90					
23-V	0.900	1.270E+02	7.0	INOR89	47-Ag	0.500	3.550E-03	17.0	XKEI90					
23-V	1.100	2.120E+02	7.0	INOR89	47-Ag	0.550	5.000E-03	17.0	XKEI90					
23-V	1.300	2.830E+02	7.0	INOR89	57-La	66.000	5.303E+01	12.0	INPP90					
23-V	1.500	3.760E+02	7.0	INOR89	58-Ce	66.000	4.446E+01	12.0	INPP90					
24-Cr	0.500	1.310E+01	7.0	INOR89	59-Pr	66.000	4.048E+01	12.0	INPP90					
24-Cr	0.700	4.000E+01	7.0	INOR89	60-Nd	66.000	4.209E+01	12.0	INPP90					
24-Cr	0.900	8.030E+01	7.0	INOR89	62-Sm	66.000	3.646E+01	12.0	INPP90					
24-Cr	1.100	1.370E+02	7.0	INOR89	63-Eu	66.000	3.323E+01	12.0	INPP90					
24-Cr	1.300	1.940E+02	7.0	INOR89	64-Gd	66.000	2.903E+01	12.0	INPP90					
24-Cr	1.500	2.770E+02	7.0	INOR89	65-Tb	66.000	2.980E+01	12.0	INPP90					
26-Fe	0.500	5.420E+00	7.0	INOR89	66-Dy	66.000	2.665E+01	12.0	INPP90					
26-Fe	0.700	1.670E+01	7.0	INOR89	67-Ho	66.000	2.617E+01	12.0	INPP90					
26-Fe	0.900	3.610E+01	7.0	INOR89	68-Er	66.000	2.550E+01	12.0	INPP90					
26-Fe	1.100	6.480E+01	7.0	INOR89	69-Tm	66.000	2.564E+01	12.0	INPP90					
26-Fe	1.300	9.310E+01	7.0	INOR89	70-Yb	66.000	2.679E+01	12.0	INPP90					
27-Co	0.075	2.800E-05	17.0	XKEI90	71-Lu	66.000	2.289E+01	12.0	INPP90					

TABLE IIb. Experimental Cross Sections for *K*-Shell X-Ray Production, Auger Electron Production, and Ionization by ^{2}H Ions

See page 84 for Explanation of Tables

TARGET	ENERGY [MeV]	SIGMA [barn]	ERROR [%]	REFERENCE	TARGET	ENERGY [MeV]	SIGMA [barn]	ERROR [%]	REFERENCE	TARGET	ENERGY [MeV]	SIGMA [barn]	ERROR [%]	REFERENCE
4-Be	0.028	5.590E+00	10.0	XNSH80	15-P	3.800	6.770E+02	20.0	XNSN81	24-Cr	0.900	2.632E+00	8.0	XNLJ82
4-Be	0.037	2.270E+01	10.0	XNSH80	15-P	4.000	7.040E+02	20.0	XNSN81	25-Mn	0.400	1.055E-01	10.0	XNLJ82
4-Be	0.047	6.430E+01	10.0	XNSH80	16-S	1.600	1.640E+02	20.0	XNSN81	25-Mn	0.500	2.495E-01	10.0	XNLJ82
4-Be	0.057	1.350E+02	10.0	XNSH80	16-S	1.800	2.150E+02	20.0	XNSN81	25-Mn	0.600	4.914E-01	9.0	XNLJ82
4-Be	0.077	3.390E+02	10.0	XNSH80	16-S	2.000	2.680E+02	20.0	XNSN81	25-Mn	0.700	8.608E-01	9.0	XNLJ82
4-Be	0.096	5.900E+02	10.0	XNSH80	16-S	2.200	2.840E+02	20.0	XNSN81	25-Mn	0.800	1.340E+00	8.0	XNLJ82
4-Be	0.116	8.950E+02	10.0	XNSH80	16-S	2.400	3.250E+02	20.0	XNSN81	25-Mn	0.900	2.000E+00	8.0	XNLJ82
6-C	0.058	1.350E+01	12.0	XGHA73H	16-S	2.600	3.590E+02	20.0	XNSN81	26-Fe	0.400	8.858E-02	9.0	XNJL80
6-C	0.058	6.579E+00	12.0	XGHA73O	16-S	2.800	4.040E+02	20.0	XNSN81	26-Fe	0.400	7.490E-02	9.0	XNLJ82
6-C	0.077	3.942E+01	12.0	XGHA73H	16-S	3.000	4.290E+02	20.0	XNSN81	26-Fe	0.500	2.032E-01	8.0	XNJL80
6-C	0.077	1.867E+01	12.0	XGHA73O	16-S	3.200	4.620E+02	20.0	XNSN81	26-Fe	0.500	1.867E-01	9.0	XNLJ82
6-C	0.087	2.856E+01	12.0	XGHA73O	16-S	3.400	5.100E+02	20.0	XNSN81	26-Fe	0.600	3.967E-01	7.0	XNJL80
6-C	0.097	8.883E+01	12.0	XGHA73H	16-S	3.600	5.340E+02	20.0	XNSN81	26-Fe	0.600	3.796E-01	8.0	XNLJ82
6-C	0.097	4.148E+01	9.0	XGHA73O	16-S	3.800	5.550E+02	20.0	XNSN81	26-Fe	0.700	6.532E-01	8.0	XNLJ82
6-C	0.100	3.200E+04	15.6	IGST73	16-S	4.000	5.590E+02	20.0	XNSN81	26-Fe	0.800	1.139E+00	7.0	XNJL80
6-C	0.106	1.261E+02	12.0	XGHA73H	17-Cl	1.600	1.190E+02	15.0	XNSN81	26-Fe	0.800	1.005E+00	7.0	XNLJ82
6-C	0.106	5.780E+01	12.0	XGHA73O	17-Cl	1.800	1.510E+02	15.0	XNSN81	26-Fe	0.900	1.498E+00	7.0	XNLJ82
6-C	0.116	1.679E+02	12.0	XGHA73H	17-Cl	2.000	1.810E+02	15.0	XNSN81	26-Fe	1.000	2.380E+00	7.0	XNJL80
6-C	0.116	7.973E+01	12.0	XGHA73O	17-Cl	2.200	2.180E+02	15.0	XNSN81	26-Fe	1.200	4.138E+00	6.0	XNJL80
6-C	0.126	2.201E+02	12.0	XGHA73H	17-Cl	2.400	2.280E+02	15.0	XNSN81	26-Fe	1.400	6.669E+00	6.0	XNJL80
6-C	0.126	1.030E+02	12.0	XGHA73O	17-Cl	2.600	2.720E+02	15.0	XNSN81	26-Fe	1.600	9.850E+00	6.0	XNJL80
6-C	0.135	2.808E+02	12.0	XGHA73H	17-Cl	2.800	2.990E+02	15.0	XNSN81	26-Fe	1.800	1.330E+01	6.0	XNJL80
6-C	0.135	1.319E+02	12.0	XGHA73O	17-Cl	3.000	3.310E+02	15.0	XNSN81	27-Co	0.400	4.831E-02	9.0	XNLJ82
6-C	0.145	3.348E+02	12.0	XGHA73H	17-Cl	3.200	3.410E+02	15.0	XNSN81	27-Co	0.500	1.219E-01	9.0	XNLJ82
6-C	0.145	1.649E+02	12.0	XGHA73O	17-Cl	3.400	3.960E+02	15.0	XNSN81	27-Co	0.600	2.481E-01	8.0	XNLJ82
6-C	0.150	9.100E+04	15.4	IGST73	17-Cl	3.600	4.340E+02	15.0	XNSN81	27-Co	0.700	4.429E-01	8.0	XNLJ82
6-C	0.300	3.500E+05	20.0	AGWA73	17-Cl	3.800	4.390E+02	15.0	XNSN81	27-Co	0.800	7.027E-01	7.0	XNLJ82
6-C	0.500	6.600E+05	19.7	AGWA73	17-Cl	4.000	4.570E+02	15.0	XNSN81	27-Co	0.900	1.010E+00	7.0	XNLJ82
6-C	0.500	6.510E+05	15.4	IGST73	18-Ar	0.135	3.818E-02	12.0	XGHA73	27-Co	1.000	1.527E+00	6.0	XNBA82
6-C	0.750	1.050E+06	20.0	AGWA73	19-K	1.600	6.130E+01	15.0	XNSN81	27-Co	2.000	1.345E+01	6.0	XNBA82
6-C	1.000	1.100E+06	20.0	AGWA73	19-K	1.800	8.050E+01	15.0	XNSN81	27-Co	2.500	2.348E+01	6.0	XNBA82
7-N	0.058	6.298E+00	11.9	XGHA73	19-K	2.000	1.010E+02	15.0	XNSN81	27-Co	3.000	3.715E+01	6.0	XNBA82
7-N	0.077	1.791E+01	12.1	XGHA73	19-K	2.200	1.230E+02	15.0	XNSN81	28-Ni	0.100	4.200E-06	50.0	XKLL80
7-N	0.097	3.901E+01	12.0	XGHA73	19-K	2.400	1.320E+02	15.0	XNSN81	28-Ni	0.110	8.700E-06	50.0	XKLL80
7-N	0.100	6.200E+03	14.5	IGST73	19-K	2.600	1.620E+02	15.0	XNSN81	28-Ni	0.120	1.900E-05	25.0	XKLL80
7-N	0.106	5.546E+01	11.9	XGHA73	19-K	2.800	1.800E+02	15.0	XNSN81	28-Ni	0.130	4.300E-05	25.0	XKLL80
7-N	0.116	7.379E+01	12.1	XGHA73	19-K	3.000	2.040E+02	15.0	XNSN81	28-Ni	0.140	7.500E-05	25.0	XKLL80
7-N	0.135	1.231E+02	11.8	XGHA73	19-K	3.200	2.130E+02	15.0	XNSN81	28-Ni	0.150	1.300E-04	25.0	XKLL80
7-N	0.145	1.462E+02	11.9	XGHA73	19-K	3.400	2.470E+02	15.0	XNSN81	28-Ni	0.160	2.000E-04	25.0	XKLL80
7-N	0.150	2.130E+04	14.1	IGST73	19-K	3.600	2.740E+02	15.0	XNSN81	28-Ni	0.180	4.400E-04	25.0	XKLL80
7-N	0.300	1.130E+05	17.7	IGST73	19-K	3.800	2.820E+02	15.0	XNSN81	28-Ni	0.200	9.300E-04	25.0	XKLL80
7-N	0.500	2.150E+05	14.0	IGST73	19-K	4.000	3.010E+02	15.0	XNSN81	28-Ni	0.220	1.600E-03	25.0	XKLL80
10-Ne	0.077	1.170E+00	12.1	XGHA73	20-Ca	0.400	6.928E-01	17.0	XNLJ82	28-Ni	0.240	2.600E-03	25.0	XKLL80
10-Ne	0.097	2.604E+00	11.9	XGHA73	20-Ca	0.500	1.524E+00	17.0	XNLJ82	28-Ni	0.260	4.000E-03	25.0	XKLL80
10-Ne	0.116	5.797E+00	12.0	XGHA73	20-Ca	0.600	2.804E+00	16.0	XNLJ82	28-Ni	0.280	6.100E-03	25.0	XKLL80
10-Ne	0.135	9.935E+00	12.0	XGHA73	20-Ca	0.700	4.727E+00	16.0	XNLJ82	28-Ni	0.300	8.300E-03	25.0	XKLL80
13-Al	0.040	1.680E-03	25.0	XKBB73	20-Ca	0.800	8.150E+00	15.0	XNLJ82	28-Ni	0.400	3.266E-02	10.0	XNLJ82
13-Al	0.044	3.250E-03	24.9	XKBB73	20-Ca	0.900	1.051E+01	15.0	XNLJ82	28-Ni	0.500	8.321E-02	9.0	XNLJ82
13-Al	0.050	5.710E-03	25.0	XKBB73	21-Sc	0.400	4.731E-01	8.0	XNJL80	28-Ni	0.600	1.743E-01	9.0	XNLJ82
13-Al	0.060	1.440E-02	25.0	XKBB73	21-Sc	0.400	4.446E-01	14.0	XNLJ82	28-Ni	0.700	3.163E-01	8.0	XNLJ82
13-Al	0.070	3.000E-02	25.0	XKBB73	21-Sc	0.500	1.015E+00	14.0	XNLJ82	28-Ni	0.800	5.216E-01	7.0	XNLJ82
13-Al	0.080	5.460E-02	24.9	XKBB73	21-Sc	0.600	1.976E+00	7.0	XNJL80	28-Ni	0.900	7.990E-01	7.0	XNLJ82
13-Al	0.090	9.000E-02	25.0	XKBB73	21-Sc	0.600	1.919E+00	13.0	XNLJ82	28-Ni	1.600	5.135E+00	10.0	XNSN81
13-Al	0.100	1.390E-01	25.2	XKBB73	21-Sc	0.700	3.249E+00	13.0	XNLJ82	28-Ni	1.800	6.774E+00	10.0	XNSN81
13-Al	0.110	2.020E-01	24.8	XKBB73	21-Sc	0.800	5.187E+00	7.0	XNJL80	28-Ni	2.000	8.370E+00	10.0	XNBB78
13-Al	0.120	2.800E-01	25.0	XKBB73	21-Sc	0.800	5.187E+00	12.0	XNLJ82	28-Ni	2.000	9.654E+00	10.0	XNSN81
13-Al	0.130	3.810E-01	24.9	XKBB73	21-Sc	0.900	7.296E+00	12.0	XNLJ82	28-Ni	2.200	1.264E+01	10.0	XNSN81
13-Al	0.140	4.980E-01	24.9	XKBB73	21-Sc	1.000	1.039E+01	6.0	XNJL80	28-Ni	2.400	1.594E+01	10.0	XNSN81
13-Al	0.160	8.050E-01	25.0	XKBB73	21-Sc	1.200	1.780E+01	6.0	XNJL80	28-Ni	2.600	1.935E+01	10.0	XNSN81
13-Al	0.180	1.220E+00	24.6	XKBB73	21-Sc	1.400	2.622E+01	6.0	XNJL80	28-Ni	2.800	2.379E+01	10.0	XNSN81
13-Al	0.180	1.182E+00	5.1	XKSH71	21-Sc	1.600	3.743E+01	6.0	XNJL80	28-Ni	3.000	2.550E+01	10.0	XNBB78
13-Al	0.200	1.790E+00	25.1	XKBB73	21-Sc	1.800	5.073E+01	6.0	XNJL80	28-Ni	3.000	2.778E+01	10.0	XNSN81
13-Al	0.200	1.681E+00	5.1	XKSH71	22-Ti	0.180	1.259E-02	5.0	XKSH71	28-Ni	3.200	3.188E+01	10.0	XNSN81
13-Al	0.220	2.540E+00	24.8	XKBB73	22-Ti	0.220	3.285E-02	5.3	XKSH71	28-Ni	3.400	3.643E+01	10.0	XNSN81
13-Al	0.240	3.500E+00	25.1	XKBB73	22-Ti	0.260	7.052E-02	5.0	XKSH71	28-Ni	3.600	4.190E+01	10.0	XNSN81
13-Al	0.260	4.790E+00	25.1	XKBB73	22-Ti	0.300	1.290E-01	4.9	XKSH71	28-Ni	3.800	4.804E+01	10.0	XNSN81
13-Al	0.280	6.420E+00	24.9	XKBB73	22-Ti	0.340	2.234E-01	4.9	XKSH71	28-Ni	4.000	4.910E+01	10.0	XNBB78
13-Al	0.280	5.069E+00	4.9	XKSH71	22-Ti	0.400	2.984E-01	13.0	XNLJ82	28-Ni	4.000	5.260E+01	10.0	XNSN81
13-Al	0.360	1.121E+01	5.1	XKSH71	22-Ti	0.500	6.962E-01	13.0	XNLJ82	28-Ni	5.000	7.700E+01	10.0	XNBB78
13-Al	2.000	5.040E+02	10.0	XNBB78	22-Ti	0.600	1.346E+00	12.0	XNLJ82	28-Ni	6.000	1.070E+02	10.0	XNBB78
13-Al	3.000	7.590E+02	10.0	XNBB78	22-Ti	0.700	2.195E+00	12.0	XNLJ82	28-Ni	8.000	1.640E+02	10.0	XNBB78
13-Al	4.000	8.890E+02	10.0	XNBB78	22-Ti	0.800	3.403E+00	11.0	XNLJ82	28-Ni	10.000	2.160E+02	10.0	XNBB78
13-Al	5.000	9.460E+02	10.0	XNBB78	22-Ti	0.900	5.017E+00	11.0	XNLJ82	28-Ni	12.000	2.580E+02	10.0	XNBB78
13-Al	6.000	9.790E+02	10.0	XNBB78	22-Ti	1.000	7.721E+00	6.0	XNBA82	28-Ni	14.000	2.960E+02	10.0	XNBB78
13-Al	8.000	9.510E+02	10.0	XNBB78	22-Ti	2.000	5.212E+01	6.0	XNBA82	28-Ni	15.000	3.080E+02	10.0	XNBB78
13-Al	10.000	9.070E+02	10.0	XNBB78	22-Ti	2.500	8.891E+01	6.0	XNBA82	28-Ni	18.000	3.370E+02	10.0	XNBB78
13-Al	12.000	8.460E+02	10.0	XNBB78	22-Ti	3.000	1.260E+02	6.0	XNBA82	28-Ni	19.500	3.530E+02	10.0	XNBB78
13-Al	14.000	8.080E+02	10.0	XNBB78	23-V	0.400	2.229E-01	11.0	XNLJ82	29-Cu	0.160	6.900E-05	12.4	XNRI81
13-Al	15.000	7.750E+02	10.0	XNBB78	23-V	0.500	5.161E-01	11.0	XNLJ82	29-Cu	0.180	2.250E-04	12.4	XNRI81
13-Al	18.000	7.060E+02	10.0	XNBB78	23-V	0.600	9.968E-01	10.0	XNLJ82	29-Cu	0.200	6.090E-04	12.4	XNRI81
13-Al	19.500	6.830E+02	10.0	XNBB78	23-V	0.700	1.736E+00	10.0	XNLJ82	29-Cu	0.240	2.030E-03	12.4	XNRI81
15-P	1.600	2.320E+02	20.0	XNSN81	23-V	0.800	2.732E+00	10.0	XNLJ82	29-Cu	0.280	4.590E-03	12.4	XNRI81
15-P	1.800	2.830E+02	20.0	XNSN81	23-V	0.900	3.770E+00	10.0	XNLJ82	29-Cu	0.320	9.180E-03	12.4	XNRI81
15-P	2.000	3.220E+02	20.0	XNSN81	23-V	1.000	5.445E+00	6.0	XNBA82	29-Cu	0.360	1.630E-02	12.4	XNRI81
15-P	2.200	3.680E+02	20.0	XNSN81	23-V	2.000	4.081E+01	6.0	XNBA82	29-Cu	0.400	2.397E-02	8.0	XNLJ82
15-P	2.400	4.290E+02	20.0	XNSN81	23-V	2.500	6.922E+01	6.0	XNBA82	29-Cu	0.400	2.580E-02	12.4	XNRI81
15-P	2.600	4.770E+02	20.0	XNSN81	23-V	3.000	1.015E+02	6.0	XNBA82	29-Cu	0.500	6.335E-02	8.0	XNLJ82
15-P	2.800	5.200E+02	20.0	XNSN81	24-Cr	0.400	1.528E-01	10.0	XNLJ82	29-Cu	0.500	6.590E-02	12.4	XNRI81
15-P	3.000	5.460E+02	20.0	XNSN81	24-Cr	0.500	3.538E-01	10.0	XNLJ82	29-Cu	0.600	1.263E-01	7.0	XNLJ82
15-P	3.200	6.130E+02	20.0	XNSN81	24-Cr	0.600	6.792E-01	9.0	XNLJ82	29-Cu	0.600	1.450E-01	12.4	XNRI81
15-P	3.400	6.310E+02	20.0	XNSN81	24-Cr	0.700	1.200E+00	9.0	XNLJ82	29-Cu	0.600	2.200E-01	12.0	XNSI57
15-P	3.600	6.700E+02	20.0	XNSN81	24-Cr	0.800	1.837E+00	8.0	XNLJ82	29-Cu	0.700	2.326E-01	7.0	XNLJ82

TABLE IIb. Experimental Cross Sections for *K*-Shell X-Ray Production, Auger Electron Production, and Ionization by ^{2}H Ions

See page 84 for Explanation of Tables

TARGET	ENERGY [MeV]	SIGMA [barn]	ERROR [%]	REFERENCE
29-Cu	0.700	4.360E-01	12.0	XNSI57
29-Cu	0.800	3.832E-01	6.0	XNLJ82
29-Cu	0.800	8.850E-01	12.0	XNSI57
29-Cu	0.900	5.803E-01	6.0	XNLJ82
29-Cu	0.900	1.210E+00	12.0	XNSI57
29-Cu	1.000	7.726E-01	6.0	XNBA82
29-Cu	1.000	1.807E+00	12.0	XNSI57
29-Cu	1.600	3.629E+00	10.0	XNSN81
29-Cu	1.800	5.421E+00	10.0	XNSN81
29-Cu	2.000	7.681E+00	6.0	XNBA82
29-Cu	2.000	7.692E+00	10.0	XNSN81
29-Cu	2.200	9.392E+00	10.0	XNSN81
29-Cu	2.400	1.175E+01	10.0	XNSN81
29-Cu	2.500	1.434E+01	6.0	XNBA82
29-Cu	2.600	1.438E+01	10.0	XNSN81
29-Cu	2.800	1.792E+01	10.0	XNSN81
29-Cu	3.000	2.333E+01	6.0	XNBA82
29-Cu	3.000	2.066E+01	10.0	XNSN81
29-Cu	3.200	2.431E+01	10.0	XNSN81
29-Cu	3.400	2.864E+01	10.0	XNSN81
29-Cu	3.600	3.298E+01	10.0	XNSN81
29-Cu	3.800	3.686E+01	10.0	XNSN81
29-Cu	4.000	3.983E+01	10.0	XNSN81
29-Cu	30.000	3.342E+02	6.0	XNHW73
29-Cu	40.000	4.112E+02	6.0	XNHW73
29-Cu	50.000	4.717E+02	59.4	XNHW73
29-Cu	60.000	5.118E+02	60.0	XNHW73
29-Cu	70.000	5.340E+02	60.0	XNHW73
29-Cu	80.000	5.385E+02	60.3	XNHW73
31-Ga	1.600	2.135E+00	10.0	XNSN81
31-Ga	1.800	3.100E+00	10.0	XNSN81
31-Ga	2.000	4.156E+00	10.0	XNSN81
31-Ga	2.200	5.464E+00	10.0	XNSN81
31-Ga	2.400	7.152E+00	10.0	XNSN81
31-Ga	2.600	9.000E+00	10.0	XNSN81
31-Ga	2.800	1.111E+01	10.0	XNSN81
31-Ga	3.000	1.297E+01	10.0	XNSN81
31-Ga	3.200	1.584E+01	10.0	XNSN81
31-Ga	3.400	1.779E+01	10.0	XNSN81
31-Ga	3.600	2.078E+01	10.0	XNSN81
31-Ga	3.800	2.273E+01	10.0	XNSN81
31-Ga	4.000	2.663E+01	10.0	XNSN81
32-Ge	1.000	3.187E-01	6.0	XNBA82
32-Ge	2.000	3.474E+00	6.0	XNBA82
32-Ge	2.500	6.833E+00	6.0	XNBA82
32-Ge	3.000	1.093E+01	6.0	XNBA82
37-Rb	0.800	1.890E-02	14.1	XNMA79
37-Rb	1.000	5.130E-02	14.3	XNMA79
37-Rb	1.200	1.055E-01	12.7	XNMA79
37-Rb	1.400	1.917E-01	13.9	XNMA79
37-Rb	1.600	3.180E-01	14.7	XNMA79
37-Rb	1.800	4.609E-01	14.5	XNMA79
37-Rb	2.000	7.148E-01	14.0	XNMA79
37-Rb	2.200	8.884E-01	15.0	XNMA79
37-Rb	2.400	1.269E+00	15.8	XNMA79
37-Rb	2.600	1.623E+00	14.4	XNMA79
38-Sr	0.700	1.037E-02	13.3	XNMA79
38-Sr	0.800	1.914E-02	14.4	XNMA79
38-Sr	1.000	4.782E-02	12.3	XNMA79
38-Sr	1.200	1.002E-01	13.8	XNMA79
38-Sr	1.400	1.928E-01	12.5	XNMA79
38-Sr	1.600	3.006E-01	14.9	XNMA79
38-Sr	1.800	4.381E-01	12.6	XNMA79
38-Sr	2.000	6.226E-01	10.0	XNMA79
38-Sr	2.200	7.947E-01	13.0	XNMA79
38-Sr	2.400	1.113E+00	12.4	XNMA79
38-Sr	2.600	1.555E+00	13.3	XNMA79
40-Zr	0.700	4.964E-03	14.7	XNMA79
40-Zr	0.800	9.636E-03	11.4	XNMA79
40-Zr	1.000	2.628E-02	13.9	XNMA79
40-Zr	1.200	5.074E-02	14.4	XNMA79
40-Zr	1.400	1.007E-01	14.5	XNMA79
40-Zr	1.600	1.591E-01	13.8	XNMA79
40-Zr	1.800	2.343E-01	14.0	XNMA79
40-Zr	2.000	3.606E-01	15.2	XNMA79
40-Zr	2.200	4.803E-01	15.2	XNMA79
40-Zr	2.400	5.906E-01	14.8	XNMA79
40-Zr	2.600	6.811E-01	15.0	XNMA79
41-Nb	0.320	5.090E-05	12.4	XNRI81
41-Nb	0.400	4.630E-04	12.4	XNRI81
41-Nb	0.480	1.350E-03	12.4	XNRI81
41-Nb	0.560	3.020E-03	12.4	XNRI81
41-Nb	0.640	5.600E-03	12.4	XNRI81
41-Nb	0.720	1.030E-02	12.4	XNRI81
41-Nb	0.800	1.520E-02	12.4	XNRI81
41-Nb	0.900	2.350E-02	12.4	XNRI81
47-Ag	0.400	3.060E-05	12.4	XNRI81
47-Ag	0.500	1.056E-04	15.0	XNJL80
47-Ag	0.500	1.820E-04	12.4	XNRI81
47-Ag	0.600	4.232E-04	17.0	XNJL80
47-Ag	0.600	5.260E-04	12.4	XNRI81
47-Ag	0.700	1.073E-03	15.0	XNJL80
47-Ag	0.700	1.330E-03	12.4	XNRI81
47-Ag	0.800	2.020E-03	14.0	XNJL80
47-Ag	0.800	2.590E-03	12.4	XNRI81
47-Ag	0.900	4.660E-03	12.4	XNRI81
47-Ag	1.000	5.832E-03	10.0	XNJL80
47-Ag	1.000	6.640E-03	12.4	XNRI81
47-Ag	1.100	1.090E-02	12.4	XNRI81
47-Ag	1.200	1.215E-02	10.0	XNJL80
47-Ag	1.200	1.640E-02	12.4	XNRI81
47-Ag	1.400	2.179E-02	10.0	XNJL80
47-Ag	1.600	3.629E-02	10.0	XNJL80
47-Ag	1.800	5.539E-02	10.0	XNJL80
48-Cd	0.900	1.008E-03	12.5	XNMA79
48-Cd	1.000	2.352E-03	14.3	XNMA79
48-Cd	1.200	6.888E-03	14.6	XNMA79
48-Cd	1.400	1.050E-02	12.0	XNMA79
48-Cd	1.600	2.411E-02	13.9	XNMA79
48-Cd	1.800	3.091E-02	14.9	XNMA79
48-Cd	2.000	4.427E-02	14.2	XNMA79
48-Cd	2.200	5.477E-02	13.8	XNMA79
48-Cd	2.400	7.896E-02	13.8	XNMA79
48-Cd	2.600	9.996E-02	12.6	XNMA79
49-In	0.600	2.023E-04	15.0	XNJL82
49-In	0.700	4.981E-04	14.0	XNJL82
49-In	0.800	1.105E-03	12.0	XNJL82
49-In	1.000	3.460E-03	10.0	XNJL82
49-In	1.200	7.378E-03	10.0	XNJL82
49-In	1.400	1.335E-02	10.0	XNJL82
49-In	1.600	2.295E-02	10.0	XNJL82
49-In	1.800	3.613E-02	10.0	XNJL82
51-Sb	0.500	4.540E-05	12.4	XNRI81
51-Sb	0.600	1.710E-04	12.4	XNRI81
51-Sb	0.700	3.121E-04	14.0	XNJL82
51-Sb	0.700	4.270E-04	12.4	XNRI81
51-Sb	0.800	6.745E-04	13.0	XNJL82
51-Sb	0.800	8.400E-04	12.4	XNRI81
51-Sb	0.900	1.630E-03	12.4	XNRI81
51-Sb	1.000	1.829E-03	14.0	XNJL82
51-Sb	1.000	2.820E-03	12.4	XNRI81
51-Sb	1.100	4.490E-03	12.4	XNRI81
51-Sb	1.200	4.413E-03	10.0	XNJL82
51-Sb	1.200	1.942E-03	13.4	XNMA79
51-Sb	1.200	6.730E-03	12.4	XNRI81
51-Sb	1.300	8.500E-03	12.4	XNRI81
51-Sb	1.400	8.271E-03	10.0	XNJL82
51-Sb	1.400	5.029E-03	13.8	XNMA79
51-Sb	1.600	1.465E-02	10.0	XNJL82
51-Sb	1.600	9.537E-03	13.6	XNMA79
51-Sb	1.800	2.272E-02	10.0	XNJL82
51-Sb	1.800	1.379E-02	12.6	XNMA79
51-Sb	2.000	1.777E-02	14.6	XNMA79
51-Sb	2.200	2.653E-02	16.3	XNMA79
51-Sb	2.400	3.390E-02	14.1	XNMA79
51-Sb	2.600	4.006E-02	13.0	XNMA79
58-Ce	1.000	3.936E-04	25.0	XNJL82
58-Ce	1.200	1.020E-03	13.0	XNJL82
58-Ce	1.400	2.305E-03	10.0	XNJL82
58-Ce	1.600	4.191E-03	10.0	XNJL82
58-Ce	1.800	6.723E-03	10.0	XNJL82
64-Gd	0.890	7.400E-05	20.3	INEA87
64-Gd	1.070	2.270E-04	15.0	INEA87
64-Gd	1.140	2.460E-04	13.0	INEA87
64-Gd	1.340	5.650E-04	15.0	INEA87
64-Gd	1.550	1.350E-03	11.1	INEA87
64-Gd	1.750	1.560E-03	17.3	INEA87
64-Gd	1.960	4.010E-03	14.0	INEA87
64-Gd	2.160	4.970E-03	12.1	INEA87
64-Gd	2.360	7.200E-03	13.9	INEA87
64-Gd	2.560	1.010E-02	18.8	INEA87
64-Gd	2.760	1.160E-02	12.9	INEA87
73-Ta	1.090	4.000E-05	22.5	INEA87
73-Ta	1.300	8.700E-05	12.6	INEA87
73-Ta	1.500	2.280E-04	12.3	INEA87
73-Ta	1.710	4.120E-04	11.4	INEA87
73-Ta	1.920	7.930E-04	12.0	INEA87
73-Ta	2.120	1.350E-03	12.6	INEA87
73-Ta	2.320	2.290E-03	17.5	INEA87
73-Ta	2.530	2.850E-03	13.7	INEA87
73-Ta	2.730	3.390E-03	12.4	INEA87
73-Ta	2.930	4.270E-03	13.6	INEA87
73-Ta	3.130	4.810E-03	12.9	INEA87
74-W	1.300	1.072E-04	10.0	XKFR85
74-W	1.440	1.885E-04	10.0	XKFR85
74-W	1.600	3.129E-04	10.0	XKFR85
74-W	1.700	4.096E-04	10.0	XKFR85
74-W	1.800	5.216E-04	10.0	XKFR85
74-W	1.900	6.508E-04	10.0	XKFR85
74-W	2.000	7.934E-04	10.0	XKFR85
74-W	2.250	1.225E-03	10.0	XKFR85
74-W	2.500	1.790E-03	10.0	XKFR85
74-W	2.750	2.479E-03	10.0	XKFR85
74-W	3.000	3.330E-03	10.0	XKFR85
74-W	3.250	4.345E-03	10.0	XKFR85
74-W	3.500	5.541E-03	10.0	XKFR85
74-W	3.750	6.957E-03	10.0	XKFR85
74-W	4.000	8.718E-03	10.0	XKFR85
79-Au	1.420	8.146E-05	8.0	XKFR85
79-Au	1.520	1.195E-04	8.0	XKFR85
79-Au	1.640	1.774E-04	8.0	XKFR85
79-Au	1.800	2.728E-04	8.0	XKFR85
79-Au	1.900	3.432E-04	8.0	XKFR85
79-Au	2.000	4.222E-04	8.0	XKFR85
79-Au	2.250	6.700E-04	8.0	XKFR85
79-Au	2.500	9.740E-04	8.0	XKFR85
79-Au	2.750	1.388E-03	8.0	XKFR85
79-Au	3.000	1.909E-03	8.0	XKFR85
79-Au	3.250	2.574E-03	8.0	XKFR85
79-Au	3.500	3.403E-03	8.0	XKFR85
79-Au	3.750	4.357E-03	8.0	XKFR85
79-Au	4.000	5.639E-03	8.0	XKFR85

TABLE IIc. Experimental Cross Sections for *K*-Shell X-Ray Production, Auger Electron Production, and Ionization by ^{3}He Ions

See page 84 for Explanation of Tables

TARGET	ENERGY [MeV]	SIGMA [barn]	ERROR [%]	REFERENCE	TARGET	ENERGY [MeV]	SIGMA [barn]	ERROR [%]	REFERENCE	TARGET	ENERGY [MeV]	SIGMA [barn]	ERROR [%]	REFERENCE
13-Al	0.045	1.240E-04	25.0	XKBB73	22-Ti	1.500	9.730E+01	6.0	INPA90	28-Ni	0.750	3.950E-01	7.0	INPA90
13-Al	0.051	3.260E-04	24.8	XKBB73	22-Ti	1.500	1.905E+01	8.2	XNSI89	28-Ni	0.900	8.870E-01	7.0	INPA90
13-Al	0.060	1.280E-03	25.0	XKBB73	22-Ti	1.507	1.916E+01	5.0	XNSI84	28-Ni	1.050	1.720E+00	7.0	INPA90
13-Al	0.069	3.970E-03	24.9	XKBB73	22-Ti	1.508	1.855E+01	5.0	XNSI84	28-Ni	1.200	2.980E+00	7.0	INPA90
13-Al	0.081	9.580E-03	24.9	XKBB73	22-Ti	1.800	1.800E+02	6.0	INPA90	28-Ni	1.350	4.600E+00	7.0	INPA90
13-Al	0.090	1.750E-02	25.1	XKBB73	22-Ti	1.800	3.574E+01	8.2	XNSI89	28-Ni	1.500	6.640E+00	7.0	INPA90
13-Al	0.099	3.290E-02	24.9	XKBB73	22-Ti	1.809	3.438E+01	5.0	XNSI84	28-Ni	1.800	1.320E+01	7.0	INPA90
13-Al	0.111	5.120E-02	25.0	XKBB73	22-Ti	1.809	3.613E+01	5.0	XNSI84	28-Ni	2.100	2.310E+01	7.0	INPA90
13-Al	0.120	7.550E-02	25.0	XKBB73	22-Ti	2.100	2.860E+02	6.0	INPA90	28-Ni	2.400	3.070E+01	7.0	INPA90
13-Al	0.141	1.450E-01	24.8	XKBB73	22-Ti	2.100	5.885E+01	8.2	XNSI89	28-Ni	2.700	4.790E+01	7.0	INPA90
13-Al	0.159	2.430E-01	25.1	XKBB73	22-Ti	2.110	6.088E+01	5.0	XNSI84	28-Ni	3.000	6.330E+01	7.0	INPA90
13-Al	0.180	3.880E-01	25.0	XKBB73	22-Ti	2.111	5.628E+01	5.0	XNSI84	28-Ni	3.300	8.070E+01	7.0	INPA90
13-Al	0.201	5.960E-01	25.0	XKBB73	22-Ti	2.400	4.050E+02	6.0	INPA90	28-Ni	3.600	1.340E+02	7.0	INPA90
13-Al	0.408	8.280E+00	25.0	XKBB73	22-Ti	2.400	8.795E+01	8.2	XNSI89	29-Cu	0.525	2.640E-02	8.2	XNSI89
13-Al	0.510	1.950E+01	25.1	XKBB73	22-Ti	2.412	8.563E+01	5.0	XNSI84	29-Cu	0.600	4.972E-02	8.2	XNSI89
13-Al	0.600	1.310E+03	9.0	INPA90	22-Ti	2.412	8.869E+01	5.0	XNSI84	29-Cu	0.603	4.939E-02	5.0	XNSI84
13-Al	0.612	3.700E+01	24.9	XKBB73	22-Ti	2.700	6.410E+02	6.0	INPA90	29-Cu	0.675	8.976E-02	8.2	XNSI89
13-Al	0.714	6.490E+01	25.0	XKBB73	22-Ti	2.700	1.252E+02	8.2	XNSI89	29-Cu	0.750	1.452E-01	8.2	XNSI89
13-Al	0.750	2.760E+03	9.0	INPA90	22-Ti	2.713	1.250E+02	5.0	XNSI84	29-Cu	0.754	1.428E-01	5.0	XNSI84
13-Al	0.816	1.010E+02	24.8	XKBB73	22-Ti	3.000	8.990E+02	6.0	INPA90	29-Cu	0.900	3.120E-01	8.2	XNSI89
13-Al	0.900	5.540E+03	9.0	INPA90	22-Ti	3.000	1.684E+02	8.2	XNSI89	29-Cu	0.905	3.075E-01	5.0	XNSI84
13-Al	1.020	2.110E+02	25.1	XKBB73	22-Ti	3.015	1.684E+02	5.0	XNSI84	29-Cu	1.050	5.720E-01	8.2	XNSI89
13-Al	1.050	9.720E+03	9.0	INPA90	22-Ti	3.300	1.170E+03	6.0	INPA90	29-Cu	1.055	5.696E-01	5.0	XNSI84
13-Al	1.119	2.800E+02	25.0	XKBB73	22-Ti	3.300	2.131E+02	8.2	XNSI89	29-Cu	1.200	9.328E-01	8.2	XNSI89
13-Al	1.200	1.230E+04	9.0	INPA90	22-Ti	3.316	2.131E+02	5.0	XNSI84	29-Cu	1.206	9.345E-01	5.0	XNSI84
13-Al	1.221	3.620E+02	24.9	XKBB73	22-Ti	3.600	1.460E+03	6.0	INPA90	29-Cu	1.500	2.116E+00	8.2	XNSI89
13-Al	1.350	1.910E+04	9.0	INPA90	22-Ti	3.600	2.803E+02	8.2	XNSI89	29-Cu	1.507	2.100E+00	5.0	XNSI84
13-Al	1.431	5.540E+02	24.9	XKBB73	22-Ti	3.618	2.803E+02	5.0	XNSI84	29-Cu	1.508	2.123E+00	5.0	XNSI84
13-Al	1.500	2.530E+04	9.0	INPA90	22-Ti	3.900	3.296E+02	8.2	XNSI89	29-Cu	1.800	4.030E+00	8.2	XNSI89
13-Al	1.629	7.650E+02	25.0	XKBB73	22-Ti	3.919	3.285E+02	5.0	XNSI84	29-Cu	1.809	3.969E+00	5.0	XNSI84
13-Al	1.800	3.800E+04	9.0	INPA90	22-Ti	4.200	4.066E+02	8.2	XNSI89	29-Cu	1.809	4.076E+00	5.0	XNSI84
13-Al	1.839	1.010E+03	24.8	XKBB73	22-Ti	4.221	4.073E+02	5.0	XNSI84	29-Cu	2.100	6.820E+00	8.2	XNSI89
13-Al	2.040	1.270E+03	25.2	XKBB73	22-Ti	4.500	4.622E+02	8.2	XNSI89	29-Cu	2.110	6.898E+00	5.0	XNSI84
13-Al	2.100	5.070E+04	9.0	INPA90	22-Ti	4.522	4.621E+02	5.0	XNSI84	29-Cu	2.111	6.720E+00	5.0	XNSI84
13-Al	2.241	1.570E+03	24.8	XKBB73	22-Ti	4.800	5.286E+02	8.2	XNSI89	29-Cu	2.400	1.082E+01	8.2	XNSI89
13-Al	2.250	8.080E+02	12.0	XNTA76	22-Ti	4.824	5.278E+02	5.0	XNSI84	29-Cu	2.412	1.019E+01	5.0	XNSI84
13-Al	2.400	6.460E+04	9.0	INPA90	24-Cr	0.525	1.851E-01	8.2	XNSI89	29-Cu	2.412	1.081E+01	5.0	XNSI84
13-Al	2.451	1.900E+03	24.7	XKBB73	24-Cr	0.600	1.300E+00	7.0	INPA90	29-Cu	2.700	1.575E+01	8.2	XNSI89
13-Al	2.649	2.220E+03	24.8	XKBB73	24-Cr	0.600	3.300E-01	8.2	XNSI89	29-Cu	2.713	1.575E+01	5.0	XNSI84
13-Al	2.850	2.580E+03	24.8	XKBB73	24-Cr	0.675	5.500E-01	8.2	XNSI89	29-Cu	3.000	2.156E+01	8.2	XNSI89
13-Al	2.961	2.760E+03	25.0	XKBB73	24-Cr	0.750	2.790E+00	7.0	INPA90	29-Cu	3.015	2.158E+01	5.0	XNSI84
13-Al	3.000	1.450E+03	11.7	XNTA76	24-Cr	0.750	7.837E-01	8.2	XNSI89	29-Cu	3.300	2.952E+01	8.2	XNSI89
13-Al	3.750	2.080E+03	12.0	XNTA76	24-Cr	0.900	5.370E+00	7.0	INPA90	29-Cu	3.316	2.955E+01	5.0	XNSI84
13-Al	4.500	2.660E+03	12.0	XNTA76	24-Cr	0.900	1.601E+00	8.2	XNSI89	29-Cu	3.600	3.731E+01	8.2	XNSI89
13-Al	6.000	3.680E+03	12.0	XNTA76	24-Cr	1.050	1.090E+01	7.0	INPA90	29-Cu	3.618	3.734E+01	5.0	XNSI84
13-Al	7.500	4.350E+03	12.0	XNTA76	24-Cr	1.050	2.805E+00	8.2	XNSI89	29-Cu	3.900	4.796E+01	8.2	XNSI89
13-Al	9.000	4.650E+03	12.0	XNTA76	24-Cr	1.200	1.930E+01	7.0	INPA90	29-Cu	3.919	4.806E+01	5.0	XNSI84
14-Si	2.250	6.280E+02	11.9	XNTA76	24-Cr	1.200	4.510E+00	8.2	XNSI89	29-Cu	4.200	5.808E+01	8.2	XNSI89
14-Si	3.000	1.150E+03	12.2	XNTA76	24-Cr	1.350	2.670E+01	7.0	INPA90	29-Cu	4.221	5.829E+01	5.0	XNSI84
14-Si	3.750	1.670E+03	12.0	XNTA76	24-Cr	1.500	3.820E+01	7.0	INPA90	29-Cu	4.500	6.732E+01	8.2	XNSI89
14-Si	4.500	2.270E+03	11.9	XNTA76	24-Cr	1.500	9.928E+00	8.2	XNSI89	29-Cu	4.522	7.031E+01	5.0	XNSI84
14-Si	6.000	3.200E+03	11.9	XNTA76	24-Cr	1.800	7.120E+01	7.0	INPA90	29-Cu	4.800	8.316E+01	8.2	XNSI89
14-Si	7.500	3.680E+03	12.0	XNTA76	24-Cr	1.800	1.837E+01	8.2	XNSI89	29-Cu	4.824	8.321E+01	5.0	XNSI84
14-Si	9.000	3.910E+03	12.0	XNTA76	24-Cr	2.100	1.160E+02	7.0	INPA90	32-Ge	0.600	2.390E-02	10.0	INPA90
15-P	0.600	1.690E+02	10.0	INPA90	24-Cr	2.100	2.998E+01	8.2	XNSI89	32-Ge	0.600	1.808E-02	8.2	XNSI89
15-P	0.750	5.090E+02	10.0	INPA90	24-Cr	2.400	1.780E+02	7.0	INPA90	32-Ge	0.750	7.380E-02	10.0	INPA90
15-P	0.900	1.090E+03	10.0	INPA90	24-Cr	2.400	4.538E+01	8.2	XNSI89	32-Ge	0.750	5.457E-02	8.2	XNSI89
15-P	1.050	1.740E+03	10.0	INPA90	24-Cr	2.700	2.630E+02	7.0	INPA90	32-Ge	0.900	1.640E-01	10.0	INPA90
15-P	1.200	2.720E+03	10.0	INPA90	24-Cr	2.700	6.683E+01	8.2	XNSI89	32-Ge	0.900	1.145E-01	8.2	XNSI89
15-P	1.350	4.030E+03	10.0	INPA90	24-Cr	3.000	3.670E+02	7.0	INPA90	32-Ge	1.050	3.950E-01	10.0	INPA90
15-P	1.500	6.170E+03	10.0	INPA90	24-Cr	3.000	9.405E+01	8.2	XNSI89	32-Ge	1.050	2.311E-01	8.2	XNSI89
15-P	1.800	9.980E+03	10.0	INPA90	24-Cr	3.300	4.570E+02	7.0	INPA90	32-Ge	1.200	6.990E-01	10.0	INPA90
15-P	2.100	1.220E+04	10.0	INPA90	24-Cr	3.300	1.155E+02	8.2	XNSI89	32-Ge	1.200	3.948E-01	8.2	XNSI89
15-P	2.400	1.780E+04	10.0	INPA90	24-Cr	3.600	5.440E+02	7.0	INPA90	32-Ge	1.350	1.080E+00	10.0	INPA90
18-Ar	0.600	2.450E+01	8.0	INPA90	24-Cr	3.600	1.576E+02	8.2	XNSI89	32-Ge	1.500	1.500E+00	10.0	INPA90
18-Ar	0.750	6.860E+01	8.0	INPA90	24-Cr	3.900	1.889E+02	8.2	XNSI89	32-Ge	1.500	9.042E-01	8.2	XNSI89
18-Ar	0.900	1.150E+02	8.0	INPA90	24-Cr	4.200	2.266E+02	8.2	XNSI89	32-Ge	1.800	2.930E+00	10.0	INPA90
18-Ar	1.050	2.430E+02	8.0	INPA90	24-Cr	4.500	2.684E+02	8.2	XNSI89	32-Ge	1.800	1.712E+00	8.2	XNSI89
18-Ar	1.200	3.650E+02	8.0	INPA90	26-Fe	0.600	4.300E-01	9.0	INPA90	32-Ge	2.100	5.430E+00	10.0	INPA90
18-Ar	1.350	6.060E+02	8.0	INPA90	26-Fe	0.750	9.070E-01	9.0	INPA90	32-Ge	2.100	2.953E+00	8.2	XNSI89
18-Ar	1.500	7.900E+02	8.0	INPA90	26-Fe	0.900	2.350E+00	9.0	INPA90	32-Ge	2.400	7.780E+00	10.0	INPA90
18-Ar	1.800	1.530E+03	8.0	INPA90	26-Fe	1.050	4.270E+00	9.0	INPA90	32-Ge	2.400	4.585E+00	8.2	XNSI89
18-Ar	2.100	2.140E+03	8.0	INPA90	26-Fe	1.200	7.450E+00	9.0	INPA90	32-Ge	2.700	1.230E+01	10.0	INPA90
18-Ar	2.400	3.170E+03	8.0	INPA90	26-Fe	1.350	1.120E+01	9.0	INPA90	32-Ge	3.000	1.580E+01	10.0	INPA90
22-Ti	0.525	4.280E-01	8.2	XNSI89	26-Fe	1.500	1.640E+01	9.0	INPA90	32-Ge	3.300	2.380E+01	10.0	INPA90
22-Ti	0.600	3.420E+00	6.0	INPA90	26-Fe	1.800	2.980E+01	9.0	INPA90	32-Ge	3.600	3.220E+01	10.0	INPA90
22-Ti	0.600	6.741E-01	8.2	XNSI89	26-Fe	2.100	4.560E+01	9.0	INPA90	34-Se	0.600	7.620E-03	15.0	INPA90
22-Ti	0.603	6.658E-01	5.0	XNSI84	26-Fe	2.400	7.130E+01	9.0	INPA90	34-Se	0.750	3.330E-02	15.0	INPA90
22-Ti	0.675	1.128E+00	8.2	XNSI89	26-Fe	2.700	1.110E+02	9.0	INPA90	34-Se	1.050	1.980E-01	15.0	INPA90
22-Ti	0.750	8.850E+00	6.0	INPA90	26-Fe	3.000	1.560E+02	9.0	INPA90	34-Se	1.200	3.080E-01	15.0	INPA90
22-Ti	0.750	1.616E+00	8.2	XNSI89	26-Fe	3.300	2.050E+02	9.0	INPA90	34-Se	1.350	5.450E-01	15.0	INPA90
22-Ti	0.754	1.577E+00	5.0	XNSI84	26-Fe	3.600	2.570E+02	9.0	INPA90	34-Se	1.500	8.430E-01	15.0	INPA90
22-Ti	0.900	1.490E+01	6.0	INPA90	27-Co	0.600	1.119E-01	8.2	XNSI89	34-Se	1.800	1.460E+00	15.0	INPA90
22-Ti	0.900	3.124E+00	8.2	XNSI89	27-Co	0.750	3.059E-01	8.2	XNSI89	34-Se	2.100	2.810E+00	15.0	INPA90
22-Ti	0.905	3.088E+00	5.0	XNSI84	27-Co	0.900	6.304E-01	8.2	XNSI89	34-Se	2.400	4.220E+00	15.0	INPA90
22-Ti	1.050	2.760E+01	6.0	INPA90	27-Co	1.050	1.167E+00	8.2	XNSI89	34-Se	2.700	6.220E+00	15.0	INPA90
22-Ti	1.050	5.521E+00	8.2	XNSI89	27-Co	1.200	1.854E+00	8.2	XNSI89	34-Se	3.000	9.330E+00	15.0	INPA90
22-Ti	1.055	5.431E+00	5.0	XNSI84	27-Co	1.500	3.954E+00	8.2	XNSI89	34-Se	3.300	1.240E+01	15.0	INPA90
22-Ti	1.200	4.630E+01	6.0	INPA90	27-Co	1.800	7.162E+00	8.2	XNSI89	34-Se	3.600	1.780E+01	15.0	INPA90
22-Ti	1.200	8.967E+00	8.2	XNSI89	27-Co	2.100	1.227E+01	8.2	XNSI89	40-Zr	1.200	3.504E-02	9.0	XNSI89
22-Ti	1.206	9.023E+00	5.0	XNSI84	27-Co	2.400	1.921E+01	8.2	XNSI89	40-Zr	1.500	9.198E-02	9.0	XNSI89
22-Ti	1.350	6.790E+01	6.0	INPA90	28-Ni	0.600	1.520E-01	7.0	INPA90	40-Zr	1.800	2.008E-01	9.0	XNSI89

TABLE IIc. Experimental Cross Sections for *K*-Shell X-Ray Production, Auger Electron Production, and Ionization by ^{3}He Ions

See page 84 for Explanation of Tables

TARGET	ENERGY [MeV]	SIGMA [barn]	ERROR [%]	REFERENCE	TARGET	ENERGY [MeV]	SIGMA [barn]	ERROR [%]	REFERENCE	TARGET	ENERGY [MeV]	SIGMA [barn]	ERROR [%]	REFERENCE
40-Zr	2.100	3.577E-01	9.0	XNSI89										
40-Zr	2.400	5.336E-01	9.0	XNSI89										
40-Zr	3.000	1.212E+00	9.0	XNSI89										
40-Zr	3.600	2.365E+00	9.0	XNSI89										
40-Zr	4.200	3.774E+00	9.0	XNSI89										
40-Zr	4.800	5.760E+00	9.0	XNSI89										
41-Nb	1.050	1.520E-02	10.0	INPA90										
41-Nb	1.200	3.640E-02	10.0	INPA90										
41-Nb	1.350	5.820E-02	10.0	INPA90										
41-Nb	1.500	8.650E-02	10.0	INPA90										
41-Nb	1.800	1.880E-01	10.0	INPA90										
41-Nb	2.400	5.490E-01	10.0	INPA90										
41-Nb	3.000	1.200E+00	10.0	INPA90										
41-Nb	3.300	1.770E+00	10.0	INPA90										
46-Pd	1.050	2.910E-03	15.0	INPA90										
46-Pd	1.200	5.050E-03	15.0	INPA90										
46-Pd	1.350	1.120E-02	15.0	INPA90										
46-Pd	1.500	1.940E-02	15.0	INPA90										
46-Pd	1.800	4.410E-02	15.0	INPA90										
46-Pd	2.100	9.530E-02	15.0	INPA90										
46-Pd	2.400	1.360E-01	15.0	INPA90										
46-Pd	2.700	2.230E-01	15.0	INPA90										
46-Pd	3.000	3.280E-01	15.0	INPA90										
46-Pd	3.300	4.350E-01	15.0	INPA90										
46-Pd	3.600	6.160E-01	15.0	INPA90										
47-Ag	1.500	1.413E-02	11.0	XNSI89										
47-Ag	1.800	2.992E-02	11.0	XNSI89										
47-Ag	2.100	6.149E-02	11.0	XNSI89										
47-Ag	2.400	1.006E-01	11.0	XNSI89										
47-Ag	2.700	2.003E-01	11.0	XNSI89										
47-Ag	3.000	2.260E-01	11.0	XNSI89										
47-Ag	3.600	4.487E-01	11.0	XNSI89										
47-Ag	4.200	7.579E-01	11.0	XNSI89										
47-Ag	4.800	1.222E+00	11.0	XNSI89										

TABLE IId. Experimental Cross Sections for *K*-Shell X-Ray Production, Auger Electron Production, and Ionization by ^{4}He Ions

See page 84 for Explanation of Tables

TARGET	ENERGY [MeV]	SIGMA [barn]	ERROR [%]	REFERENCE	TARGET	ENERGY [MeV]	SIGMA [barn]	ERROR [%]	REFERENCE	TARGET	ENERGY [MeV]	SIGMA [barn]	ERROR [%]	REFERENCE
6-C	0.116	5.319E+00	12.0	XGHA73H	7-N	0.400	4.860E+02	24.9	XGLA76	12-Mg	1.000	2.889E+02	23.9	XKSE69
6-C	0.116	4.505E+00	12.0	XGHA73O	7-N	0.500	1.550E+05	14.8	IGST75	12-Mg	1.000	1.650E+02	12.0	XNYU91A
6-C	0.135	7.400E+03	15.0	IGST73	7-N	0.500	1.020E+03	24.5	XGLA76	12-Mg	1.180	3.850E+02	12.0	XNOR89
6-C	0.135	1.010E+01	12.0	XGHA73H	7-N	0.600	4.600E+05	20.9	AGME87	12-Mg	1.500	5.009E+02	12.2	XKSE69
6-C	0.135	7.361E+00	12.0	XGHA73O	7-N	0.600	3.040E+05	15.0	IGST73	12-Mg	1.500	6.230E+02	12.0	XNYU91A
6-C	0.136	8.300E+03	15.1	IGST75	7-N	0.600	3.000E+05	15.0	IGST75	12-Mg	2.000	1.250E+03	12.0	XNYU91A
6-C	0.150	1.080E+04	15.0	IGST73	7-N	0.600	1.740E+03	24.7	XGLA76	12-Mg	2.500	1.730E+03	4.6	XKSE69
6-C	0.152	1.550E+04	14.8	IGST75	7-N	0.700	2.710E+03	25.1	XGLA76	12-Mg	2.500	2.020E+03	12.0	XNYU91A
6-C	0.152	2.260E+01	24.8	XGLA76	7-N	0.800	3.660E+03	24.9	XGLA76	12-Mg	3.000	2.390E+03	6.0	XKSE69
6-C	0.175	2.130E+04	15.0	IGST73	7-N	1.000	1.440E+06	16.9	AGME87	12-Mg	3.000	2.760E+03	12.0	XNYU91A
6-C	0.176	2.380E+04	15.1	IGST75	7-N	1.000	6.600E+05	21.2	IGKB76	12-Mg	3.500	3.154E+03	5.9	XKSE69
6-C	0.176	4.130E+01	24.9	XGLA76	7-N	1.000	9.740E+05	30.0	IGMK76	12-Mg	3.500	3.360E+03	12.0	XNYU91A
6-C	0.200	3.400E+04	15.0	IGST73	7-N	1.000	5.340E+03	24.9	XGLA76	12-Mg	4.000	3.551E+03	3.0	XKSE69
6-C	0.200	4.520E+04	15.0	IGST75	7-N	1.200	2.040E+06	17.4	AGME87	12-Mg	4.000	3.890E+03	12.0	XNYU91A
6-C	0.200	6.620E+01	24.9	XGLA76	7-N	1.200	6.450E+03	25.0	XGLA76	12-Mg	4.500	4.055E+03	3.9	XKSE69
6-C	0.200	9.000E-02	80.0	XKTE70	7-N	1.400	2.300E+06	17.0	AGME87	12-Mg	4.500	4.310E+03	12.0	XNYU91A
6-C	0.240	1.530E-01	80.0	XKTE70	7-N	1.400	1.300E+06	23.1	IGKB76	12-Mg	5.000	4.611E+03	3.4	XKSE69
6-C	0.250	8.600E+04	15.0	IGST73	7-N	1.600	2.420E+06	16.6	AGME87	12-Mg	5.000	4.680E+03	12.0	XNYU91A
6-C	0.252	9.000E+04	15.6	IGST75	7-N	1.600	8.150E+03	25.0	XGLA76	12-Mg	5.500	4.900E+03	12.0	XNYU91A
6-C	0.252	1.890E+02	24.9	XGLA76	7-N	1.800	2.510E+06	16.4	AGME87	12-Mg	6.000	5.350E+03	12.0	XNYU91A
6-C	0.280	2.340E-01	80.0	XKTE70	7-N	1.800	1.700E+06	17.6	IGKB76	12-Mg	6.500	5.530E+03	12.0	XNYU91A
6-C	0.300	1.620E+05	15.0	IGST73	7-N	2.000	2.040E+06	29.9	IGMK76	12-Mg	7.000	5.730E+03	12.0	XNYU91A
6-C	0.300	1.720E+05	15.1	IGST75	7-N	2.000	9.810E+03	25.0	XGLA76	12-Mg	7.500	5.900E+03	12.0	XNYU91A
6-C	0.300	3.440E+02	25.0	XGLA76	7-N	2.200	1.900E+06	21.1	IGKB76	12-Mg	8.000	6.020E+03	12.0	XNYU91A
6-C	0.320	3.420E-01	80.0	XKTE70	7-N	2.600	2.000E+06	20.0	IGKB76	13-Al	0.060	2.100E-04	24.8	XKBB73
6-C	0.360	5.140E-01	80.0	XKTE70	7-N	3.000	2.520E+06	30.2	IGMK76	13-Al	0.068	7.350E-04	25.0	XKBB73
6-C	0.400	4.880E+05	15.0	IGST73	7-N	4.000	2.180E+06	29.8	IGMK76	13-Al	0.080	1.750E-03	25.1	XKBB73
6-C	0.400	4.910E+05	15.1	IGST75	7-N	5.000	1.870E+06	29.9	IGMK76	13-Al	0.088	4.060E-03	24.9	XKBB73
6-C	0.400	1.030E+03	25.2	XGLA76	8-O	0.600	1.260E+03	80.0	XKKW77	13-Al	0.100	7.650E-03	25.0	XKBB73
6-C	0.400	7.660E-01	80.0	XKTE70	8-O	0.900	2.900E+03	80.0	XKKW77	13-Al	0.120	2.100E-02	24.8	XKBB73
6-C	0.440	1.170E+00	80.0	XKTE70	8-O	1.000	2.100E+05	23.8	IGKB76	13-Al	0.140	4.770E-02	24.9	XKBB73
6-C	0.450	1.110E+06	20.8	AGME87	8-O	1.200	4.340E+03	80.0	XKKW77	13-Al	0.160	8.420E-02	24.9	XKBB73
6-C	0.480	1.800E+00	80.0	XKTE70	8-O	1.400	4.100E+05	22.0	IGKB76	13-Al	0.180	1.500E-01	25.3	XKBB73
6-C	0.500	9.300E+05	15.1	IGST75	8-O	1.500	5.450E+03	80.0	XKKW77	13-Al	0.200	2.410E-01	24.9	XKBB73
6-C	0.520	2.620E+00	80.0	XKTE70	8-O	1.800	5.500E+05	23.6	IGKB76	13-Al	0.408	3.250E+00	24.9	XKBB73
6-C	0.560	3.700E+00	80.0	XKTE70	8-O	1.800	6.430E+03	80.0	XKKW77	13-Al	0.500	7.600E+00	12.0	XNYU91A
6-C	0.600	1.870E+06	18.6	AGME87	8-O	2.200	7.200E+05	23.6	IGKB76	13-Al	0.508	6.600E+00	25.0	XKBB73
6-C	0.600	1.850E+06	20.0	AGWA73	8-O	2.600	7.900E+05	22.8	IGKB76	13-Al	0.612	1.250E+01	24.8	XKBB73
6-C	0.600	1.370E+06	15.0	IGST73	9-F	0.400	7.190E+01	15.0	XKKW80	13-Al	0.712	2.300E+01	24.8	XKBB73
6-C	0.600	1.320E+06	15.2	IGST75	9-F	0.600	2.000E+02	15.0	XKKW80	13-Al	0.750	2.390E+01	12.0	XNYU91A
6-C	0.600	3.090E+03	24.9	XGLA76	9-F	0.800	4.010E+02	15.0	XKKW80	13-Al	0.816	3.780E+01	24.9	XKBB73
6-C	0.600	5.400E+00	80.0	XKTE70	9-F	1.000	8.000E+04	23.0	AGKB79	13-Al	0.900	5.400E+01	19.0	XNOR89
6-C	0.640	7.740E+00	80.0	XKTE70	9-F	1.000	2.100E+04	30.0	IGMK76	13-Al	1.000	1.357E+02	84.2	XKSE69
6-C	0.680	1.080E+01	80.0	XKTE70	9-F	1.000	7.010E+02	15.0	XKKW80	13-Al	1.000	1.000E+02	15.0	XNMK74
6-C	0.700	2.610E+06	17.7	AGME87	9-F	1.200	1.070E+03	15.0	XKKW80	13-Al	1.000	6.990E+01	12.0	XNYU91A
6-C	0.720	1.350E+01	80.0	XKTE70	9-F	1.250	1.300E+05	23.0	AGKB79	13-Al	1.020	8.470E+01	25.0	XKBB73
6-C	0.760	1.620E+01	80.0	XKTE70	9-F	1.400	1.700E+05	47.1	IGKB76	13-Al	1.120	1.170E+02	24.8	XKBB73
6-C	0.800	3.150E+06	17.3	AGME87	9-F	1.400	1.620E+03	15.0	XKKW80	13-Al	1.180	1.450E+02	10.0	XNOR89
6-C	0.800	5.260E+03	24.9	XGLA76	9-F	1.500	2.000E+05	23.0	AGKB79	13-Al	1.220	1.540E+02	24.7	XKBB73
6-C	0.800	1.890E+01	80.0	XKTE70	9-F	1.600	1.980E+03	15.0	XKKW80	13-Al	1.300	1.900E+02	11.0	XNOR89
6-C	1.000	4.070E+06	15.7	AGME87	9-F	1.750	2.400E+05	23.0	AGKB79	13-Al	1.428	2.480E+02	25.0	XKBB73
6-C	1.000	3.340E+06	20.1	AGWA73	9-F	1.800	2.400E+05	50.0	IGKB76	13-Al	1.500	5.926E+02	13.9	XKSE69
6-C	1.000	2.500E+06	24.0	IGKB76	9-F	1.800	2.520E+03	15.0	XKKW80	13-Al	1.500	4.410E+02	15.0	XNMK74
6-C	1.000	7.000E+03	25.0	XGLA76	9-F	2.000	3.400E+05	23.0	AGKB79	13-Al	1.500	2.650E+02	12.0	XNYU91A
6-C	1.200	4.690E+06	16.0	AGME87	9-F	2.000	1.330E+05	30.1	IGMK76	13-Al	1.628	3.620E+02	24.9	XKBB73
6-C	1.200	8.090E+03	25.0	XGLA76	9-F	2.000	3.050E+03	15.0	XKKW80	13-Al	1.840	4.950E+02	25.1	XKBB73
6-C	1.400	4.970E+06	15.8	AGME87	9-F	2.200	3.400E+05	50.0	IGKB76	13-Al	2.000	9.639E+02	7.8	XKSE69
6-C	1.400	3.200E+06	25.0	IGKB76	9-F	2.250	3.900E+05	23.0	AGKB79	13-Al	2.000	9.420E+02	15.0	XNMK74
6-C	1.500	5.200E+06	19.2	AGWA73	9-F	2.500	4.400E+05	23.0	AGKB79	13-Al	2.000	5.890E+02	12.0	XNYU91A
6-C	1.600	4.900E+06	16.1	AGME87	9-F	2.600	3.800E+05	50.0	IGKB76	13-Al	2.040	6.460E+02	24.9	XKBB73
6-C	1.600	8.890E+03	25.0	XGLA76	9-F	3.000	2.400E+05	30.0	IGMK76	13-Al	2.240	8.200E+02	25.0	XKBB73
6-C	1.800	5.170E+06	16.0	AGME87	9-F	4.000	3.400E+05	30.0	IGMK76	13-Al	2.348	9.150E+02	25.0	XKBB73
6-C	1.800	3.500E+06	22.9	IGKB76	9-F	5.000	3.890E+05	30.1	IGMK76	13-Al	2.448	1.020E+03	24.5	XKBB73
6-C	2.000	4.940E+06	18.5	AGME87	10-Ne	0.097	1.364E-01	12.5	XGHA73	13-Al	2.500	1.521E+03	4.7	XKSE69
6-C	2.000	4.960E+06	20.0	AGWA73	10-Ne	0.116	3.503E-01	11.9	XGHA73	13-Al	2.500	1.570E+03	15.3	XNMK74
6-C	2.000	9.860E+03	24.9	XGLA76	10-Ne	0.135	6.123E-01	11.9	XGHA73	13-Al	2.500	9.940E+02	12.0	XNYU91A
6-C	2.200	3.600E+06	25.0	IGKB76	10-Ne	0.200	6.900E+01	14.5	IGST75	13-Al	2.548	1.140E+03	24.6	XKBB73
6-C	2.400	9.820E+03	24.9	XGLA76	10-Ne	0.200	1.380E+00	24.6	XGLA76	13-Al	2.648	1.250E+03	24.8	XKBB73
6-C	2.600	3.500E+06	22.9	IGKB76	10-Ne	0.252	3.660E+00	24.9	XGLA76	13-Al	2.848	1.490E+03	24.8	XKBB73
7-N	0.097	2.110E+00	12.0	XGHA73	10-Ne	0.300	3.900E+02	15.4	IGST75	13-Al	2.900	2.449E+02	5.0	XKKO68
7-N	0.116	4.277E+00	12.1	XGHA73	10-Ne	0.300	7.100E+00	24.9	XGLA76	13-Al	2.980	1.590E+03	25.2	XKBB73
7-N	0.124	4.650E+00	24.9	XGLA76	10-Ne	0.400	1.230E+03	14.6	IGST75	13-Al	3.000	1.988E+03	5.4	XKSE69
7-N	0.135	9.500E+02	15.0	IGST73	10-Ne	0.400	2.190E+01	25.1	XGLA76	13-Al	3.000	2.190E+03	15.1	XNMK74
7-N	0.135	8.319E+00	11.9	XGHA73	10-Ne	0.500	2.420E+03	14.9	IGST75	13-Al	3.000	1.470E+03	12.0	XNYU91A
7-N	0.136	1.180E+03	15.3	IGST75	10-Ne	0.500	5.400E+01	25.0	XGLA76	13-Al	3.060	1.680E+03	25.0	XKBB73
7-N	0.150	1.430E+03	15.0	IGST73	10-Ne	0.600	5.120E+03	14.3	IGST75	13-Al	3.160	1.760E+03	25.0	XKBB73
7-N	0.152	1.820E+03	14.8	IGST75	10-Ne	0.600	1.120E+02	25.0	XGLA76	13-Al	3.260	1.850E+03	24.9	XKBB73
7-N	0.152	1.060E+01	24.5	XGLA76	10-Ne	0.800	3.090E+02	24.9	XGLA76	13-Al	3.500	2.931E+03	4.6	XKSE69
7-N	0.175	3.400E+03	15.0	IGST73	10-Ne	1.000	7.400E+04	30.0	IGMK76	13-Al	3.500	2.890E+03	14.9	XNMK74
7-N	0.176	3.740E+03	15.0	IGST75	10-Ne	1.000	6.170E+02	25.0	XGLA76	13-Al	3.500	1.770E+03	12.0	XNYU91A
7-N	0.176	1.900E+01	24.7	XGLA76	10-Ne	1.200	9.870E+02	25.0	XGLA76	13-Al	3.550	3.541E+02	5.0	XKKO68
7-N	0.200	5.600E+03	15.0	IGST73	10-Ne	1.400	4.500E+04	26.7	IGKB76	13-Al	3.668	2.180E+03	24.8	XKBB73
7-N	0.200	5.900E+03	15.3	IGST75	10-Ne	1.800	1.100E+05	27.3	IGKB76	13-Al	3.900	4.463E+02	4.8	XKKO68
7-N	0.200	3.140E+01	24.8	XGLA76	10-Ne	2.000	2.670E+05	30.0	IGMK76	13-Al	4.000	2.390E+03	40.0	XKBM78
7-N	0.250	1.150E+04	15.0	IGST73	10-Ne	2.200	1.300E+05	23.1	IGKB76	13-Al	4.000	2.170E+03	10.0	XNBB78
7-N	0.252	1.250E+04	15.2	IGST75	10-Ne	2.600	1.700E+05	23.5	IGKB76	13-Al	4.000	3.520E+03	15.1	XNMK74
7-N	0.252	8.030E+01	25.0	XGLA76	10-Ne	3.000	3.700E+05	30.0	IGMK76	13-Al	4.000	2.140E+03	12.0	XNYU91A
7-N	0.300	2.340E+04	15.0	IGST73	10-Ne	4.000	4.400E+05	30.0	IGMK76	13-Al	4.080	2.440E+03	25.0	XKBB73
7-N	0.300	2.830E+04	15.2	IGST75	10-Ne	5.000	4.700E+05	30.0	IGMK76	13-Al	4.080	2.428E+03	5.3	XKSE69
7-N	0.300	1.480E+02	25.0	XGLA76	12-Mg	0.500	1.390E+01	12.0	XNYU91A	13-Al	4.450	5.748E+02	5.0	XKKO68
7-N	0.400	8.200E+04	15.0	IGST73	12-Mg	0.750	4.730E+01	12.0	XNYU91A	13-Al	4.500	4.820E+03	5.9	XKSE69
7-N	0.400	7.900E+04	15.2	IGST75	12-Mg	0.900	1.600E+02	13.0	XNOR89	13-Al	4.500	4.180E+03	15.1	XNMK74

TABLE IId. Experimental Cross Sections for *K*-Shell X-Ray Production, Auger Electron Production, and Ionization by ^{4}He Ions

See page 84 for Explanation of Tables

TARGET	ENERGY [MeV]	SIGMA [barn]	ERROR [%]	REFERENCE	TARGET	ENERGY [MeV]	SIGMA [barn]	ERROR [%]	REFERENCE	TARGET	ENERGY [MeV]	SIGMA [barn]	ERROR [%]	REFERENCE
13-Al	4.500	2.470E+03	12.0	XNYU91A	21-Sc	1.400	8.493E+00	13.0	XNLJ82	22-Ti	3.120	8.460E+01	15.0	XNSO76
13-Al	4.688	2.760E+03	25.0	XKBB73	21-Sc	1.600	1.378E+01	13.0	XNLJ82	22-Ti	3.200	9.080E+01	15.0	XNSO76
13-Al	4.800	6.676E+02	4.8	XKKO68	21-Sc	1.800	2.033E+01	13.0	XNLJ82	22-Ti	3.320	1.000E+02	15.0	XNSO76
13-Al	5.000	5.069E+03	5.6	XKSE69	21-Sc	2.000	3.002E+01	30.4	XNLD73	22-Ti	3.400	1.110E+02	15.3	XNSO76
13-Al	5.000	4.830E+03	14.9	XNMK74	21-Sc	3.000	1.216E+02	29.7	XNLD73	22-Ti	3.520	1.190E+02	15.1	XNSO76
13-Al	5.000	2.800E+03	12.0	XNYU91A	21-Sc	4.000	2.945E+02	30.3	XNLD73	22-Ti	3.550	4.073E+01	4.8	XKKO68
13-Al	5.300	8.354E+02	5.1	XKKO68	21-Sc	5.000	5.092E+02	30.2	XNLD73	22-Ti	3.600	1.230E+02	15.4	XNSO76
13-Al	5.500	3.080E+03	12.0	XNYU91A	21-Sc	6.000	7.790E+02	29.3	XNLD73	22-Ti	3.720	1.340E+02	14.9	XNSO76
13-Al	5.508	3.100E+03	24.8	XKBB73	21-Sc	7.000	1.083E+03	29.8	XNLD73	22-Ti	3.800	1.460E+02	15.1	XNSO76
13-Al	5.920	3.170E+03	24.9	XKBB73	21-Sc	8.000	1.273E+03	29.9	XNLD73	22-Ti	3.900	5.278E+01	5.0	XKKO68
13-Al	6.000	3.300E+03	10.0	XNBB78	21-Sc	9.000	1.558E+03	30.5	XNLD73	22-Ti	3.920	1.580E+02	15.2	XNSO76
13-Al	6.000	3.370E+03	12.0	XNYU91A	21-Sc	10.000	1.729E+03	29.7	XNLD73	22-Ti	4.000	1.790E+02	40.0	XKBM78
13-Al	6.500	3.610E+03	12.0	XNYU91A	21-Sc	11.000	2.071E+03	30.3	XNLD73	22-Ti	4.000	2.059E+02	29.8	XNLD73
13-Al	7.000	3.890E+03	12.0	XNYU91A	21-Sc	12.000	2.413E+03	29.9	XNLD73	22-Ti	4.000	1.690E+02	14.8	XNSO76
13-Al	7.500	4.120E+03	12.0	XNYU91A	21-Sc	20.000	2.846E+03	10.0	XNIS82	22-Ti	4.120	1.750E+02	14.9	XNSO76
13-Al	8.000	3.910E+03	10.0	XNBB78	21-Sc	24.000	3.209E+03	10.0	XNIS82	22-Ti	4.200	1.900E+02	15.3	XNSO76
13-Al	8.000	4.430E+03	12.0	XNYU91A	21-Sc	28.000	3.323E+03	10.0	XNIS82	22-Ti	4.320	2.040E+02	15.2	XNSO76
13-Al	10.000	4.150E+03	10.0	XNBB78	22-Ti	0.190	3.960E-04	23.0	XNBG80	22-Ti	4.400	2.030E+02	15.3	XNSO76
13-Al	12.000	4.280E+03	10.0	XNBB78	22-Ti	0.210	8.540E-04	22.0	XNBG80	22-Ti	4.450	7.643E+01	4.9	XKKO68
13-Al	16.000	4.120E+03	10.0	XNBB78	22-Ti	0.235	1.890E-03	21.0	XNBG80	22-Ti	4.800	1.010E+02	5.0	XKKO68
13-Al	20.000	3.930E+03	10.0	XNBB78	22-Ti	0.255	3.210E-03	20.0	XNBG80	22-Ti	5.000	3.679E+02	29.8	XNLD73
13-Al	24.000	3.580E+03	10.0	XNBB78	22-Ti	0.300	8.620E-03	20.0	XNBG80	22-Ti	5.300	1.211E+02	5.1	XKKO68
13-Al	28.000	3.380E+03	10.0	XNBB78	22-Ti	0.350	1.930E-02	20.0	XNBG80	22-Ti	6.000	5.825E+02	30.1	XNLD73
13-Al	30.000	3.300E+03	10.0	XNBB78	22-Ti	0.450	6.570E-02	20.0	XNBG80	22-Ti	7.000	7.665E+02	31.4	XNLD73
14-Si	0.700	1.680E+01	10.0	XNOR89	22-Ti	0.500	1.210E-01	9.1	XNMD75A	22-Ti	8.000	9.417E+02	30.2	XNLD73
14-Si	0.900	3.090E+01	10.0	XNOR89	22-Ti	0.550	1.590E-01	20.0	XNBG80	22-Ti	9.000	1.117E+03	29.4	XNLD73
14-Si	1.180	1.010E+02	10.0	XNOR89	22-Ti	0.600	2.610E-01	8.8	XNMD75A	22-Ti	10.000	1.358E+03	30.6	XNLD73
14-Si	1.300	1.300E+02	10.0	XNOR89	22-Ti	0.650	3.150E-01	20.0	XNBG80	22-Ti	10.000	1.029E+03	10.0	XNPE79
14-Si	2.900	1.429E+02	4.9	XKKO68	22-Ti	0.700	4.910E-01	9.0	XNMD75A	22-Ti	11.000	1.555E+03	29.6	XNLD73
14-Si	3.550	2.040E+02	5.1	XKKO68	22-Ti	0.700	4.815E-01	8.2	XNSI89	22-Ti	12.000	1.708E+03	29.5	XNLD73
14-Si	3.900	2.472E+02	4.9	XKKO68	22-Ti	0.750	5.560E-01	20.0	XNBG80	22-Ti	12.000	1.380E+03	10.0	XNPE79
14-Si	4.450	3.088E+02	5.0	XKKO68	22-Ti	0.800	7.801E-01	13.0	XNLJ82	22-Ti	14.000	1.643E+03	10.0	XNPE79
14-Si	4.800	3.539E+02	5.0	XKKO68	22-Ti	0.800	8.480E-01	9.0	XNMD75A	22-Ti	16.000	1.862E+03	10.0	XNPE79
14-Si	5.300	4.089E+02	5.1	XKKO68	22-Ti	0.800	8.239E-01	8.2	XNSI89	22-Ti	18.000	2.212E+03	10.0	XNPE79
15-P	0.700	7.390E+00	10.0	XNOR89	22-Ti	0.800	7.884E-01	5.0	XNSI84	22-Ti	20.000	2.364E+03	10.0	XNIS82
15-P	0.900	1.910E+01	10.0	XNOR89	22-Ti	0.900	1.330E+00	9.0	XNMD75A	22-Ti	20.000	2.321E+03	10.0	XNPE79
15-P	1.180	6.690E+01	10.0	XNOR89	22-Ti	0.900	1.260E+00	10.0	XNOR89	22-Ti	22.000	2.497E+03	10.0	XNPE79
15-P	1.300	7.630E+01	10.0	XNOR89	22-Ti	0.900	1.209E+00	8.2	XNSI89	22-Ti	24.000	2.721E+03	10.0	XNIS82
17-Cl	0.800	1.242E+01	15.0	XNAV81	22-Ti	1.000	1.812E+00	13.0	XNLJ82	22-Ti	24.000	2.650E+03	10.0	XNPE79
17-Cl	1.000	2.798E+01	15.0	XNAV81	22-Ti	1.000	2.000E+00	9.0	XNMD75A	22-Ti	26.000	2.716E+03	10.0	XNPE79
17-Cl	1.200	4.718E+01	15.0	XNAV81	22-Ti	1.000	1.832E+00	8.2	XNSI89	22-Ti	28.000	2.877E+03	10.0	XNIS82
17-Cl	1.400	7.564E+01	15.0	XNAV81	22-Ti	1.000	2.630E+00	14.8	XNSO76	22-Ti	28.000	2.803E+03	10.0	XNPE79
17-Cl	1.600	1.041E+02	15.0	XNAV81	22-Ti	1.001	1.822E+00	5.0	XNSI84	22-Ti	30.000	2.957E+03	10.0	XNPE79
17-Cl	1.800	1.518E+02	15.0	XNAV81	22-Ti	1.100	2.870E+00	9.1	XNMD75A	23-V	0.500	7.400E-02	9.1	XNMD75A
17-Cl	2.000	2.187E+02	15.0	XNAV81	22-Ti	1.120	3.370E+00	15.1	XNSO76	23-V	0.600	1.810E-01	8.8	XNMD75A
17-Cl	2.200	2.540E+02	15.0	XNAV81	22-Ti	1.180	3.270E+00	10.0	XNOR89	23-V	0.700	3.360E-01	8.9	XNMD75A
17-Cl	2.400	2.970E+02	15.0	XNAV81	22-Ti	1.200	3.337E+00	12.0	XNLJ82	23-V	0.800	5.389E-01	12.0	XNLJ82
17-Cl	2.600	4.546E+02	15.0	XNAV81	22-Ti	1.200	3.830E+00	8.9	XNMD75A	23-V	0.800	5.830E-01	8.9	XNMD75A
17-Cl	2.800	5.042E+02	15.0	XNAV81	22-Ti	1.200	3.574E+00	8.2	XNSI89	23-V	0.900	9.470E-01	9.0	XNMD75A
18-Ar	0.300	4.150E-02	25.1	XGLA76	22-Ti	1.201	3.657E+00	5.0	XNSI84	23-V	0.900	7.900E-01	10.0	XNOR89
18-Ar	0.500	4.590E-01	25.1	XGLA76	22-Ti	1.300	5.150E+00	8.9	XNMD75A	23-V	1.000	1.470E+00	29.9	XKMK74
18-Ar	0.800	3.110E+00	25.1	XGLA76	22-Ti	1.300	4.950E+00	11.0	XNOR89	23-V	1.000	1.275E+00	11.0	XNLJ82
18-Ar	1.000	6.600E+00	25.0	XGLA76	22-Ti	1.320	5.430E+00	14.9	XNSO76	23-V	1.000	1.390E+00	9.4	XNMD75A
18-Ar	1.200	1.200E+01	25.0	XGLA76	22-Ti	1.400	5.724E+00	12.0	XNLJ82	23-V	1.000	1.210E+00	14.9	XNSO76
20-Ca	0.800	2.412E+00	15.0	XNAV81	22-Ti	1.400	6.570E+00	9.0	XNMD75A	23-V	1.100	2.070E+00	9.2	XNMD75A
20-Ca	0.800	1.506E+00	17.0	XNLJ82	22-Ti	1.400	6.292E+00	8.2	XNSI89	23-V	1.180	2.230E+00	10.0	XNOR89
20-Ca	0.900	2.770E+00	10.0	XNOR89	22-Ti	1.401	6.066E+00	5.0	XNSI84	23-V	1.200	2.419E+00	10.0	XNLJ82
20-Ca	1.000	5.004E+00	15.0	XNAV81	22-Ti	1.500	8.760E+00	30.0	XNLD73	23-V	1.200	2.690E+00	8.9	XNMD75A
20-Ca	1.000	3.342E+00	17.0	XNLJ82	22-Ti	1.500	8.420E+00	9.0	XNMD75A	23-V	1.300	3.660E+00	9.0	XNMD75A
20-Ca	1.180	6.910E+00	10.0	XNOR89	22-Ti	1.520	7.800E+00	15.0	XNSO76	23-V	1.300	3.300E+00	10.0	XNOR89
20-Ca	1.200	1.004E+01	15.0	XNAV81	22-Ti	1.600	9.149E+00	11.0	XNLJ82	23-V	1.400	4.200E+00	10.0	XNLJ82
20-Ca	1.200	6.683E+00	16.0	XNLJ82	22-Ti	1.600	1.030E+01	8.7	XNMD75A	23-V	1.400	4.900E+00	9.0	XNMD75A
20-Ca	1.300	1.000E+01	11.0	XNOR89	22-Ti	1.600	1.034E+01	8.2	XNSI89	23-V	1.500	4.300E+00	30.0	XKMK74
20-Ca	1.400	1.760E+01	15.0	XNAV81	22-Ti	1.601	9.592E+00	5.0	XNSI84	23-V	1.500	6.020E+00	9.0	XNMD75A
20-Ca	1.400	1.126E+01	16.0	XNLJ82	22-Ti	1.700	1.290E+01	9.3	XNMD75A	23-V	1.600	7.059E+00	10.0	XNLJ82
20-Ca	1.500	1.793E+01	30.0	XNLD73	22-Ti	1.720	1.150E+01	14.8	XNSO76	23-V	1.600	7.590E+00	9.0	XNMD75A
20-Ca	1.600	2.901E+01	15.0	XNAV81	22-Ti	1.800	1.326E+01	11.0	XNLJ82	23-V	1.700	9.340E+00	9.0	XNMD75A
20-Ca	1.600	1.760E+01	15.0	XNLJ82	22-Ti	1.800	1.470E+01	8.8	XNMD75A	23-V	1.800	1.032E+01	10.0	XNLJ82
20-Ca	1.800	4.271E+01	15.0	XNAV81	22-Ti	1.900	1.820E+01	8.8	XNMD75A	23-V	1.800	1.130E+01	8.8	XNMD75A
20-Ca	1.800	2.445E+01	15.0	XNLJ82	22-Ti	1.920	1.830E+01	14.8	XNSO76	23-V	1.900	1.370E+01	8.8	XNMD75A
20-Ca	2.000	6.145E+01	15.0	XNAV81	22-Ti	2.000	2.387E+01	30.3	XNLD73	23-V	2.000	9.790E+00	30.0	XKMK74
20-Ca	2.000	5.216E+01	30.0	XNLD73	22-Ti	2.000	2.170E+01	9.2	XNMD75A	23-V	2.000	1.620E+01	9.3	XNMD75A
20-Ca	2.200	8.623E+01	15.0	XNAV81	22-Ti	2.000	2.089E+01	8.2	XNSI89	23-V	2.100	1.900E+01	8.9	XNMD75A
20-Ca	2.400	1.053E+02	15.0	XNAV81	22-Ti	2.001	2.056E+01	5.0	XNSI84	23-V	2.200	2.130E+01	8.9	XNMD75A
20-Ca	2.600	1.408E+02	15.0	XNAV81	22-Ti	2.100	2.400E+01	9.2	XNMD75A	23-V	2.200	1.970E+01	15.2	XNSO76
20-Ca	2.800	1.695E+02	15.0	XNAV81	22-Ti	2.120	2.280E+01	14.9	XNSO76	23-V	2.300	2.600E+01	8.8	XNMD75A
20-Ca	3.000	1.826E+02	30.4	XNLD73	22-Ti	2.200	2.850E+01	9.1	XNMD75A	23-V	2.320	2.260E+01	15.0	XNSO76
20-Ca	4.000	3.977E+02	29.9	XNLD73	22-Ti	2.200	3.060E+01	14.1	XNSO76	23-V	2.400	2.870E+01	9.1	XNMD75A
20-Ca	5.000	6.846E+02	31.0	XNLD73	22-Ti	2.300	3.220E+01	9.0	XNMD75A	23-V	2.400	2.590E+01	15.1	XNSO76
20-Ca	6.000	9.454E+02	29.3	XNLD73	22-Ti	2.320	3.490E+01	14.9	XNSO76	23-V	2.500	1.620E+01	30.2	XKMK74
20-Ca	7.000	1.190E+03	30.1	XNLD73	22-Ti	2.400	3.700E+01	8.9	XNMD75A	23-V	2.520	2.960E+01	14.9	XNSO76
20-Ca	8.000	1.434E+03	30.7	XNLD73	22-Ti	2.400	3.724E+01	8.2	XNSI89	23-V	2.600	2.960E+01	14.9	XNSO76
20-Ca	9.000	1.663E+03	30.4	XNLD73	22-Ti	2.400	4.020E+01	14.9	XNSO76	23-V	2.720	3.700E+01	15.1	XNSO76
20-Ca	10.000	1.777E+03	30.3	XNLD73	22-Ti	2.401	3.701E+01	5.0	XNSI84	23-V	2.800	4.150E+01	14.9	XNSO76
20-Ca	11.000	1.989E+03	30.3	XNLD73	22-Ti	2.520	4.510E+01	15.1	XNSO76	23-V	2.920	4.540E+01	15.0	XNSO76
20-Ca	12.000	2.070E+03	29.9	XNLD73	22-Ti	2.600	4.670E+01	15.0	XNSO76	23-V	3.000	2.730E+01	30.0	XKMK74
20-Ca	20.000	3.179E+03	10.0	XNIS82	22-Ti	2.720	5.670E+01	15.0	XNSO76	23-V	3.000	5.000E+01	15.0	XNSO76
20-Ca	24.000	3.358E+03	10.0	XNIS82	22-Ti	2.800	6.380E+01	15.0	XNSO76	23-V	3.120	5.510E+01	15.1	XNSO76
20-Ca	28.000	3.505E+03	10.0	XNIS82	22-Ti	2.900	2.365E+01	4.6	XKKO68	23-V	3.200	5.930E+01	15.0	XNSO76
21-Sc	0.800	1.068E+00	14.0	XNLJ82	22-Ti	2.920	7.010E+01	15.0	XNSO76	23-V	3.320	6.460E+01	15.0	XNSO76
21-Sc	1.000	2.527E+00	14.0	XNLJ82	22-Ti	3.000	9.198E+01	31.0	XNLD73	23-V	3.400	6.860E+01	15.0	XNSO76
21-Sc	1.200	4.617E+00	13.0	XNLJ82	22-Ti	3.000	7.750E+01	15.0	XNSO76	23-V	3.500	4.000E+01	30.0	XKMK74

TABLE IId. Experimental Cross Sections for *K*-Shell X-Ray Production, Auger Electron Production, and Ionization by ^{4}He Ions

See page 84 for Explanation of Tables

TARGET	ENERGY [MeV]	SIGMA [barn]	ERROR [%]	REFERENCE	TARGET	ENERGY [MeV]	SIGMA [barn]	ERROR [%]	REFERENCE	TARGET	ENERGY [MeV]	SIGMA [barn]	ERROR [%]	REFERENCE
23-V	3.520	7.710E+01	15.0	XNSO76	25-Mn	1.520	2.500E+00	15.2	XNSO76	26-Fe	3.000	1.930E+01	15.0	XNSO76
23-V	3.600	8.370E+01	15.1	XNSO76	25-Mn	1.600	3.568E+00	8.0	XNLJ82	26-Fe	3.120	2.150E+01	14.9	XNSO76
23-V	3.720	9.010E+01	15.0	XNSO76	25-Mn	1.720	3.540E+00	15.0	XNSO76	26-Fe	3.200	2.340E+01	15.0	XNSO76
23-V	3.800	9.700E+01	15.1	XNSO76	25-Mn	1.800	5.352E+00	8.0	XNLJ82	26-Fe	3.320	2.650E+01	15.1	XNSO76
23-V	3.920	1.050E+02	15.2	XNSO76	25-Mn	1.920	6.270E+00	15.0	XNSO76	26-Fe	3.400	2.990E+01	15.1	XNSO76
23-V	4.000	5.820E+01	30.1	XKMK74	25-Mn	2.000	8.258E+00	30.0	XNLD73	26-Fe	3.500	3.030E+01	14.9	XNMK74
23-V	4.000	1.120E+02	15.2	XNSO76	25-Mn	2.200	1.080E+01	14.8	XNSO76	26-Fe	3.520	3.330E+01	15.0	XNSO76
23-V	4.120	1.110E+02	15.3	XNSO76	25-Mn	2.320	1.230E+01	14.6	XNSO76	26-Fe	3.550	1.950E+01	5.0	XKKO68
23-V	4.200	1.250E+02	15.2	XNSO76	25-Mn	2.400	1.350E+01	14.8	XNSO76	26-Fe	3.600	3.590E+01	15.0	XNSO76
23-V	4.320	1.360E+02	14.7	XNSO76	25-Mn	2.520	1.610E+01	14.9	XNSO76	26-Fe	3.720	3.890E+01	14.9	XNSO76
23-V	4.400	1.380E+02	15.2	XNSO76	25-Mn	2.600	1.590E+01	15.1	XNSO76	26-Fe	3.800	4.140E+01	15.0	XNSO76
23-V	4.500	7.590E+01	30.0	XKMK74	25-Mn	2.720	1.980E+01	15.2	XNSO76	26-Fe	3.900	2.491E+01	5.0	XKKO68
23-V	20.000	2.044E+03	10.0	XNIS82	25-Mn	2.800	2.220E+01	14.9	XNSO76	26-Fe	3.920	4.770E+01	15.1	XNSO76
23-V	24.000	2.381E+03	10.0	XNIS82	25-Mn	2.920	2.470E+01	15.0	XNSO76	26-Fe	4.000	6.330E+01	40.0	XKBM78
23-V	28.000	2.678E+03	10.0	XNIS82	25-Mn	3.000	3.391E+01	29.6	XNLD73	26-Fe	4.000	4.310E+01	15.1	XNMK74
24-Cr	0.500	5.300E-02	9.1	XNMD75A	25-Mn	3.000	2.880E+01	14.9	XNSO76	26-Fe	4.000	4.940E+01	15.0	XNSO76
24-Cr	0.600	1.250E-01	8.8	XNMD75A	25-Mn	3.120	3.010E+01	15.0	XNSO76	26-Fe	4.120	5.210E+01	15.0	XNSO76
24-Cr	0.700	2.380E-01	8.8	XNMD75A	25-Mn	3.200	3.190E+01	15.0	XNSO76	26-Fe	4.200	5.660E+01	15.0	XNSO76
24-Cr	0.700	1.880E-01	11.0	XNOR89	25-Mn	3.320	3.530E+01	15.0	XNSO76	26-Fe	4.320	6.000E+01	15.0	XNSO76
24-Cr	0.800	4.019E-01	10.0	XNLJ82	25-Mn	3.400	3.840E+01	15.1	XNSO76	26-Fe	4.400	6.120E+01	15.0	XNSO76
24-Cr	0.800	4.250E-01	8.9	XNMD75A	25-Mn	3.520	4.250E+01	15.1	XNSO76	26-Fe	4.450	3.574E+01	4.9	XKKO68
24-Cr	0.900	1.100E+00	9.1	XNKL76	25-Mn	3.600	4.600E+01	15.0	XNSO76	26-Fe	4.500	5.680E+01	15.0	XNMK74
24-Cr	0.900	6.610E-01	8.9	XNMD75A	25-Mn	3.720	4.940E+01	15.0	XNSO76	26-Fe	4.800	4.407E+01	4.7	XKKO68
24-Cr	0.900	5.400E-01	11.0	XNOR89	25-Mn	3.800	5.430E+01	14.9	XNSO76	26-Fe	5.000	8.410E+01	15.0	XNMK74
24-Cr	1.000	9.169E-01	10.0	XNLJ82	25-Mn	3.920	5.570E+01	15.1	XNSO76	26-Fe	5.300	5.795E+01	4.8	XKKO68
24-Cr	1.000	1.030E+00	8.7	XNMD75A	25-Mn	4.000	8.541E+01	30.1	XNLD73	26-Fe	10.000	4.510E+02	10.0	XNPE79
24-Cr	1.100	2.100E+00	14.3	XNKL76	25-Mn	4.000	6.200E+01	15.0	XNSO76	26-Fe	12.000	6.590E+02	10.0	XNPE79
24-Cr	1.100	1.450E+00	9.0	XNMD75A	25-Mn	4.120	6.740E+01	15.0	XNSO76	26-Fe	14.000	8.740E+02	10.0	XNPE79
24-Cr	1.180	1.440E+00	10.0	XNOR89	25-Mn	4.200	7.190E+01	15.0	XNSO76	26-Fe	16.000	1.051E+03	10.0	XNPE79
24-Cr	1.200	1.797E+00	9.0	XNLJ82	25-Mn	4.320	7.760E+01	14.9	XNSO76	26-Fe	18.000	1.343E+03	10.0	XNPE79
24-Cr	1.200	1.960E+00	9.2	XNMD75A	25-Mn	4.400	8.360E+01	15.0	XNSO76	26-Fe	20.000	1.353E+03	10.0	XNPE79
24-Cr	1.300	4.200E+00	14.3	XNKL76	25-Mn	5.000	1.633E+02	30.8	XNLD73	26-Fe	22.000	1.562E+03	10.0	XNPE79
24-Cr	1.300	2.670E+00	9.0	XNMD75A	25-Mn	6.000	2.606E+02	30.1	XNLD73	26-Fe	24.000	1.704E+03	10.0	XNPE79
24-Cr	1.300	2.110E+00	10.0	XNOR89	25-Mn	7.000	3.768E+02	30.0	XNLD73	26-Fe	26.000	1.804E+03	10.0	XNPE79
24-Cr	1.400	3.056E+00	9.0	XNLJ82	25-Mn	8.000	5.087E+02	30.2	XNLD73	26-Fe	28.000	1.853E+03	10.0	XNPE79
24-Cr	1.400	3.510E+00	9.1	XNMD75A	25-Mn	9.000	6.154E+02	30.1	XNLD73	26-Fe	30.000	2.301E+03	6.0	XNHW73
24-Cr	1.500	5.800E+00	13.8	XNKL76	25-Mn	10.000	7.410E+02	30.1	XNLD73	26-Fe	30.000	1.846E+03	10.0	XNPE79
24-Cr	1.500	6.796E+00	29.9	XNLD73	25-Mn	11.100	9.043E+02	29.9	XNLD73	26-Fe	40.000	2.582E+03	6.0	XNHW73
24-Cr	1.500	4.240E+00	9.0	XNMD75A	25-Mn	12.000	1.017E+03	29.9	XNLD73	26-Fe	50.000	2.752E+03	6.1	XNHW73
24-Cr	1.600	4.896E+00	8.0	XNLJ82	26-Fe	0.235	1.108E-04	23.0	XNBG80	26-Fe	60.000	2.811E+03	6.0	XNHW73
24-Cr	1.600	5.240E+00	9.0	XNMD75A	26-Fe	0.255	2.380E-04	22.0	XNBG80	26-Fe	70.000	2.800E+03	5.9	XNHW73
24-Cr	1.700	1.000E+01	12.0	XNKL76	26-Fe	0.300	8.130E-04	21.0	XNBG80	26-Fe	80.000	2.700E+03	6.0	XNHW73
24-Cr	1.700	6.970E+00	9.0	XNMD75A	26-Fe	0.350	2.480E-03	20.0	XNBG80	27-Co	0.700	7.600E-02	21.0	XNOR89
24-Cr	1.800	6.962E+00	8.0	XNLJ82	26-Fe	0.450	1.095E-02	20.0	XNBG80	27-Co	0.800	1.362E-01	9.0	XNLJ82
24-Cr	1.800	8.280E+00	9.1	XNMD75A	26-Fe	0.500	2.100E-02	9.0	XNMD75A	27-Co	0.900	2.150E-01	15.0	XNOR89
24-Cr	1.900	1.400E+01	10.0	XNKL76	26-Fe	0.550	3.040E-02	20.0	XNBG80	27-Co	1.000	3.298E-01	9.0	XNLJ82
24-Cr	1.900	9.920E+00	9.0	XNMD75A	26-Fe	0.600	4.400E-02	9.0	XNMD75A	27-Co	1.000	3.990E-01	15.0	XNSO76
24-Cr	2.000	1.523E+01	29.6	XNLD73	26-Fe	0.650	6.320E-02	20.0	XNBG80	27-Co	1.120	6.190E-01	15.0	XNSO76
24-Cr	2.000	1.150E+01	8.7	XNMD75A	26-Fe	0.700	8.700E-02	9.0	XNMD75A	27-Co	1.180	6.160E-01	10.0	XNOR89
24-Cr	2.100	2.100E+01	11.9	XNKL76	26-Fe	0.700	7.900E-02	19.0	XNBG80	27-Co	1.200	6.771E-01	8.0	XNLJ82
24-Cr	2.100	1.350E+01	8.9	XNMD75A	26-Fe	0.750	1.240E-01	20.0	XNBG80	27-Co	1.300	1.070E+00	10.0	XNOR89
24-Cr	2.200	1.580E+01	8.9	XNMD75A	26-Fe	0.800	2.059E-01	9.0	XNLJ82	27-Co	1.320	9.750E-01	15.0	XNSO76
24-Cr	2.300	2.570E+01	12.1	XNKL76	26-Fe	0.800	1.690E-01	9.0	XNMD75A	27-Co	1.400	1.215E+00	8.0	XNLJ82
24-Cr	2.300	1.830E+01	8.7	XNMD75A	26-Fe	0.900	2.820E-01	9.0	XNMD75A	27-Co	1.520	1.470E+00	15.0	XNSO76
24-Cr	2.400	1.990E+01	9.0	XNMD75A	26-Fe	0.900	2.910E-01	10.0	XNOR89	27-Co	1.600	1.907E+00	7.0	XNLJ82
24-Cr	2.500	3.340E+01	12.0	XNKL76	26-Fe	1.000	4.856E-01	9.0	XNLJ82	27-Co	1.720	2.120E+00	15.1	XNSO76
24-Cr	2.700	4.720E+01	12.1	XNKL76	26-Fe	1.000	4.280E-01	9.0	XNMD75A	27-Co	1.800	2.807E+00	7.0	XNLJ82
24-Cr	2.900	1.509E+01	5.0	XKKO68	26-Fe	1.000	4.140E-01	15.0	XNMK74	27-Co	1.920	3.390E+00	15.0	XNSO76
24-Cr	2.900	5.430E+01	12.0	XNKL76	26-Fe	1.000	4.530E-01	15.0	XNSO76	27-Co	2.200	5.310E+00	15.1	XNSO76
24-Cr	3.000	5.302E+01	29.8	XNLD73	26-Fe	1.100	6.170E-01	9.0	XNMD75A	27-Co	2.320	6.000E+00	15.0	XNSO76
24-Cr	3.100	7.480E+01	12.2	XNKL76	26-Fe	1.180	9.200E-01	10.0	XNOR89	27-Co	2.400	6.800E+00	15.0	XNSO76
24-Cr	3.300	8.890E+01	9.6	XNKL76	26-Fe	1.200	1.012E+00	8.0	XNLJ82	27-Co	2.520	7.830E+00	14.9	XNSO76
24-Cr	3.500	1.140E+02	12.3	XNKL76	26-Fe	1.200	8.450E-01	9.0	XNMD75A	27-Co	2.600	9.690E+00	15.0	XNSO76
24-Cr	3.550	2.561E+01	5.0	XKKO68	26-Fe	1.300	1.130E+00	9.0	XNMD75A	27-Co	2.720	9.860E+00	15.0	XNSO76
24-Cr	3.700	1.220E+02	12.3	XNKL76	26-Fe	1.300	1.890E+00	10.0	XNOR89	27-Co	2.800	1.110E+01	15.3	XNSO76
24-Cr	3.900	3.412E+01	5.0	XKKO68	26-Fe	1.400	1.618E+00	8.0	XNLJ82	27-Co	2.920	1.230E+01	14.6	XNSO76
24-Cr	3.900	1.560E+02	12.2	XNKL76	26-Fe	1.400	1.490E+00	9.0	XNMD75A	27-Co	3.000	1.360E+01	14.7	XNSO76
24-Cr	4.000	1.100E+02	30.8	XNLD73	26-Fe	1.500	1.860E+00	9.0	XNMD75A	27-Co	3.120	1.480E+01	14.9	XNSO76
24-Cr	4.100	1.920E+02	12.0	XNKL76	26-Fe	1.500	1.760E+00	14.8	XNMK74	27-Co	3.200	1.620E+01	14.8	XNSO76
24-Cr	4.450	5.076E+01	5.0	XKKO68	26-Fe	1.600	2.695E+00	8.0	XNLJ82	27-Co	3.320	1.790E+01	15.1	XNSO76
24-Cr	4.800	5.809E+01	4.9	XKKO68	26-Fe	1.600	2.320E+00	9.0	XNMD75A	27-Co	3.400	1.970E+01	15.2	XNSO76
24-Cr	5.000	2.059E+02	30.1	XNLD73	26-Fe	1.700	2.870E+00	9.0	XNMD75A	27-Co	3.520	2.160E+01	14.8	XNSO76
24-Cr	5.300	8.122E+01	4.9	XKKO68	26-Fe	1.800	4.070E+00	7.0	XNLJ82	27-Co	3.600	2.310E+01	15.2	XNSO76
24-Cr	6.000	3.271E+02	30.2	XNLD73	26-Fe	1.800	3.420E+00	9.0	XNMD75A	27-Co	3.720	2.520E+01	15.1	XNSO76
24-Cr	7.000	4.625E+02	29.9	XNLD73	26-Fe	1.900	4.320E+00	9.0	XNMD75A	27-Co	3.800	2.720E+01	15.1	XNSO76
24-Cr	8.000	6.007E+02	30.0	XNLD73	26-Fe	2.000	4.890E+00	9.0	XNMD75A	27-Co	3.920	2.950E+01	14.9	XNSO76
24-Cr	9.000	7.473E+02	29.8	XNLD73	26-Fe	2.000	4.860E+00	15.0	XNMK74	27-Co	4.000	3.220E+01	14.9	XNSO76
24-Cr	10.000	8.855E+02	29.9	XNLD73	26-Fe	2.100	5.740E+00	9.0	XNMD75A	27-Co	4.120	3.400E+01	15.0	XNSO76
24-Cr	11.000	9.870E+02	31.4	XNLD73	26-Fe	2.200	6.740E+00	9.0	XNMD75A	27-Co	4.200	3.690E+01	14.9	XNSO76
24-Cr	12.000	1.156E+03	29.3	XNLD73	26-Fe	2.200	7.480E+00	15.0	XNSO76	27-Co	4.320	3.920E+01	15.1	XNSO76
24-Cr	20.000	1.858E+03	10.0	XNIS82	26-Fe	2.300	7.500E+00	9.0	XNMD75A	27-Co	4.400	4.350E+01	14.9	XNSO76
24-Cr	24.000	2.185E+03	10.0	XNIS82	26-Fe	2.320	8.660E+00	15.0	XNSO76	28-Ni	0.200	6.900E-06	50.0	XKLL80
24-Cr	28.000	2.185E+03	10.0	XNIS82	26-Fe	2.400	8.440E+00	9.0	XNMD75A	28-Ni	0.210	1.100E-05	50.0	XKLL80
25-Mn	0.800	2.698E-01	10.0	XNLJ82	26-Fe	2.400	9.860E+00	15.0	XNSO76	28-Ni	0.220	1.900E-05	25.0	XKLL80
25-Mn	1.000	7.442E-01	30.0	XNLD73	26-Fe	2.500	1.020E+01	14.7	XNMK74	28-Ni	0.240	4.700E-05	25.0	XKLL80
25-Mn	1.000	6.730E-01	10.0	XNLJ82	26-Fe	2.520	1.100E+01	15.5	XNSO76	28-Ni	0.260	8.500E-05	25.0	XKLL80
25-Mn	1.000	8.140E-01	15.0	XNSO76	26-Fe	2.600	1.170E+01	15.4	XNSO76	28-Ni	0.280	1.500E-04	25.0	XKLL80
25-Mn	1.120	1.020E+00	14.7	XNSO76	26-Fe	2.720	1.410E+01	14.9	XNSO76	28-Ni	0.300	2.500E-04	25.0	XKLL80
25-Mn	1.200	1.315E+00	9.0	XNLJ82	26-Fe	2.800	1.590E+01	15.1	XNSO76	28-Ni	0.570	2.100E-02	9.0	XNMD75A
25-Mn	1.320	1.660E+00	15.1	XNSO76	26-Fe	2.900	1.156E+01	5.1	XKKO68	28-Ni	0.670	5.100E-02	9.0	XNMD75A
25-Mn	1.400	2.204E+00	9.0	XNLJ82	26-Fe	2.920	1.790E+01	15.1	XNSO76	28-Ni	0.700	5.300E-02	30.0	XNOR89
25-Mn	1.500	2.920E+00	30.1	XNLD73	26-Fe	3.000	1.940E+01	14.9	XNMK74	28-Ni	0.770	8.100E-02	9.0	XNMD75A

TABLE IId. Experimental Cross Sections for *K*-Shell X-Ray Production, Auger Electron Production, and Ionization by ^{4}He Ions

See page 84 for Explanation of Tables

TARGET	ENERGY [MeV]	SIGMA [barn]	ERROR [%]	REFERENCE	TARGET	ENERGY [MeV]	SIGMA [barn]	ERROR [%]	REFERENCE	TARGET	ENERGY [MeV]	SIGMA [barn]	ERROR [%]	REFERENCE
28-Ni	0.800	9.439E-02	10.0	XNLJ82	29-Cu	0.700	2.270E-02	12.0	XNSI57	29-Cu	2.500	4.210E+00	15.0	XNMK74
28-Ni	0.870	1.380E-01	9.0	XNMD75A	29-Cu	0.700	3.238E-02	8.2	XNSI89	29-Cu	2.520	4.220E+00	14.9	XNSO76
28-Ni	0.900	1.560E-01	21.0	XNOR89	29-Cu	0.710	3.310E-02	32.0	XKBR78	29-Cu	2.600	8.680E+00	13.8	XNKL76
28-Ni	0.970	2.110E-01	9.0	XNMD75A	29-Cu	0.720	3.690E-02	12.4	XNRI81	29-Cu	2.600	4.700E+00	15.1	XNSO76
28-Ni	1.000	2.422E-01	9.0	XNLJ82	29-Cu	0.770	4.840E-02	20.0	XNBG80	29-Cu	2.720	5.450E+00	15.0	XNSO76
28-Ni	1.000	2.070E-01	15.0	XNSO76	29-Cu	0.800	7.964E-02	15.0	XNBH82	29-Cu	2.800	1.120E+01	13.4	XNKL76
28-Ni	1.070	3.210E-01	9.0	XNMD75A	29-Cu	0.800	6.291E-02	9.0	XNLJ82	29-Cu	2.800	6.120E+00	15.0	XNSO76
28-Ni	1.170	4.490E-01	9.0	XNMD75A	29-Cu	0.800	5.700E-02	9.0	XNMD75A	29-Cu	2.900	3.578E+00	5.0	XKKO68
28-Ni	1.180	4.560E-01	14.0	XNOR89	29-Cu	0.800	5.810E-02	12.4	XNRI81	29-Cu	2.920	6.840E+00	15.1	XNSO76
28-Ni	1.200	5.216E-01	9.0	XNLJ82	29-Cu	0.800	3.720E-02	12.0	XNSI57	29-Cu	3.000	1.370E+01	13.1	XNKL76
28-Ni	1.280	6.340E-01	9.0	XNMD75A	29-Cu	0.800	6.600E-02	8.2	XNSI89	29-Cu	3.000	8.150E+00	15.0	XNMK74
28-Ni	1.300	6.800E-01	10.0	XNOR89	29-Cu	0.800	6.631E-02	5.0	XNSI84	29-Cu	3.000	7.670E+00	15.0	XNSO76
28-Ni	1.380	8.350E-01	9.0	XNMD75A	29-Cu	0.900	1.263E-01	15.0	XNBH82	29-Cu	3.120	8.320E+00	15.0	XNSO76
28-Ni	1.400	9.936E-01	8.0	XNLJ82	29-Cu	0.900	1.600E-01	12.5	XNKL76	29-Cu	3.200	1.820E+01	13.2	XNKL76
28-Ni	1.480	1.070E+00	9.0	XNMD75A	29-Cu	0.900	9.700E-02	9.0	XNMD75A	29-Cu	3.200	8.990E+00	15.0	XNSO76
28-Ni	1.580	1.390E+00	9.0	XNMD75A	29-Cu	0.900	1.030E-01	19.0	XNOR89	29-Cu	3.320	9.800E+00	15.0	XNSO76
28-Ni	1.600	1.610E+00	7.0	XNLJ82	29-Cu	0.900	7.120E-02	12.0	XNSI57	29-Cu	3.400	2.110E+01	13.3	XNKL76
28-Ni	1.680	1.730E+00	9.0	XNMD75A	29-Cu	0.900	1.008E-01	8.2	XNSI89	29-Cu	3.400	1.080E+01	14.8	XNSO76
28-Ni	1.780	2.100E+00	9.0	XNMD75A	29-Cu	0.910	9.070E-02	32.0	XKBR78	29-Cu	3.500	1.320E+01	15.2	XNMK74
28-Ni	1.800	2.381E+00	7.0	XNLJ82	29-Cu	1.000	2.165E-01	15.0	XNBH82	29-Cu	3.520	1.210E+01	14.9	XNSO76
28-Ni	1.880	2.520E+00	9.0	XNMD75A	29-Cu	1.000	1.600E-01	12.5	XNKL76	29-Cu	3.550	6.408E+00	4.9	XKKO68
28-Ni	1.980	3.030E+00	9.0	XNMD75A	29-Cu	1.000	1.670E-01	9.0	XNLJ82	29-Cu	3.600	2.820E+01	12.4	XNKL76
28-Ni	2.080	3.580E+00	9.0	XNMD75A	29-Cu	1.000	1.570E-01	9.0	XNMD75A	29-Cu	3.600	1.350E+01	14.8	XNSO76
28-Ni	2.180	4.290E+00	9.0	XNMD75A	29-Cu	1.000	1.400E-01	15.0	XNMK74	29-Cu	3.720	1.460E+01	15.1	XNSO76
28-Ni	2.200	3.950E+00	14.9	XNSO76	29-Cu	1.000	1.420E-01	12.4	XNRI81	29-Cu	3.800	3.220E+01	13.0	XNKL76
28-Ni	2.280	5.060E+00	9.0	XNMD75A	29-Cu	1.000	1.038E-01	12.0	XNSI57	29-Cu	3.800	1.590E+01	15.1	XNSO76
28-Ni	2.320	4.520E+00	15.0	XNSO76	29-Cu	1.000	1.720E-01	8.2	XNSI89	29-Cu	3.900	8.722E+00	5.1	XKKO68
28-Ni	2.380	5.830E+00	9.0	XNMD75A	29-Cu	1.000	2.010E-01	14.9	XNSO76	29-Cu	3.920	1.670E+01	15.0	XNSO76
28-Ni	2.400	4.970E+00	15.1	XNSO76	29-Cu	1.001	1.713E-01	5.0	XNSI84	29-Cu	4.000	3.690E+01	13.0	XNKL76
28-Ni	2.480	6.620E+00	9.0	XNMD75A	29-Cu	1.100	3.137E-01	15.0	XNBH82	29-Cu	4.000	1.940E+01	14.9	XNMK74
28-Ni	2.520	5.800E+00	15.0	XNSO76	29-Cu	1.100	2.340E-01	9.0	XNMD75A	29-Cu	4.000	1.850E+01	15.1	XNSO76
28-Ni	2.600	6.660E+00	15.0	XNSO76	29-Cu	1.110	1.930E-01	32.0	XKBR78	29-Cu	4.120	1.970E+01	15.2	XNSO76
28-Ni	2.720	7.320E+00	15.0	XNSO76	29-Cu	1.120	2.700E-01	15.2	XNSO76	29-Cu	4.200	2.150E+01	14.9	XNSO76
28-Ni	2.800	8.210E+00	15.0	XNSO76	29-Cu	1.180	3.060E-01	15.0	XNOR89	29-Cu	4.320	2.270E+01	15.0	XNSO76
28-Ni	2.900	8.363E+00	5.0	XKKO68	29-Cu	1.200	4.488E-01	15.0	XNBH82	29-Cu	4.400	2.570E+01	15.2	XNSO76
28-Ni	2.920	9.080E+00	15.0	XNSO76	29-Cu	1.200	5.100E-01	13.7	XNKL76	29-Cu	4.450	1.268E+01	4.9	XKKO68
28-Ni	3.000	1.020E+01	14.7	XNSO76	29-Cu	1.200	3.500E-01	8.0	XNLJ82	29-Cu	4.500	2.770E+01	15.2	XNMK74
28-Ni	3.120	1.100E+01	15.5	XNSO76	29-Cu	1.200	3.330E-01	9.0	XNMD75A	29-Cu	4.800	1.682E+01	5.0	XKKO68
28-Ni	3.200	1.200E+01	15.0	XNSO76	29-Cu	1.200	2.730E-01	12.4	XNRI81	29-Cu	5.000	3.680E+01	14.9	XNMK74
28-Ni	3.320	1.340E+01	14.9	XNSO76	29-Cu	1.200	3.634E-01	8.2	XNSI89	29-Cu	5.200	6.400E+01	7.7	XNBA77
28-Ni	3.400	1.460E+01	15.1	XNSO76	29-Cu	1.201	3.627E-01	5.0	XNSI84	29-Cu	5.300	2.688E+01	5.0	XKKO68
28-Ni	3.520	1.610E+01	14.9	XNSO76	29-Cu	1.300	6.336E-01	15.0	XNBH82	29-Cu	10.000	2.360E+02	10.0	XNPE79
28-Ni	3.550	1.271E+01	4.9	XKKO68	29-Cu	1.300	4.680E-01	9.0	XNMD75A	29-Cu	12.000	3.150E+02	10.0	XNFB74
28-Ni	3.600	1.760E+01	14.8	XNSO76	29-Cu	1.300	4.950E-01	10.0	XNOR89	29-Cu	12.000	3.650E+02	10.0	XNPE79
28-Ni	3.720	1.920E+01	15.1	XNSO76	29-Cu	1.310	3.790E-01	32.0	XKBR78	29-Cu	14.000	4.900E+02	10.0	XNPE79
28-Ni	3.800	2.090E+01	14.8	XNSO76	29-Cu	1.320	4.640E-01	15.1	XNSO76	29-Cu	16.000	6.360E+02	10.0	XNPE79
28-Ni	3.900	1.735E+01	5.0	XKKO68	29-Cu	1.400	7.876E-01	15.0	XNBH82	29-Cu	16.500	6.610E+02	6.5	XNBA77
28-Ni	3.920	2.170E+01	15.2	XNSO76	29-Cu	1.400	9.200E-01	13.0	XNKL76	29-Cu	18.000	8.140E+02	10.0	XNPE79
28-Ni	4.000	2.430E+01	10.0	XNBB78	29-Cu	1.400	6.069E-01	8.0	XNLJ82	29-Cu	20.000	9.300E+02	10.0	XNFB74
28-Ni	4.000	2.430E+01	14.8	XNSO76	29-Cu	1.400	6.120E-01	9.0	XNMD75A	29-Cu	20.000	8.370E+02	10.0	XNPE79
28-Ni	4.120	2.610E+01	14.9	XNSO76	29-Cu	1.400	6.600E-01	8.2	XNSI89	29-Cu	22.000	9.750E+02	10.0	XNPE79
28-Ni	4.200	2.810E+01	14.9	XNSO76	29-Cu	1.401	6.586E-01	5.0	XNSI84	29-Cu	24.000	1.086E+03	10.0	XNPE79
28-Ni	4.320	2.970E+01	15.2	XNSO76	29-Cu	1.410	4.920E-01	32.0	XKBR78	29-Cu	26.000	1.184E+03	10.0	XNPE79
28-Ni	4.400	3.240E+01	15.1	XNSO76	29-Cu	1.500	6.350E-01	32.0	XKBR78	29-Cu	27.000	1.080E+03	7.4	XNBA77
28-Ni	4.450	2.099E+01	4.9	XKKO68	29-Cu	1.500	1.021E+00	15.0	XNBH82	29-Cu	28.000	1.335E+03	10.0	XNPE79
28-Ni	4.800	2.480E+01	5.0	XKKO68	29-Cu	1.500	8.160E-01	9.0	XNMD75A	29-Cu	30.000	1.380E+03	5.5	XNHW73
28-Ni	5.200	5.690E+01	10.2	XNBA77	29-Cu	1.500	6.560E-01	14.9	XNMK74	29-Cu	30.000	1.415E+03	10.0	XNPE79
28-Ni	5.300	3.209E+01	5.0	XKKO68	29-Cu	1.520	7.090E-01	15.0	XNSO76	29-Cu	40.000	1.682E+03	6.1	XNHW73
28-Ni	6.000	8.130E+01	10.0	XNBB78	29-Cu	1.600	1.192E+00	15.0	XNBH82	29-Cu	50.000	1.909E+03	6.1	XNHW73
28-Ni	8.000	1.780E+02	10.0	XNBB78	29-Cu	1.600	1.480E+00	19.6	XNKL76	29-Cu	60.000	2.051E+03	6.1	XNHW73
28-Ni	10.000	2.820E+02	10.0	XNBB78	29-Cu	1.600	1.001E+00	7.0	XNLJ82	29-Cu	70.000	2.132E+03	6.1	XNHW73
28-Ni	12.000	4.110E+02	10.0	XNBB78	29-Cu	1.600	1.020E+00	9.0	XNMD75A	29-Cu	80.000	2.149E+03	6.0	XNHW73
28-Ni	16.000	6.160E+02	10.0	XNBB78	29-Cu	1.600	1.030E+00	8.2	XNSI89	29-Cu	90.000	2.100E+03	5.9	XNHW73
28-Ni	16.500	6.620E+02	10.4	XNBA77	29-Cu	1.601	1.028E+00	5.0	XNSI84	29-Cu	100.000	2.051E+03	6.1	XNHW73
28-Ni	20.000	8.770E+02	10.0	XNBB78	29-Cu	1.700	1.417E+00	15.0	XNBH82	30-Zn	0.900	7.000E-02	30.0	XNOR89
28-Ni	24.000	1.080E+03	10.0	XNBB78	29-Cu	1.700	1.220E+00	9.0	XNMD75A	30-Zn	1.000	1.040E-01	15.4	XNSO76
28-Ni	27.000	1.180E+03	10.2	XNBA77	29-Cu	1.710	1.050E+00	32.0	XKBR78	30-Zn	1.160	2.170E-01	9.2	XNMD75A
28-Ni	28.000	1.250E+03	10.0	XNBB78	29-Cu	1.720	1.020E+00	14.7	XNSO76	30-Zn	1.180	2.310E-01	17.0	XNOR89
28-Ni	30.000	1.300E+03	10.0	XNBB78	29-Cu	1.800	1.698E+00	15.0	XNBH82	30-Zn	1.300	3.670E-01	12.0	XNOR89
28-Ni	30.000	1.569E+03	6.1	XNHW73	29-Cu	1.800	2.360E+00	13.1	XNKL76	30-Zn	1.380	4.030E-01	8.9	XNMD75A
28-Ni	40.000	1.900E+03	6.1	XNHW73	29-Cu	1.800	1.506E+00	6.0	XNLJ82	30-Zn	1.500	6.850E-01	50.3	XNLD73
28-Ni	50.000	2.161E+03	5.9	XNHW73	29-Cu	1.800	1.590E+00	9.0	XNMD75A	30-Zn	1.560	6.600E-01	8.9	XNMD75A
28-Ni	60.000	2.281E+03	6.0	XNHW73	29-Cu	1.900	1.710E+00	9.0	XNMD75A	30-Zn	1.760	1.030E+00	8.7	XNMD75A
28-Ni	70.000	2.339E+03	6.0	XNHW73	29-Cu	1.920	1.850E+00	15.1	XNSO76	30-Zn	1.960	1.540E+00	9.1	XNMD75A
28-Ni	80.000	2.302E+03	5.9	XNHW73	29-Cu	2.000	3.300E+00	13.0	XNKL76	30-Zn	2.000	1.916E+00	50.0	XNLD73
28-Ni	90.000	2.240E+03	6.1	XNHW73	29-Cu	2.000	2.220E+00	9.0	XNMD75A	30-Zn	2.160	2.120E+00	9.0	XNMD75A
28-Ni	100.000	2.140E+03	6.0	XNHW73	29-Cu	2.000	1.940E+00	14.9	XNMK74	30-Zn	2.200	2.110E+00	15.2	XNSO76
29-Cu	0.320	1.580E-04	12.4	XNRI81	29-Cu	2.000	2.266E+00	8.2	XNSI89	30-Zn	2.320	2.380E+00	15.1	XNSO76
29-Cu	0.325	2.540E-04	23.0	XNBG80	29-Cu	2.001	2.283E+00	5.0	XNSI84	30-Zn	2.360	2.960E+00	9.1	XNMD75A
29-Cu	0.360	5.830E-04	12.4	XNRI81	29-Cu	2.100	2.690E+00	9.0	XNMD75A	30-Zn	2.400	2.980E+00	13.4	XNSO76
29-Cu	0.370	7.520E-04	22.0	XNBG80	29-Cu	2.120	2.810E+00	14.9	XNSO76	30-Zn	2.460	3.550E+00	9.0	XNMD75A
29-Cu	0.400	1.120E-03	12.4	XNRI81	29-Cu	2.200	4.650E+00	13.3	XNKL76	30-Zn	2.520	3.050E+00	15.1	XNSO76
29-Cu	0.420	1.840E-03	21.0	XNBG80	29-Cu	2.200	3.120E+00	9.0	XNMD75A	30-Zn	2.600	3.390E+00	15.0	XNSO76
29-Cu	0.470	3.690E-03	20.0	XNBG80	29-Cu	2.200	3.020E+00	14.9	XNSO76	30-Zn	2.720	3.940E+00	15.0	XNSO76
29-Cu	0.480	3.920E-03	12.4	XNRI81	29-Cu	2.300	3.600E+00	9.0	XNMD75A	30-Zn	2.800	4.440E+00	15.1	XNSO76
29-Cu	0.560	9.990E-03	12.4	XNRI81	29-Cu	2.320	3.380E+00	15.1	XNSO76	30-Zn	2.920	5.010E+00	15.0	XNSO76
29-Cu	0.570	1.120E-02	20.0	XNBG80	29-Cu	2.400	6.250E+00	12.8	XNKL76	30-Zn	3.000	8.095E+00	50.3	XNLD73
29-Cu	0.600	1.100E-02	9.0	XNMD75A	29-Cu	2.400	4.250E+00	9.0	XNMD75A	30-Zn	3.000	5.580E+00	15.1	XNSO76
29-Cu	0.640	2.070E-02	12.4	XNRI81	29-Cu	2.400	4.132E+00	8.2	XNSI89	30-Zn	3.120	6.240E+00	15.1	XNSO76
29-Cu	0.670	2.620E-02	20.0	XNBG80	29-Cu	2.400	3.740E+00	15.0	XNSO76	30-Zn	3.200	6.420E+00	15.0	XNSO76
29-Cu	0.700	2.800E-02	9.0	XNMD75A	29-Cu	2.401	4.187E+00	5.0	XNSI84	30-Zn	3.320	7.110E+00	15.0	XNSO76
29-Cu	0.700	3.300E-02	30.0	XNOR89	29-Cu	2.500	4.750E+00	9.0	XNMD75A	30-Zn	3.400	7.740E+00	15.0	XNSO76

TABLE IId. Experimental Cross Sections for *K*-Shell X-Ray Production, Auger Electron Production, and Ionization by ^{4}He Ions

See page 84 for Explanation of Tables

TARGET	ENERGY [MeV]	SIGMA [barn]	ERROR [%]	REFERENCE
30-Zn	3.520	8.440E+00	15.0	XNSO76
30-Zn	3.600	9.420E+00	15.0	XNSO76
30-Zn	3.720	1.040E+01	15.4	XNSO76
30-Zn	3.800	1.130E+01	15.0	XNSO76
30-Zn	3.920	1.140E+01	14.9	XNSO76
30-Zn	4.000	1.964E+01	51.2	XNLD73
30-Zn	4.000	1.230E+01	14.6	XNSO76
30-Zn	4.120	1.430E+01	14.7	XNSO76
30-Zn	4.200	1.540E+01	14.9	XNSO76
30-Zn	4.320	1.650E+01	15.2	XNSO76
30-Zn	4.400	1.810E+01	14.9	XNSO76
30-Zn	5.000	4.024E+01	50.0	XNLD73
30-Zn	6.000	6.275E+01	49.6	XNLD73
30-Zn	7.000	9.436E+01	49.7	XNLD73
30-Zn	8.000	1.341E+02	50.0	XNLD73
30-Zn	9.000	1.820E+02	50.0	XNLD73
30-Zn	10.000	2.251E+02	48.9	XNLD73
30-Zn	11.000	2.826E+02	50.8	XNLD73
30-Zn	12.000	3.353E+02	50.0	XNLD73
31-Ga	1.292	2.030E-01	8.9	XNMD75A
31-Ga	1.488	3.660E-01	9.0	XNMD75A
31-Ga	1.692	6.110E-01	9.0	XNMD75A
31-Ga	1.900	9.410E-01	9.0	XNMD75A
31-Ga	2.100	1.360E+00	8.8	XNMD75A
31-Ga	2.300	1.910E+00	8.9	XNMD75A
31-Ga	2.500	2.210E+00	9.0	XNMD75A
32-Ge	0.800	4.189E-02	15.0	XNBH82
32-Ge	0.800	2.488E-02	8.2	XNSI89
32-Ge	0.900	5.992E-02	15.0	XNBH82
32-Ge	1.000	8.400E-02	15.0	XNBH82
32-Ge	1.000	6.848E-02	8.2	XNSI89
32-Ge	1.000	6.760E-02	14.9	XNSO76
32-Ge	1.100	1.113E-01	15.0	XNBH82
32-Ge	1.120	9.080E-02	15.0	XNSO76
32-Ge	1.200	1.626E-01	15.0	XNBH82
32-Ge	1.200	1.493E-01	8.2	XNSI89
32-Ge	1.300	2.210E-01	15.0	XNBH82
32-Ge	1.320	1.610E-01	14.9	XNSO76
32-Ge	1.400	2.878E-01	15.0	XNBH82
32-Ge	1.400	2.755E-01	8.2	XNSI89
32-Ge	1.500	3.654E-01	15.0	XNBH82
32-Ge	1.520	2.620E-01	14.9	XNSO76
32-Ge	1.600	4.360E-01	15.0	XNBH82
32-Ge	1.600	4.532E-01	8.2	XNSI89
32-Ge	1.700	5.725E-01	15.0	XNBH82
32-Ge	1.720	3.890E-01	14.9	XNSO76
32-Ge	1.800	7.062E-01	15.0	XNBH82
32-Ge	1.920	6.300E-01	15.1	XNSO76
32-Ge	2.000	1.033E+00	8.2	XNSI89
32-Ge	2.120	1.050E+00	15.2	XNSO76
32-Ge	2.200	1.110E+00	15.3	XNSO76
32-Ge	2.320	1.300E+00	15.4	XNSO76
32-Ge	2.400	1.867E+00	8.2	XNSI89
32-Ge	2.400	1.480E+00	14.9	XNSO76
32-Ge	2.520	1.660E+00	15.1	XNSO76
32-Ge	2.600	1.830E+00	14.8	XNSO76
32-Ge	2.720	2.170E+00	15.2	XNSO76
32-Ge	2.800	2.420E+00	14.9	XNSO76
32-Ge	2.920	2.730E+00	15.0	XNSO76
32-Ge	3.000	2.960E+00	14.9	XNSO76
32-Ge	3.120	3.280E+00	14.9	XNSO76
32-Ge	3.200	3.680E+00	14.9	XNSO76
32-Ge	3.320	4.080E+00	15.0	XNSO76
32-Ge	3.400	4.540E+00	15.0	XNSO76
32-Ge	3.520	5.170E+00	15.1	XNSO76
32-Ge	3.600	5.560E+00	14.9	XNSO76
32-Ge	3.720	5.910E+00	15.1	XNSO76
32-Ge	3.800	6.490E+00	14.9	XNSO76
32-Ge	3.920	6.920E+00	15.0	XNSO76
32-Ge	4.000	7.590E+00	15.0	XNSO76
32-Ge	4.120	8.040E+00	15.0	XNSO76
32-Ge	4.200	8.720E+00	15.0	XNSO76
32-Ge	4.320	9.170E+00	15.0	XNSO76
32-Ge	4.400	1.000E+01	15.0	XNSO76
32-Ge	5.200	3.470E+01	7.5	XNBA77
32-Ge	16.500	4.470E+02	6.9	XNBA77
32-Ge	27.000	7.750E+02	7.1	XNBA77
33-As	1.180	9.600E-02	21.0	XNOR89
33-As	1.300	1.370E-01	19.0	XNOR89
33-As	1.360	1.130E-01	8.8	XNMD75A
33-As	1.572	2.210E-01	9.0	XNMD75A
33-As	1.772	3.540E-01	9.0	XNMD75A
33-As	1.872	4.620E-01	9.1	XNMD75A
33-As	1.972	5.760E-01	9.0	XNMD75A
33-As	2.072	6.780E-01	9.0	XNMD75A
33-As	2.172	8.660E-01	9.0	XNMD75A
33-As	2.372	1.190E+00	9.2	XNMD75A
33-As	2.472	1.400E+00	9.3	XNMD75A
33-As	3.000	4.876E+00	30.2	XNLD73
33-As	4.000	1.015E+01	30.2	XNLD73
33-As	5.000	2.098E+01	48.6	XNLD73
33-As	6.000	3.912E+01	30.4	XNLD73
33-As	7.000	6.124E+01	29.6	XNLD73
33-As	8.000	8.845E+01	30.1	XNLD73
33-As	9.000	1.117E+02	29.9	XNLD73
33-As	10.000	1.559E+02	30.2	XNLD73
33-As	11.000	1.809E+02	30.1	XNLD73
33-As	12.000	2.211E+02	30.8	XNLD73
34-Se	1.000	3.030E-02	14.9	XNSO76
34-Se	1.100	4.820E-02	15.0	XNSO76
34-Se	1.180	6.100E-02	9.0	XNMD75A
34-Se	1.320	7.360E-02	14.9	XNSO76
34-Se	1.372	1.130E-01	8.8	XNMD75A
34-Se	1.500	1.252E-01	30.0	XNLD73
34-Se	1.520	1.210E-01	14.9	XNSO76
34-Se	1.580	1.920E-01	8.9	XNMD75A
34-Se	1.720	1.860E-01	15.1	XNSO76
34-Se	1.780	3.040E-01	8.9	XNMD75A
34-Se	1.920	2.950E-01	14.9	XNSO76
34-Se	1.980	4.950E-01	9.1	XNMD75A
34-Se	2.000	3.993E-01	29.9	XNLD73
34-Se	2.120	3.860E-01	15.0	XNSO76
34-Se	2.200	6.650E-01	15.0	XNSO76
34-Se	2.320	8.010E-01	15.0	XNSO76
34-Se	2.400	8.870E-01	15.0	XNSO76
34-Se	2.500	8.940E-01	30.0	XNLD73
34-Se	2.520	9.430E-01	15.0	XNSO76
34-Se	2.600	1.120E+00	15.2	XNSO76
34-Se	2.720	1.300E+00	15.4	XNSO76
34-Se	2.800	1.480E+00	14.9	XNSO76
34-Se	2.920	1.620E+00	14.8	XNSO76
34-Se	3.000	2.324E+00	30.8	XNLD73
34-Se	3.000	1.760E+00	14.8	XNSO76
34-Se	3.120	1.890E+00	14.8	XNSO76
34-Se	3.200	2.110E+00	15.2	XNSO76
34-Se	3.320	2.620E+00	14.9	XNSO76
34-Se	3.400	3.150E+00	14.9	XNSO76
34-Se	3.520	2.840E+00	15.1	XNSO76
34-Se	3.600	3.410E+00	15.0	XNSO76
34-Se	3.720	3.560E+00	14.9	XNSO76
34-Se	3.800	3.870E+00	15.0	XNSO76
34-Se	3.920	4.260E+00	15.0	XNSO76
34-Se	4.000	6.139E+00	30.1	XNLD73
34-Se	4.000	4.920E+00	15.0	XNSO76
34-Se	4.120	5.080E+00	15.0	XNSO76
34-Se	4.200	5.900E+00	15.1	XNSO76
34-Se	4.320	5.760E+00	14.9	XNSO76
34-Se	4.400	5.810E+00	15.0	XNSO76
34-Se	5.000	1.228E+01	30.1	XNLD73
34-Se	6.000	2.086E+01	31.4	XNLD73
34-Se	7.000	3.218E+01	29.6	XNLD73
34-Se	8.000	4.410E+01	29.7	XNLD73
34-Se	9.000	6.020E+01	29.7	XNLD73
34-Se	10.000	7.986E+01	29.9	XNLD73
34-Se	10.000	8.940E+01	10.0	XNPE79
34-Se	11.000	9.596E+01	29.8	XNLD73
34-Se	12.000	1.168E+02	30.1	XNLD73
34-Se	12.000	1.250E+02	10.0	XNPE79
34-Se	14.000	1.790E+02	10.0	XNPE79
34-Se	16.000	2.440E+02	10.0	XNPE79
34-Se	18.000	3.220E+02	10.0	XNPE79
34-Se	20.000	3.700E+02	10.0	XNPE79
34-Se	22.000	4.410E+02	10.0	XNPE79
34-Se	24.000	5.070E+02	10.0	XNPE79
34-Se	26.000	5.780E+02	10.0	XNPE79
34-Se	28.000	6.200E+02	10.0	XNPE79
34-Se	30.000	6.790E+02	10.0	XNPE79
37-Rb	1.172	2.000E-02	9.0	XNMD75A
37-Rb	1.372	3.800E-02	8.9	XNMD75A
37-Rb	1.570	7.300E-02	9.0	XNMD75A
37-Rb	1.770	1.210E-01	9.0	XNMD75A
37-Rb	1.970	1.830E-01	9.0	XNMD75A
37-Rb	2.170	2.610E-01	9.0	XNMD75A
37-Rb	2.370	3.840E-01	9.0	XNMD75A
38-Sr	1.000	4.500E-03	9.0	XNMD75A
38-Sr	1.200	1.400E-02	9.0	XNMD75A
38-Sr	1.400	3.600E-02	9.0	XNMD75A
38-Sr	1.600	7.700E-02	9.0	XNMD75A
38-Sr	1.800	1.080E-01	9.0	XNMD75A
38-Sr	2.000	1.640E-01	9.0	XNMD75A
38-Sr	2.100	1.900E-01	9.0	XNMD75A
38-Sr	2.200	2.490E-01	9.0	XNMD75A
38-Sr	2.400	3.190E-01	9.0	XNMD75A
38-Sr	2.500	3.600E-01	9.0	XNMD75A
39-Y	0.870	2.200E-03	9.0	XNMD75A
39-Y	0.970	3.900E-03	9.0	XNMD75A
39-Y	1.070	7.000E-03	9.0	XNMD75A
39-Y	1.170	1.100E-02	9.0	XNMD75A
39-Y	1.270	1.600E-02	9.0	XNMD75A
39-Y	1.370	2.300E-02	9.0	XNMD75A
39-Y	1.470	3.400E-02	9.0	XNMD75A
39-Y	1.500	3.180E-02	29.9	XKMK74
39-Y	1.570	4.200E-02	9.0	XNMD75A
39-Y	1.670	5.700E-02	9.0	XNMD75A
39-Y	1.770	7.400E-02	9.0	XNMD75A
39-Y	1.880	9.000E-02	9.0	XNMD75A
39-Y	1.980	1.130E-01	9.0	XNMD75A
39-Y	2.000	1.130E-01	30.1	XKMK74
39-Y	2.080	1.310E-01	9.0	XNMD75A
39-Y	2.180	1.620E-01	9.0	XNMD75A
39-Y	2.280	1.860E-01	9.0	XNMD75A
39-Y	2.380	2.240E-01	9.0	XNMD75A
39-Y	2.480	2.640E-01	9.0	XNMD75A
39-Y	2.500	2.780E-01	29.9	XKMK74
39-Y	3.000	7.820E-01	25.0	XKAN78
39-Y	3.000	5.680E-01	29.9	XKMK74
39-Y	3.500	1.000E+00	30.0	XKMK74
39-Y	4.000	2.060E+00	25.0	XKAN78
39-Y	4.000	1.710E+00	29.8	XKMK74
39-Y	4.500	2.560E+00	30.1	XKMK74
39-Y	5.000	4.190E+00	25.0	XKAN78
39-Y	6.000	7.110E+00	25.0	XKAN78
39-Y	8.000	1.640E+01	25.0	XKAN78
39-Y	10.000	3.130E+01	25.0	XKAN78
39-Y	10.000	2.630E+01	10.0	XNPE79
39-Y	12.000	5.260E+01	25.0	XKAN78
39-Y	12.000	3.600E+01	10.0	XNFB74
39-Y	12.000	4.480E+01	10.0	XNPE79
39-Y	14.000	6.400E+01	10.0	XNPE79
39-Y	16.000	9.240E+01	10.0	XNPE79
39-Y	18.000	1.210E+02	10.0	XNPE79
39-Y	20.000	1.090E+02	10.0	XNFB74
39-Y	20.000	1.490E+02	10.0	XNPE79
39-Y	22.000	1.920E+02	10.0	XNPE79
39-Y	24.000	2.280E+02	10.0	XNPE79
39-Y	26.000	2.600E+02	10.0	XNPE79
39-Y	28.000	2.770E+02	10.0	XNPE79
39-Y	30.000	3.060E+02	10.0	XNPE79
40-Zr	30.000	2.270E+02	6.1	XNHW73
40-Zr	40.000	3.168E+02	6.0	XNHW73
40-Zr	50.000	4.183E+02	5.9	XNHW73
40-Zr	60.000	5.168E+02	5.9	XNHW73
40-Zr	70.000	5.869E+02	6.0	XNHW73
40-Zr	80.000	6.220E+02	6.0	XNHW73
41-Nb	0.640	1.400E-04	12.4	XNRI81
41-Nb	0.800	1.140E-03	12.4	XNRI81
41-Nb	0.960	2.970E-03	12.4	XNRI81
41-Nb	1.040	5.400E-03	14.8	XNWL76
41-Nb	1.120	7.370E-03	12.4	XNRI81
41-Nb	1.240	1.100E-02	18.2	XNWL76
41-Nb	1.280	1.440E-02	12.4	XNRI81
41-Nb	1.440	2.570E-02	12.4	XNRI81
41-Nb	1.440	2.300E-02	8.7	XNWL76
41-Nb	1.600	4.040E-02	12.4	XNRI81
41-Nb	1.680	4.300E-02	9.3	XNWL76
41-Nb	1.800	7.020E-02	12.4	XNRI81
41-Nb	1.880	8.500E-02	8.2	XNWL76
41-Nb	2.080	1.000E-01	10.0	XNWL76
41-Nb	2.280	1.400E-01	7.1	XNWL76
41-Nb	2.480	2.000E-01	10.0	XNWL76
41-Nb	2.720	2.800E-01	7.1	XNWL76
41-Nb	30.000	1.892E+02	5.9	XNHW73
41-Nb	40.000	2.753E+02	6.0	XNHW73
41-Nb	50.000	3.658E+02	5.9	XNHW73
41-Nb	60.000	4.548E+02	5.9	XNHW73
41-Nb	70.000	5.221E+02	6.0	XNHW73
41-Nb	80.000	5.550E+02	6.1	XNHW73
42-Mo	1.000	2.000E-03	30.0	XKMK74
42-Mo	1.000	2.555E-03	15.0	XNBH82
42-Mo	1.040	3.300E-03	15.2	XNWL76
42-Mo	1.100	3.527E-03	15.0	XNBH82
42-Mo	1.200	5.707E-03	15.0	XNBH82
42-Mo	1.240	8.000E-03	11.3	XNWL76
42-Mo	1.300	9.180E-03	15.0	XNBH82
42-Mo	1.400	1.193E-02	15.0	XNBH82
42-Mo	1.440	1.700E-02	11.8	XNWL76
42-Mo	1.500	1.830E-02	30.1	XKMK74
42-Mo	1.500	1.614E-02	15.0	XNBH82
42-Mo	1.600	2.670E-02	15.0	XNBH82
42-Mo	1.680	3.100E-02	9.7	XNWL76
42-Mo	1.700	3.458E-02	15.0	XNBH82
42-Mo	1.800	3.848E-02	15.0	XNBH82
42-Mo	1.880	4.900E-02	10.2	XNWL76
42-Mo	2.000	5.810E-02	29.9	XKMK74
42-Mo	2.080	7.800E-02	10.3	XNWL76
42-Mo	2.280	1.100E-01	9.1	XNWL76
42-Mo	2.480	1.600E-01	12.5	XNWL76
42-Mo	2.500	1.430E-01	30.1	XKMK74
42-Mo	2.900	8.175E-01	4.7	XKKO68
42-Mo	3.000	3.290E-01	25.0	XKAN78
42-Mo	3.000	2.930E-01	30.0	XKMK74
42-Mo	3.500	5.350E-01	29.9	XKMK74
42-Mo	3.550	1.207E+00	5.1	XKKO68
42-Mo	3.900	1.398E+00	4.9	XKKO68
42-Mo	4.000	9.170E-01	25.0	XKAN78
42-Mo	4.000	8.860E-01	30.0	XKMK74

TABLE IId. Experimental Cross Sections for *K*-Shell X-Ray Production, Auger Electron Production, and Ionization by ^{4}He Ions

See page 84 for Explanation of Tables

TARGET	ENERGY [MeV]	SIGMA [barn]	ERROR [%]	REFERENCE
42-Mo	4.450	1.681E+00	5.0	XKKO68
42-Mo	4.500	1.380E+00	29.7	XKMK74
42-Mo	4.800	1.849E+00	5.0	XKKO68
42-Mo	5.000	1.830E+00	25.0	XKAN78
42-Mo	5.300	2.063E+00	4.8	XKKO68
42-Mo	6.000	3.360E+00	25.0	XKAN78
42-Mo	8.000	7.640E+00	25.0	XKAN78
42-Mo	10.000	1.530E+01	25.0	XKAN78
42-Mo	12.000	2.290E+01	25.0	XKAN78
45-Rh	1.040	1.200E-03	16.7	XNWL76
45-Rh	1.240	3.300E-03	12.1	XNWL76
45-Rh	1.440	6.600E-03	10.6	XNWL76
45-Rh	1.680	1.300E-02	7.7	XNWL76
45-Rh	1.880	2.200E-02	9.1	XNWL76
45-Rh	2.080	3.700E-02	10.8	XNWL76
45-Rh	2.280	4.500E-02	8.9	XNWL76
45-Rh	2.480	7.400E-02	8.1	XNWL76
45-Rh	2.720	9.900E-02	7.1	XNWL76
45-Rh	30.000	9.523E+01	5.9	XNHW73
45-Rh	40.000	1.420E+02	6.3	XNHW73
45-Rh	50.000	2.009E+02	6.0	XNHW73
45-Rh	52.000	2.098E+02	15.0	XNLA62
45-Rh	60.000	2.703E+02	6.0	XNHW73
45-Rh	70.000	2.736E+02	7.1	XNHW73
45-Rh	80.000	3.567E+02	6.1	XNHW73
46-Pd	30.000	8.002E+01	6.0	XNHW73
46-Pd	40.000	1.188E+02	6.0	XNHW73
46-Pd	50.000	1.704E+02	6.3	XNHW73
46-Pd	60.000	2.318E+02	6.0	XNHW73
46-Pd	70.000	2.809E+02	6.1	XNHW73
46-Pd	80.000	3.153E+02	6.0	XNHW73
47-Ag	0.600	4.560E-06	12.4	XNRI81
47-Ag	0.800	1.100E-04	12.4	XNRI81
47-Ag	0.800	2.100E-04	14.3	XNWL76
47-Ag	1.000	5.100E-04	12.4	XNRI81
47-Ag	1.000	5.400E-04	11.1	XNWL76
47-Ag	1.110	6.130E-04	36.0	XKBR78
47-Ag	1.200	6.789E-04	15.0	XNBH82
47-Ag	1.200	1.550E-03	12.4	XNRI81
47-Ag	1.200	1.500E-03	6.7	XNWL76
47-Ag	1.300	1.429E-03	15.0	XNBH82
47-Ag	1.310	1.550E-03	36.0	XKBR78
47-Ag	1.400	2.277E-03	15.0	XNBH82
47-Ag	1.400	3.680E-03	12.4	XNRI81
47-Ag	1.400	3.600E-03	8.3	XNWL76
47-Ag	1.410	2.210E-03	36.0	XKBR78
47-Ag	1.500	3.140E-03	36.0	XKBR78
47-Ag	1.500	2.975E-03	15.0	XNBH82
47-Ag	1.500	2.010E-03	14.9	XNMK74
47-Ag	1.600	4.163E-03	15.0	XNBH82
47-Ag	1.600	6.730E-03	12.4	XNRI81
47-Ag	1.600	6.300E-03	9.5	XNWL76
47-Ag	1.700	5.044E-03	15.0	XNBH82
47-Ag	1.710	6.250E-03	36.0	XKBR78
47-Ag	1.800	6.498E-03	15.0	XNBH82
47-Ag	1.800	1.270E-02	12.4	XNRI81
47-Ag	1.800	1.200E-02	8.3	XNWL76
47-Ag	2.000	9.170E-03	15.0	XNMK74
47-Ag	2.000	2.000E-02	12.4	XNRI81
47-Ag	2.000	2.000E-02	10.0	XNWL76
47-Ag	2.200	3.140E-02	12.4	XNRI81
47-Ag	2.200	2.600E-02	7.7	XNWL76
47-Ag	2.400	4.630E-02	12.4	XNRI81
47-Ag	2.400	4.300E-02	9.3	XNWL76
47-Ag	2.500	2.820E-02	14.9	XNMK74
47-Ag	3.000	9.960E-02	25.0	XKAN78
47-Ag	3.000	6.460E-02	15.0	XNMK74
47-Ag	3.500	1.200E-01	15.0	XNMK74
47-Ag	4.000	2.740E-01	25.0	XKAN78
47-Ag	4.000	1.660E-01	15.1	XNMK74
47-Ag	4.500	2.540E-01	15.0	XNMK74
47-Ag	5.000	5.890E-01	25.0	XKAN78
47-Ag	5.000	4.030E-01	14.9	XNMK74
47-Ag	5.200	9.100E-01	6.6	XNBA77
47-Ag	6.000	1.080E+00	25.0	XKAN78
47-Ag	8.000	2.570E+00	25.0	XKAN78
47-Ag	10.000	5.150E+00	25.0	XKAN78
47-Ag	12.000	8.300E+00	25.0	XKAN78
47-Ag	16.500	2.140E+01	7.5	XNBA77
47-Ag	27.000	5.110E+01	10.4	XNBA77
47-Ag	27.000	5.644E+01	14.7	XNLA62
47-Ag	30.000	6.798E+01	6.0	XNHW73
47-Ag	34.000	8.383E+01	14.9	XNLA62
47-Ag	39.000	1.013E+02	14.8	XNLA62
47-Ag	40.000	1.054E+02	6.3	XNHW73
47-Ag	44.000	1.112E+02	14.9	XNLA62
47-Ag	50.000	1.477E+02	6.2	XNHW73
47-Ag	52.000	1.353E+02	14.7	XNLA62
47-Ag	60.000	2.067E+02	6.0	XNHW73
47-Ag	70.000	2.523E+02	5.9	XNHW73
47-Ag	80.000	2.847E+02	6.1	XNHW73
48-Cd	1.300	6.216E-04	11.5	XNMA79
48-Cd	1.400	1.260E-03	10.0	XNMA79
48-Cd	1.420	1.260E-03	10.0	XNMA79
48-Cd	1.440	1.369E-03	12.3	XNMA79
48-Cd	1.450	1.529E-03	11.0	XNMA79
48-Cd	1.460	1.663E-03	10.1	XNMA79
48-Cd	1.480	1.688E-03	10.0	XNMA79
48-Cd	1.500	1.680E-03	10.0	XNMA79
48-Cd	1.530	1.957E-03	8.6	XNMA79
48-Cd	1.550	2.117E-03	7.9	XNMA79
48-Cd	1.600	2.898E-03	11.6	XNMA79
48-Cd	1.700	4.091E-03	11.3	XNMA79
48-Cd	1.800	4.981E-03	11.8	XNMA79
48-Cd	1.900	6.745E-03	11.8	XNMA79
48-Cd	2.000	9.408E-03	8.9	XNMA79
48-Cd	2.100	1.201E-02	10.5	XNMA79
48-Cd	2.200	1.378E-02	9.1	XNMA79
48-Cd	2.300	1.588E-02	10.6	XNMA79
48-Cd	2.400	2.453E-02	5.8	XNAV83
48-Cd	2.400	1.982E-02	10.6	XNMA79
48-Cd	2.600	3.318E-02	5.6	XNAV83
48-Cd	2.600	2.738E-02	10.7	XNMA79
48-Cd	2.800	3.704E-02	7.0	XNAV83
48-Cd	2.800	3.688E-02	11.4	XNMA79
49-In	12.000	5.400E+00	10.0	XNFB74
49-In	20.000	2.020E+01	10.0	XNFB74
50-Sn	1.000	1.700E-04	17.6	XNWL76
50-Sn	1.200	5.300E-04	11.3	XNWL76
50-Sn	1.400	1.500E-03	6.7	XNWL76
50-Sn	1.500	5.210E-04	15.0	XNMK74
50-Sn	1.600	2.800E-03	10.7	XNWL76
50-Sn	1.800	4.600E-03	8.7	XNWL76
50-Sn	2.000	3.410E-03	15.0	XNMK74
50-Sn	2.000	8.500E-03	9.4	XNWL76
50-Sn	2.200	1.300E-02	7.7	XNWL76
50-Sn	2.400	1.108E-02	6.2	XNAV83
50-Sn	2.400	1.900E-02	10.5	XNWL76
50-Sn	2.500	9.880E-03	15.0	XNMK74
50-Sn	2.600	1.589E-02	6.5	XNAV83
50-Sn	2.800	2.156E-02	6.4	XNAV83
50-Sn	3.000	4.810E-02	25.0	XKAN78
50-Sn	3.000	2.500E-02	14.8	XNMK74
50-Sn	3.500	4.970E-02	14.9	XNMK74
50-Sn	4.000	1.460E-01	25.0	XKAN78
50-Sn	4.000	1.400E-01	30.1	XNLD73
50-Sn	4.000	7.150E-02	15.0	XNMK74
50-Sn	4.500	1.120E-01	15.2	XNMK74
50-Sn	5.000	3.180E-01	25.0	XKAN78
50-Sn	5.000	3.350E-01	30.8	XNLD73
50-Sn	5.000	1.620E-01	14.8	XNMK74
50-Sn	5.200	3.700E-01	10.8	XNBA77
50-Sn	6.000	5.840E-01	25.0	XKAN78
50-Sn	6.000	6.528E-01	30.3	XNLD73
50-Sn	7.000	1.057E+00	30.1	XNLD73
50-Sn	8.000	1.460E+00	25.0	XKAN78
50-Sn	8.000	1.589E+00	30.3	XNLD73
50-Sn	9.000	2.680E+00	12.0	XNDO81B
50-Sn	9.000	2.345E+00	30.0	XNLD73
50-Sn	10.000	2.830E+00	25.0	XKAN78
50-Sn	10.000	3.092E+00	30.6	XNLD73
50-Sn	11.000	4.123E+00	29.2	XNLD73
50-Sn	12.000	4.550E+00	25.0	XKAN78
50-Sn	15.500	1.160E+01	10.0	XNDO81B
50-Sn	16.500	1.510E+01	9.9	XNBA77
50-Sn	16.500	1.370E+01	11.0	XNDO81B
50-Sn	17.500	1.590E+01	10.0	XNDO81B
50-Sn	18.000	1.720E+01	10.0	XNDO81B
50-Sn	18.500	1.800E+01	11.0	XNDO81B
50-Sn	19.010	1.980E+01	11.0	XNDO81B
50-Sn	19.500	2.090E+01	11.0	XNDO81B
50-Sn	20.500	2.420E+01	10.0	XNDO81B
50-Sn	21.500	2.650E+01	10.0	XNDO81B
50-Sn	22.500	2.900E+01	10.0	XNDO81B
50-Sn	30.000	4.097E+01	6.1	XNHW73
50-Sn	40.000	6.494E+01	6.0	XNHW73
50-Sn	50.000	9.964E+01	6.0	XNHW73
50-Sn	52.000	8.934E+01	15.4	XNLA62
50-Sn	60.000	1.400E+02	6.1	XNHW73
50-Sn	70.000	1.701E+02	6.1	XNHW73
50-Sn	80.000	1.959E+02	6.1	XNHW73
50-Sn	90.000	2.350E+02	6.5	XNDO81B
50-Sn	100.000	2.570E+02	11.0	XNDO81B
50-Sn	110.000	2.690E+02	11.0	XNDO81B
50-Sn	120.000	2.620E+02	11.0	XNDO81B
50-Sn	130.000	3.010E+02	11.0	XNDO81B
50-Sn	140.000	3.040E+02	11.0	XNDO81B
50-Sn	155.000	3.080E+02	11.0	XNDO81B
51-Sb	0.800	2.030E-05	12.4	XNRI81
51-Sb	1.000	1.380E-04	12.4	XNRI81
51-Sb	1.200	4.460E-04	12.4	XNRI81
51-Sb	1.200	4.700E-04	19.1	XNWL76
51-Sb	1.400	8.794E-05	11.7	XNMA79
51-Sb	1.400	1.180E-03	12.4	XNRI81
51-Sb	1.400	1.300E-03	15.4	XNWL76
51-Sb	1.500	3.023E-04	11.4	XNMA79
51-Sb	1.580	5.496E-04	11.9	XNMA79
51-Sb	1.600	7.007E-04	9.8	XNMA79
51-Sb	1.600	2.500E-03	12.4	XNRI81
51-Sb	1.600	2.500E-03	12.0	XNWL76
51-Sb	1.620	7.420E-04	9.3	XNMA79
51-Sb	1.640	7.901E-04	8.7	XNMA79
51-Sb	1.680	1.085E-03	9.5	XNMA79
51-Sb	1.700	1.099E-03	9.4	XNMA79
51-Sb	1.800	1.525E-03	11.3	XNMA79
51-Sb	1.800	4.850E-03	12.4	XNRI81
51-Sb	1.800	4.600E-03	10.9	XNWL76
51-Sb	1.900	2.418E-03	11.4	XNMA79
51-Sb	2.000	2.700E-03	11.5	XNMA79
51-Sb	2.000	7.590E-03	12.4	XNRI81
51-Sb	2.000	7.300E-03	9.6	XNWL76
51-Sb	2.100	3.579E-03	11.5	XNMA79
51-Sb	2.200	4.198E-03	11.5	XNMA79
51-Sb	2.200	1.170E-02	12.4	XNRI81
51-Sb	2.200	1.200E-02	8.3	XNWL76
51-Sb	2.300	5.201E-03	10.6	XNMA79
51-Sb	2.400	6.018E-03	11.4	XNMA79
51-Sb	2.400	1.700E-02	12.4	XNRI81
51-Sb	2.400	1.700E-02	11.8	XNWL76
51-Sb	2.600	9.206E-03	11.2	XNMA79
51-Sb	2.800	1.189E-02	14.5	XNMA79
51-Sb	5.000	3.208E-01	29.7	XNLD73
51-Sb	6.000	5.549E-01	29.7	XNLD73
51-Sb	7.000	9.450E-01	30.3	XNLD73
51-Sb	8.000	1.396E+00	29.8	XNLD73
51-Sb	9.000	2.020E+00	30.0	XNLD73
51-Sb	10.000	2.861E+00	30.3	XNLD73
51-Sb	11.000	3.728E+00	30.2	XNLD73
51-Sb	12.000	4.769E+00	29.1	XNLD73
52-Te	2.200	7.417E-03	6.6	XNAV83
52-Te	2.400	9.454E-03	6.9	XNAV83
52-Te	2.600	1.353E-02	7.8	XNAV83
52-Te	2.800	1.589E-02	7.2	XNAV83
52-Te	30.000	2.800E+01	5.9	XNHW73
52-Te	40.000	4.804E+01	6.0	XNHW73
52-Te	50.000	7.604E+01	6.0	XNHW73
52-Te	60.000	1.094E+02	5.6	XNHW73
52-Te	70.000	1.409E+02	6.2	XNHW73
52-Te	80.000	1.619E+02	5.9	XNHW73
53-I	1.000	6.600E-05	10.6	XNWL76
53-I	1.200	2.500E-04	8.0	XNWL76
53-I	1.400	8.100E-04	11.1	XNWL76
53-I	1.600	1.700E-03	11.8	XNWL76
53-I	1.800	2.600E-03	11.5	XNWL76
53-I	2.000	4.300E-03	9.3	XNWL76
53-I	2.200	6.500E-03	9.2	XNWL76
53-I	2.400	1.111E-02	8.7	XNAV83
53-I	2.400	1.000E-02	10.0	XNWL76
53-I	2.600	1.482E-02	7.7	XNAV83
56-Ba	1.400	2.500E-04	24.0	XNWL76
56-Ba	1.600	5.000E-04	14.0	XNWL76
56-Ba	1.800	1.000E-03	20.0	XNWL76
56-Ba	2.000	1.800E-03	16.7	XNWL76
56-Ba	2.200	3.000E-03	13.3	XNWL76
56-Ba	2.400	4.235E-03	11.1	XNAV83
56-Ba	2.400	4.900E-03	12.2	XNWL76
56-Ba	2.600	5.712E-03	8.4	XNAV83
58-Ce	1.400	1.100E-04	18.2	XNWL76
58-Ce	1.600	3.100E-04	22.6	XNWL76
58-Ce	1.800	5.900E-04	16.9	XNWL76
58-Ce	2.000	1.200E-03	8.3	XNWL76
58-Ce	2.200	2.000E-03	20.0	XNWL76
58-Ce	2.400	3.000E-03	20.0	XNWL76
58-Ce	52.000	5.001E+01	14.9	XNLA62
59-Pr	3.000	5.580E-03	25.0	XKAN78
59-Pr	4.000	2.100E-02	25.0	XKAN78
59-Pr	5.000	4.760E-02	25.0	XKAN78
59-Pr	6.000	9.150E-02	25.0	XKAN78
59-Pr	8.000	2.290E-01	25.0	XKAN78
59-Pr	10.000	4.940E-01	25.0	XKAN78
59-Pr	12.000	8.050E-01	25.0	XKAN78
60-Nd	12.000	8.000E-01	10.0	XNFB74
60-Nd	20.000	3.400E+00	10.0	XNFB74
60-Nd	30.000	7.268E+00	6.1	XNHW73
60-Nd	40.000	1.500E+01	6.1	XNHW73
60-Nd	50.000	2.604E+01	6.0	XNHW73
60-Nd	60.000	3.901E+01	5.9	XNHW73
60-Nd	70.000	5.198E+01	6.0	XNHW73
60-Nd	80.000	6.504E+01	5.9	XNHW73
62-Sm	3.000	4.270E-03	25.0	XKAN78
62-Sm	4.000	1.480E-02	25.0	XKAN78
62-Sm	5.000	3.430E-02	25.0	XKAN78
62-Sm	6.000	6.400E-02	25.0	XKAN78
62-Sm	8.000	1.580E-01	25.0	XKAN78
62-Sm	10.000	3.160E-01	25.0	XKAN78
62-Sm	12.000	5.100E-01	25.0	XKAN78

TABLE IId. Experimental Cross Sections for *K*-Shell X-Ray Production, Auger Electron Production, and Ionization by ^{4}He Ions

See page 84 for Explanation of Tables

TARGET	ENERGY [MeV]	SIGMA [barn]	ERROR [%]	REFERENCE
64-Gd	1.200	1.300E-05	15.4	XNWL76
64-Gd	1.400	4.400E-05	15.9	XNWL76
64-Gd	1.600	1.000E-04	10.0	XNWL76
64-Gd	1.800	1.900E-04	15.8	XNWL76
64-Gd	2.000	3.100E-04	12.9	XNWL76
64-Gd	2.200	4.700E-04	10.6	XNWL76
64-Gd	3.290	6.538E-03	14.3	XNZE84
64-Gd	4.360	2.017E-02	10.2	XNZE84
64-Gd	5.380	4.932E-02	12.7	XNZE84
64-Gd	6.350	7.351E-02	7.2	XNZE84
64-Gd	7.860	1.476E-01	6.3	XNZE84
64-Gd	9.410	2.653E-01	6.0	XNZE84
64-Gd	10.890	3.811E-01	6.4	XNZE84
64-Gd	12.910	6.678E-01	6.0	XNZE84
64-Gd	13.100	7.480E-01	5.0	XNSU84
64-Gd	14.970	1.009E+00	5.6	XNZE84
64-Gd	15.000	1.122E+00	5.0	XNSU84
64-Gd	18.100	1.833E+00	5.1	XNSU84
64-Gd	20.100	2.412E+00	5.0	XNSU84
64-Gd	24.800	4.553E+00	7.2	XNSU84
64-Gd	30.000	3.904E+00	6.0	XNHW73
64-Gd	40.000	8.004E+00	6.0	XNHW73
64-Gd	50.000	1.504E+01	6.2	XNHW73
64-Gd	60.000	2.298E+01	6.1	XNHW73
64-Gd	70.000	3.204E+01	6.1	XNHW73
64-Gd	80.000	3.904E+01	6.0	XNHW73
67-Ho	3.000	1.790E-03	25.0	XKAN78
67-Ho	4.000	8.580E-03	25.0	XKAN78
67-Ho	5.000	1.980E-02	25.0	XKAN78
67-Ho	6.000	3.490E-02	25.0	XKAN78
67-Ho	8.000	7.730E-02	25.0	XKAN78
67-Ho	9.000	1.710E-01	15.0	XNDO81B
67-Ho	10.000	1.700E-01	25.0	XKAN78
67-Ho	12.000	2.730E-01	25.0	XKAN78
67-Ho	15.500	8.020E-01	15.0	XNDO81B
67-Ho	16.500	9.520E-01	15.0	XNDO81B
67-Ho	17.500	1.150E+00	15.0	XNDO81B
67-Ho	18.500	1.300E+00	14.0	XNDO81B
67-Ho	19.500	1.530E+00	15.0	XNDO81B
67-Ho	20.500	1.790E+00	15.0	XNDO81B
67-Ho	21.500	1.980E+00	15.0	XNDO81B
67-Ho	22.500	2.240E+00	15.0	XNDO81B
67-Ho	40.000	8.580E+00	10.0	XNDE77
67-Ho	50.000	1.550E+01	9.7	XNDE77
67-Ho	60.000	1.770E+01	10.2	XNDE77
67-Ho	80.000	2.940E+01	9.9	XNDE77
67-Ho	110.000	4.250E+01	9.9	XNDE77
69-Tm	9.000	1.310E-01	14.0	XNDO81B
69-Tm	12.000	3.000E-01	10.0	XNFB74
69-Tm	15.500	5.830E-01	15.0	XNDO81B
69-Tm	16.500	6.890E-01	15.0	XNDO81B
69-Tm	17.500	7.750E-01	15.0	XNDO81B
69-Tm	18.500	9.410E-01	15.0	XNDO81B
69-Tm	19.500	1.120E+00	14.0	XNDO81B
69-Tm	20.000	1.090E+00	10.0	XNFB74
69-Tm	20.500	1.290E+00	15.0	XNDO81B
69-Tm	21.500	1.540E+00	15.0	XNDO81B
69-Tm	22.500	1.680E+00	15.0	XNDO81B
69-Tm	30.000	2.105E+00	5.9	XNHW73
69-Tm	40.000	6.180E+00	10.0	XNDE77
69-Tm	40.000	4.200E+00	6.1	XNHW73
69-Tm	50.000	9.940E+00	10.0	XNDE77
69-Tm	50.000	7.897E+00	6.0	XNHW73
69-Tm	60.000	1.430E+01	9.8	XNDE77
69-Tm	60.000	1.299E+01	5.8	XNHW73
69-Tm	70.000	1.801E+01	5.8	XNHW73
69-Tm	80.000	2.330E+01	9.9	XNDE77
69-Tm	80.000	2.304E+01	6.2	XNHW73
69-Tm	110.000	3.530E+01	9.9	XNDE77
71-Lu	40.000	5.050E+00	9.9	XNDE77
71-Lu	50.000	8.710E+00	10.0	XNDE77
71-Lu	60.000	1.240E+01	9.7	XNDE77
71-Lu	80.000	2.000E+01	10.0	XNDE77
71-Lu	110.000	3.170E+01	10.1	XNDE77
72-Hf	3.000	1.140E-03	25.0	XKAN78
72-Hf	4.000	4.290E-03	25.0	XKAN78
72-Hf	5.000	1.050E-02	25.0	XKAN78
72-Hf	6.000	1.910E-02	25.0	XKAN78
72-Hf	8.000	4.670E-02	25.0	XKAN78
72-Hf	10.000	9.540E-02	25.0	XKAN78
72-Hf	12.000	1.620E-01	25.0	XKAN78
73-Ta	3.000	6.690E-04	25.0	XKAN78
73-Ta	4.000	3.250E-03	25.0	XKAN78
73-Ta	5.000	8.510E-03	25.0	XKAN78
73-Ta	6.000	1.630E-02	25.0	XKAN78
73-Ta	8.000	4.020E-02	25.0	XKAN78
73-Ta	10.000	8.220E-02	25.0	XKAN78
73-Ta	12.000	1.430E-01	25.0	XKAN78
73-Ta	30.000	1.396E+00	6.2	XNHW73
73-Ta	40.000	2.870E+00	10.1	XNDE77
73-Ta	40.000	2.801E+00	6.1	XNHW73
73-Ta	50.000	5.296E+00	6.0	XNHW73
73-Ta	52.000	5.602E+00	15.0	XNLA62
73-Ta	60.000	8.760E+00	10.0	XNDE77
73-Ta	60.000	8.700E+00	6.0	XNHW73
73-Ta	70.000	1.205E+01	6.3	XNHW73
73-Ta	80.000	1.510E+01	9.9	XNDE77
73-Ta	80.000	1.597E+01	6.0	XNHW73
73-Ta	110.000	2.450E+01	9.8	XNDE77
74-W	2.200	9.149E-05	25.0	XKFR85
74-W	2.400	1.608E-04	25.0	XKFR85
74-W	2.600	2.536E-04	25.0	XKFR85
74-W	2.880	4.249E-04	25.0	XKFR85
74-W	3.200	6.804E-04	25.0	XKFR85
74-W	3.400	8.776E-04	25.0	XKFR85
74-W	3.440	1.378E-03	21.5	XNZE84
74-W	3.600	1.072E-03	25.0	XKFR85
74-W	3.800	1.359E-03	25.0	XKFR85
74-W	4.000	1.665E-03	25.0	XKFR85
74-W	4.530	4.106E-03	14.7	XNZE84
74-W	5.380	8.996E-03	14.9	XNZE84
74-W	6.460	2.010E-02	8.1	XNZE84
74-W	7.970	4.019E-02	7.4	XNZE84
74-W	9.500	7.130E-02	6.8	XNZE84
74-W	10.980	1.129E-01	6.8	XNZE84
74-W	12.990	1.818E-01	6.8	XNZE84
74-W	40.000	3.820E+00	9.9	XNDE77
74-W	50.000	7.420E+00	10.0	XNDE77
74-W	52.000	5.397E+00	15.1	XNLA62
74-W	60.000	9.100E+00	10.0	XNDE77
74-W	80.000	1.490E+01	10.1	XNDE77
74-W	110.000	2.510E+01	10.0	XNDE77
75-Re	4.000	2.970E-03	25.0	XKAN78
75-Re	5.000	7.100E-03	25.0	XKAN78
75-Re	6.000	1.340E-02	25.0	XKAN78
75-Re	8.000	3.360E-02	25.0	XKAN78
75-Re	10.000	6.520E-02	25.0	XKAN78
75-Re	12.000	1.050E-01	25.0	XKAN78
78-Pt	3.000	2.600E-04	25.0	XKAN78
78-Pt	4.000	1.350E-03	25.0	XKAN78
78-Pt	5.000	3.950E-03	25.0	XKAN78
78-Pt	6.000	8.670E-03	25.0	XKAN78
78-Pt	8.000	2.500E-02	25.0	XKAN78
78-Pt	10.000	4.720E-02	25.0	XKAN78
78-Pt	12.000	8.190E-02	25.0	XKAN78
78-Pt	40.000	1.970E+00	10.2	XNDE77
78-Pt	52.000	3.101E+00	14.9	XNLA62
78-Pt	60.000	5.450E+00	9.9	XNDE77
78-Pt	80.000	9.240E+00	10.0	XNDE77
78-Pt	110.000	1.550E+01	9.7	XNDE77
79-Au	2.600	7.114E-05	25.0	XKFR85
79-Au	2.840	1.282E-04	25.0	XKFR85
79-Au	3.040	1.957E-04	25.0	XKFR85
79-Au	3.280	3.085E-04	25.0	XKFR85
79-Au	3.600	5.620E-04	25.0	XKFR85
79-Au	3.800	7.625E-04	25.0	XKFR85
79-Au	4.000	1.350E-03	25.0	XKAN78
79-Au	4.000	1.099E-03	25.0	XKFR85
79-Au	5.000	3.470E-03	25.0	XKAN78
79-Au	6.000	7.330E-03	25.0	XKAN78
79-Au	8.000	1.930E-02	25.0	XKAN78
79-Au	9.000	3.730E-02	12.0	XNDO81B
79-Au	10.000	4.050E-02	25.0	XKAN78
79-Au	12.000	6.750E-02	25.0	XKAN78
79-Au	16.500	2.350E-01	11.0	XNDO81B
79-Au	17.500	2.700E-01	13.0	XNDO81B
79-Au	18.000	2.920E-01	11.0	XNDO81B
79-Au	18.500	2.930E-01	12.0	XNDO81B
79-Au	19.500	3.740E-01	17.0	XNDO81B
79-Au	20.500	4.040E-01	13.0	XNDO81B
79-Au	21.500	5.090E-01	13.0	XNDO81B
79-Au	22.500	5.400E-01	13.0	XNDO81B
79-Au	30.000	8.801E-01	6.0	XNHW73
79-Au	40.000	2.040E+00	9.8	XNDE77
79-Au	40.000	1.697E+00	10.8	XNHW73
79-Au	50.000	3.580E+00	10.1	XNDE77
79-Au	50.000	2.998E+00	6.1	XNHW73
79-Au	52.000	2.796E+00	15.2	XNLA62
79-Au	60.000	5.240E+00	9.9	XNDE77
79-Au	60.000	4.897E+00	5.9	XNHW73
79-Au	70.000	6.999E+00	6.1	XNHW73
79-Au	80.000	8.500E+00	10.0	XNDE77
79-Au	80.000	9.303E+00	6.0	XNHW73
79-Au	90.000	1.050E+01	14.0	XNDO81B
79-Au	100.000	1.190E+01	14.0	XNDO81B
79-Au	110.000	1.440E+01	9.7	XNDE77
79-Au	110.000	1.330E+01	15.0	XNDO81B
79-Au	120.000	1.400E+01	15.0	XNDO81B
79-Au	130.000	1.830E+01	14.0	XNDO81B
79-Au	140.000	1.930E+01	15.0	XNDO81B
79-Au	155.000	2.060E+01	14.0	XNDO81B
82-Pb	4.000	8.810E-04	25.0	XKAN78
82-Pb	5.000	2.320E-03	25.0	XKAN78
82-Pb	6.000	4.840E-03	25.0	XKAN78
82-Pb	8.000	1.360E-02	25.0	XKAN78
82-Pb	9.000	2.500E-02	14.0	XNDO81B
82-Pb	10.000	3.000E-02	25.0	XKAN78
82-Pb	12.000	5.030E-02	25.0	XKAN78
82-Pb	12.000	5.700E-02	10.0	XNFB74
82-Pb	17.500	1.870E-01	13.0	XNDO81B
82-Pb	18.500	2.060E-01	12.0	XNDO81B
82-Pb	19.500	2.440E-01	12.0	XNDO81B
82-Pb	20.000	2.250E-01	10.0	XNFB74
82-Pb	20.500	2.880E-01	12.0	XNDO81B
82-Pb	21.500	3.170E-01	12.0	XNDO81B
82-Pb	22.500	3.490E-01	12.0	XNDO81B
82-Pb	40.000	1.420E+00	9.9	XNDE77
82-Pb	50.000	2.360E+00	12.0	XNDO81B
82-Pb	52.000	1.897E+00	14.8	XNLA62
82-Pb	60.000	3.260E+00	10.1	XNDE77
82-Pb	60.000	3.260E+00	12.0	XNDO81B
82-Pb	70.000	4.560E+00	12.0	XNDO81B
82-Pb	80.000	5.470E+00	10.1	XNDE77
82-Pb	80.000	5.910E+00	12.0	XNDO81B
82-Pb	90.000	7.590E+00	15.0	XNDO81B
82-Pb	100.000	9.780E+00	15.0	XNDO81B
82-Pb	110.000	9.680E+00	10.0	XNDE77
82-Pb	110.000	1.110E+01	15.0	XNDO81B
82-Pb	120.000	1.190E+01	15.0	XNDO81B
82-Pb	130.000	1.460E+01	15.0	XNDO81B
82-Pb	140.000	1.600E+01	15.0	XNDO81B
82-Pb	155.000	1.880E+01	15.0	XNDO81B
83-Bi	4.000	7.800E-04	25.0	XKAN78
83-Bi	5.000	2.200E-03	25.0	XKAN78
83-Bi	6.000	4.700E-03	25.0	XKAN78
83-Bi	8.000	1.400E-02	25.0	XKAN78
83-Bi	10.000	2.900E-02	25.0	XKAN78
83-Bi	12.000	4.800E-02	25.0	XKAN78
83-Bi	17.500	1.300E-01	12.0	XNDO81B
83-Bi	18.500	1.740E-01	12.0	XNDO81B
83-Bi	19.500	2.590E-01	13.0	XNDO81B
83-Bi	20.500	2.950E-01	12.0	XNDO81B
83-Bi	21.500	2.970E-01	12.0	XNDO81B
83-Bi	22.500	3.210E-01	12.0	XNDO81B
83-Bi	40.000	1.320E+00	9.8	XNDE77
83-Bi	60.000	3.050E+00	9.8	XNDE77
83-Bi	80.000	4.870E+00	10.1	XNDE77
83-Bi	110.000	8.730E+00	10.0	XNDE77
90-Th	4.000	3.300E-04	25.0	XKAN78
90-Th	4.390	4.865E-04	8.6	XNZE84
90-Th	5.000	9.100E-04	25.0	XKAN78
90-Th	5.400	1.463E-03	6.7	XNZE84
90-Th	5.990	2.301E-03	11.4	XNZE84
90-Th	6.000	2.100E-03	25.0	XKAN78
90-Th	6.140	2.886E-03	8.1	XNZE84
90-Th	7.450	5.489E-03	6.9	XNZE84
90-Th	7.510	6.065E-03	10.5	XNZE84
90-Th	8.000	6.900E-03	25.0	XKAN78
90-Th	8.650	1.082E-02	7.2	XNZE84
90-Th	8.870	1.121E-02	9.6	XNZE84
90-Th	9.740	1.823E-02	10.7	XNZE84
90-Th	10.000	1.600E-02	25.0	XKAN78
90-Th	11.480	2.847E-02	11.0	XNZE84
90-Th	11.550	3.071E-02	8.9	XNZE84
90-Th	11.910	3.032E-02	9.6	XNZE84
90-Th	12.000	2.600E-02	25.0	XKAN78
90-Th	13.460	4.524E-02	11.6	XNZE84
90-Th	15.140	5.948E-02	10.7	XNZE84
92-U	4.000	2.240E-04	25.0	XKAN78
92-U	5.000	6.930E-04	25.0	XKAN78
92-U	6.000	1.760E-03	25.0	XKAN78
92-U	8.000	5.950E-03	25.0	XKAN78
92-U	10.000	1.170E-02	25.0	XKAN78
92-U	12.000	2.240E-02	25.0	XKAN78

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